Signal/background separation · round 1

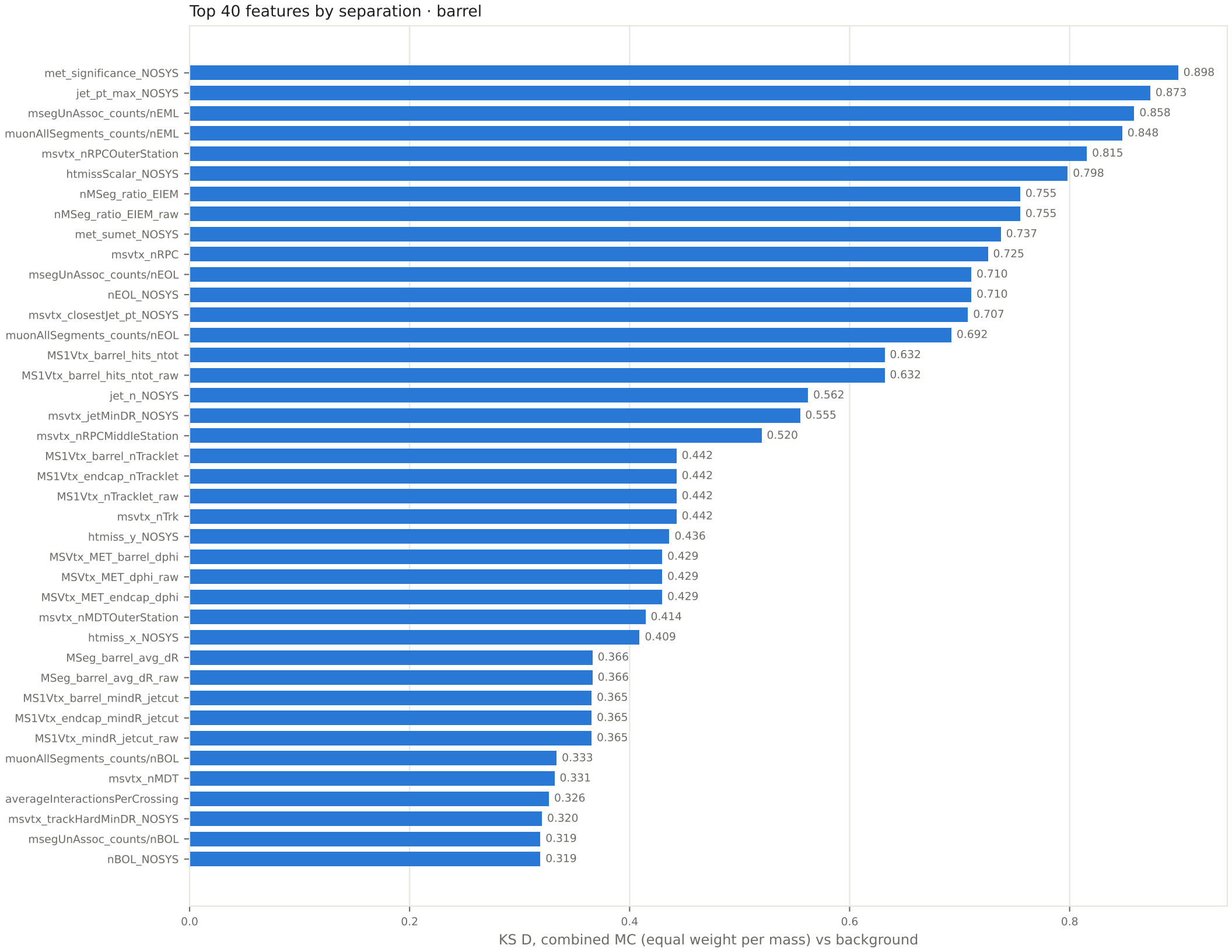
```
date        2026-10-25 04:19
data        C:\Users\natha\OneDrive\Desktop\0_mL_ATLAS\0_Sig_BG_09_21_KJ\kj_092126\run3_csv_for_nathan
background  DataHTMissLow_train_data24_cleaned.csv  rows 36,059
signal      HSS_mH125_mS15_ct580_cleaned.csv  rows 570
signal      HSS_mH125_mS55_ct5320_cleaned.csv  rows 423
signal      HSS_mH200_mS50_ct1255_cleaned.csv  rows 1,520
signal      HSS_mH400_mS100_ct1610_cleaned.csv  rows 3,792
signal      HSS_mH600_mS150_ct3310_cleaned.csv  rows 4,490
signal      HSS_mH1000_mS275_ct2399_cleaned.csv  rows 4,188

columns common to all 7 files (ALL7)   230
features scored                        191
hard-excluded                          13
user-excluded (exclude_features.txt)   0
non-numeric / constant                 6 / 20
not common to all 7 (never scored)     112

region column   ('msvtx_inBarrel_NOSYS', 'flag', 'msvtx_inEndcaps_NOSYS')
events per region (after region selection):
  barrel  bkg   1,089  mH125_mS15 159  mH125_mS55 142  mH200_mS50 449  mH400_mS100 1,144  mH600_mS150 1,608  mH1000_mS275 1,485
  endcap  bkg  34,970  mH125_mS15 411  mH125_mS55 281  mH200_mS50 1,071  mH400_mS100 2,648  mH600_mS150 2,882  mH1000_mS275 2,703
  all     bkg  36,059  mH125_mS15 570  mH125_mS55 423  mH200_mS50 1,520  mH400_mS100 3,792  mH600_mS150 4,490  mH1000_mS275 4,188

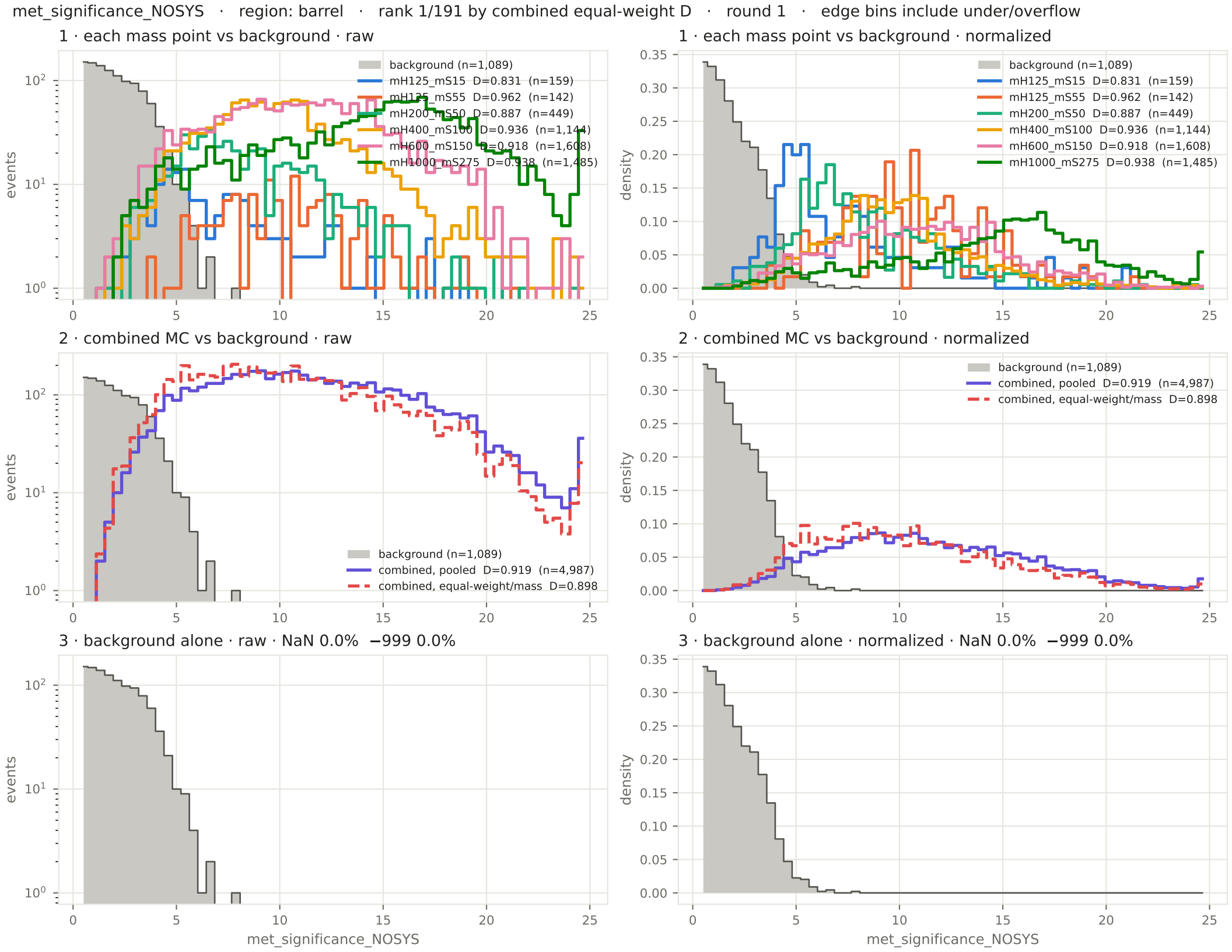
metric   two-sample KS D = sup|F_s - F_b|, unweighted events; NaN and -999 removed per feature.
         'equal' weights each mass point 1/N_m so small points count as much as large ones.
         p-values not reported (N ~ 1e3-1e4 makes them ~0 for any real difference).

this summary: per region -> ranking, top-40 table, top 15 feature pages.
all features: r1_<region>_all_features.pdf
```

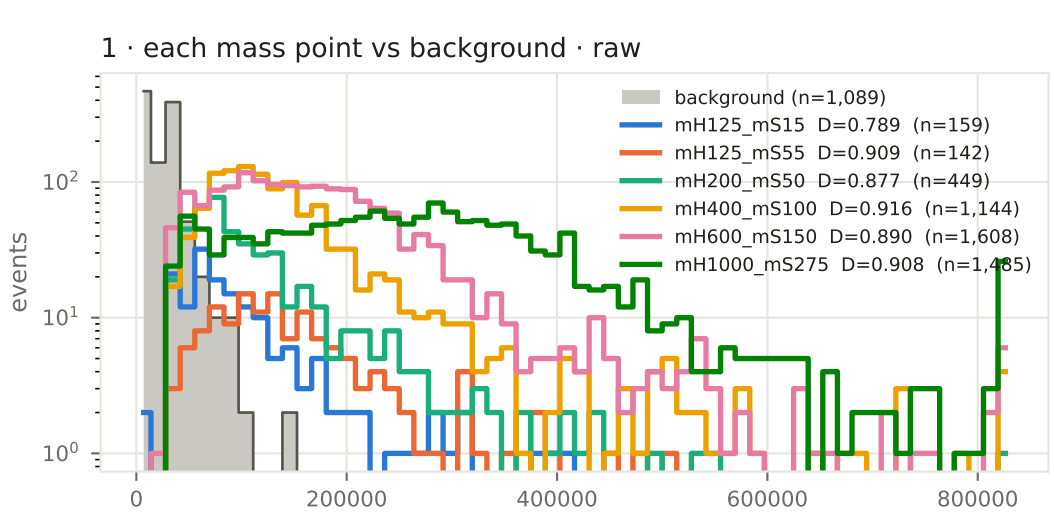

Top 40 features by separation · barrel
met_significance_NOSYS 0.898
jet_pt_max_NOSYS 0.873
msegUnAssoc_counts/nEML 0.858
muonAllSegments_counts/nEML 0.848
msvtx_nRPCOuterStation 0.815
htmissScalar_NOSYS 0.798
nMSeg_ratio_EIEM 0.755
nMSeg_ratio_EIEM_raw 0.755
met_sumet_NOSYS 0.737
msvtx_nRPC 0.725
msegUnAssoc_counts/nEOL 0.710
nEOL_NOSYS 0.710
msvtx_closestJet_pt_NOSYS 0.707
muonAllSegments_counts/nEOL 0.692
MS1Vtx_barrel_hits_ntot 0.632
MS1Vtx_barrel_hits_ntot_raw 0.632
jet_n_NOSYS 0.562
msvtx_jetMinDR_NOSYS 0.555
msvtx_nRPCMiddleStation 0.520
MS1Vtx_barrel_nTracklet 0.442
MS1Vtx_endcap_nTracklet 0.442
MS1Vtx_nTracklet_raw 0.442
msvtx_nTrk 0.442
htmiss_y_NOSYS 0.436
MSVtx_MET_barrel_dphi 0.429
MSVtx_MET_dphi_raw 0.429
MSVtx_MET_endcap_dphi 0.429
msvtx_nMDTOuterStation 0.414
htmiss_x_NOSYS 0.409
MSeg_barrel_avg_dR 0.366
MSeg_barrel_avg_dR_raw 0.366
MS1Vtx_barrel_mindR_jetcut 0.365
MS1Vtx_endcap_mindR_jetcut 0.365
MS1Vtx_mindR_jetcut_raw 0.365
muonAllSegments_counts/nBOL 0.333
msvtx_nMDT 0.331
averageInteractionsPerCrossing 0.326
msvtx_trackHardMinDR_NOSYS 0.320
msegUnAssoc_counts/nBOL 0.319
nBOL_NOSYS 0.319
0.0
0.2
0.4
0.6
0.8
KS D, combined MC (equal weight per mass) vs background

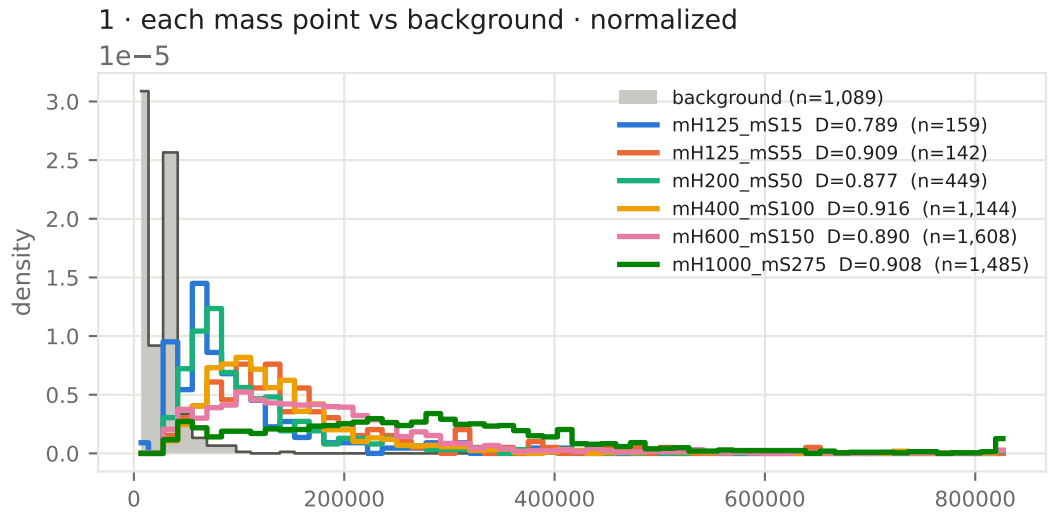
Round 1 · barrel · top 40 by combined equal-weight KS D

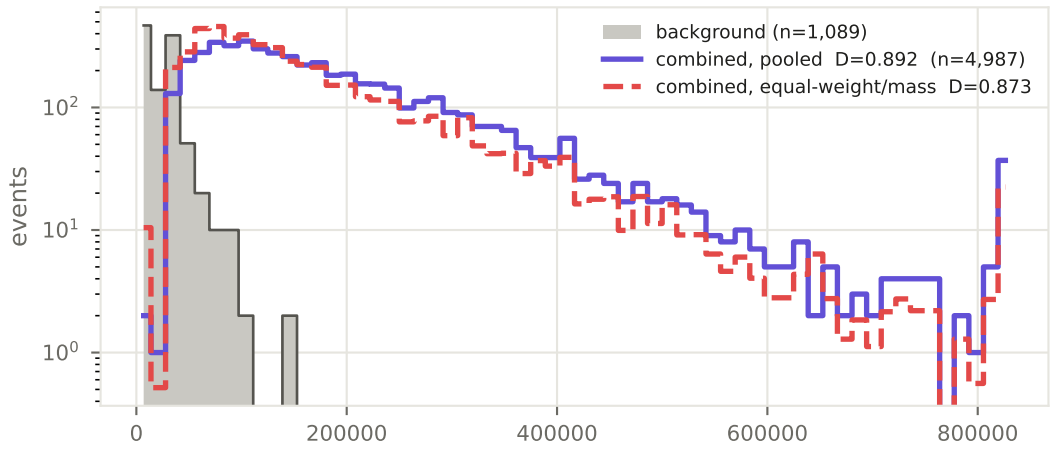
#	feature	D_equal	D_pooled	D_min	D_max	bkgNaN	bkg-999
1	met_significance_NOSYS	0.898	0.919	0.831	0.962	0.0%	0.0%
2	jet_pt_max_NOSYS	0.873	0.892	0.789	0.916	0.0%	0.0%
3	msegUnAssoc_counts/nEML	0.858	0.843	0.821	0.894	0.0%	0.0%
4	muonAllSegments_counts/nEML	0.848	0.828	0.799	0.901	0.0%	0.0%
5	msvtx_nRPCOuterStation	0.815	0.843	0.740	0.881	0.0%	0.0%
6	htmissScalar_NOSYS	0.798	0.818	0.798	0.872	0.0%	0.0%
7	nMSeg_ratio_EIEM	0.755	0.746	0.727	0.805	0.0%	0.0%
8	nMSeg_ratio_EIEM_raw	0.755	0.746	0.727	0.805	0.0%	0.0%
9	met_sumet_NOSYS	0.737	0.783	0.588	0.852	0.0%	0.0%
10	msvtx_nRPC	0.725	0.787	0.647	0.850	0.0%	0.0%
11	msegUnAssoc_counts/nEOL	0.710	0.651	0.580	0.844	0.0%	0.0%
12	nEOL_NOSYS	0.710	0.651	0.580	0.844	0.0%	0.0%
13	msvtx_closestJet_pt_NOSYS	0.707	0.728	0.639	0.740	0.0%	0.0%
14	muonAllSegments_counts/nEOL	0.692	0.632	0.563	0.821	0.0%	0.0%
15	MS1Vtx_barrel_hits_ntot	0.632	0.727	0.405	0.834	0.0%	0.0%
16	MS1Vtx_barrel_hits_ntot_raw	0.632	0.727	0.405	0.834	0.0%	0.0%
17	jet_n_NOSYS	0.562	0.611	0.458	0.691	0.0%	0.0%
18	msvtx_jetMinDR_NOSYS	0.555	0.583	0.510	0.622	0.0%	0.0%
19	msvtx_nRPCMiddleStation	0.520	0.611	0.314	0.691	0.0%	0.0%
20	MS1Vtx_barrel_nTracklet	0.442	0.458	0.418	0.475	0.0%	0.0%
21	MS1Vtx_endcap_nTracklet	0.442	0.458	0.418	0.475	0.0%	0.0%
22	MS1Vtx_nTracklet_raw	0.442	0.458	0.418	0.475	0.0%	0.0%
23	msvtx_nTrk	0.442	0.458	0.418	0.475	0.0%	0.0%
24	htmiss_y_NOSYS	0.436	0.437	0.374	0.470	0.0%	0.0%
25	MSVtx_MET_barrel_dphi	0.429	0.407	0.332	0.620	0.0%	0.0%
26	MSVtx_MET_dphi_raw	0.429	0.407	0.332	0.620	0.0%	0.0%
27	MSVtx_MET_endcap_dphi	0.429	0.407	0.332	0.620	0.0%	0.0%
28	msvtx_nMDTOuterStation	0.414	0.565	0.080	0.710	0.0%	0.0%
29	htmiss_x_NOSYS	0.409	0.421	0.363	0.458	0.0%	0.0%
30	MSeg_barrel_avg_dR	0.366	0.497	0.101	0.645	0.0%	0.0%
31	MSeg_barrel_avg_dR_raw	0.366	0.497	0.101	0.645	0.0%	0.0%
32	MS1Vtx_barrel_mindR_jetcut	0.365	0.391	0.307	0.432	0.0%	0.0%
33	MS1Vtx_endcap_mindR_jetcut	0.365	0.391	0.307	0.432	0.0%	0.0%
34	MS1Vtx_mindR_jetcut_raw	0.365	0.391	0.307	0.432	0.0%	0.0%
35	muonAllSegments_counts/nBOL	0.333	0.425	0.151	0.517	0.0%	0.0%
36	msvtx_nMDT	0.331	0.500	0.161	0.687	0.0%	0.0%
37	averageInteractionsPerCrossing	0.326	0.324	0.307	0.367	0.0%	0.0%
38	msvtx_trackHardMinDR_NOSYS	0.320	0.346	0.273	0.393	0.0%	0.0%
39	msegUnAssoc_counts/nBOL	0.319	0.408	0.142	0.508	0.0%	0.0%
40	nBOL_NOSYS	0.319	0.408	0.142	0.508	0.0%	0.0%

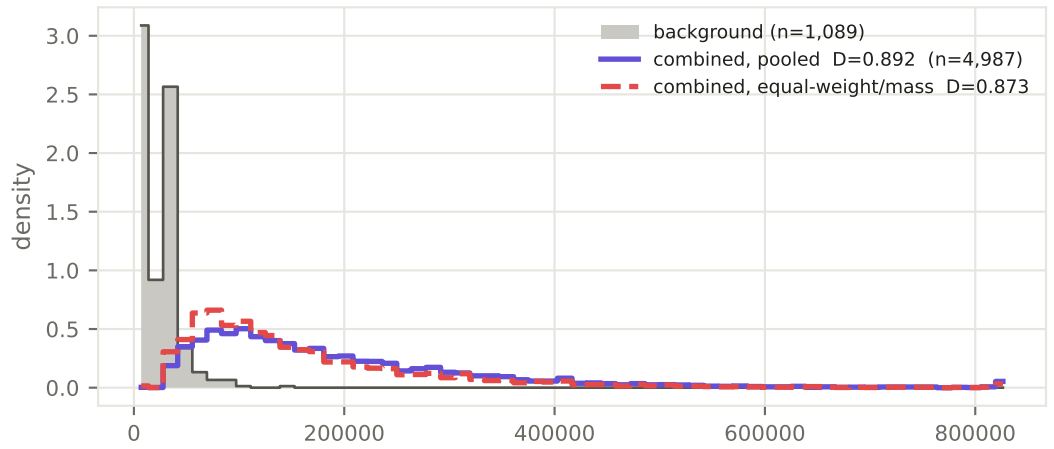

met_significance_NOSYS · region: barrel · rank 1/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
events
density
met_significance_NOSYS
background (n=1,089)
mH125_mS15 D=0.831 (n=159)
mH125_mS55 D=0.962 (n=142)
mH200_mS50 D=0.887 (n=449)
mH400_mS100 D=0.936 (n=1,144)
mH600_mS150 D=0.918 (n=1,608)
mH1000_mS275 D=0.938 (n=1,485)
combined, pooled D=0.919 (n=4,987)
combined, equal-weight/mass D=0.898

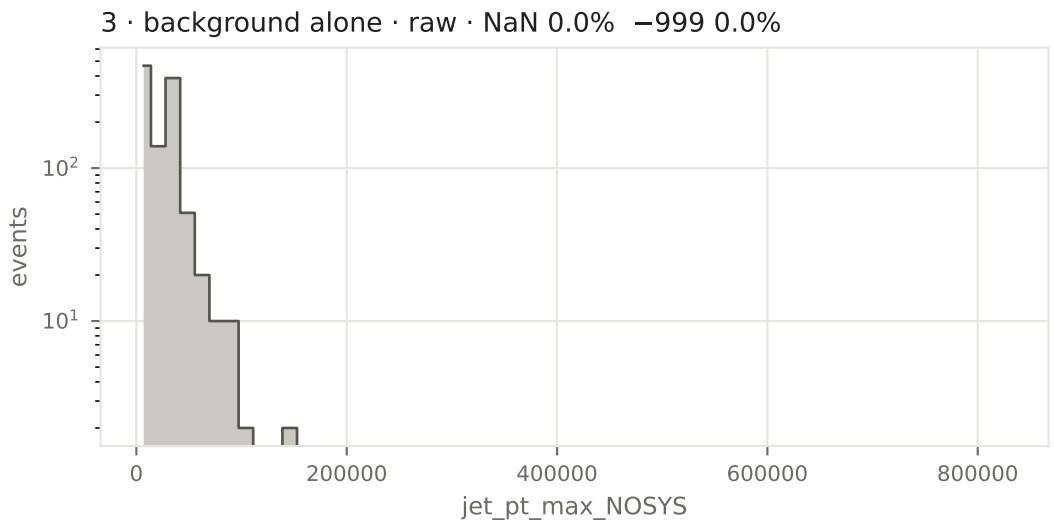
jet_pt_max_NOSYS · region: barrel · rank 2/191 by combined equal-weight D · round 1 · edge bins include under/overflow

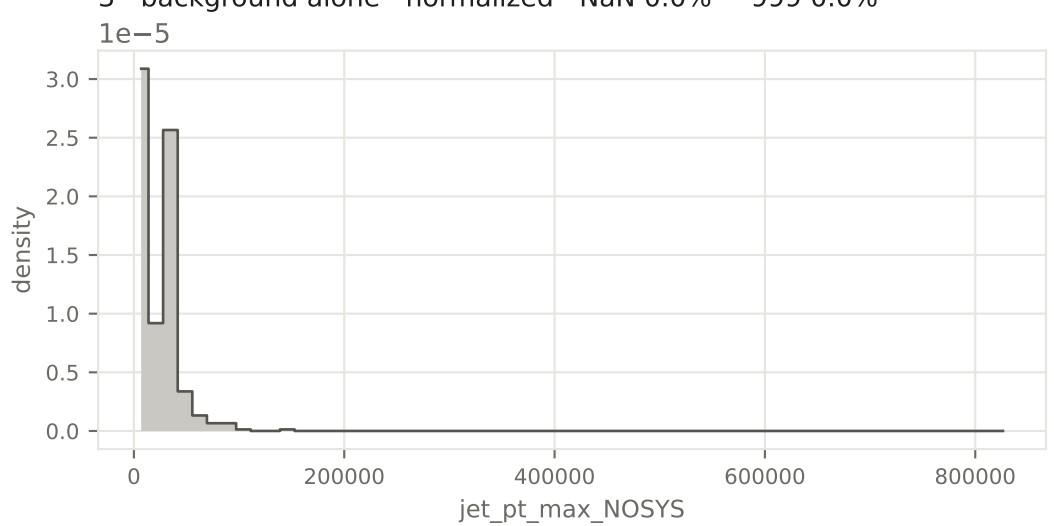


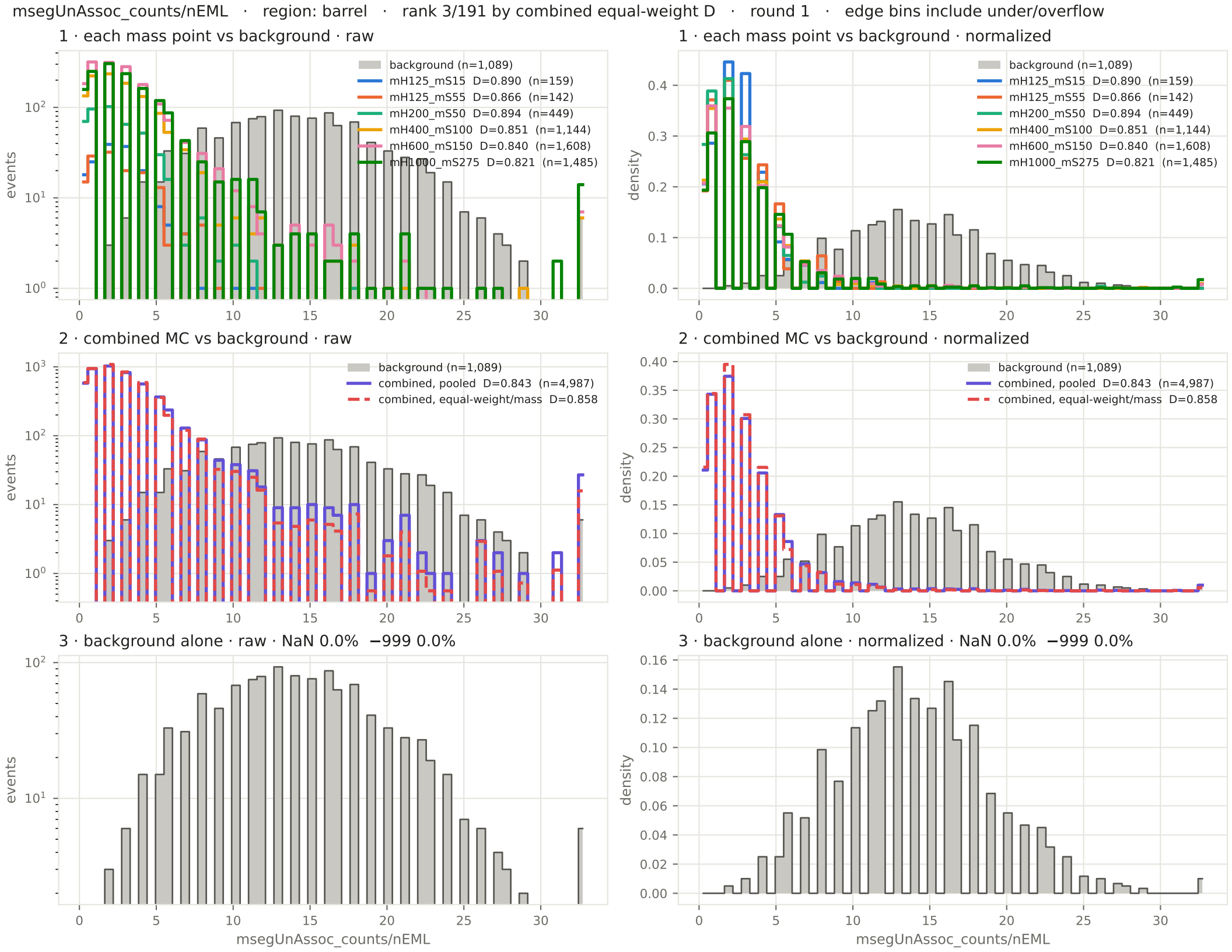


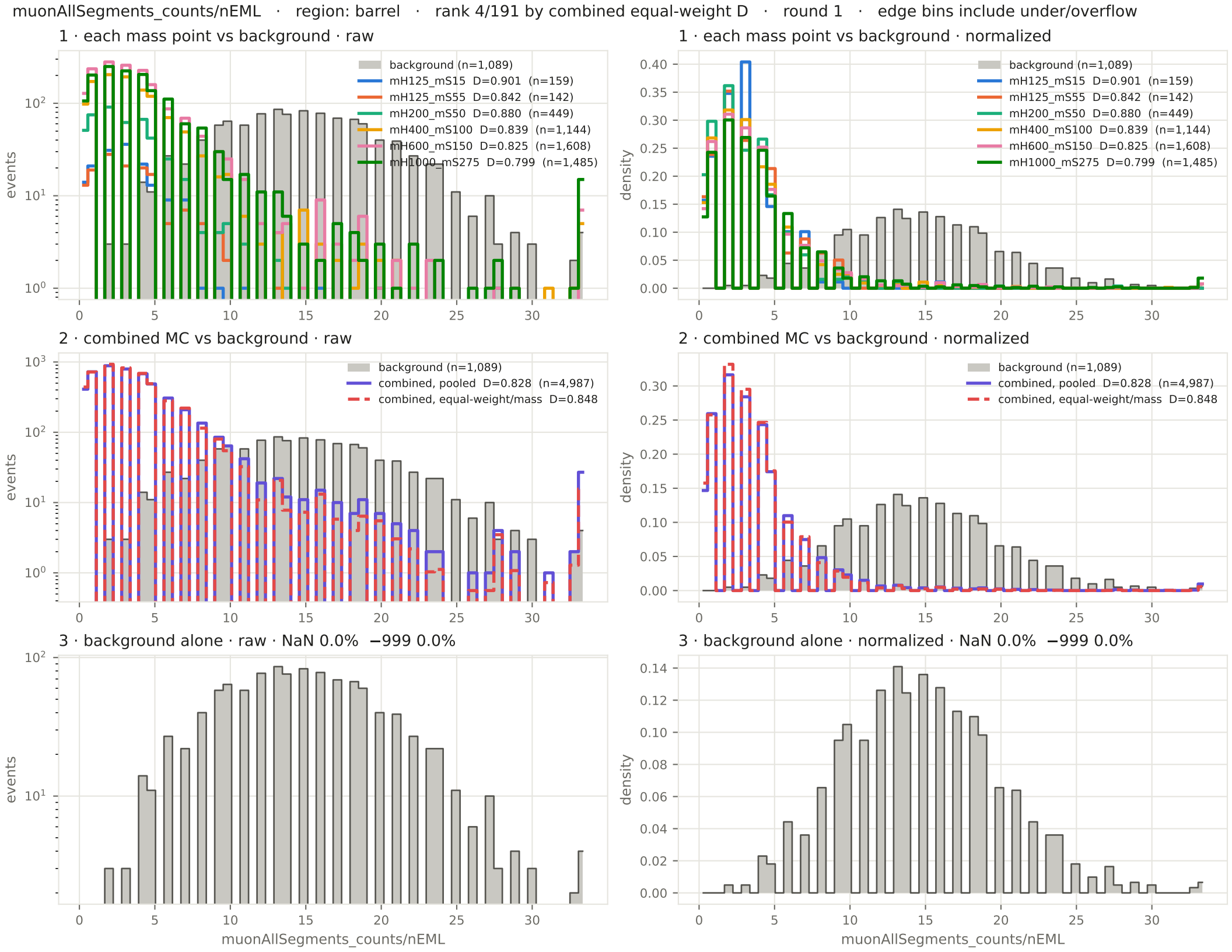




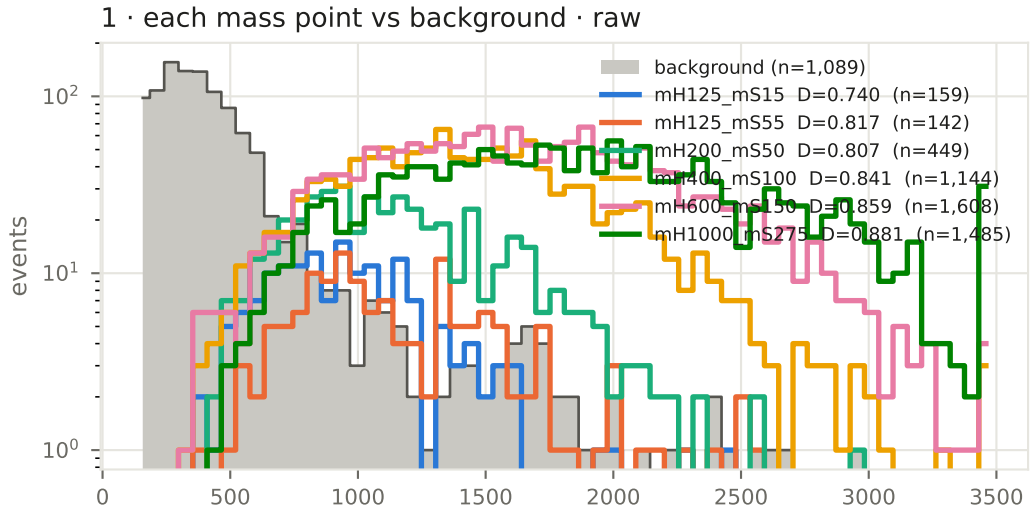


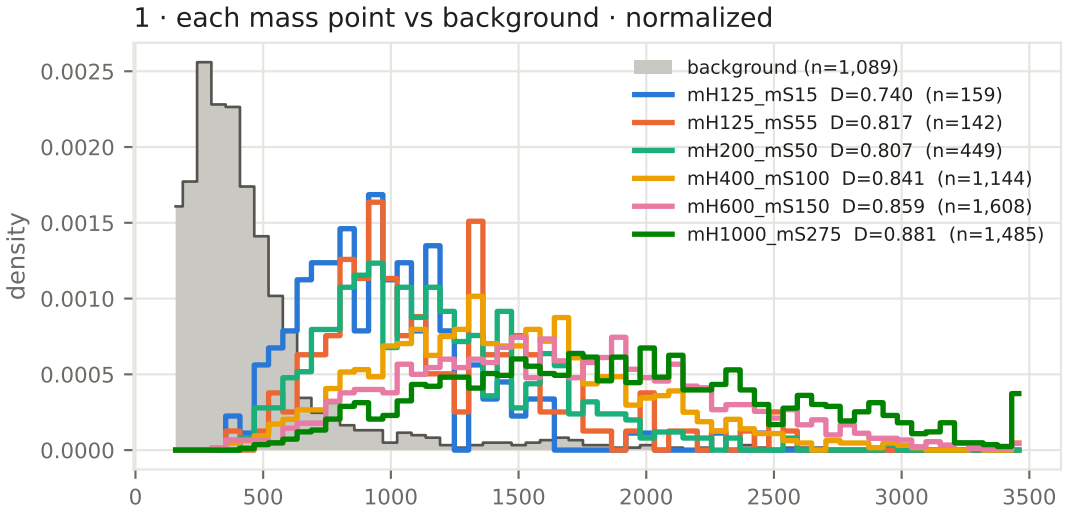


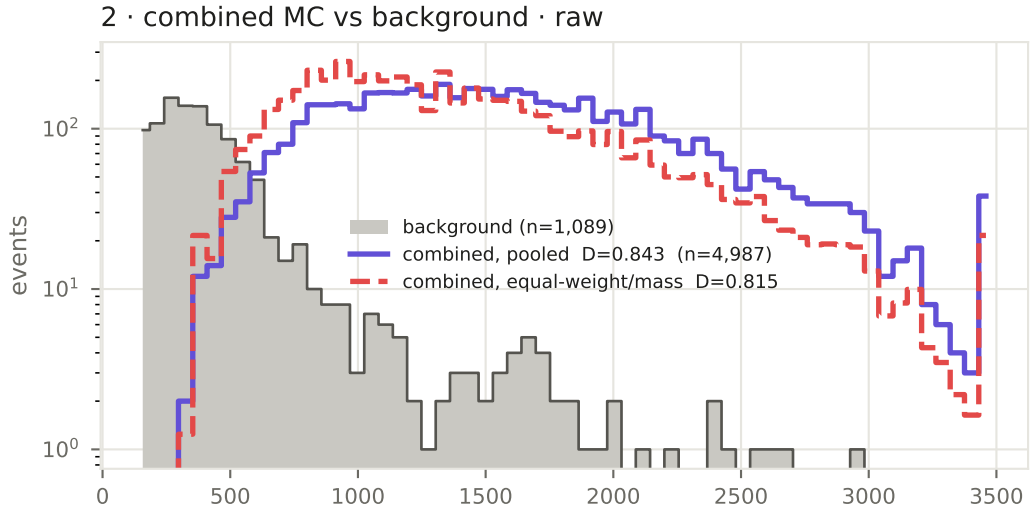

msegUnAssoc_counts/nEML · region: barrel · rank 3/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
background (n=1,089)
mH125_mS15 D=0.890 (n=159)
mH125_mS55 D=0.866 (n=142)
mH200_mS50 D=0.894 (n=449)
mH400_mS100 D=0.851 (n=1,144)
mH600_mS150 D=0.840 (n=1,608)
mH1000_mS275 D=0.821 (n=1,485)
combined, pooled D=0.843 (n=4,987)
combined, equal-weight/mass D=0.858
events
density
msegUnAssoc_counts/nEML

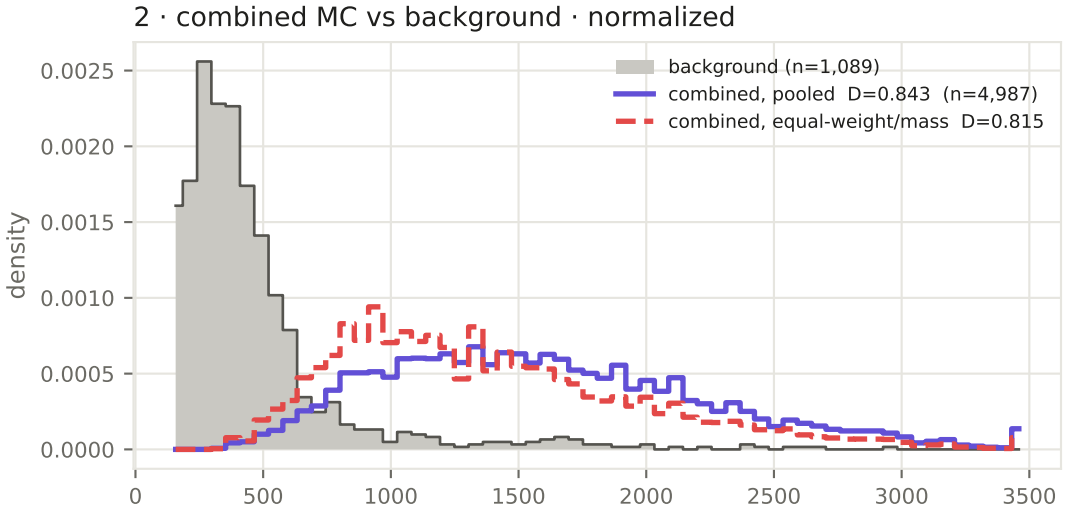

muonAllSegments_counts/nEML · region: barrel · rank 4/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
background (n=1,089)
mH125_mS15 D=0.901 (n=159)
mH125_mS55 D=0.842 (n=142)
mH200_mS50 D=0.880 (n=449)
mH400_mS100 D=0.839 (n=1,144)
mH600_mS150 D=0.825 (n=1,608)
mH1000_mS275 D=0.799 (n=1,485)
combined, pooled D=0.828 (n=4,987)
combined, equal-weight/mass D=0.848
events
density
muonAllSegments_counts/nEML

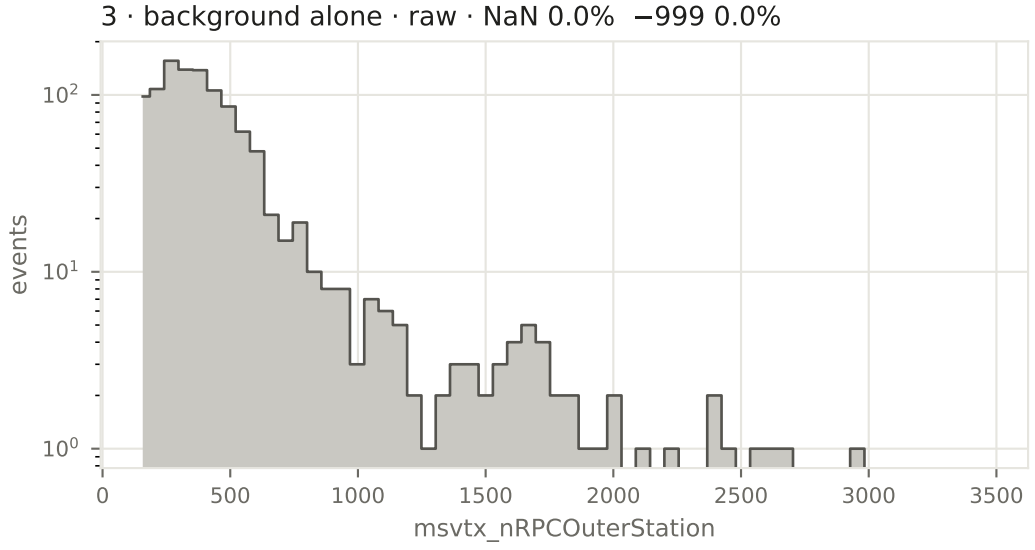
msvtx_nRPCOuterStation · region: barrel · rank 5/191 by combined equal-weight D · round 1 · edge bins include under/overflow

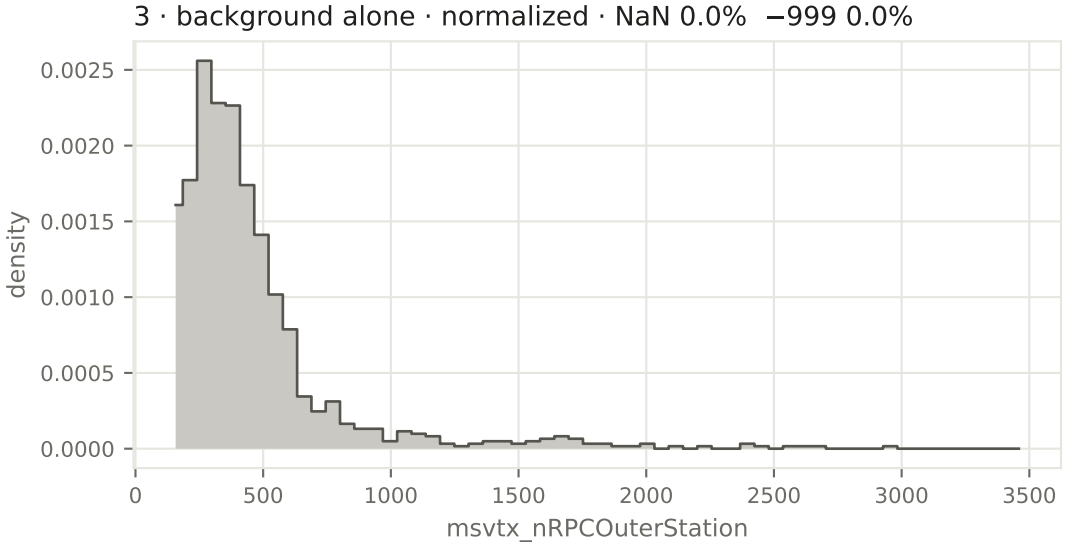






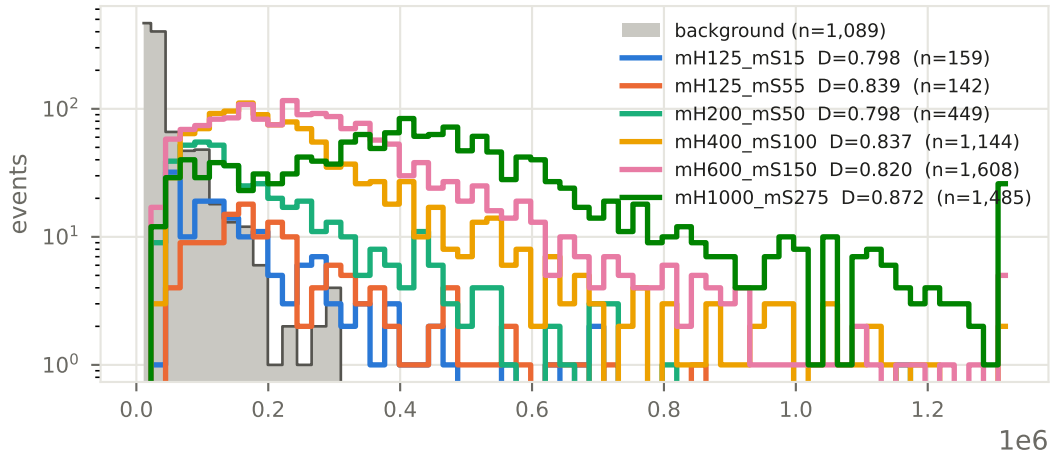




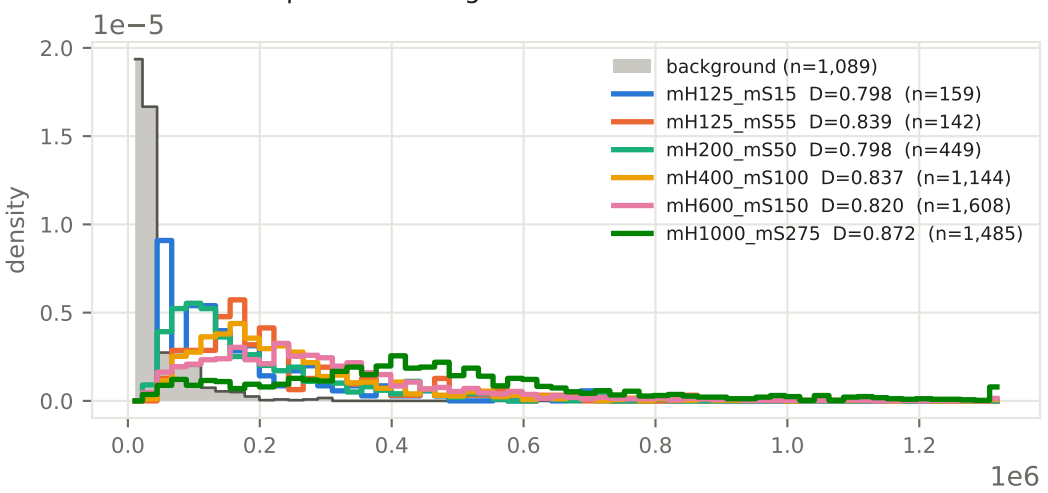


htmissScalar_NOSYS · region: barrel · rank 6/191 by combined equal-weight D · round 1 · edge bins include under/overflow

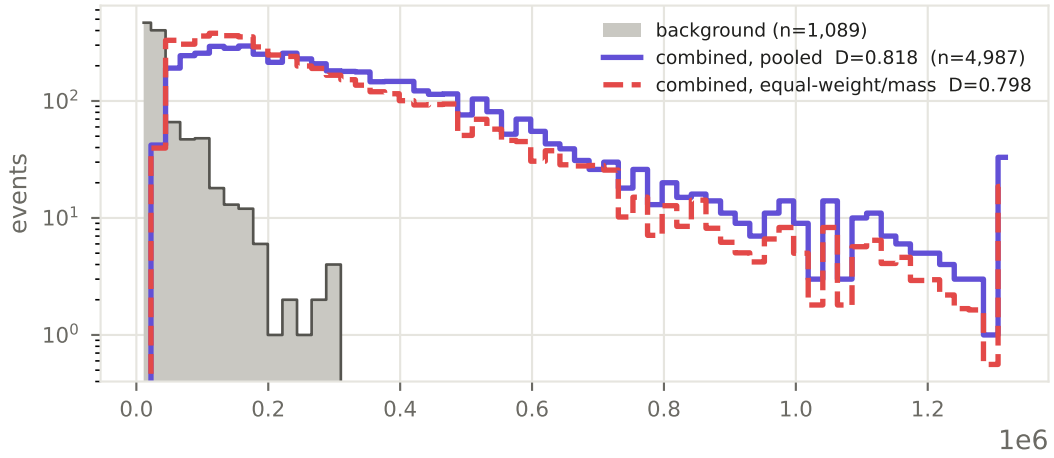
1 · each mass point vs background · raw



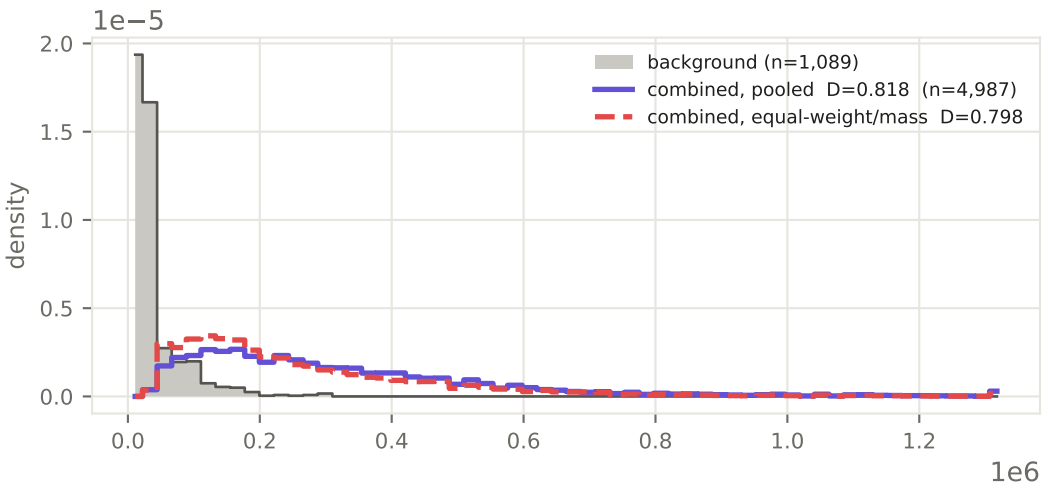
1 · each mass point vs background · normalized



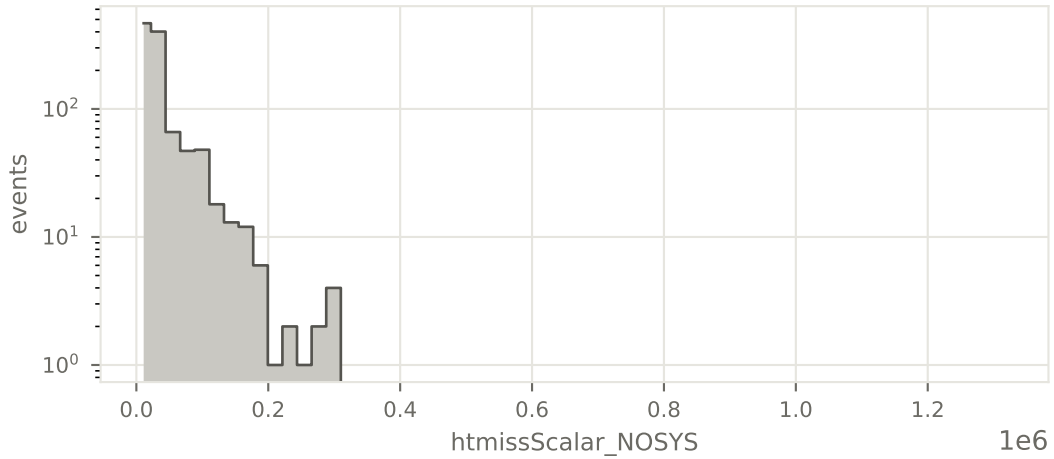
2 · combined MC vs background · raw



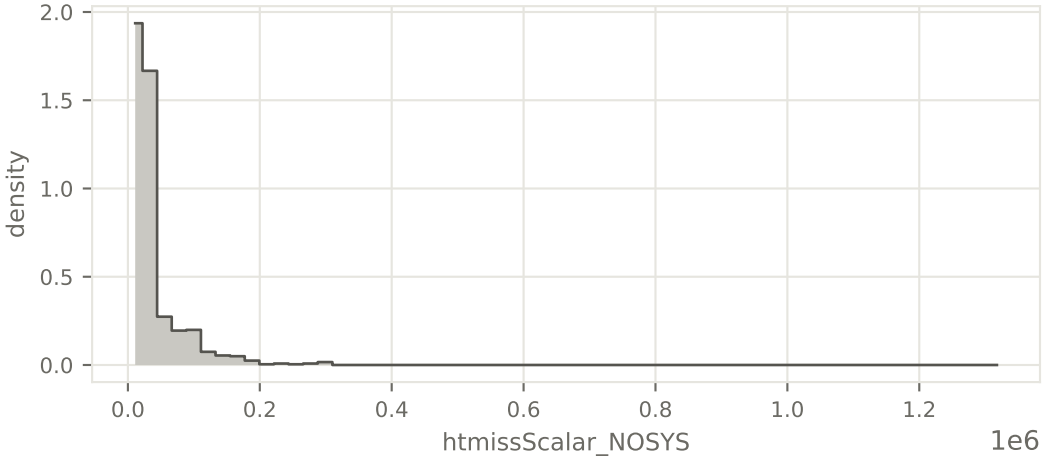
2 · combined MC vs background · normalized

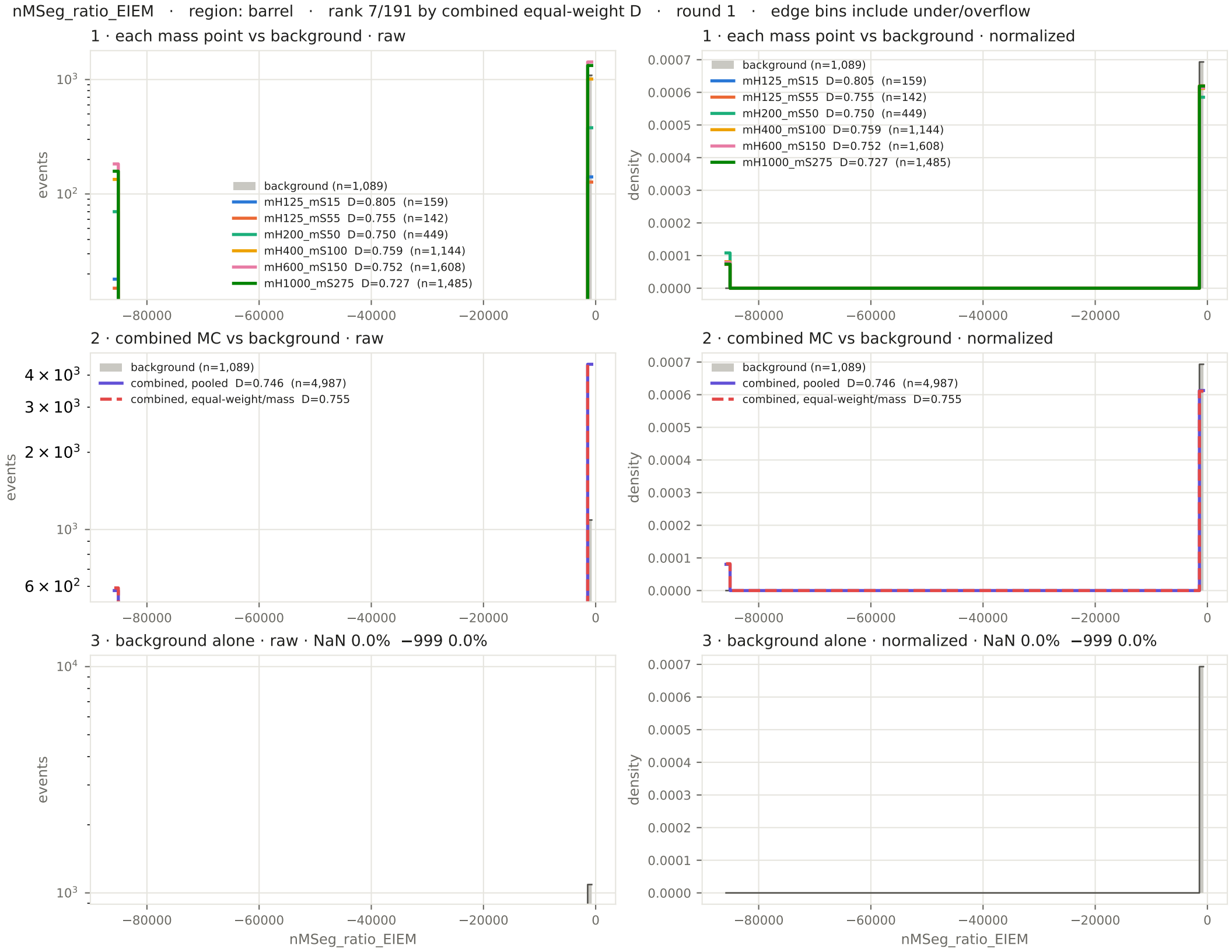


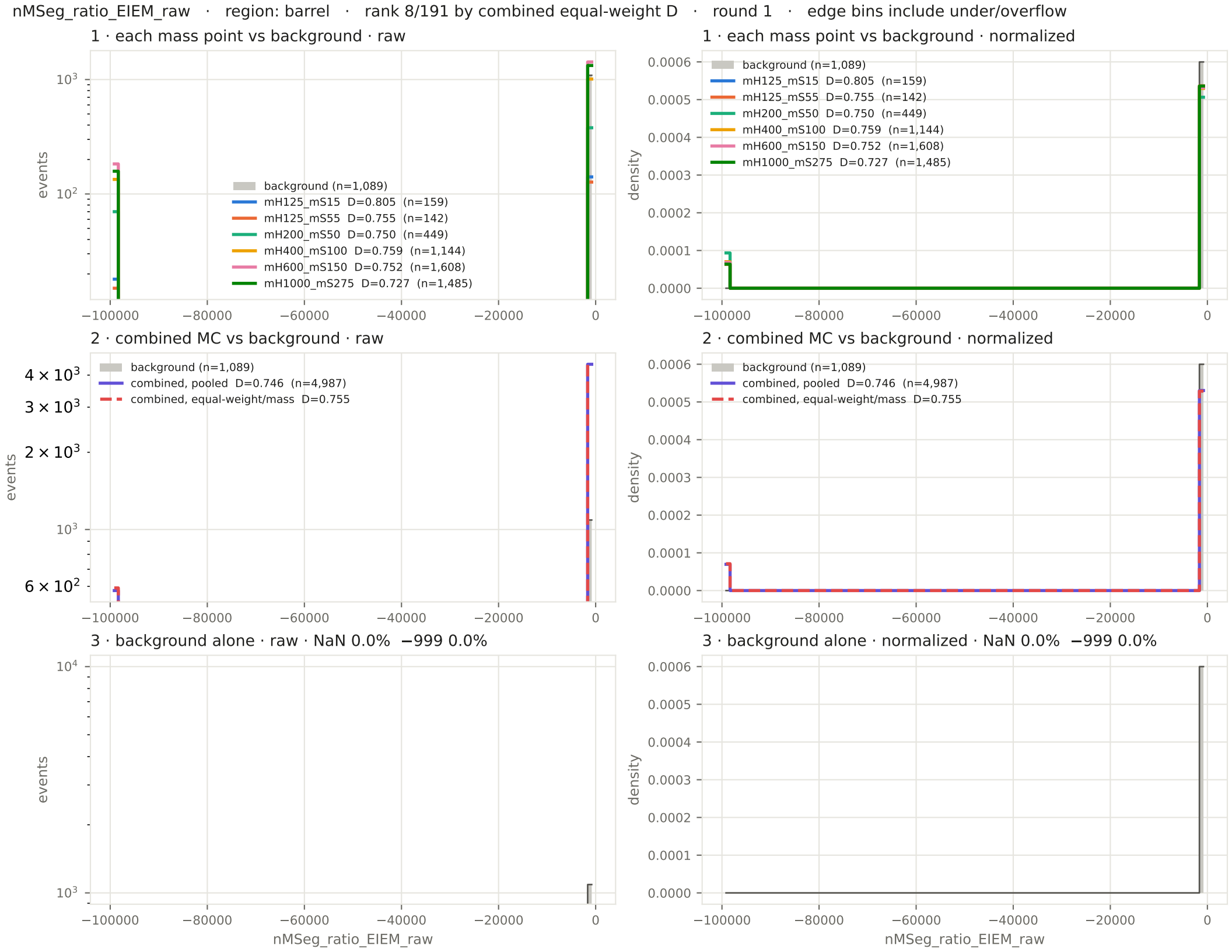
3 · background alone · raw · NaN 0.0% −999 0.0%



3 · background alone · normalized · NaN 0.0% −999 0.0%

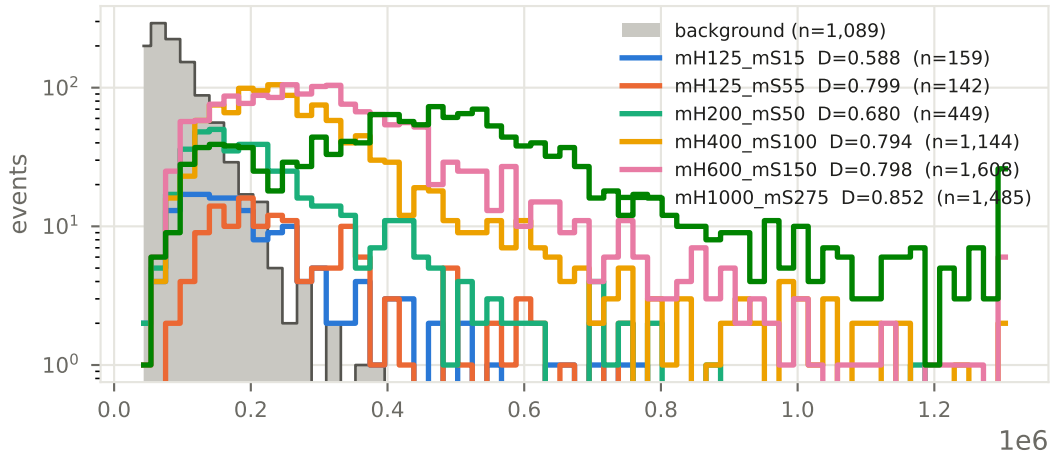



nMSeg_ratio_EIEM · region: barrel · rank 7/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
events
density
nMSeg_ratio_EIEM
background (n=1,089)
mH125_mS15 D=0.805 (n=159)
mH125_mS55 D=0.755 (n=142)
mH200_mS50 D=0.750 (n=449)
mH400_mS100 D=0.759 (n=1,144)
mH600_mS150 D=0.752 (n=1,608)
mH1000_mS275 D=0.727 (n=1,485)
combined, pooled D=0.746 (n=4,987)
combined, equal-weight/mass D=0.755

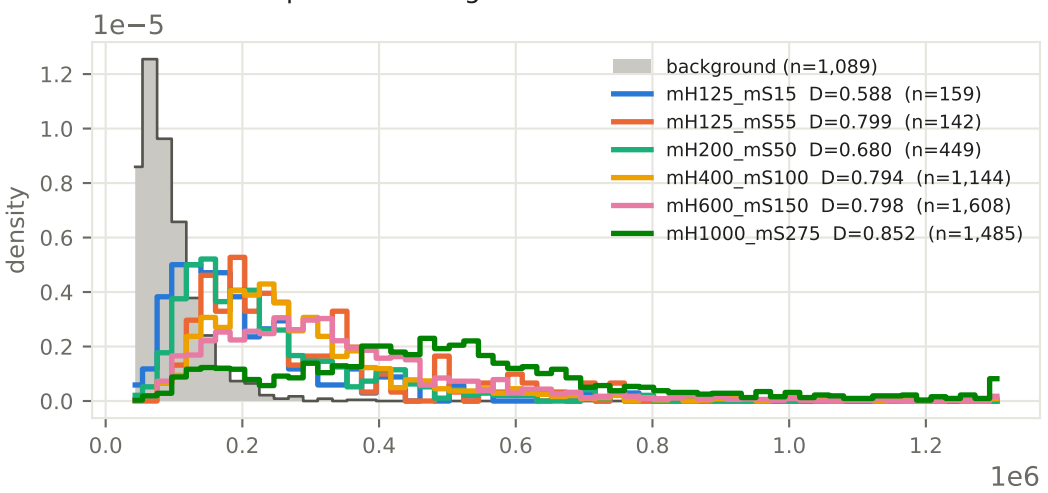

nMSeg_ratio_EIEM_raw · region: barrel · rank 8/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
events
density
nMSeg_ratio_EIEM_raw
background (n=1,089)
mH125_mS15 D=0.805 (n=159)
mH125_mS55 D=0.755 (n=142)
mH200_mS50 D=0.750 (n=449)
mH400_mS100 D=0.759 (n=1,144)
mH600_mS150 D=0.752 (n=1,608)
mH1000_mS275 D=0.727 (n=1,485)
combined, pooled D=0.746 (n=4,987)
combined, equal-weight/mass D=0.755

met_sumet_NOSYS · region: barrel · rank 9/191 by combined equal-weight D · round 1 · edge bins include under/overflow

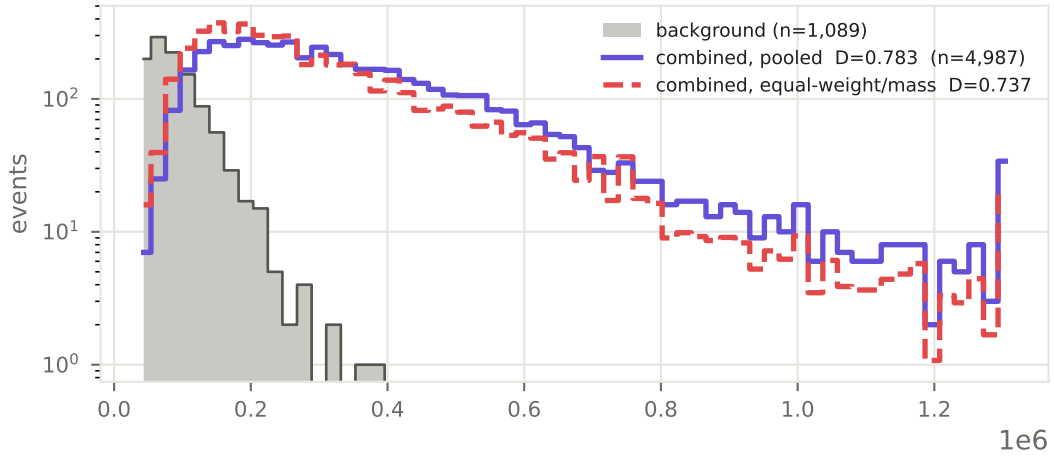
1 · each mass point vs background · raw



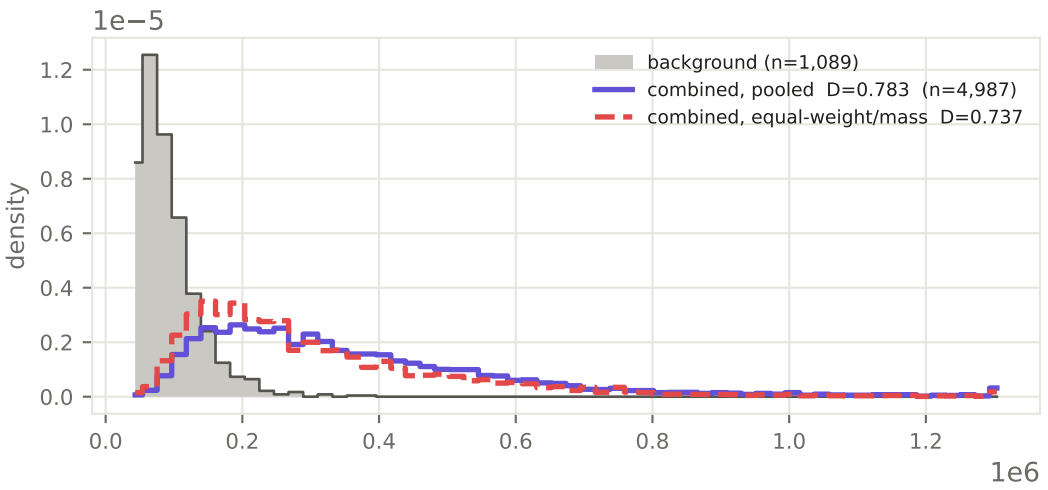
1 · each mass point vs background · normalized



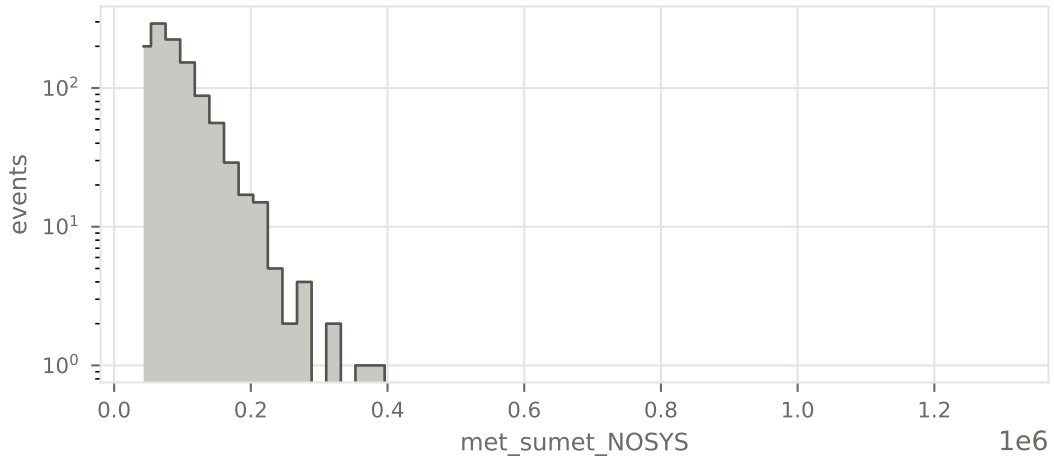
2 · combined MC vs background · raw



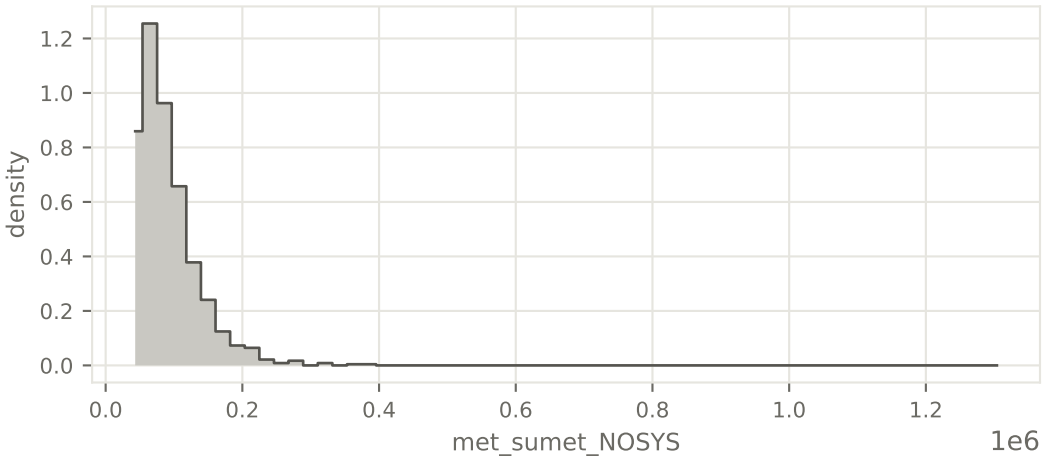
2 · combined MC vs background · normalized



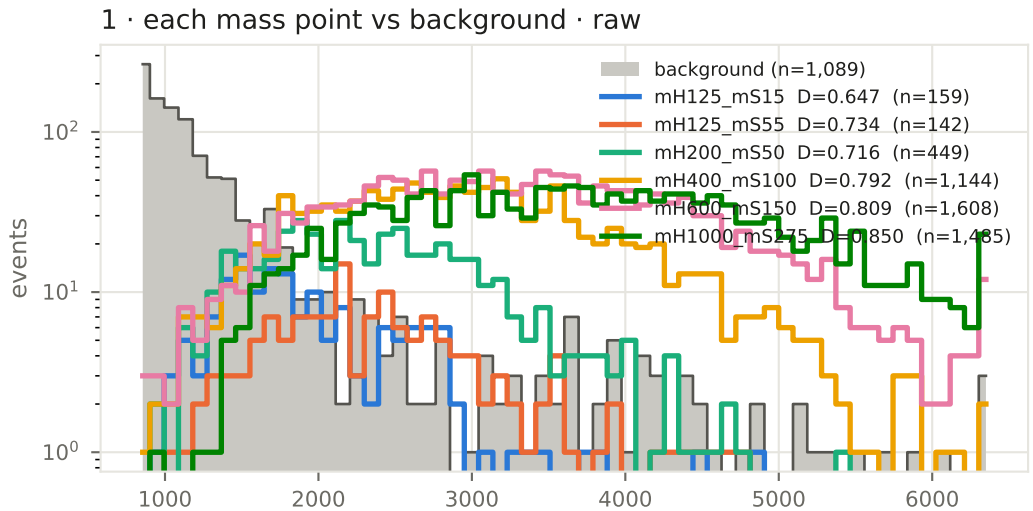
3 · background alone · raw · NaN 0.0% −999 0.0%

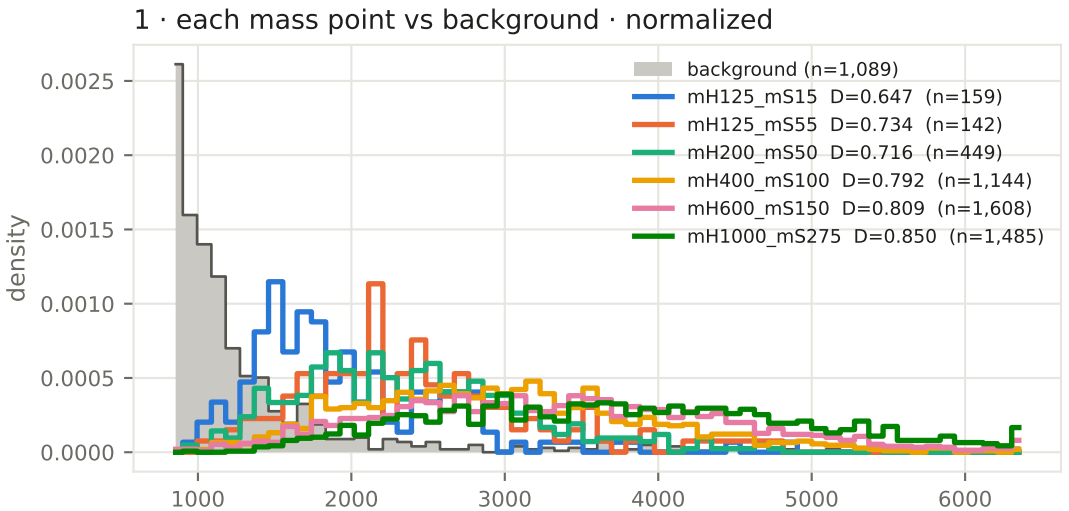


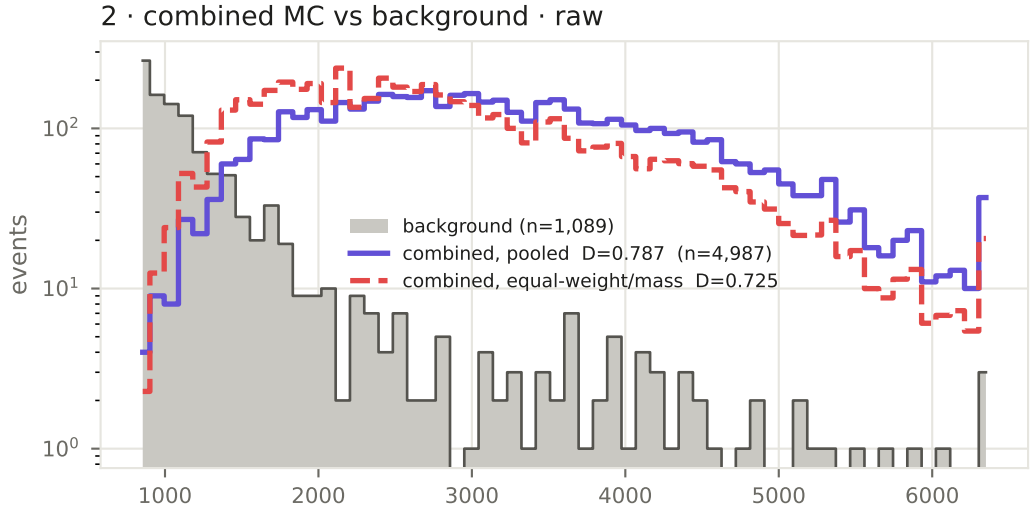
3 · background alone · normalized · NaN 0.0% −999 0.0%

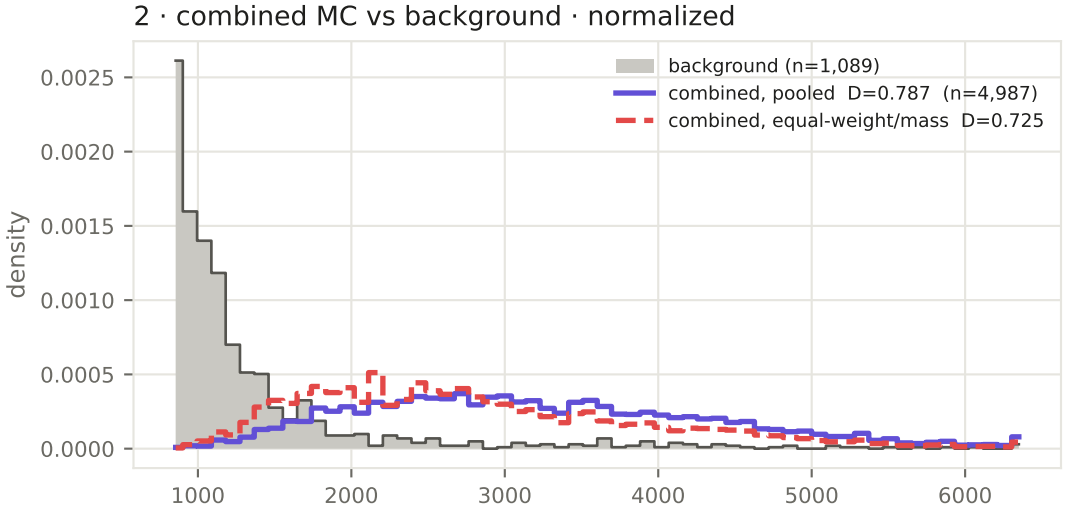


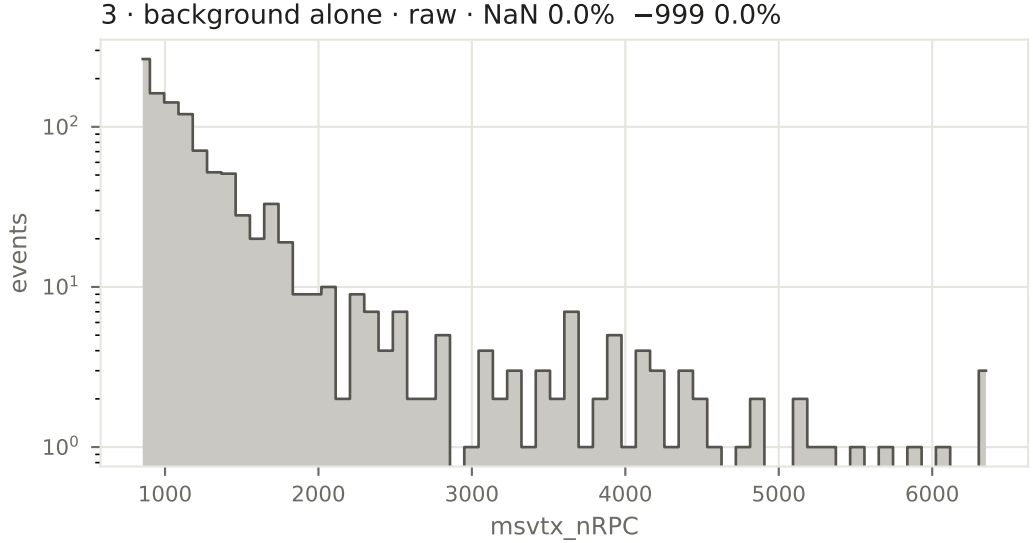
msvtx_nRPC · region: barrel · rank 10/191 by combined equal-weight D · round 1 · edge bins include under/overflow

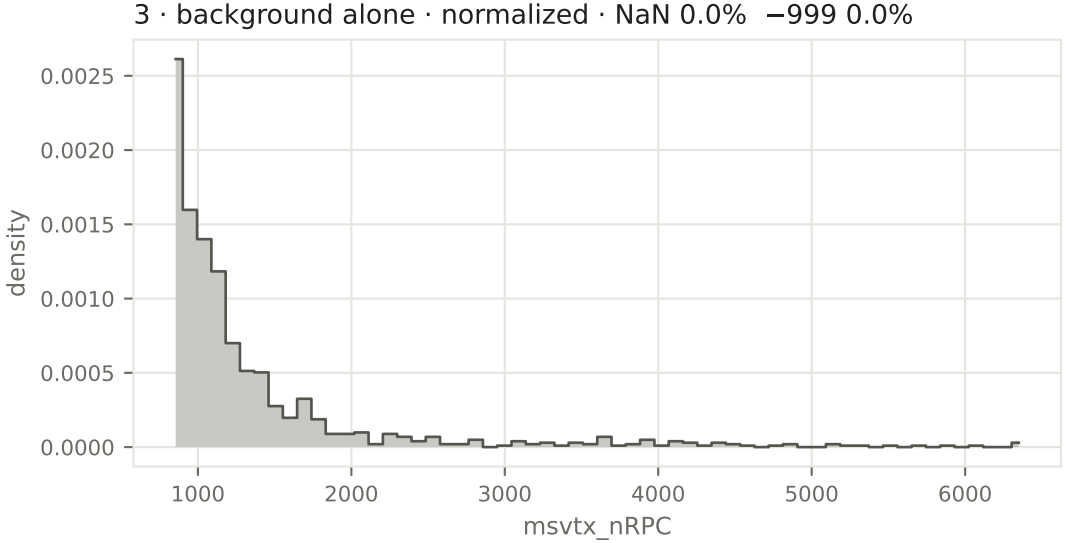


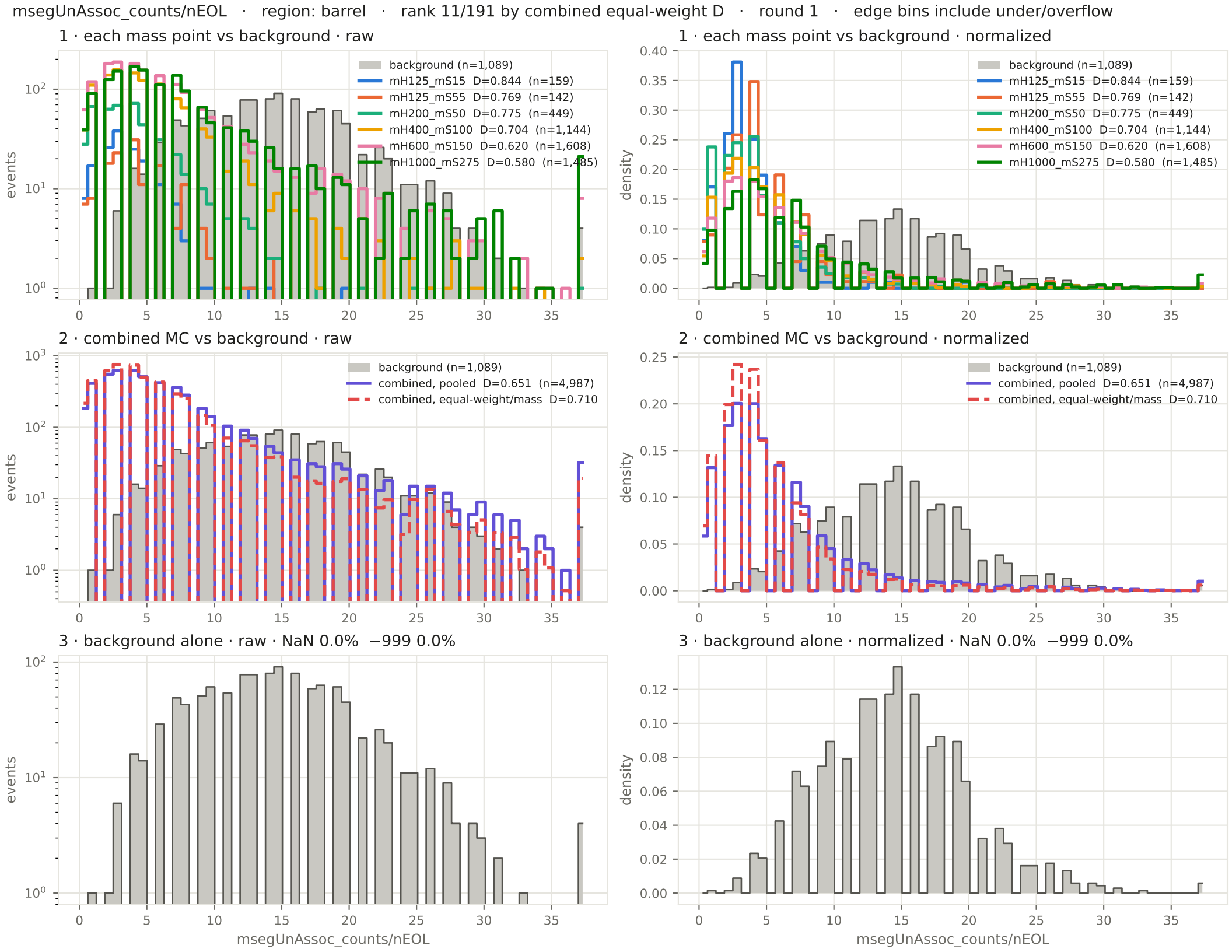




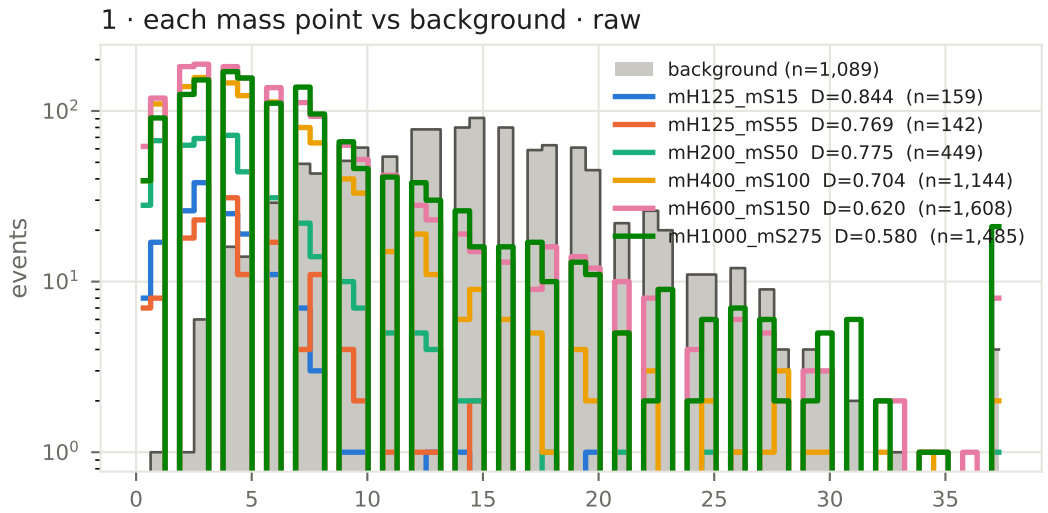


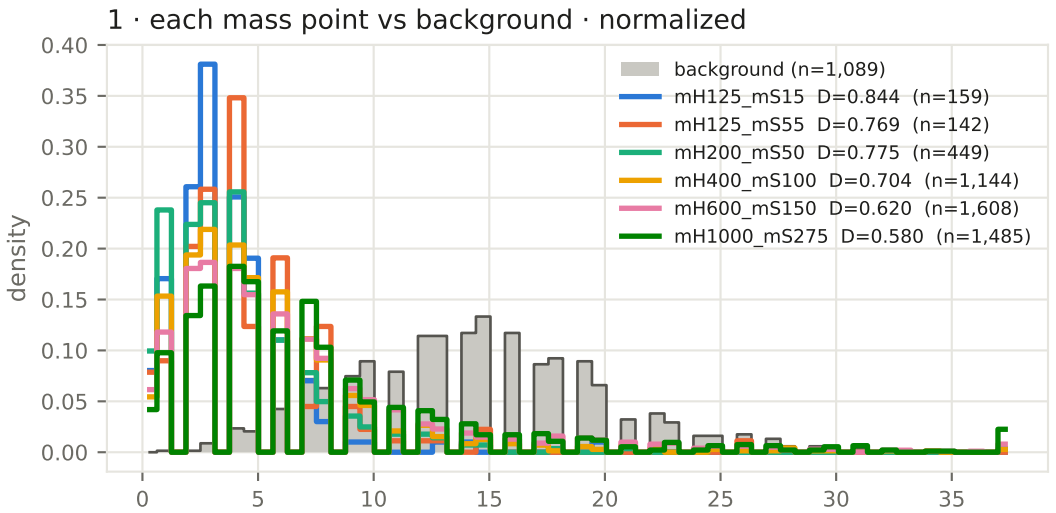


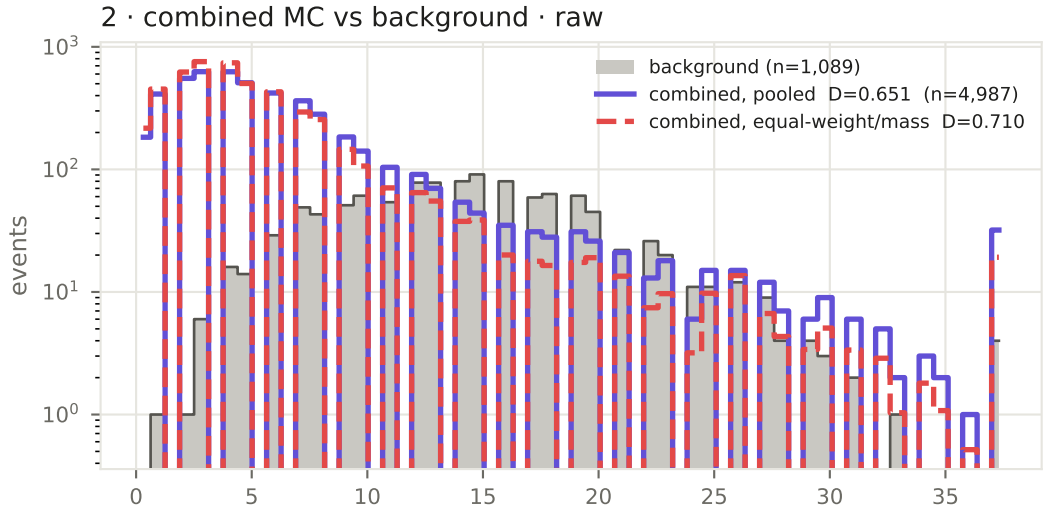


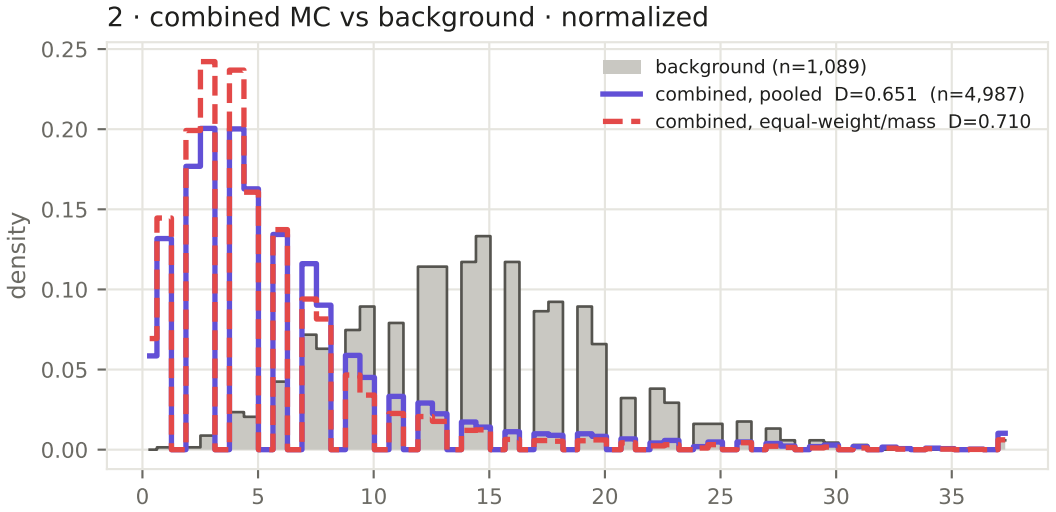

msegUnAssoc_counts/nEOL · region: barrel · rank 11/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
background (n=1,089)
mH125_mS15 D=0.844 (n=159)
mH125_mS55 D=0.769 (n=142)
mH200_mS50 D=0.775 (n=449)
mH400_mS100 D=0.704 (n=1,144)
mH600_mS150 D=0.620 (n=1,608)
mH1000_mS275 D=0.580 (n=1,485)
combined, pooled D=0.651 (n=4,987)
combined, equal-weight/mass D=0.710
events
density
msegUnAssoc_counts/nEOL

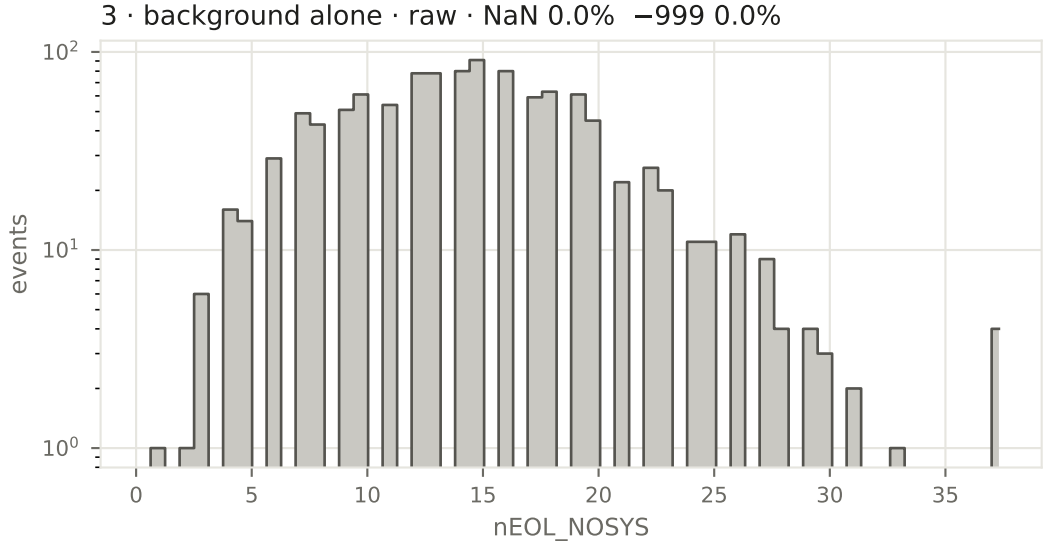
nEOL_NOSYS · region: barrel · rank 12/191 by combined equal-weight D · round 1 · edge bins include under/overflow

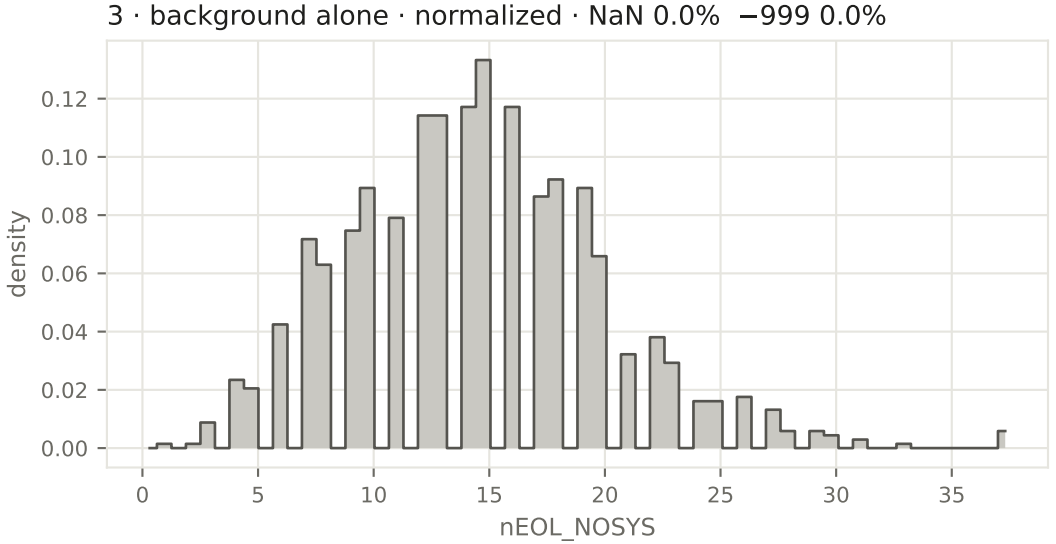






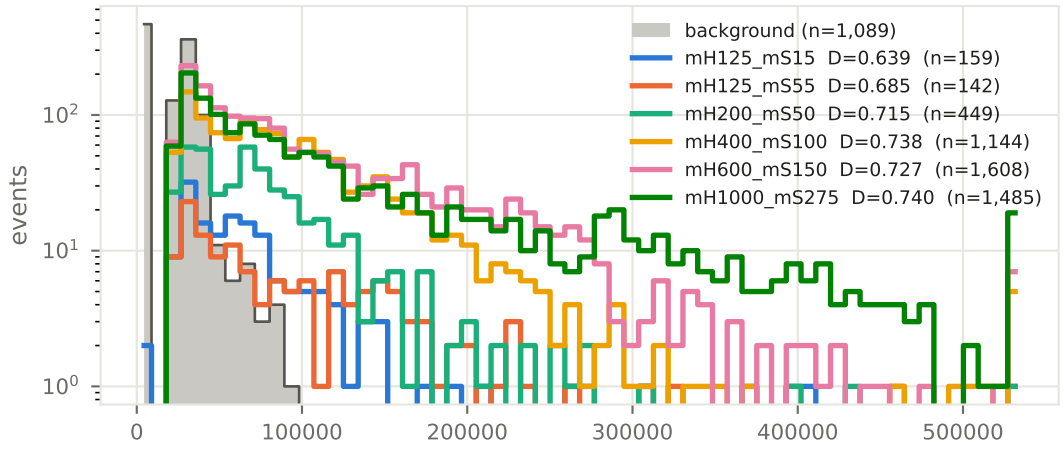




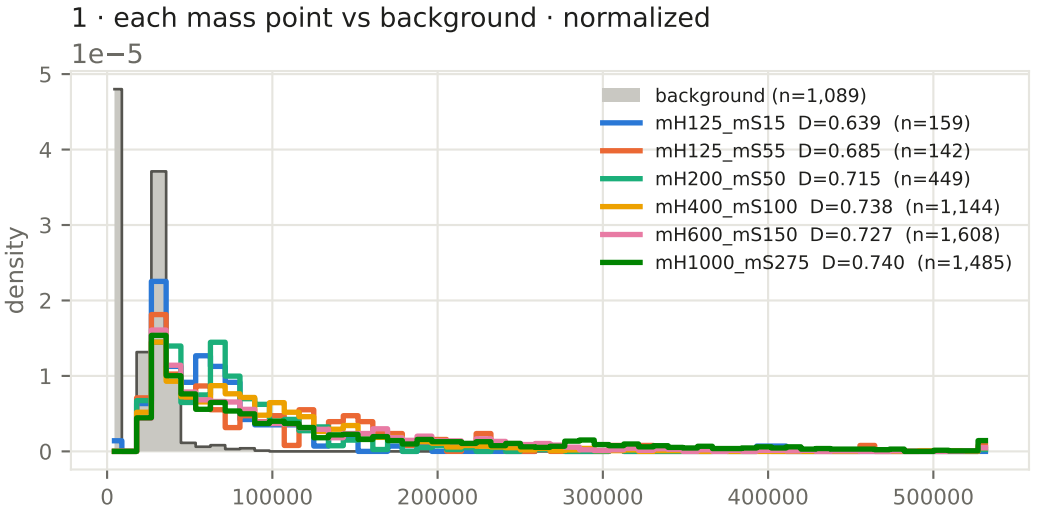


msvtx_closestJet_pt_NOSYS · region: barrel · rank 13/191 by combined equal-weight D · round 1 · edge bins include under/overflow

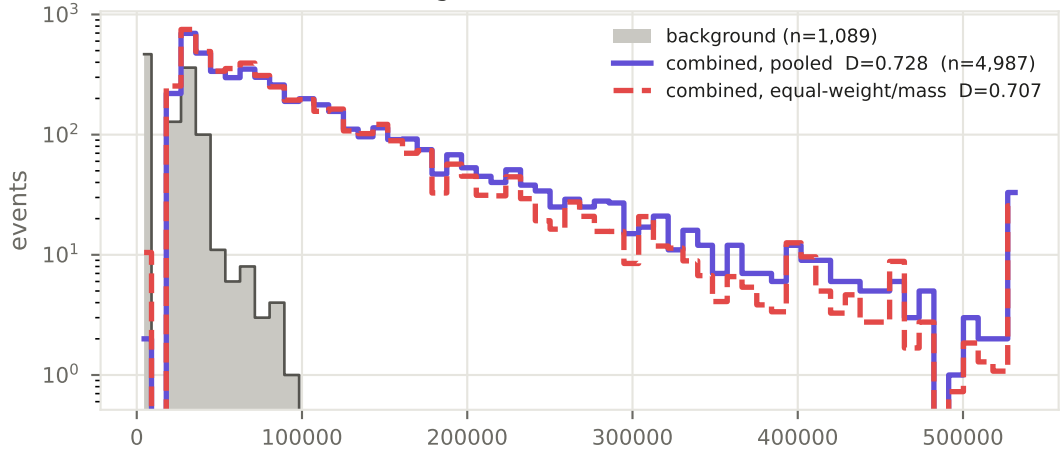
1 · each mass point vs background · raw



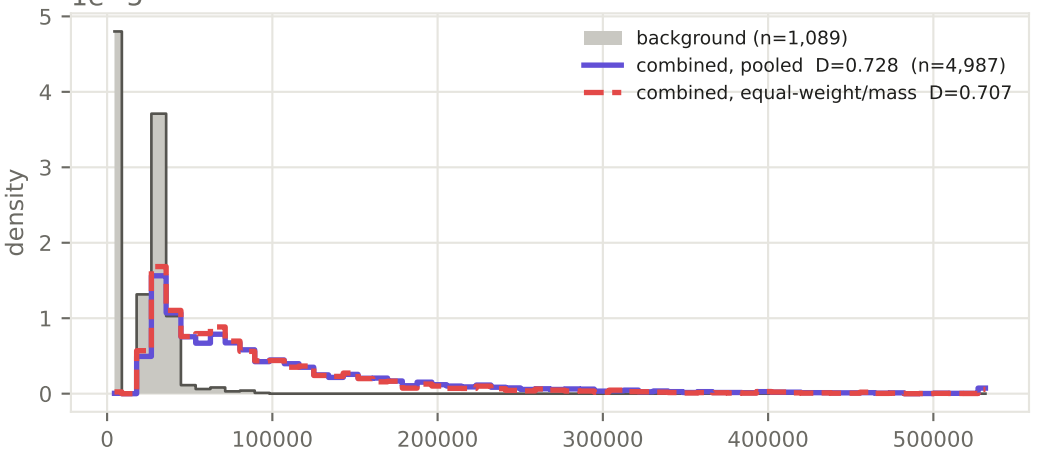
1 · each mass point vs background · normalized



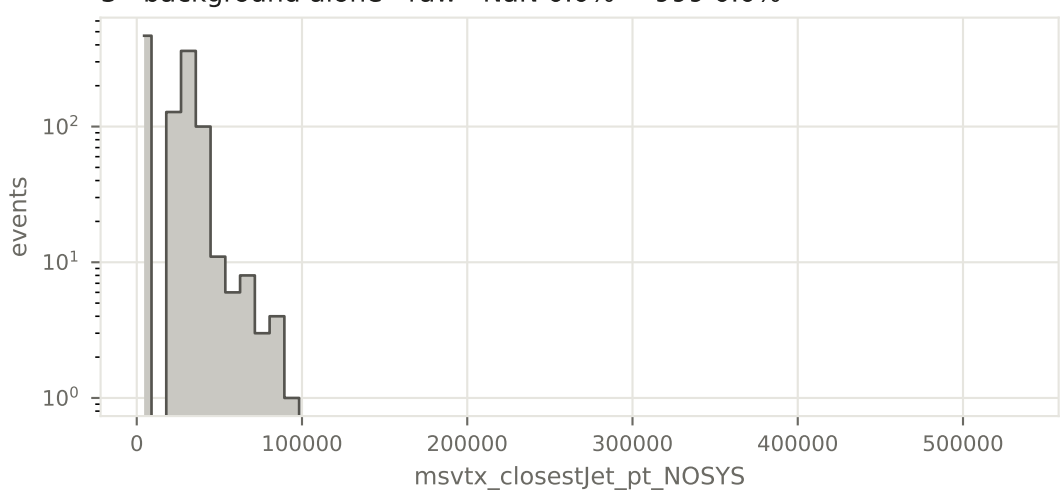
2 · combined MC vs background · raw



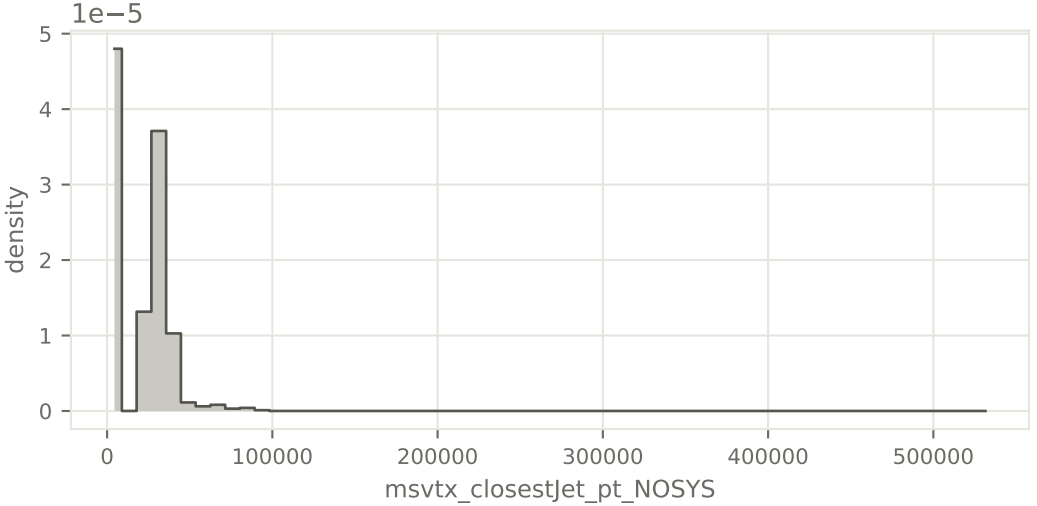
2 · combined MC vs background · normalized



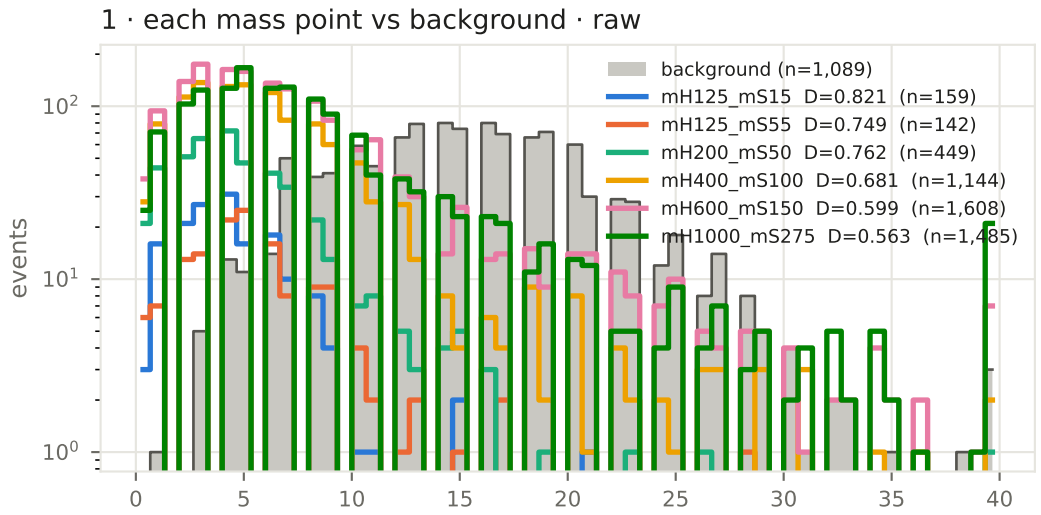
3 · background alone · raw · NaN 0.0% −999 0.0%

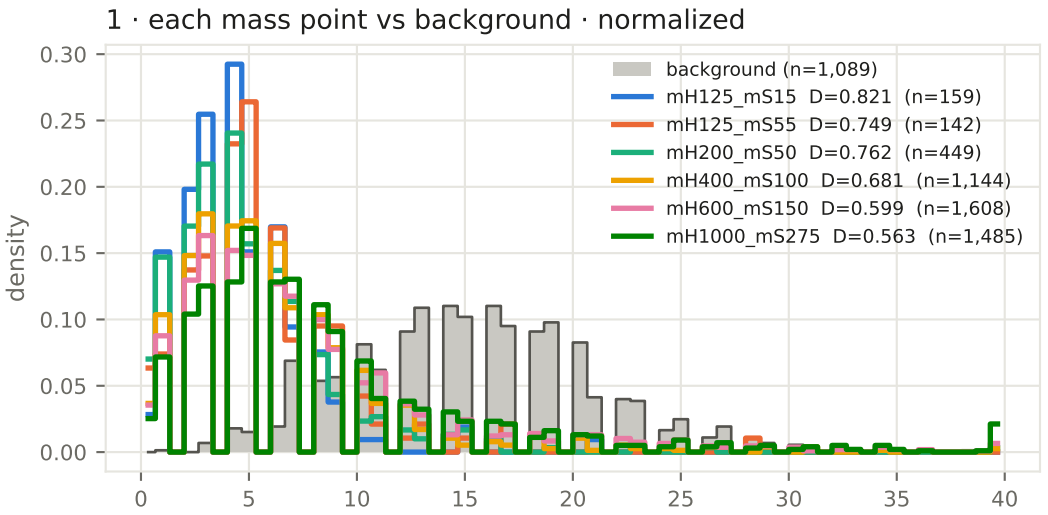


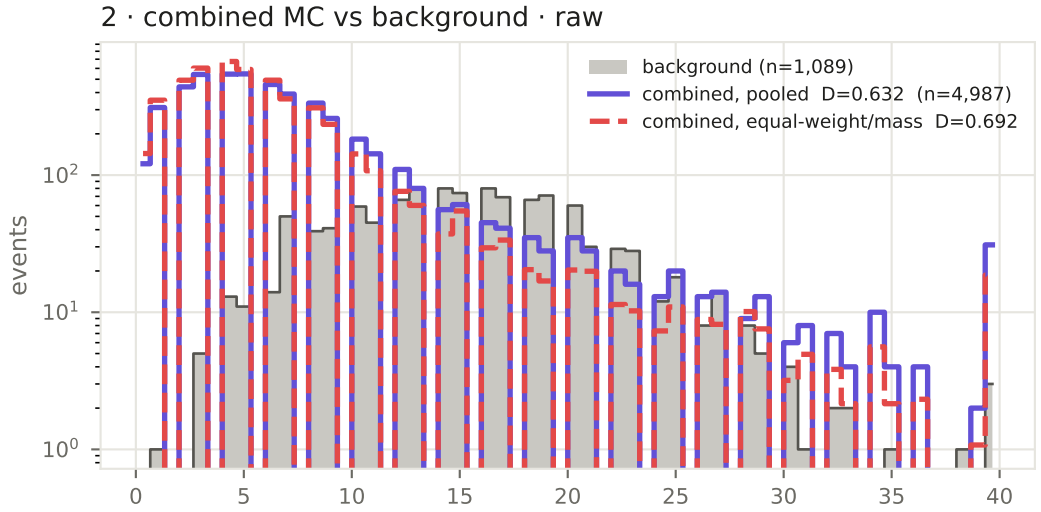
3 · background alone · normalized · NaN 0.0% −999 0.0%

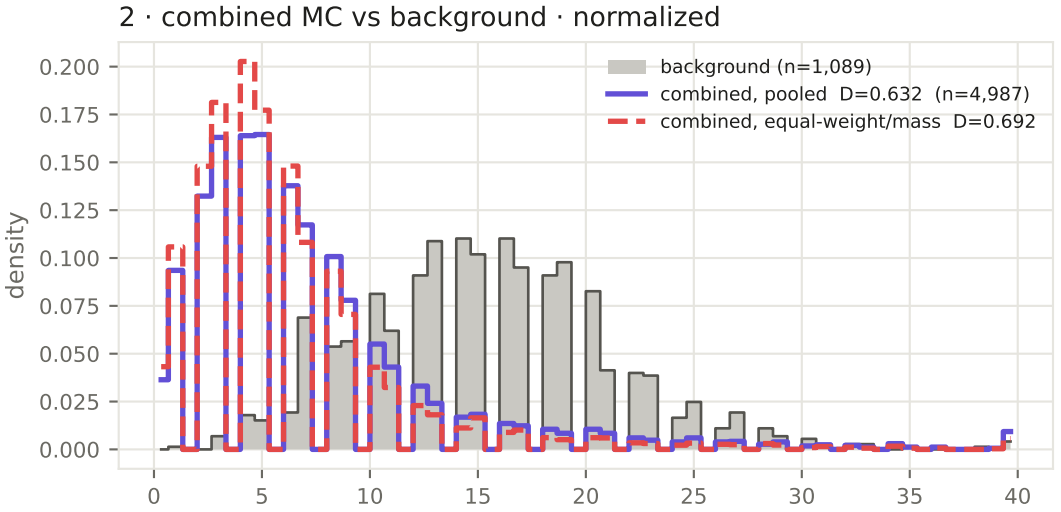


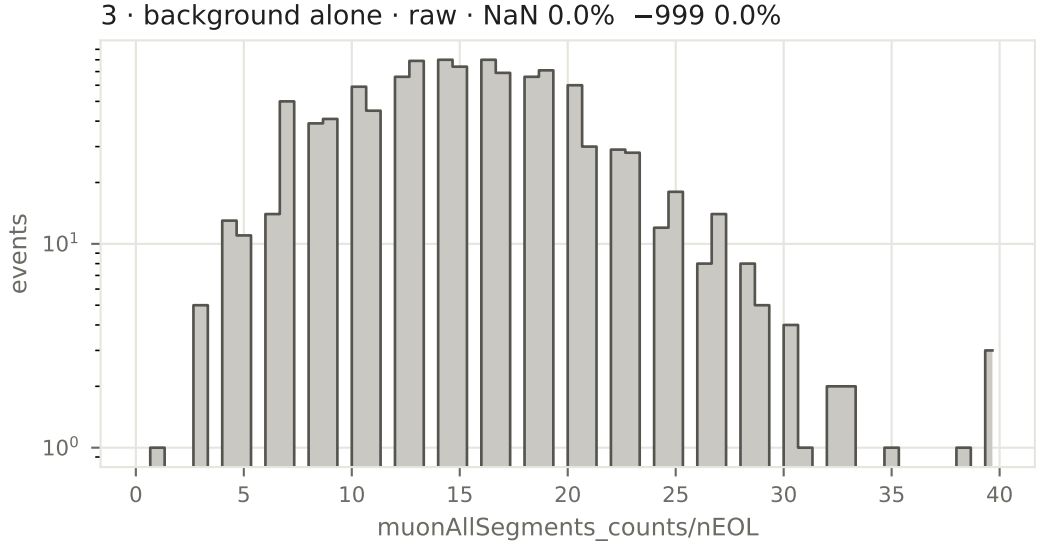
muonAllSegments_counts/nEOL · region: barrel · rank 14/191 by combined equal-weight D · round 1 · edge bins include under/overflow

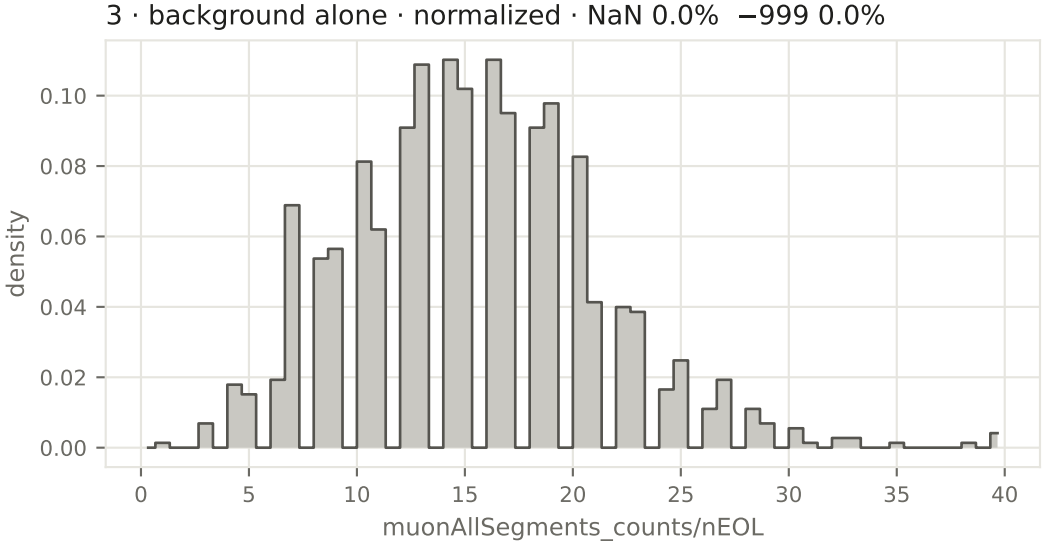






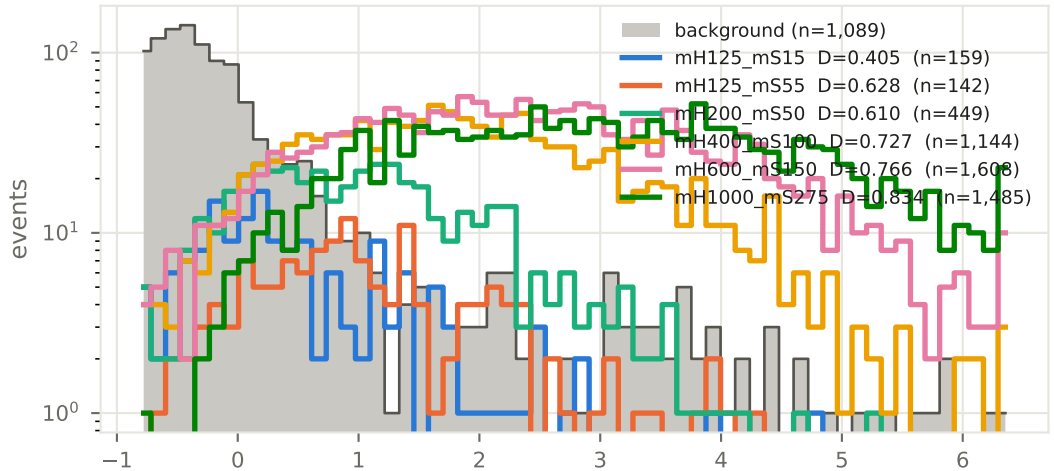




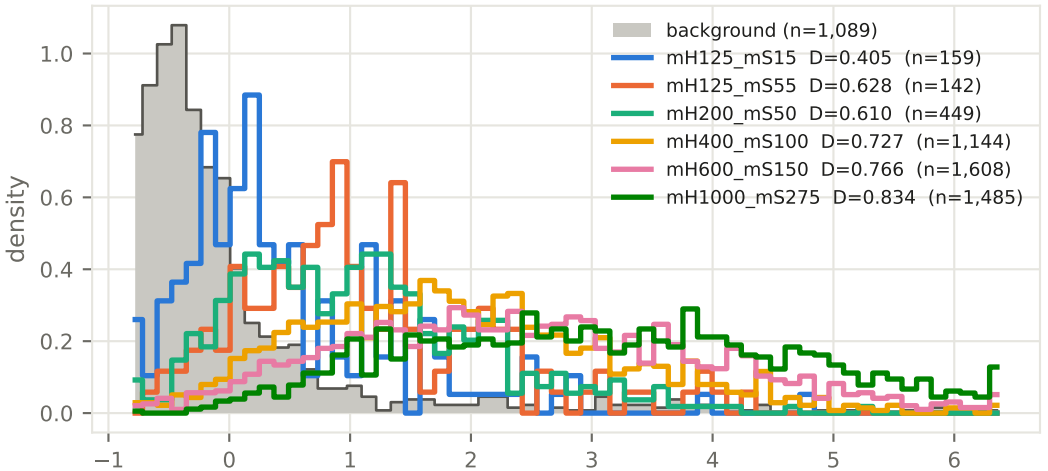


MS1Vtx_barrel_hits_ntot · region: barrel · rank 15/191 by combined equal-weight D · round 1 · edge bins include under/overflow

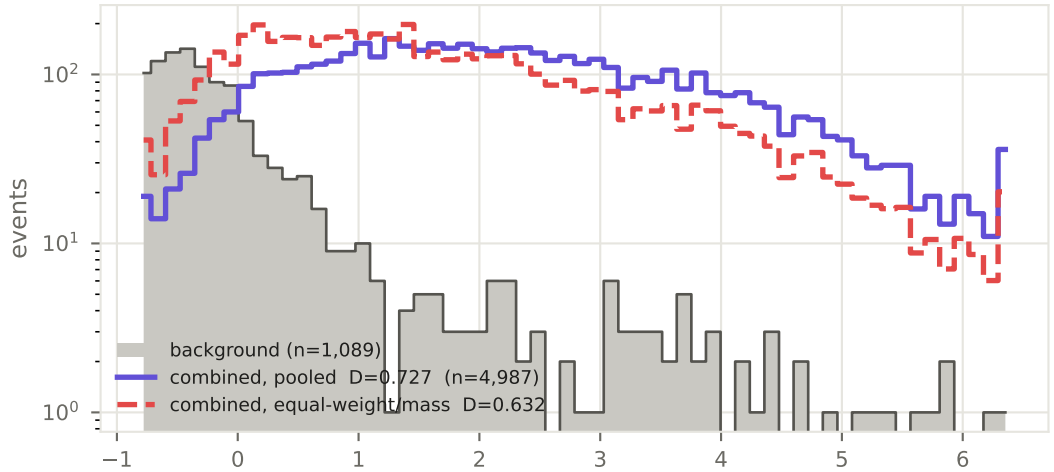
1 · each mass point vs background · raw



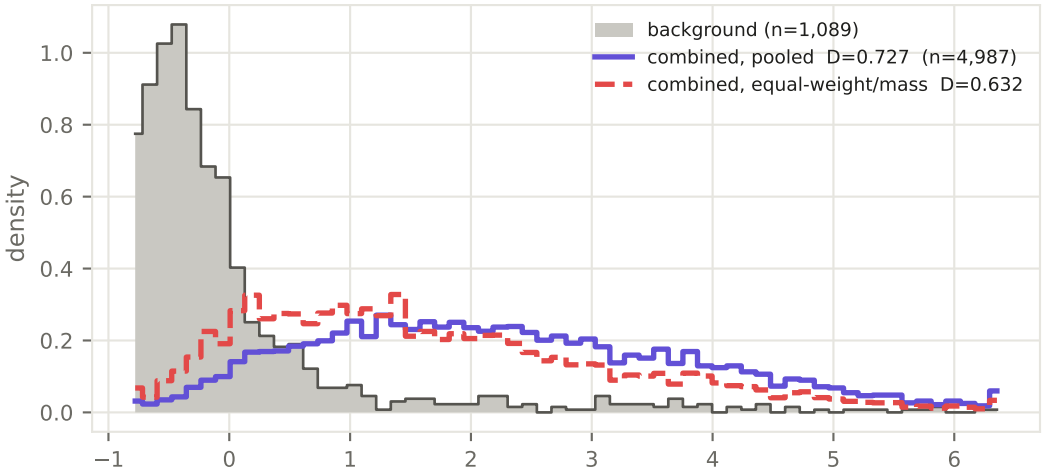
1 · each mass point vs background · normalized



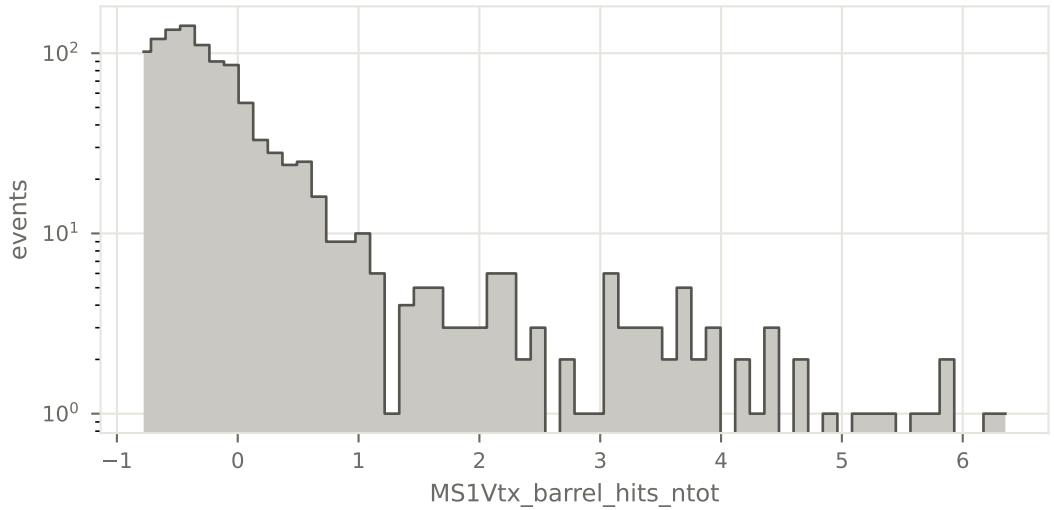
2 · combined MC vs background · raw



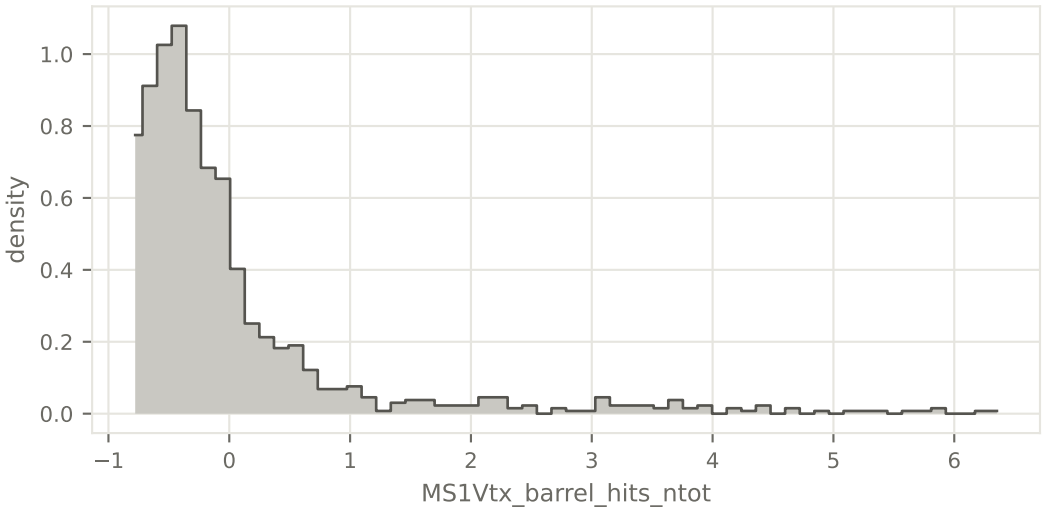
2 · combined MC vs background · normalized

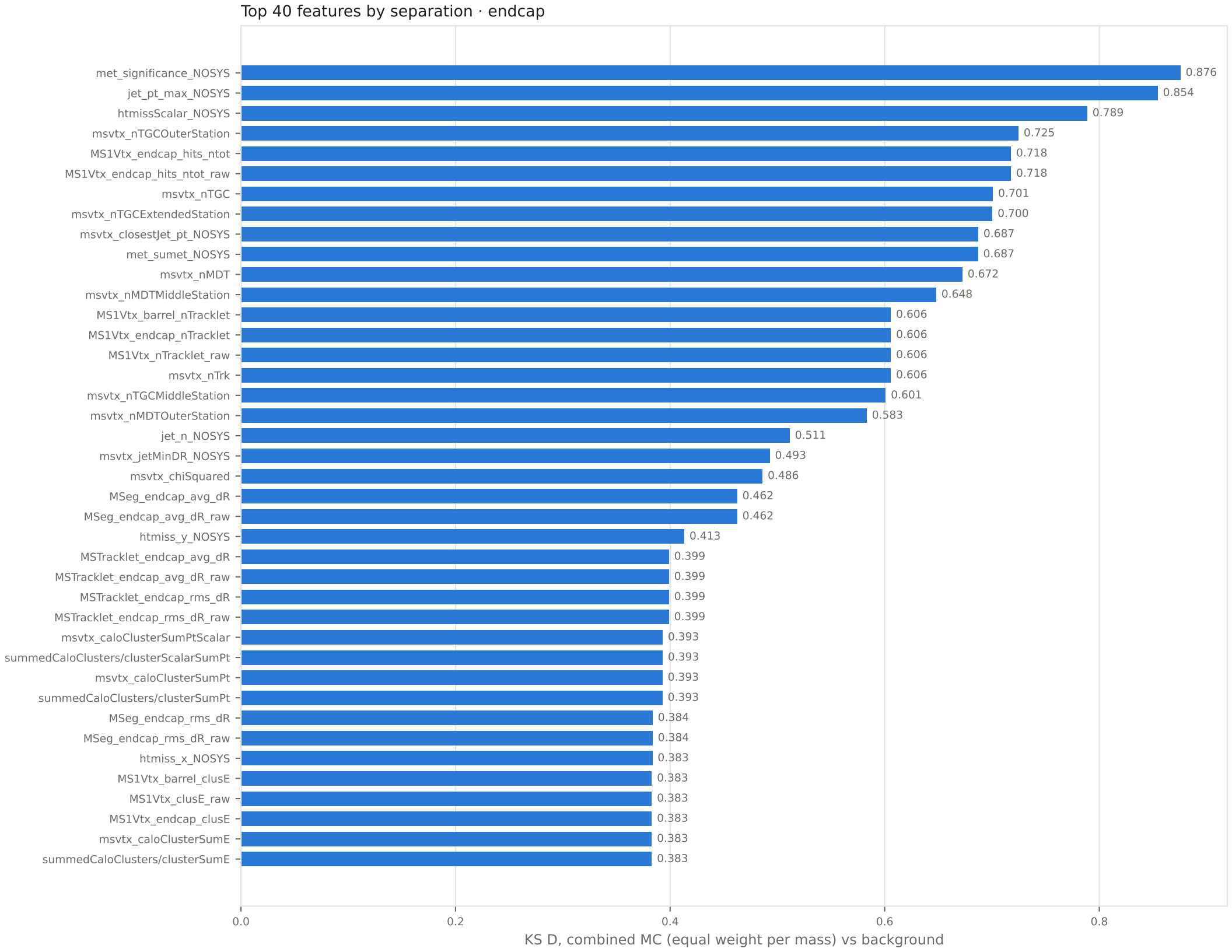


3 · background alone · raw · NaN 0.0% −999 0.0%



3 · background alone · normalized · NaN 0.0% −999 0.0%

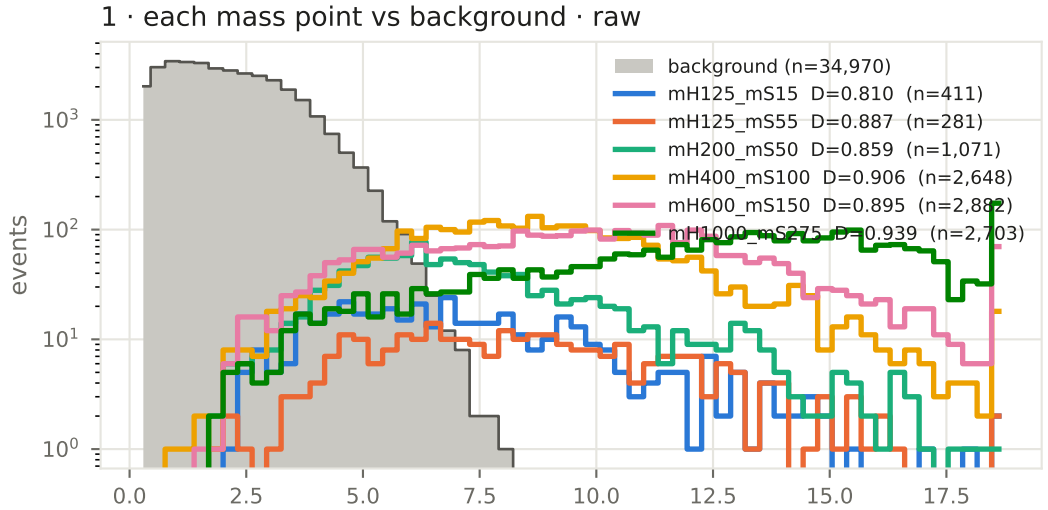


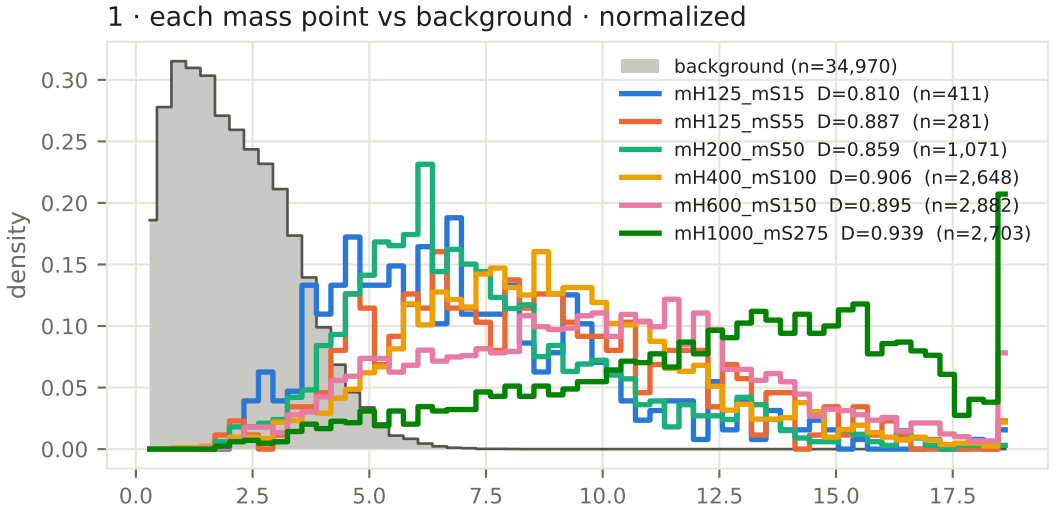

Top 40 features by separation · endcap
met_significance_NOSYS 0.876
jet_pt_max_NOSYS 0.854
htmissScalar_NOSYS 0.789
msvtx_nTGCOuterStation 0.725
MS1Vtx_endcap_hits_ntot 0.718
MS1Vtx_endcap_hits_ntot_raw 0.718
msvtx_nTGC 0.701
msvtx_nTGCExtendedStation 0.700
msvtx_closestJet_pt_NOSYS 0.687
met_sumet_NOSYS 0.687
msvtx_nMDT 0.672
msvtx_nMDTMiddleStation 0.648
MS1Vtx_barrel_nTracklet 0.606
MS1Vtx_endcap_nTracklet 0.606
MS1Vtx_nTracklet_raw 0.606
msvtx_nTrk 0.606
msvtx_nTGCMiddleStation 0.601
msvtx_nMDTOuterStation 0.583
jet_n_NOSYS 0.511
msvtx_jetMinDR_NOSYS 0.493
msvtx_chiSquared 0.486
MSeg_endcap_avg_dR 0.462
MSeg_endcap_avg_dR_raw 0.462
htmiss_y_NOSYS 0.413
MSTracklet_endcap_avg_dR 0.399
MSTracklet_endcap_avg_dR_raw 0.399
MSTracklet_endcap_rms_dR 0.399
MSTracklet_endcap_rms_dR_raw 0.399
msvtx_caloClusterSumPtScalar 0.393
summedCaloClusters/clusterScalarSumPt 0.393
msvtx_caloClusterSumPt 0.393
summedCaloClusters/clusterSumPt 0.393
MSeg_endcap_rms_dR 0.384
MSeg_endcap_rms_dR_raw 0.384
htmiss_x_NOSYS 0.383
MS1Vtx_barrel_clusE 0.383
MS1Vtx_clusE_raw 0.383
MS1Vtx_endcap_clusE 0.383
msvtx_caloClusterSumE 0.383
summedCaloClusters/clusterSumE 0.383
0.0
0.2
0.4
0.6
0.8
KS D, combined MC (equal weight per mass) vs background

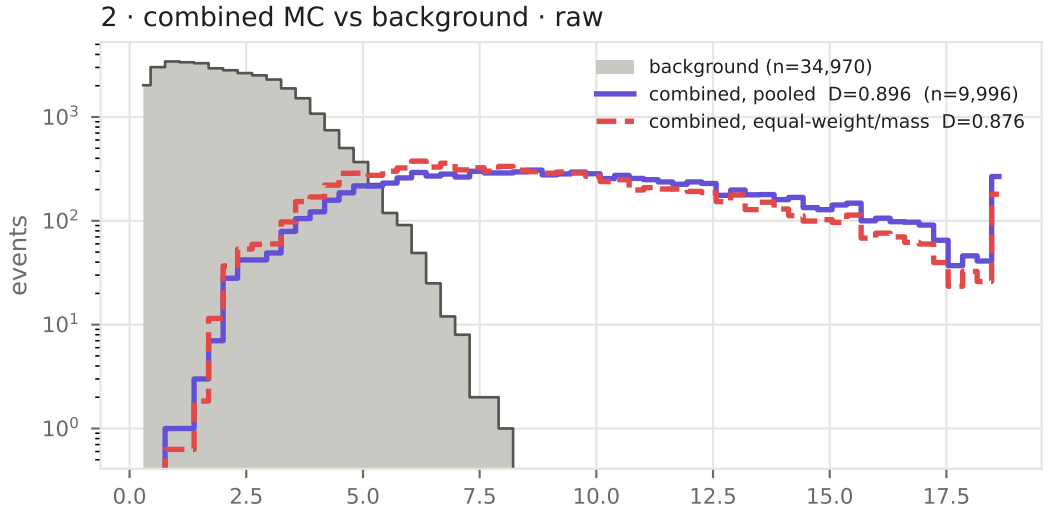
Round 1 · endcap · top 40 by combined equal-weight KS D

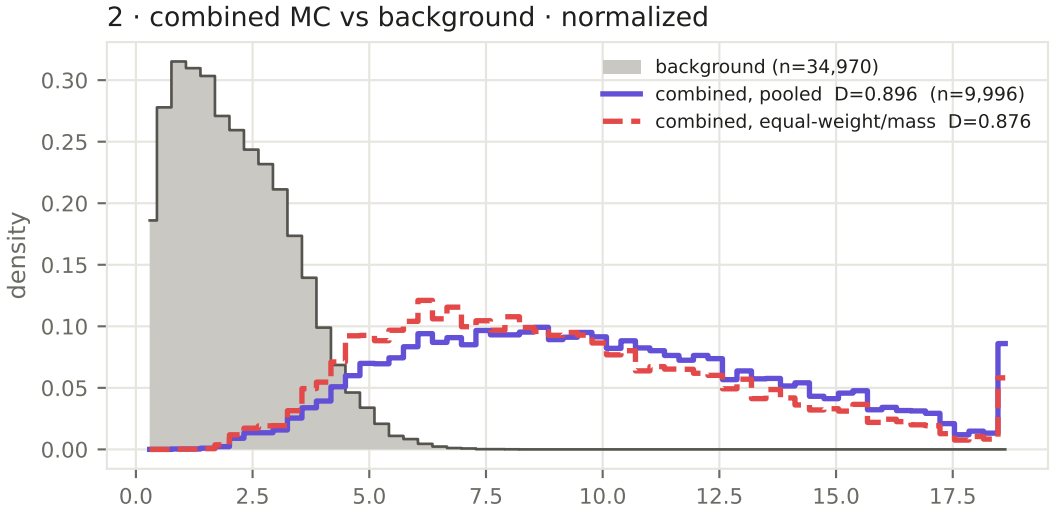
#	feature	D_equal	D_pooled	D_min	D_max	bkgNaN	bkg-999
1	met_significance_NOSYS	0.876	0.896	0.810	0.939	0.0%	0.0%
2	jet_pt_max_NOSYS	0.854	0.874	0.785	0.920	0.0%	0.0%
3	htmissScalar_NOSYS	0.789	0.789	0.789	0.868	0.0%	0.0%
4	msvtx_nTGCOuterStation	0.725	0.785	0.612	0.839	0.0%	0.0%
5	MS1Vtx_endcap_hits_ntot	0.718	0.794	0.553	0.864	0.0%	0.0%
6	MS1Vtx_endcap_hits_ntot_raw	0.718	0.794	0.553	0.864	0.0%	0.0%
7	msvtx_nTGC	0.701	0.773	0.590	0.840	0.0%	0.0%
8	msvtx_nTGCExtendedStation	0.700	0.766	0.585	0.828	0.0%	0.0%
9	msvtx_closestJet_pt_NOSYS	0.687	0.697	0.647	0.715	0.0%	0.0%
10	met_sumet_NOSYS	0.687	0.741	0.563	0.859	0.0%	0.0%
11	msvtx_nMDT	0.672	0.759	0.491	0.846	0.0%	0.0%
12	msvtx_nMDTMiddleStation	0.648	0.736	0.504	0.827	0.0%	0.0%
13	MS1Vtx_barrel_nTracklet	0.606	0.607	0.551	0.643	0.0%	0.0%
14	MS1Vtx_endcap_nTracklet	0.606	0.607	0.551	0.643	0.0%	0.0%
15	MS1Vtx_nTracklet_raw	0.606	0.607	0.551	0.643	0.0%	0.0%
16	msvtx_nTrk	0.606	0.607	0.551	0.643	0.0%	0.0%
17	msvtx_nTGCMiddleStation	0.601	0.683	0.471	0.755	0.0%	0.0%
18	msvtx_nMDTOuterStation	0.583	0.636	0.465	0.686	0.0%	0.0%
19	jet_n_NOSYS	0.511	0.556	0.423	0.647	0.0%	0.0%
20	msvtx_jetMinDR_NOSYS	0.493	0.512	0.468	0.538	0.0%	0.0%
21	msvtx_chiSquared	0.486	0.576	0.582	1.000	0.0%	0.0%
22	MSeg_endcap_avg_dR	0.462	0.541	0.235	0.667	0.0%	0.0%
23	MSeg_endcap_avg_dR_raw	0.462	0.541	0.235	0.667	0.0%	0.0%
24	htmiss_y_NOSYS	0.413	0.417	0.368	0.449	0.0%	0.0%
25	MSTracklet_endcap_avg_dR	0.399	0.431	0.297	0.502	0.0%	0.0%
26	MSTracklet_endcap_avg_dR_raw	0.399	0.431	0.297	0.502	0.0%	0.0%
27	MSTracklet_endcap_rms_dR	0.399	0.431	0.297	0.502	0.0%	0.0%
28	MSTracklet_endcap_rms_dR_raw	0.399	0.431	0.297	0.502	0.0%	0.0%
29	msvtx_caloClusterSumPtScalar	0.393	0.394	0.389	0.404	0.0%	0.0%
30	summedCaloClusters/clusterScalarSumPt	0.393	0.394	0.389	0.404	0.0%	0.0%
31	msvtx_caloClusterSumPt	0.393	0.393	0.388	0.405	0.0%	0.0%
32	summedCaloClusters/clusterSumPt	0.393	0.393	0.388	0.405	0.0%	0.0%
33	MSeg_endcap_rms_dR	0.384	0.457	0.180	0.594	0.0%	0.0%
34	MSeg_endcap_rms_dR_raw	0.384	0.457	0.180	0.594	0.0%	0.0%
35	htmiss_x_NOSYS	0.383	0.398	0.354	0.438	0.0%	0.0%
36	MS1Vtx_barrel_clusE	0.383	0.386	0.372	0.396	0.0%	0.0%
37	MS1Vtx_clusE_raw	0.383	0.386	0.372	0.396	0.0%	0.0%
38	MS1Vtx_endcap_clusE	0.383	0.386	0.372	0.396	0.0%	0.0%
39	msvtx_caloClusterSumE	0.383	0.386	0.372	0.396	0.0%	0.0%
40	summedCaloClusters/clusterSumE	0.383	0.386	0.372	0.396	0.0%	0.0%

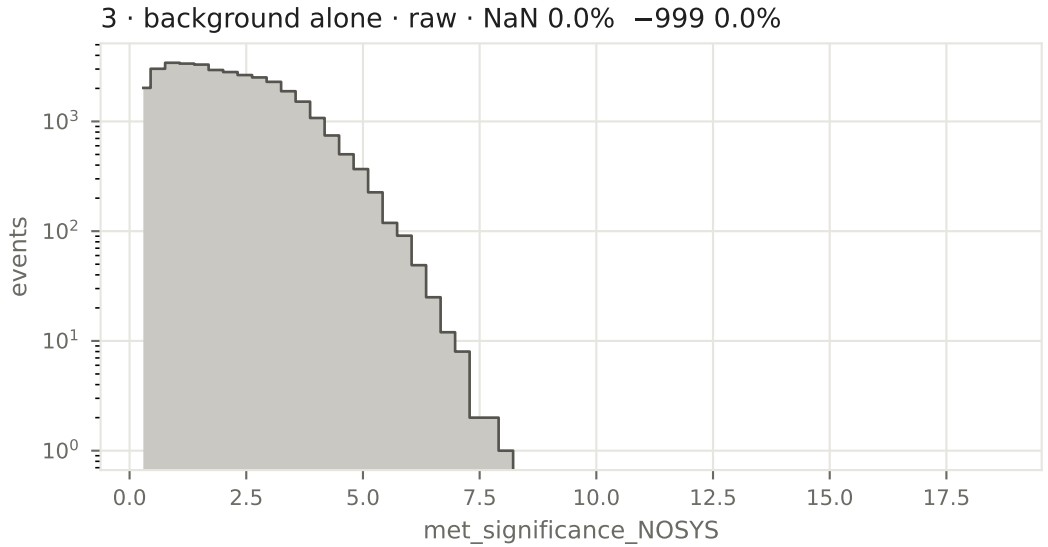
met_significance_NOSYS · region: endcap · rank 1/191 by combined equal-weight D · round 1 · edge bins include under/overflow

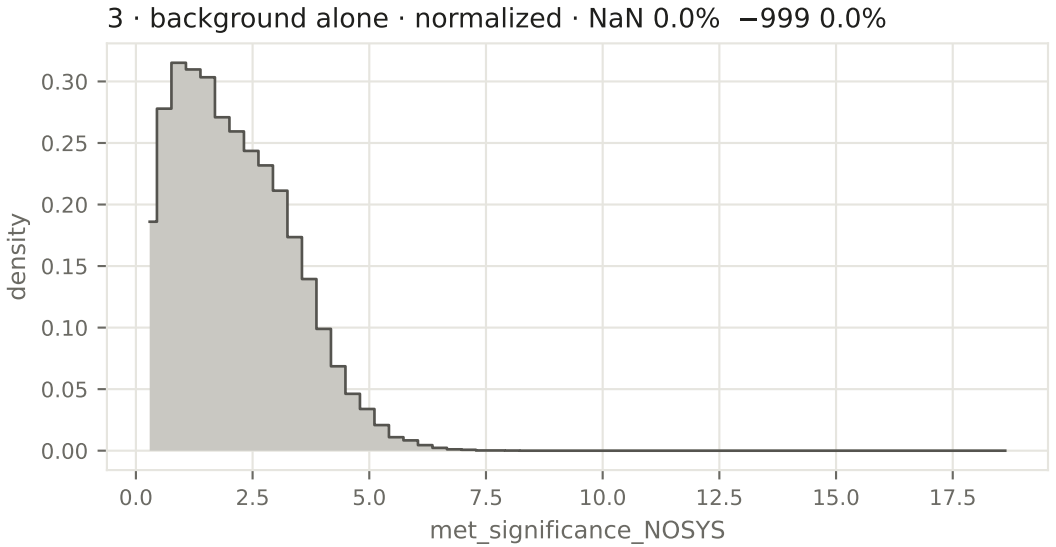






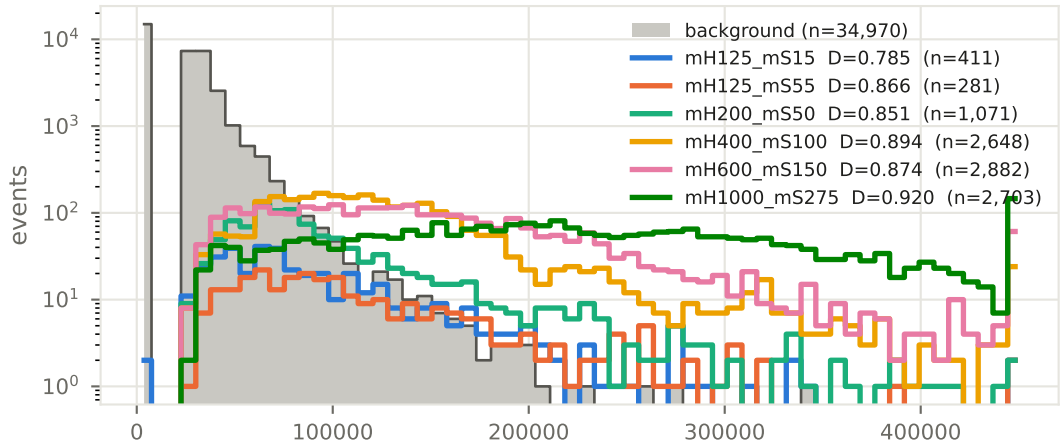




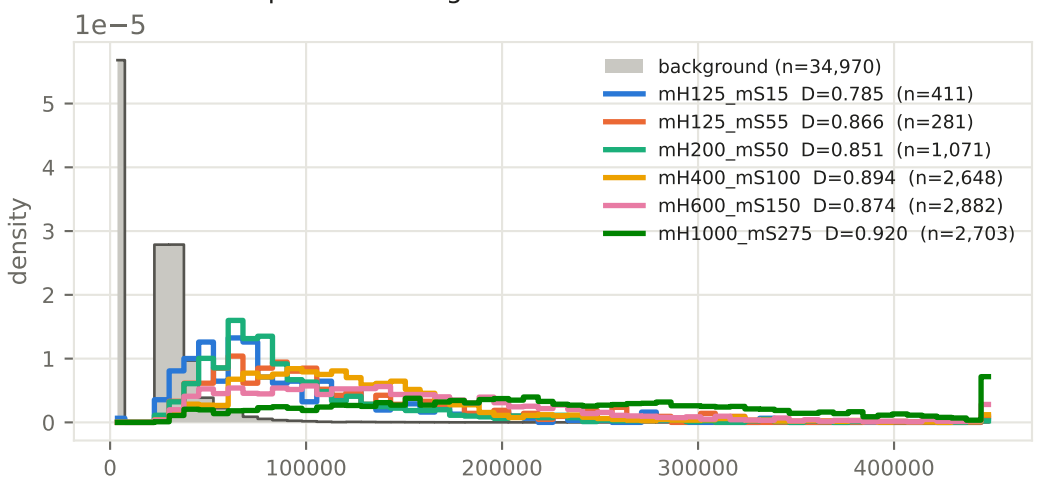


jet_pt_max_NOSYS · region: endcap · rank 2/191 by combined equal-weight D · round 1 · edge bins include under/overflow

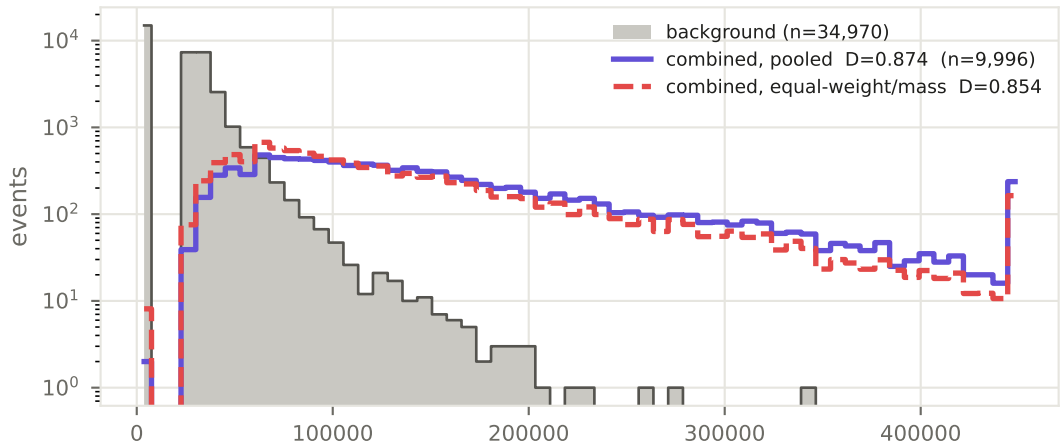
1 · each mass point vs background · raw



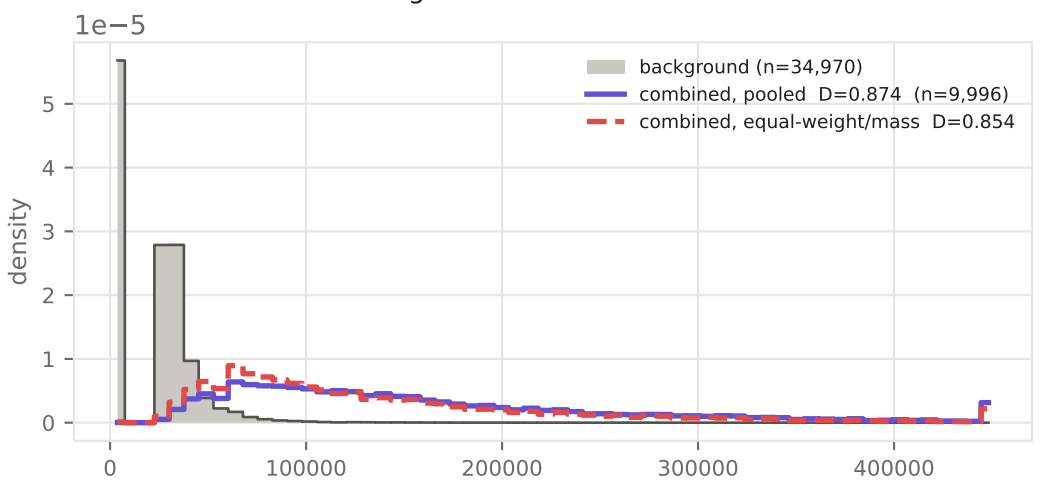
1 · each mass point vs background · normalized



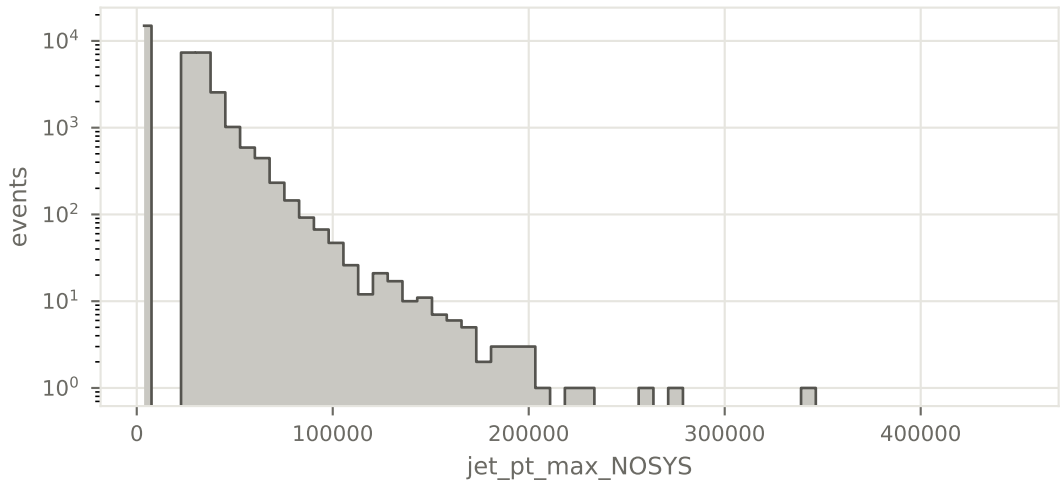
2 · combined MC vs background · raw



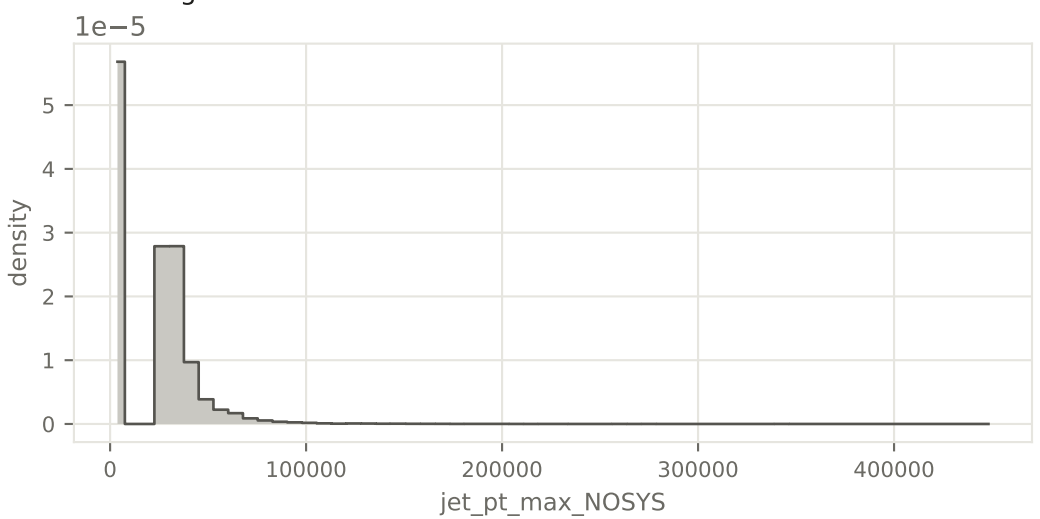
2 · combined MC vs background · normalized



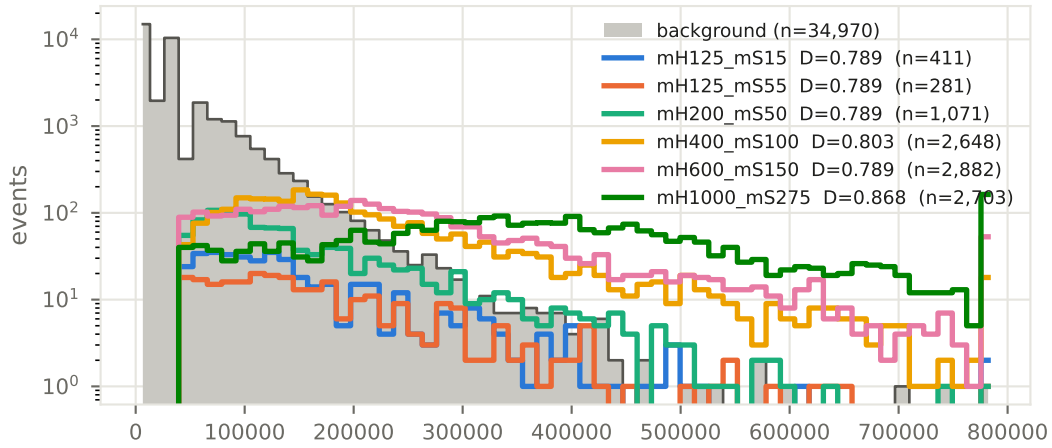
3 · background alone · raw · NaN 0.0% −999 0.0%

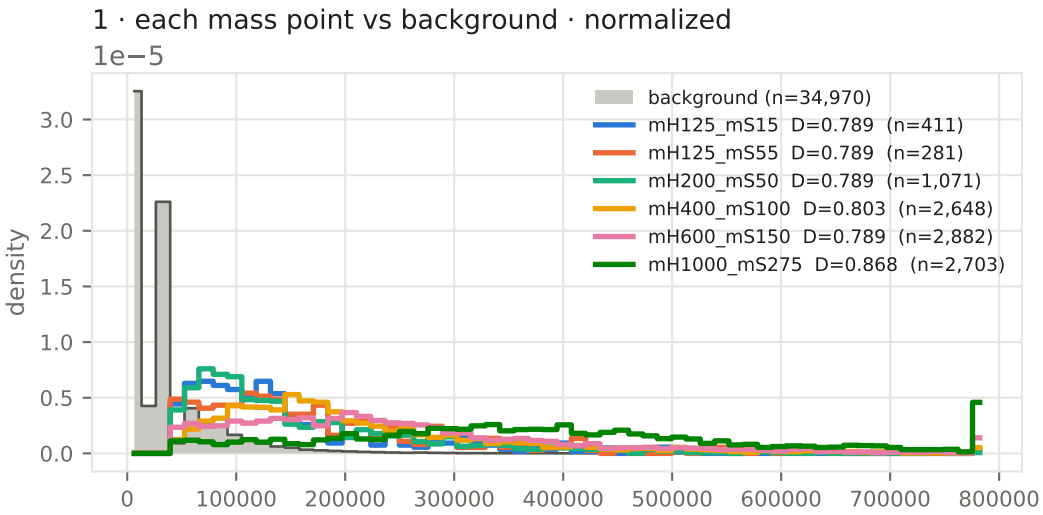


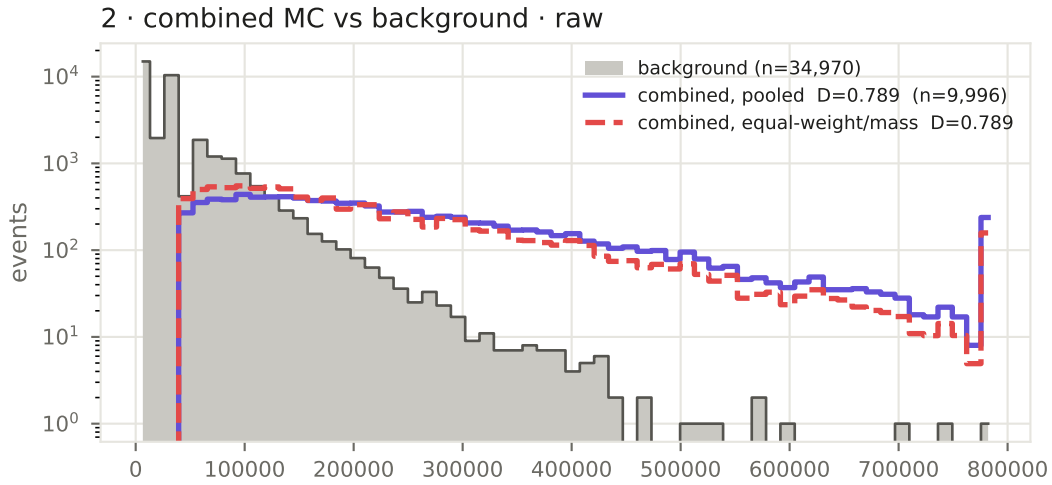
3 · background alone · normalized · NaN 0.0% −999 0.0%

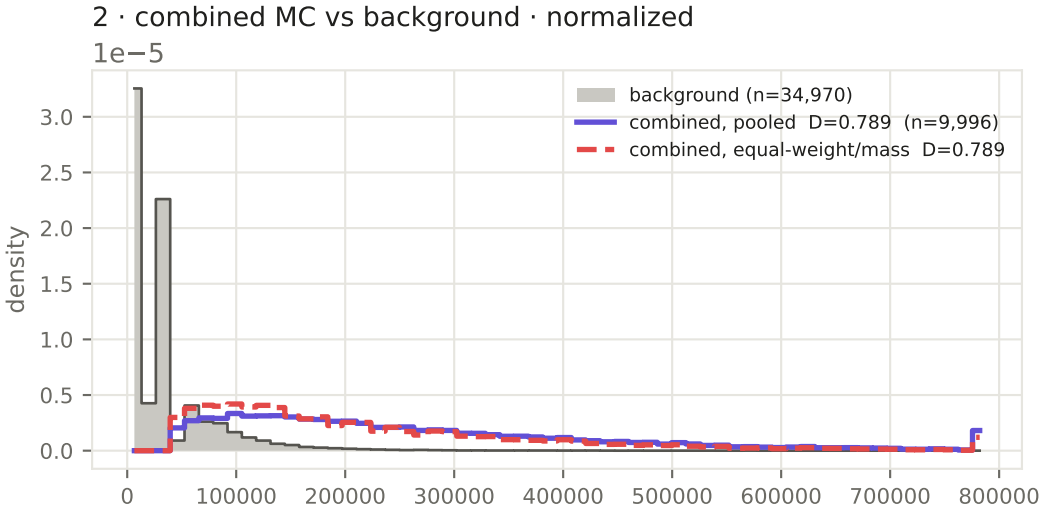


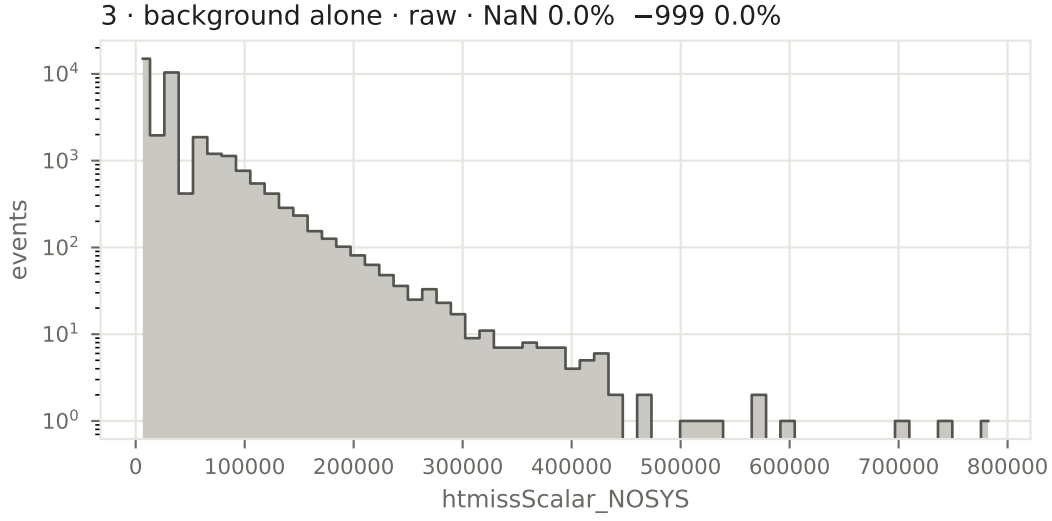
htmissScalar_NOSYS · region: endcap · rank 3/191 by combined equal-weight D · round 1 · edge bins include under/overflow

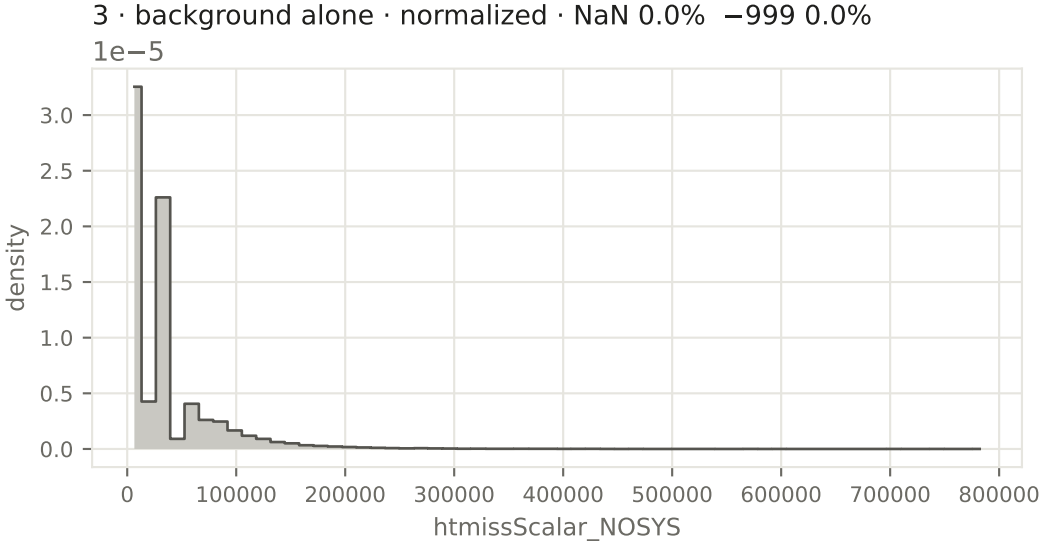




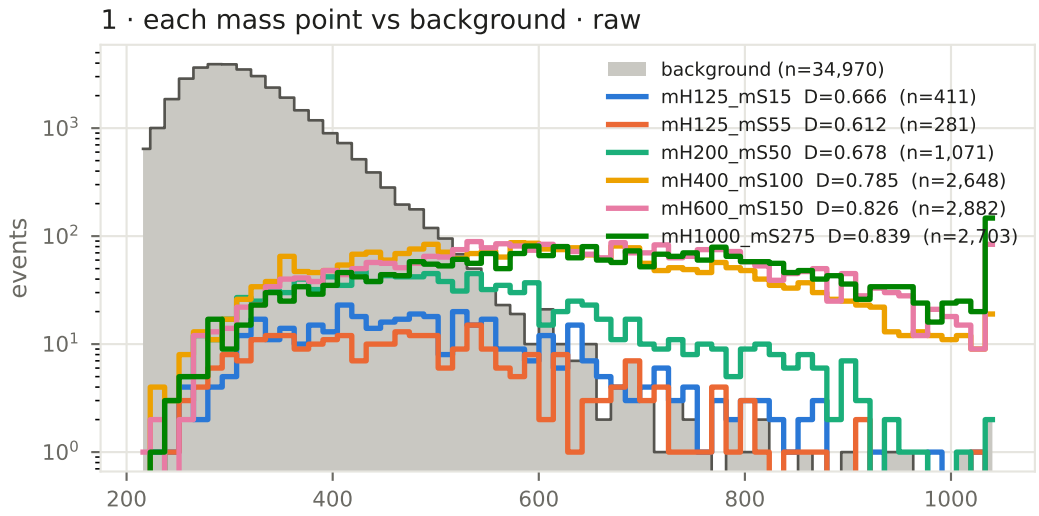


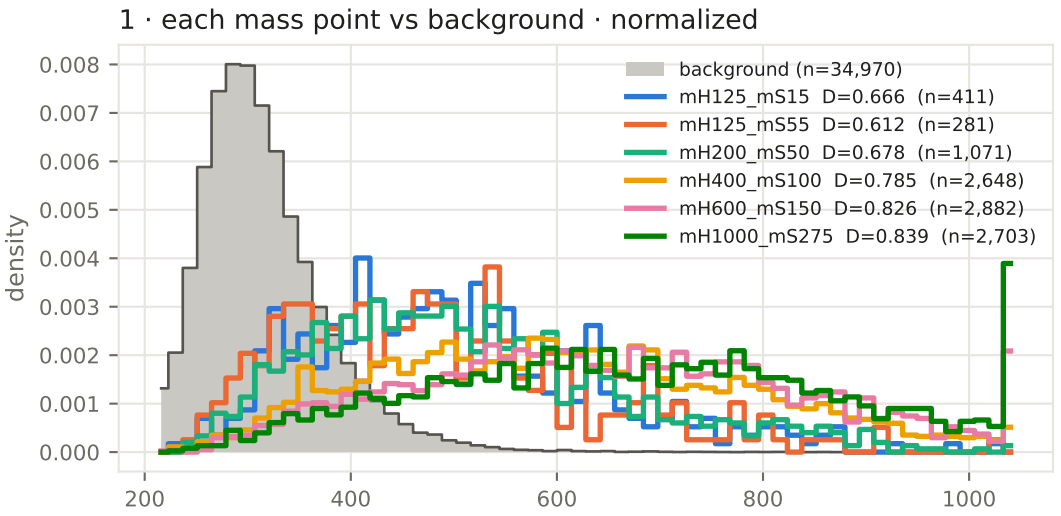


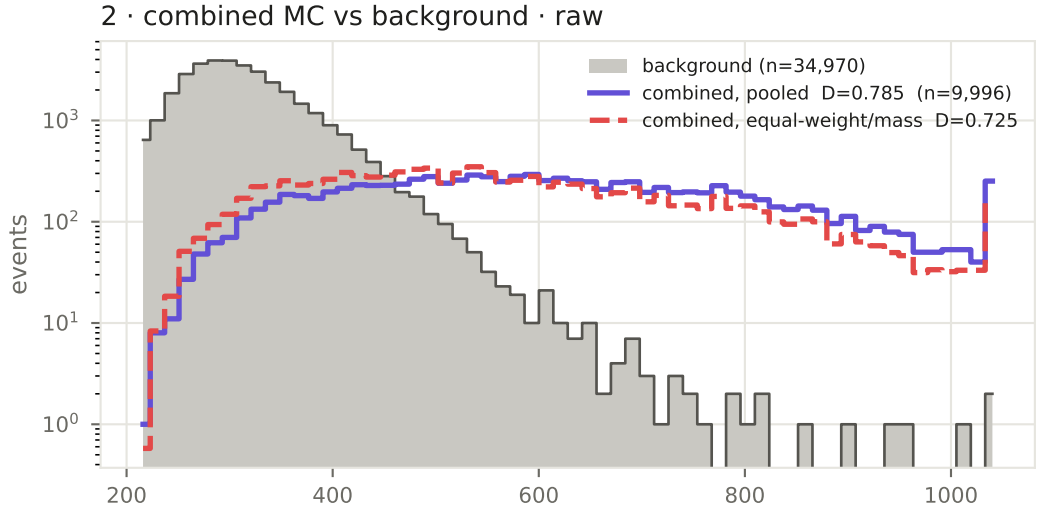


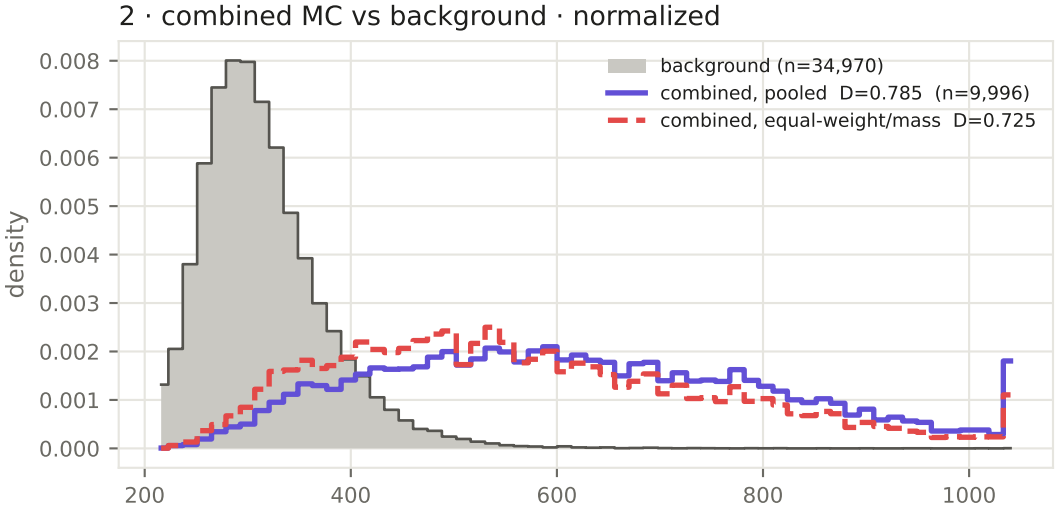


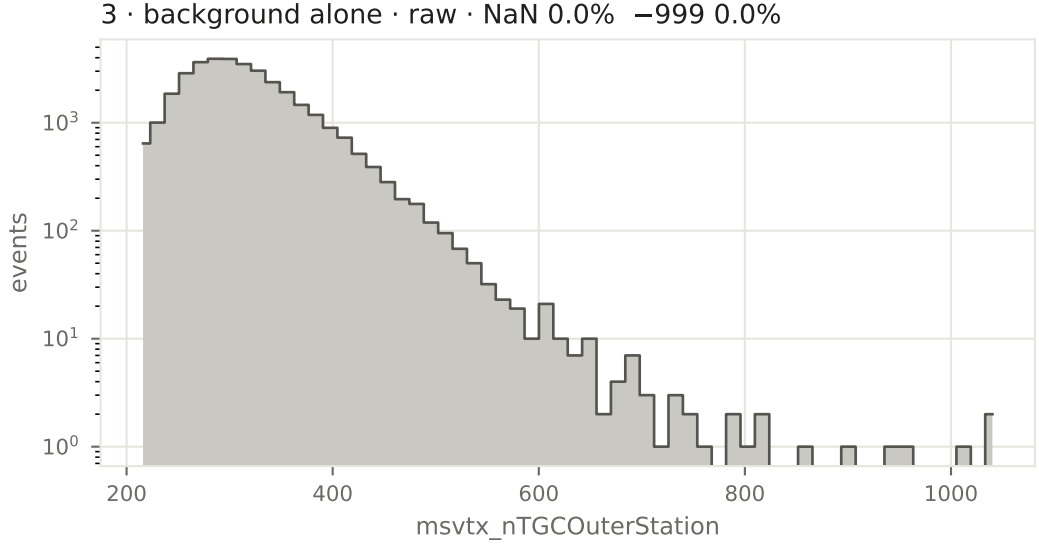
msvtx_nTGCOuterStation · region: endcap · rank 4/191 by combined equal-weight D · round 1 · edge bins include under/overflow

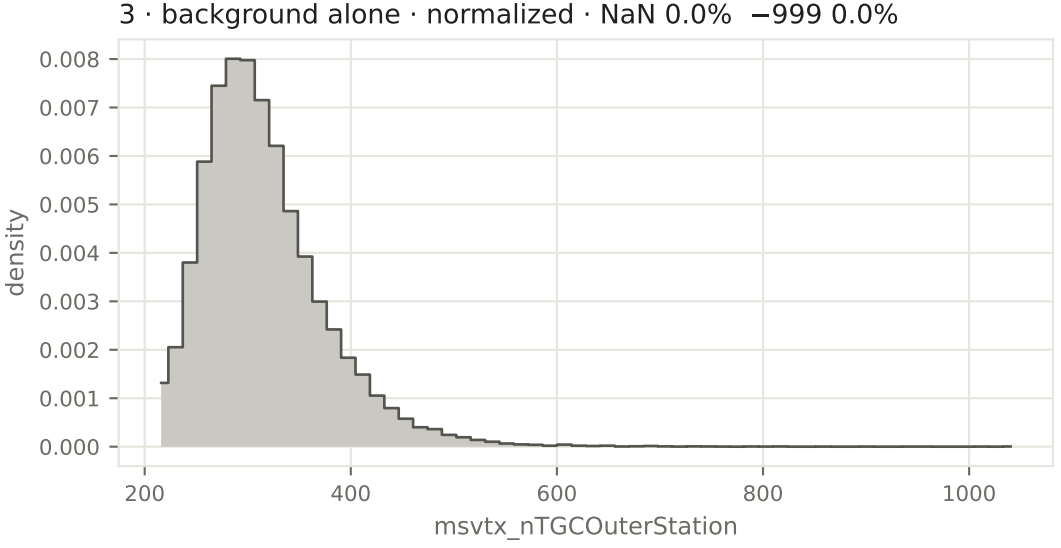




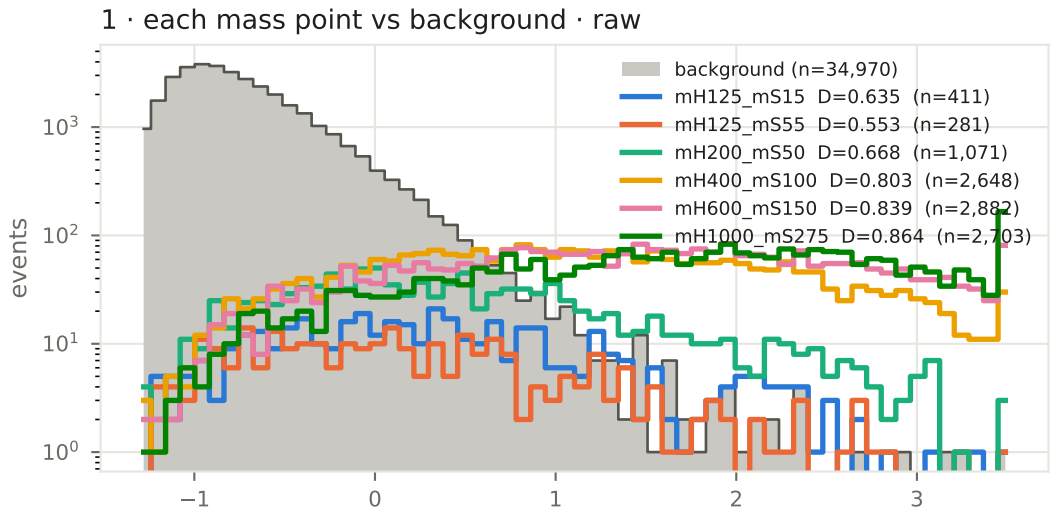


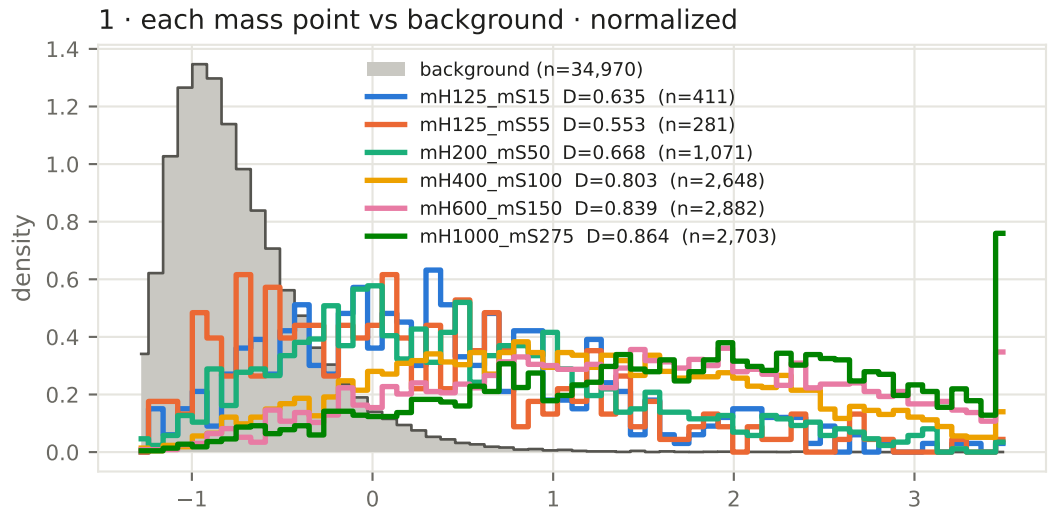


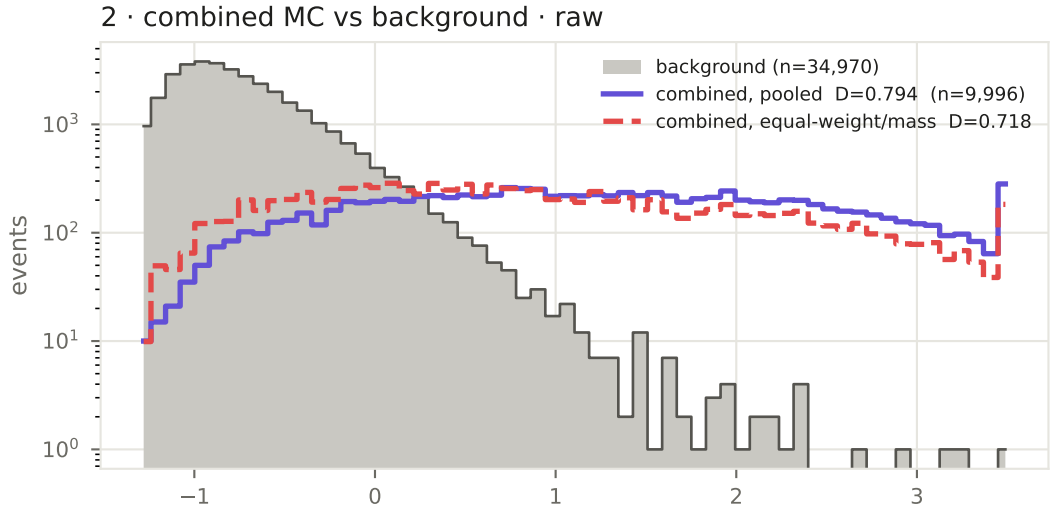


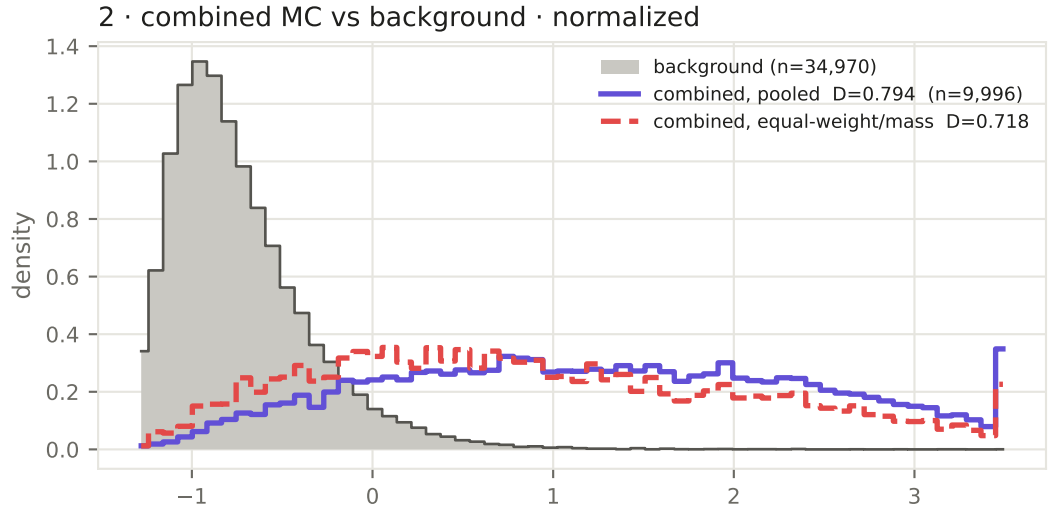


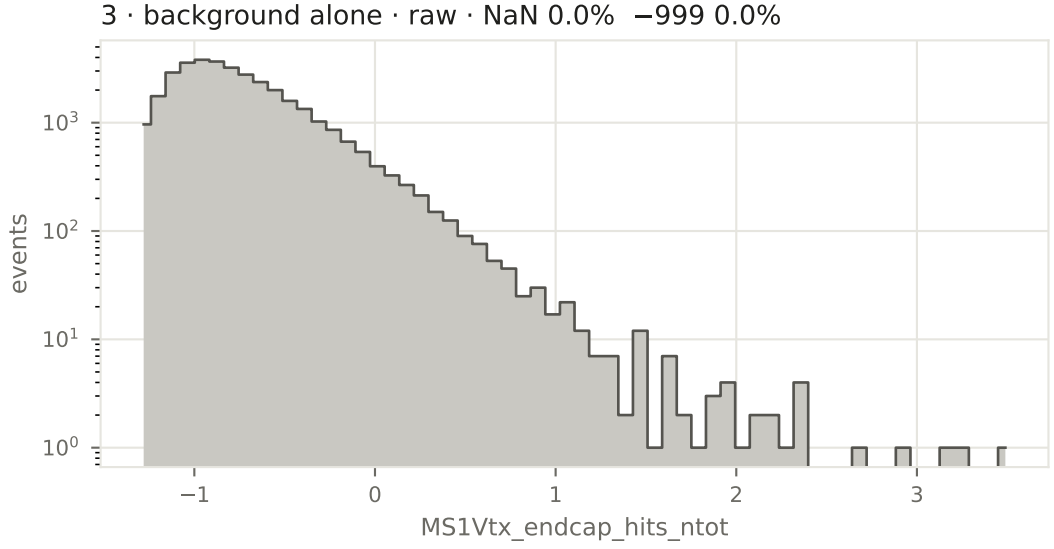
MS1Vtx_endcap_hits_ntot · region: endcap · rank 5/191 by combined equal-weight D · round 1 · edge bins include under/overflow

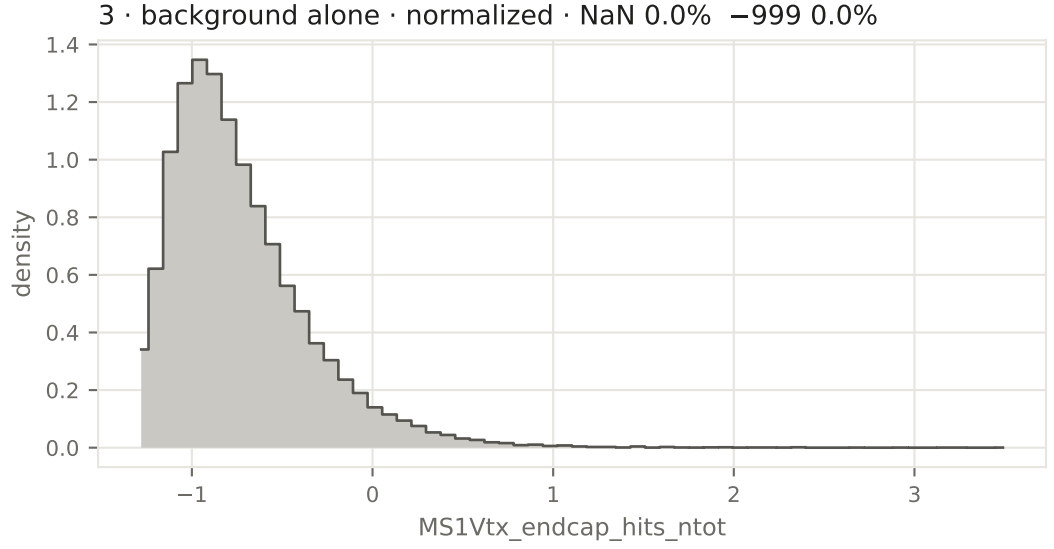




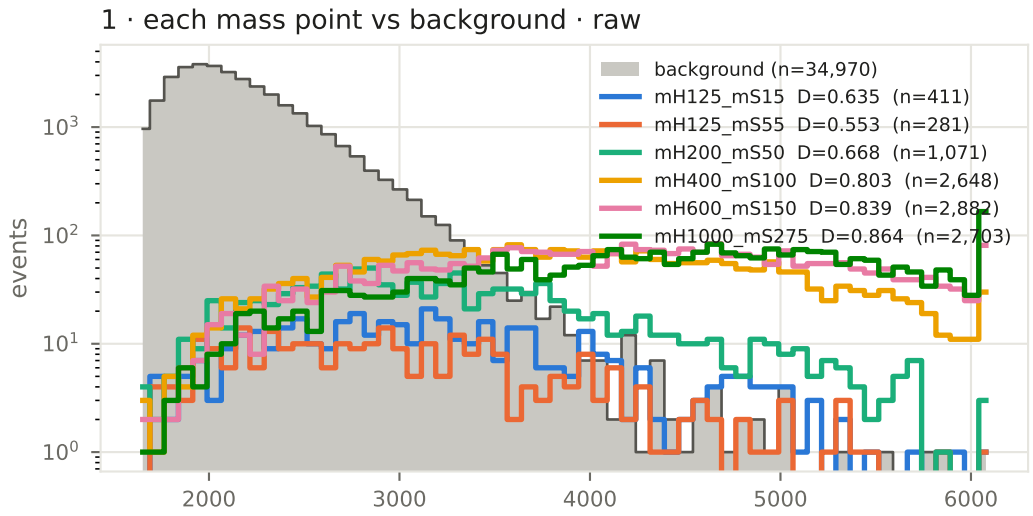


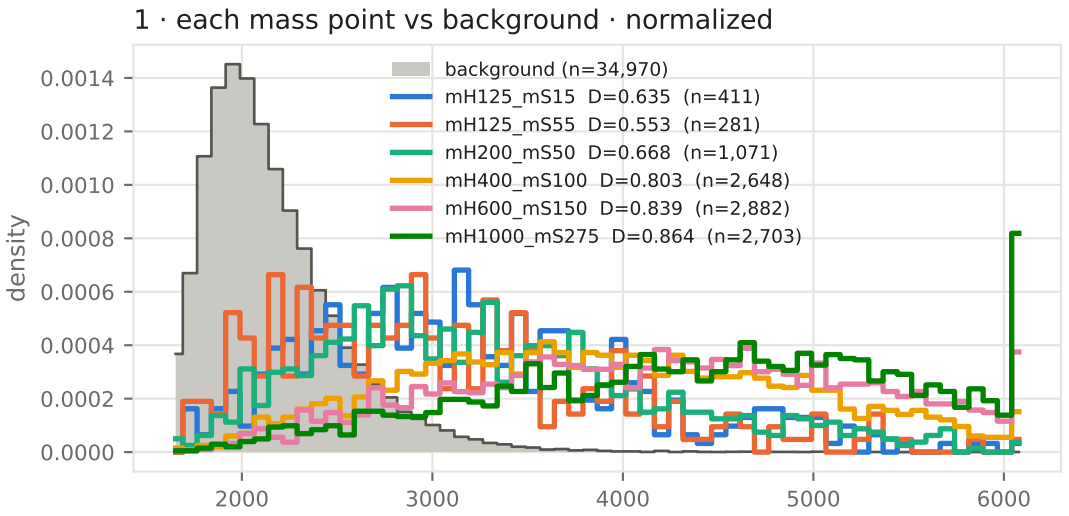


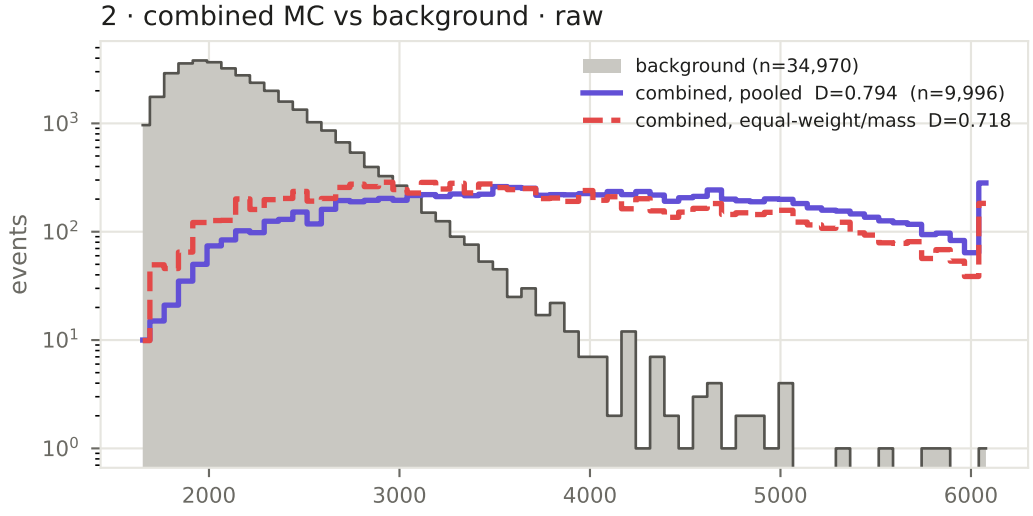


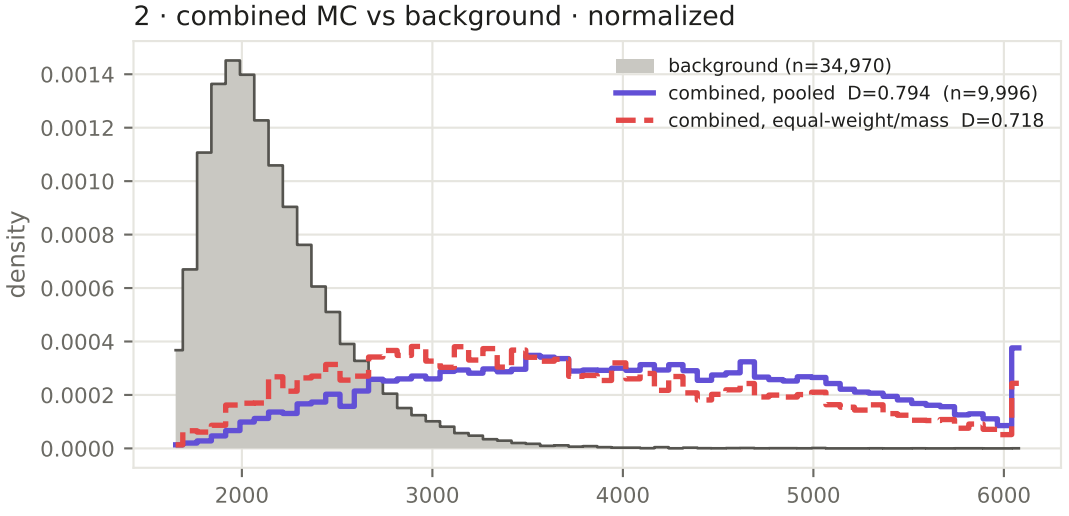


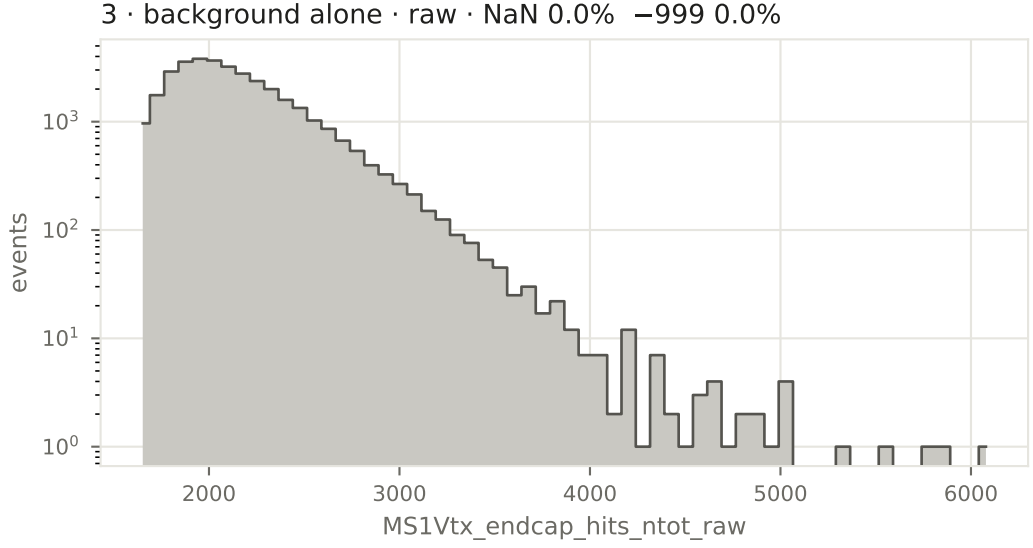
MS1Vtx_endcap_hits_ntot_raw · region: endcap · rank 6/191 by combined equal-weight D · round 1 · edge bins include under/overflow

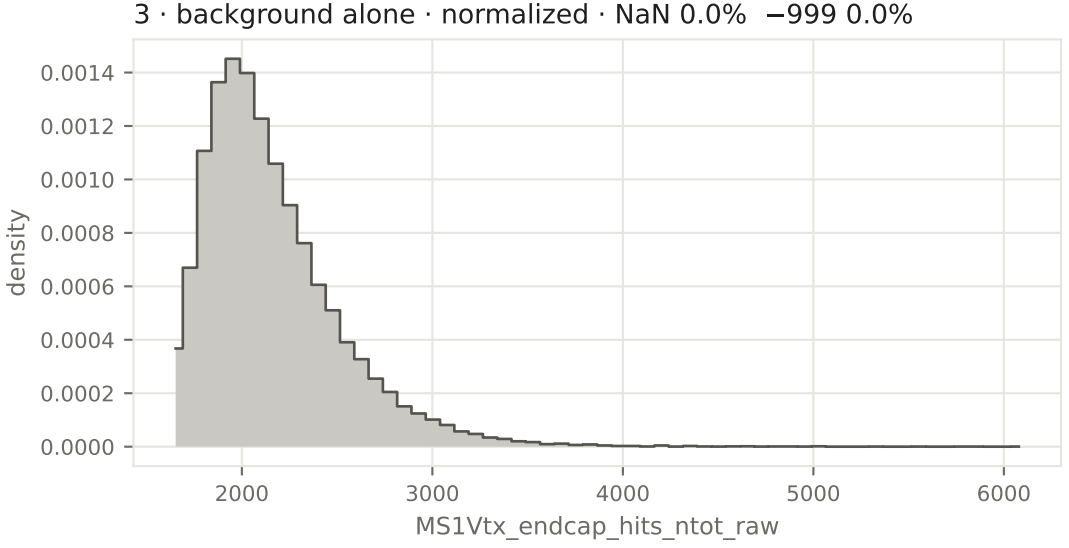




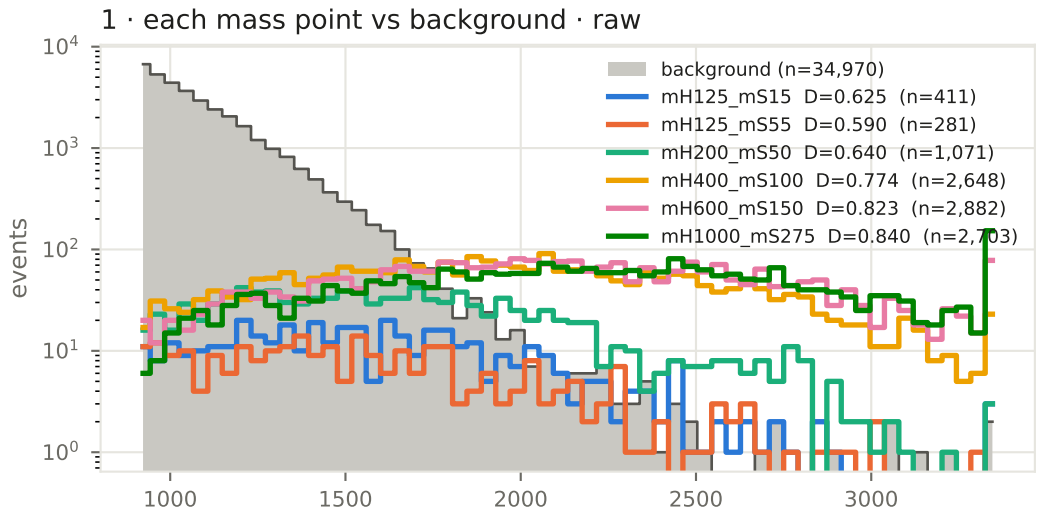


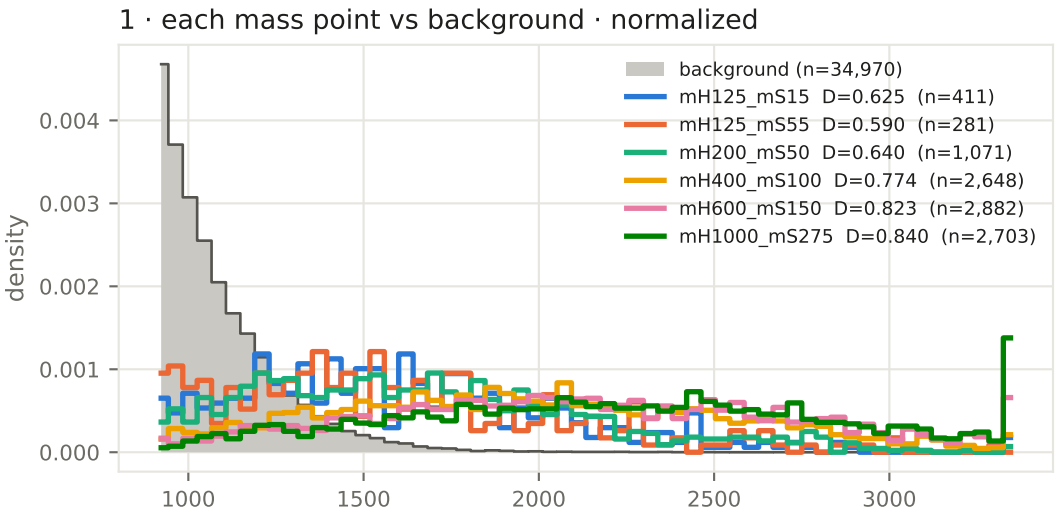


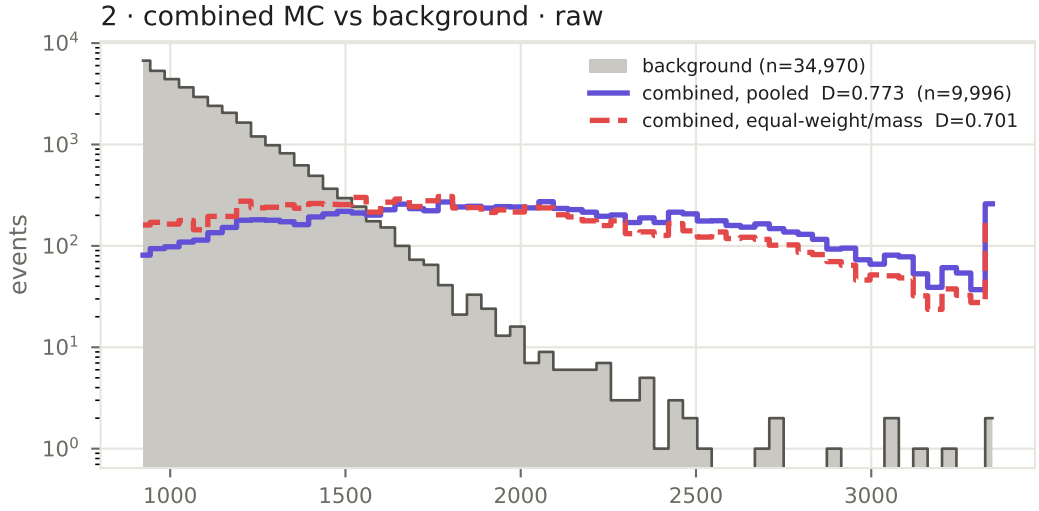


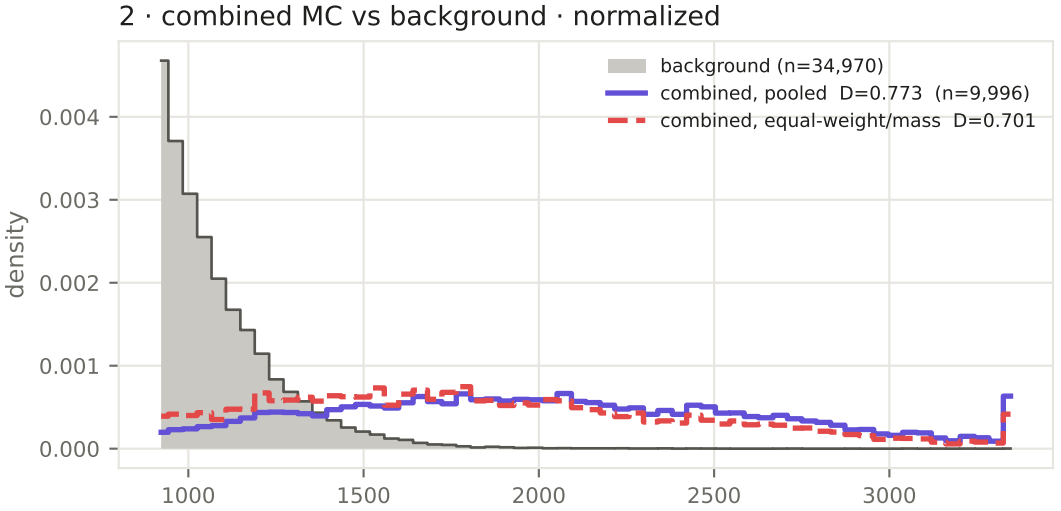


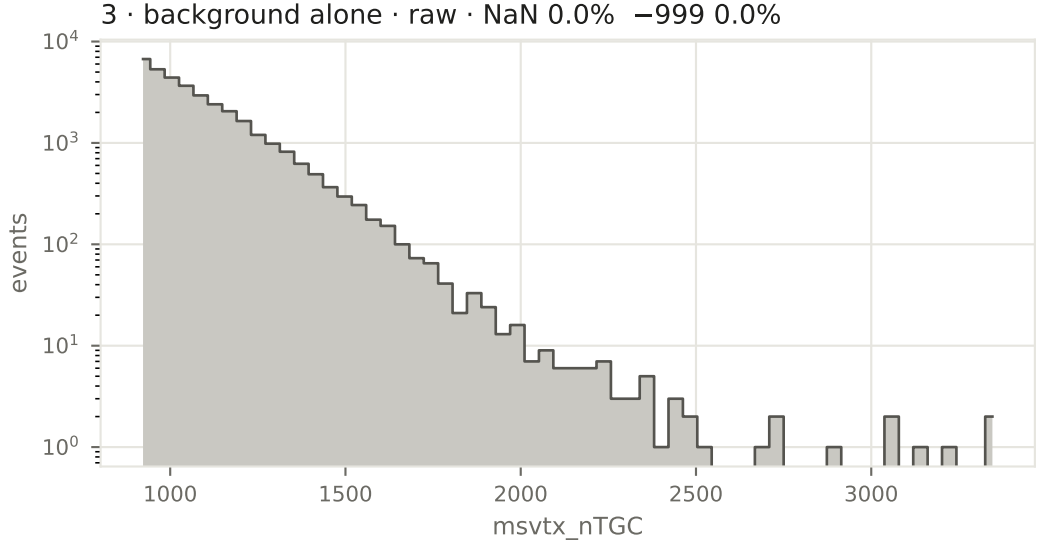
msvtx_nTGC · region: endcap · rank 7/191 by combined equal-weight D · round 1 · edge bins include under/overflow

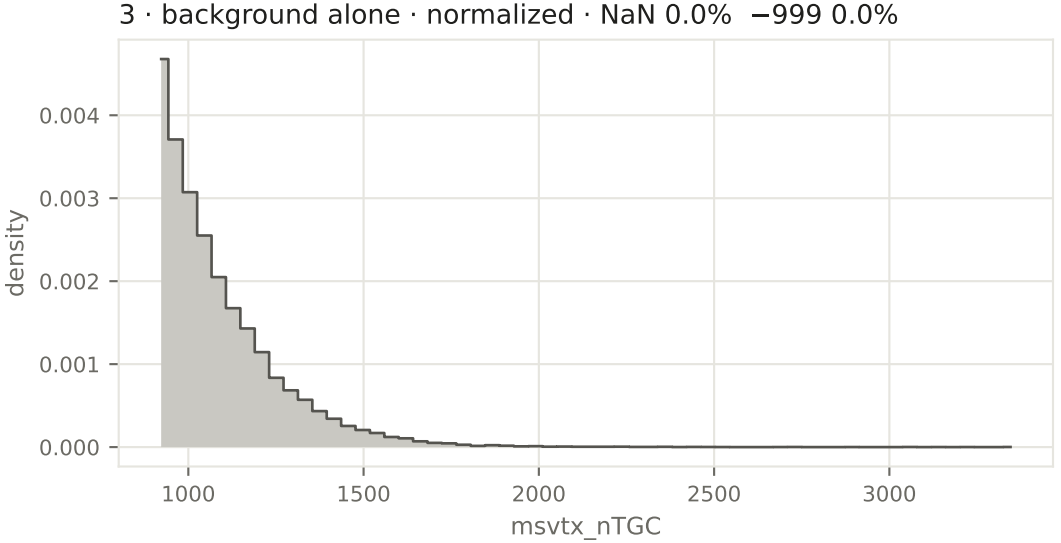






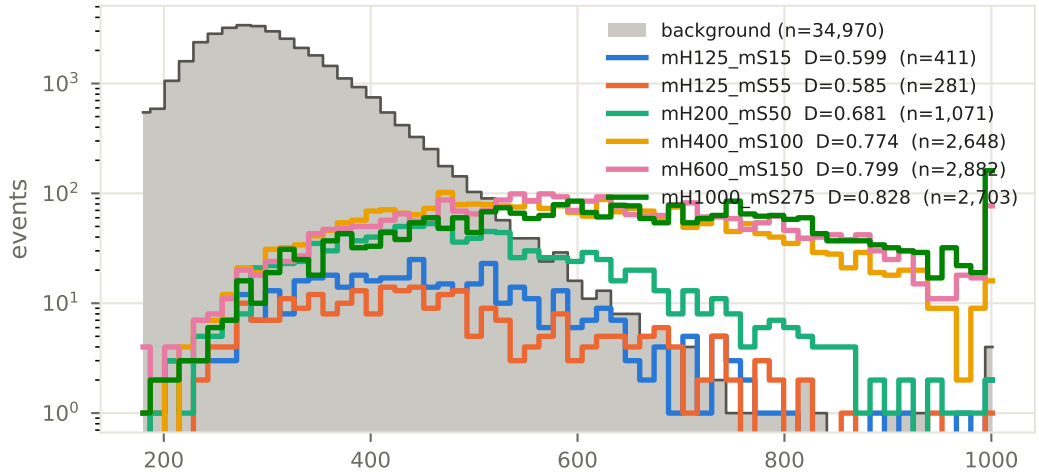




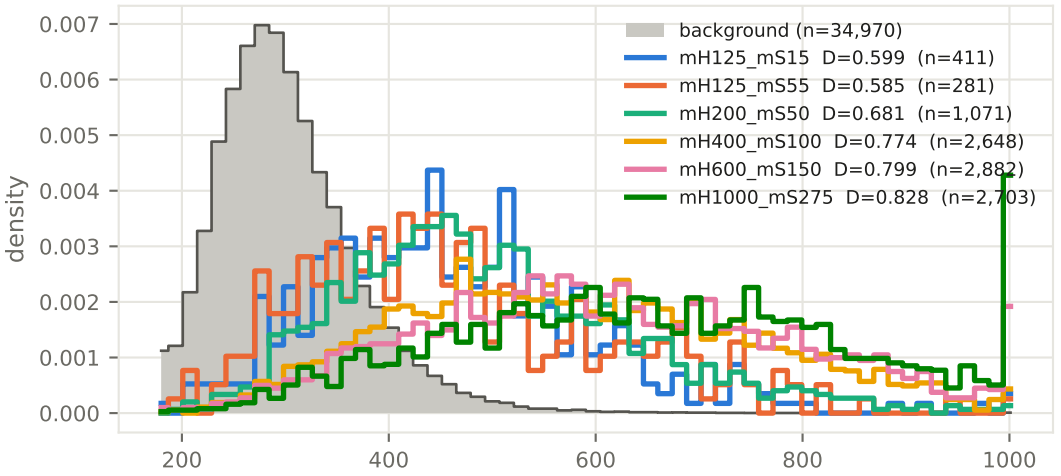


msvtx_nTGCExtendedStation · region: endcap · rank 8/191 by combined equal-weight D · round 1 · edge bins include under/overflow

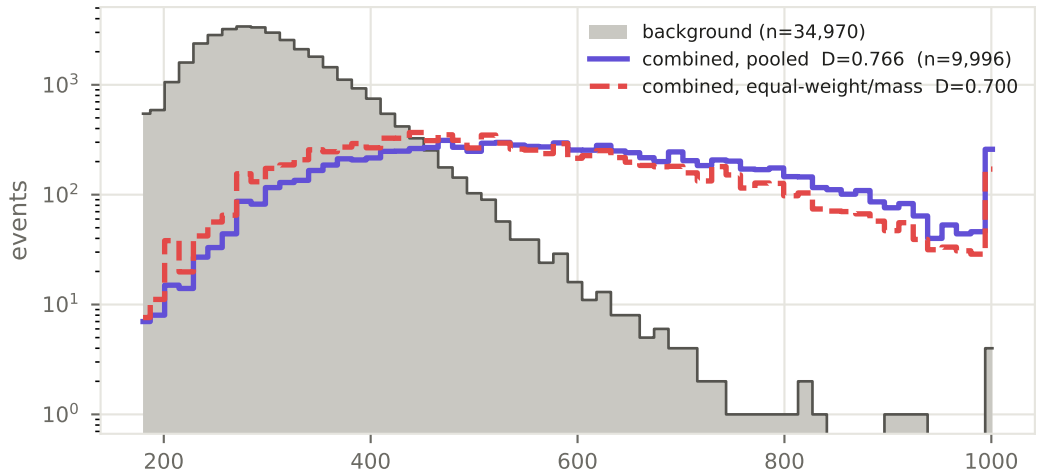
1 · each mass point vs background · raw



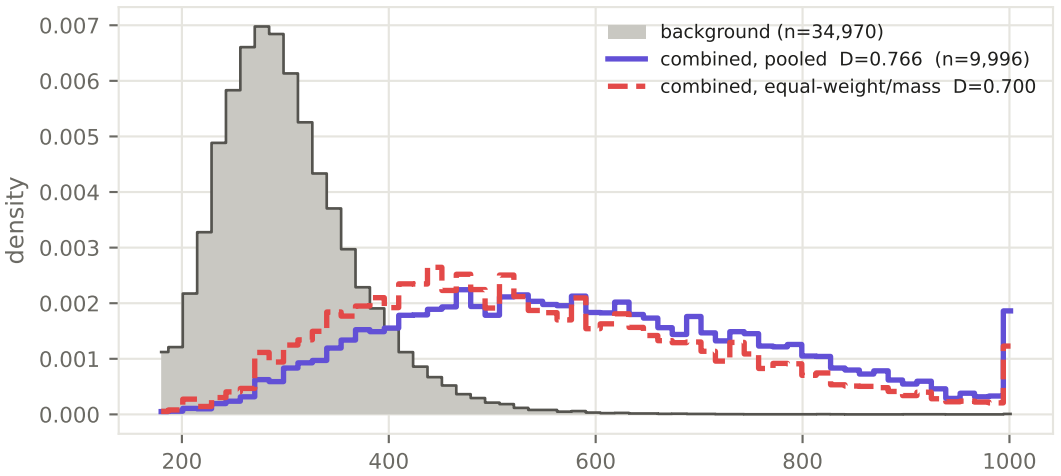
1 · each mass point vs background · normalized



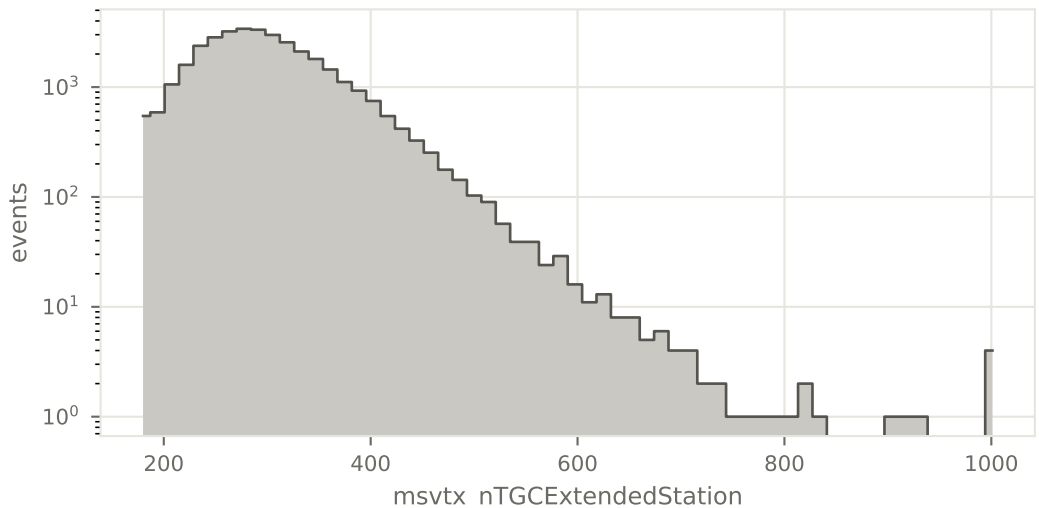
2 · combined MC vs background · raw



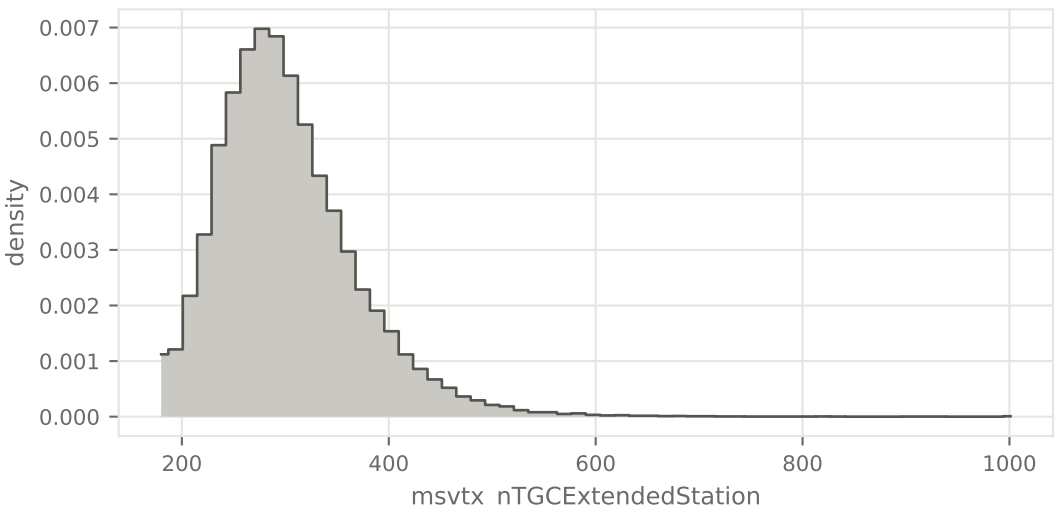
2 · combined MC vs background · normalized



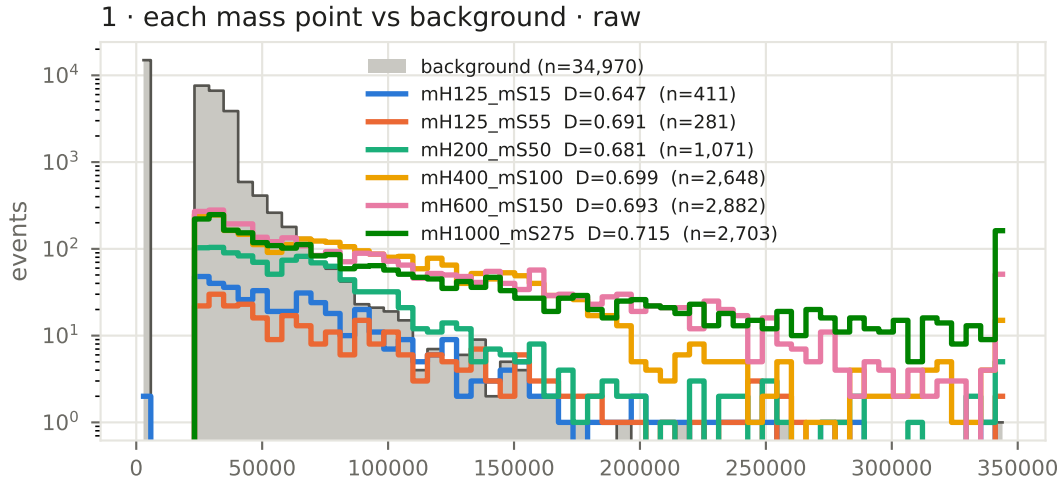
3 · background alone · raw · NaN 0.0% −999 0.0%

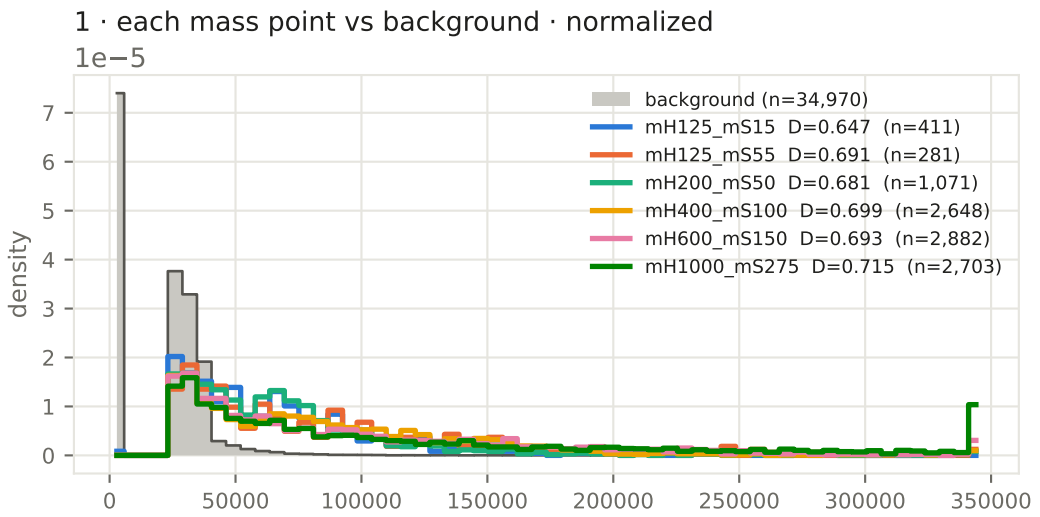


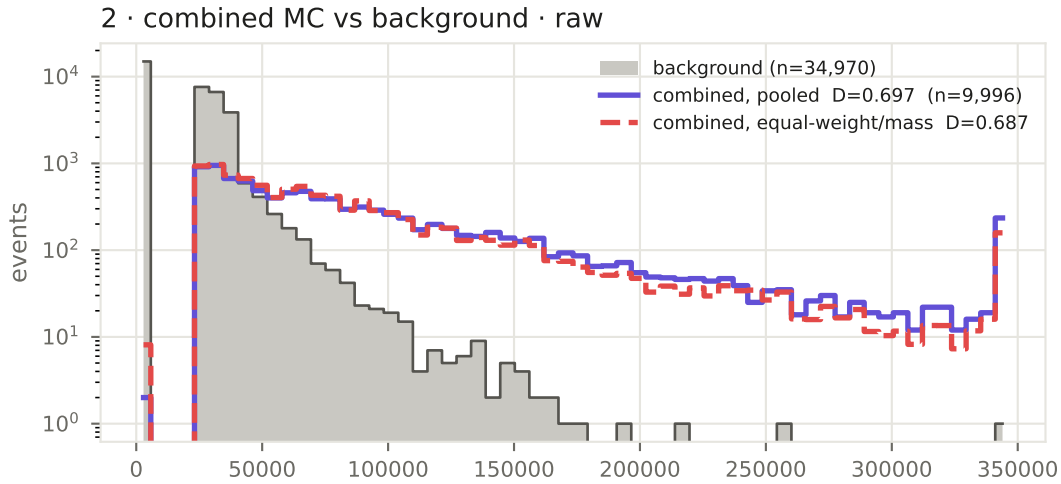
3 · background alone · normalized · NaN 0.0% −999 0.0%

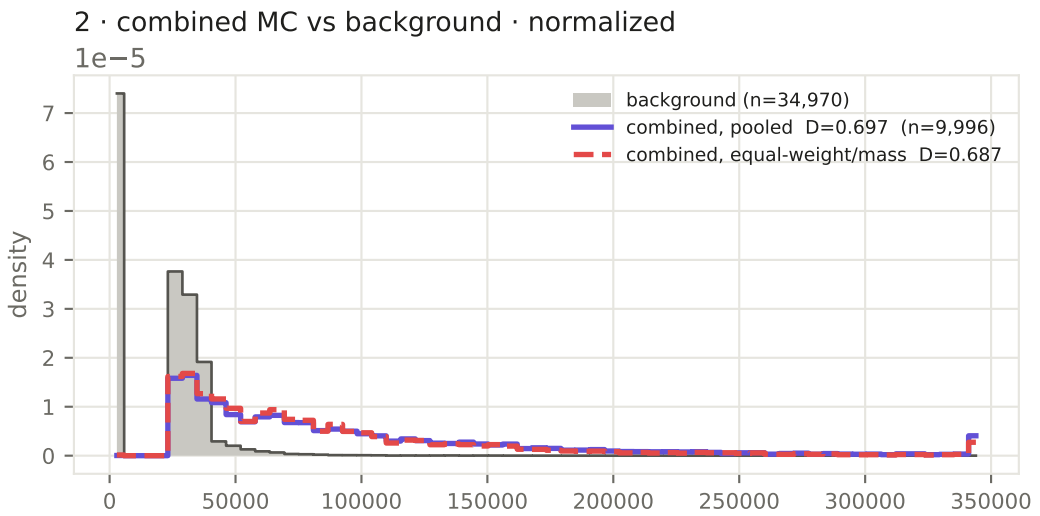


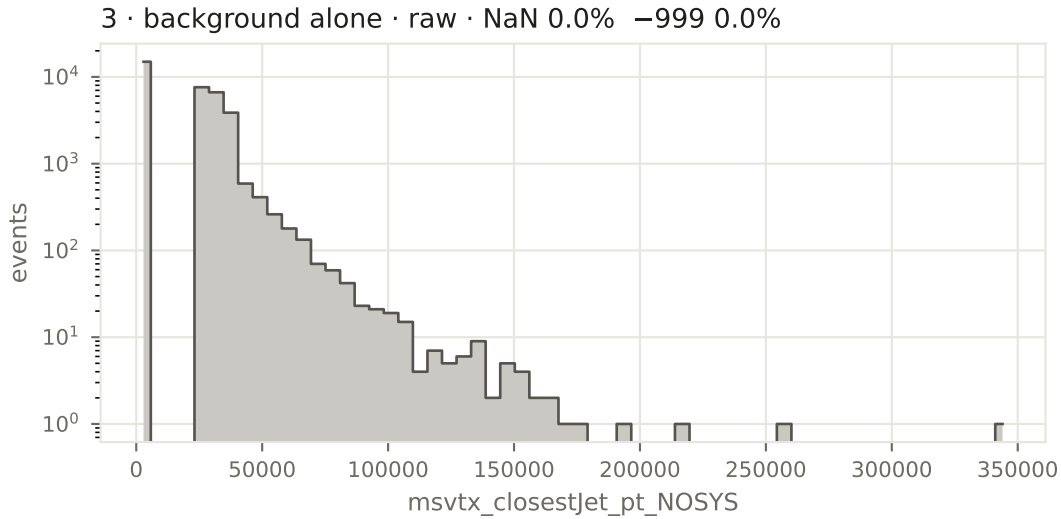
msvtx_closestJet_pt_NOSYS · region: endcap · rank 9/191 by combined equal-weight D · round 1 · edge bins include under/overflow

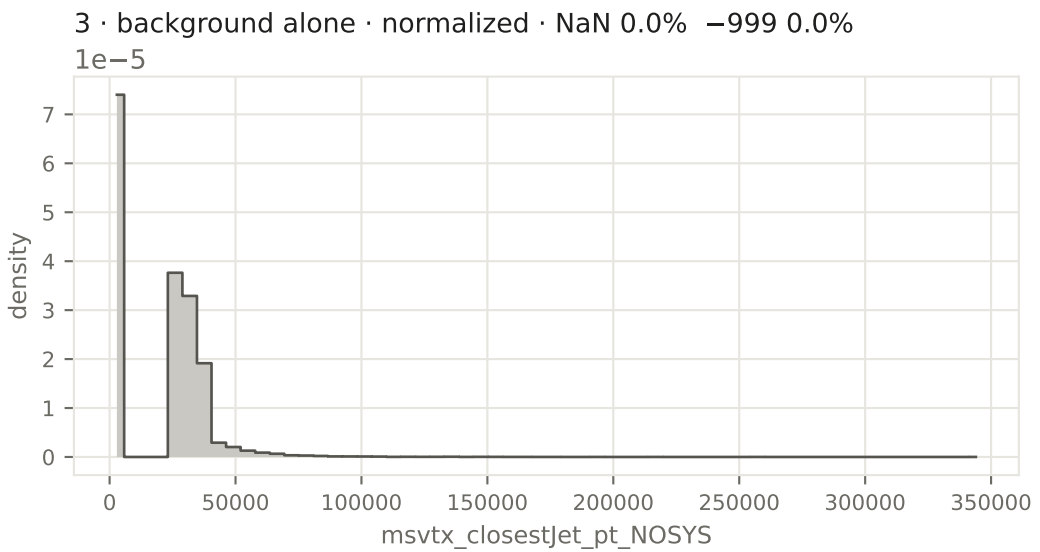




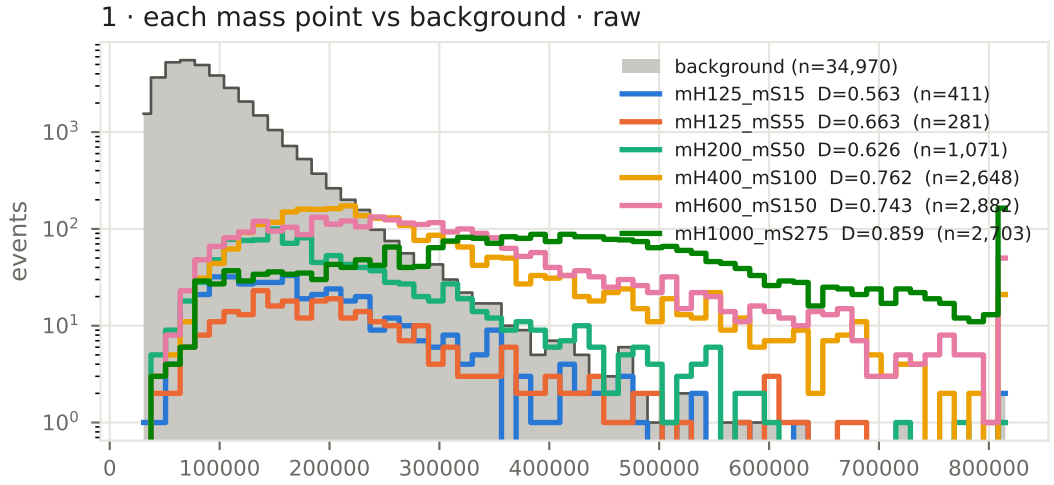


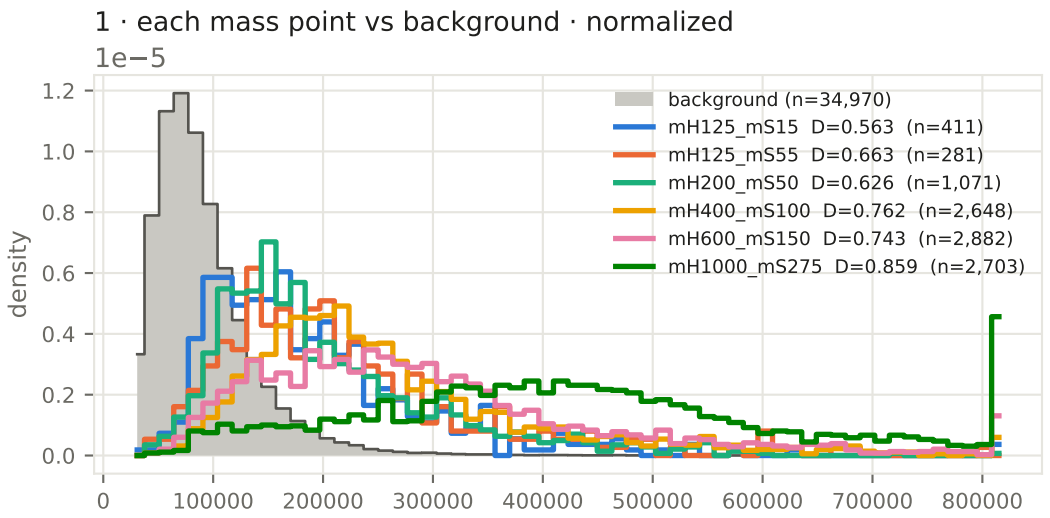


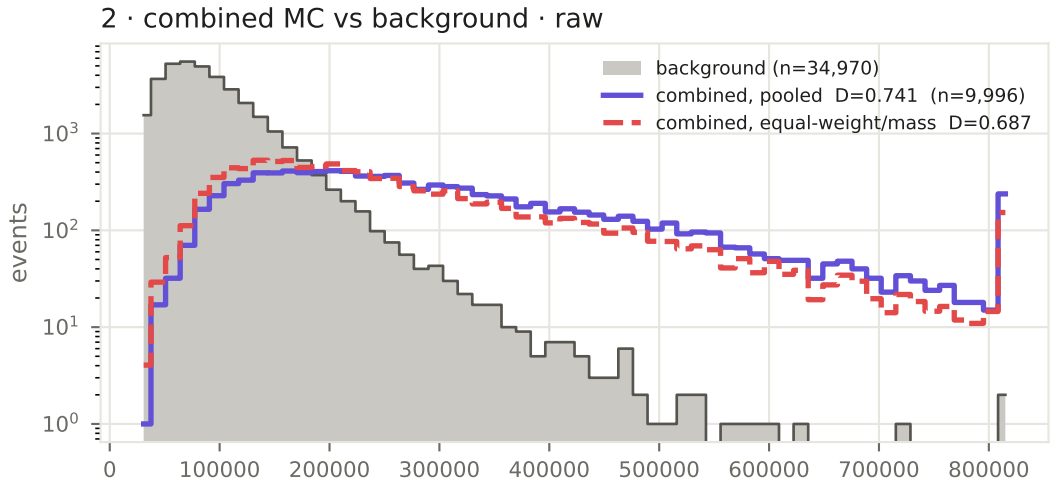


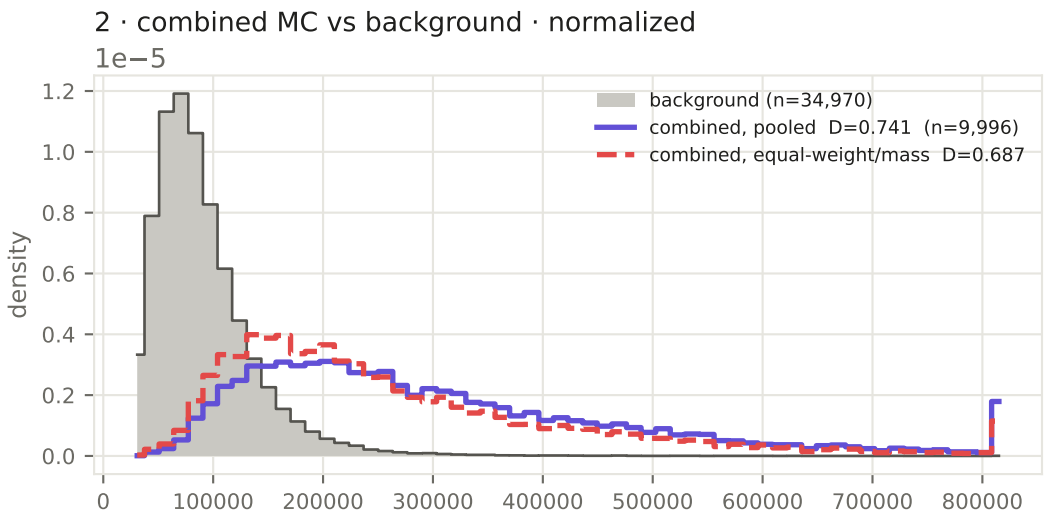


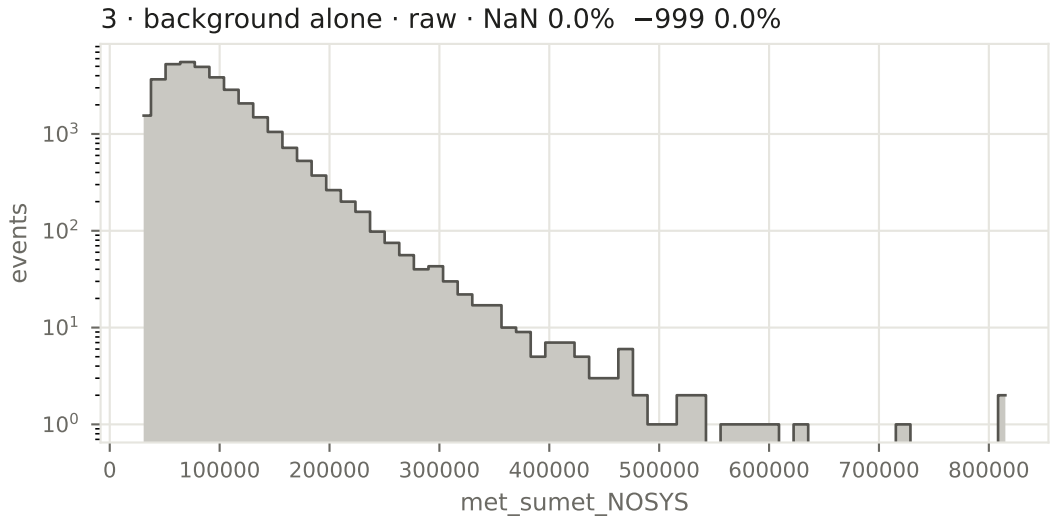
met_sumet_NOSYS · region: endcap · rank 10/191 by combined equal-weight D · round 1 · edge bins include under/overflow

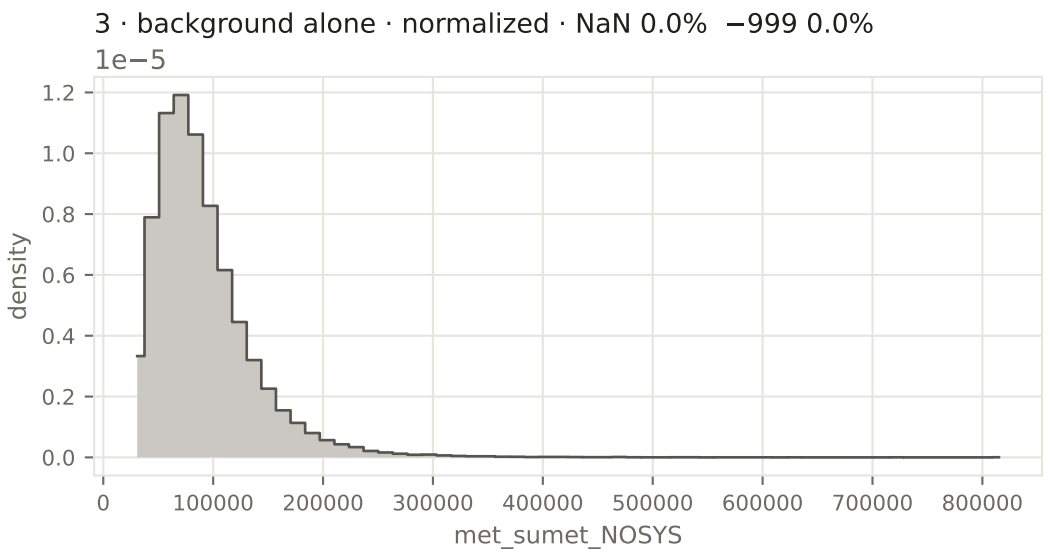




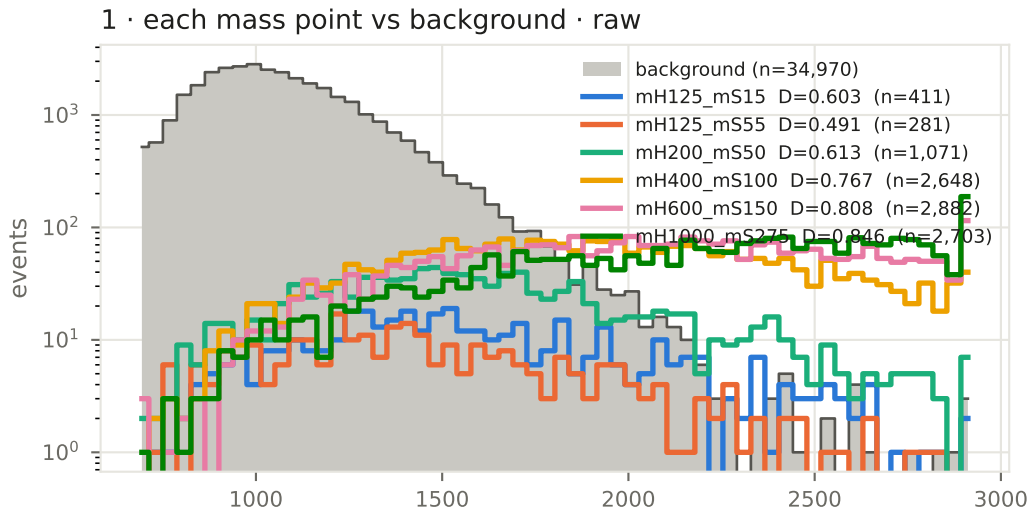


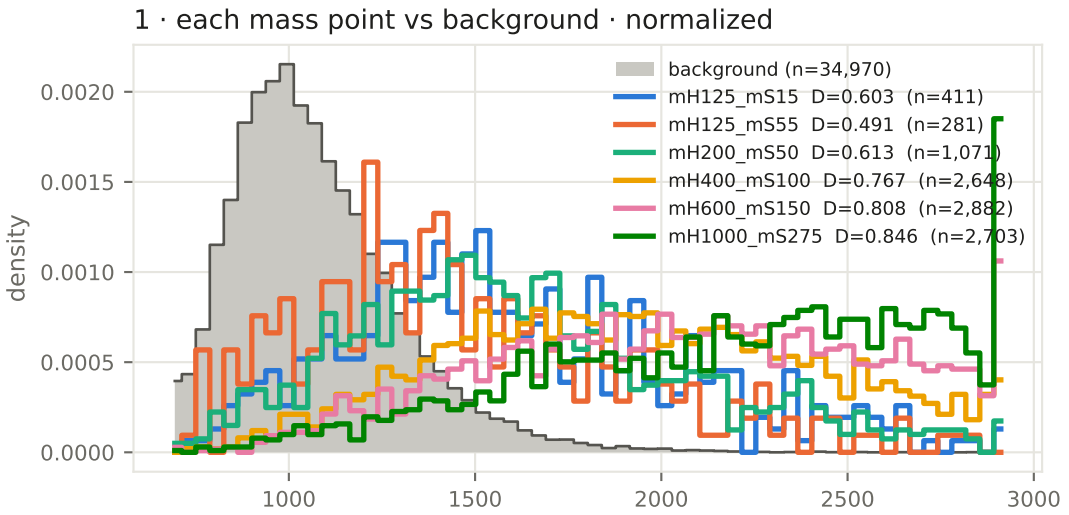


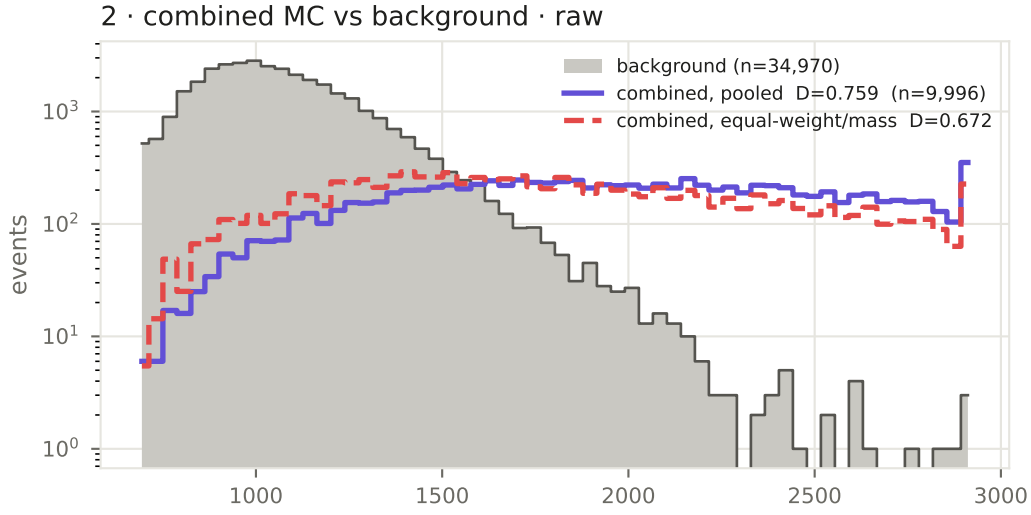


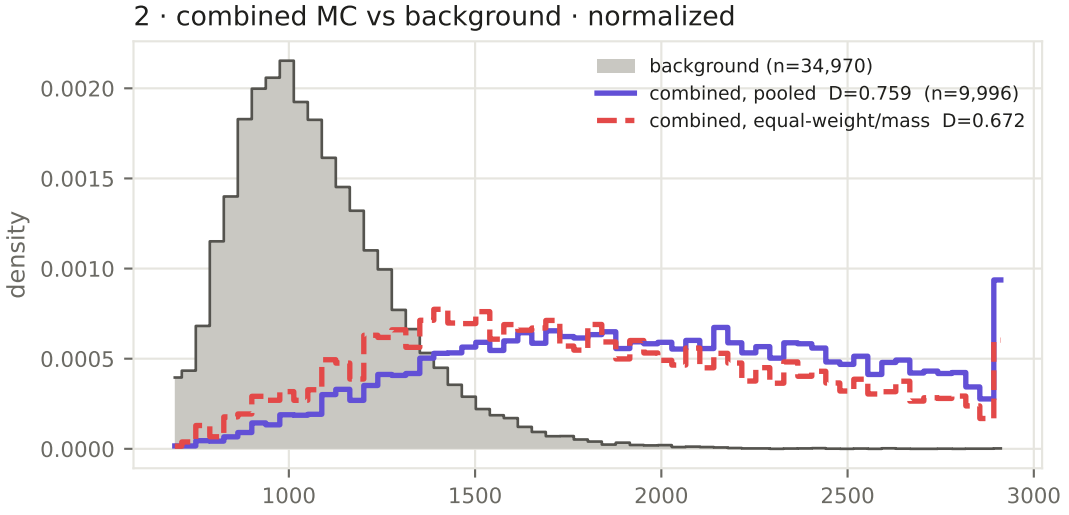


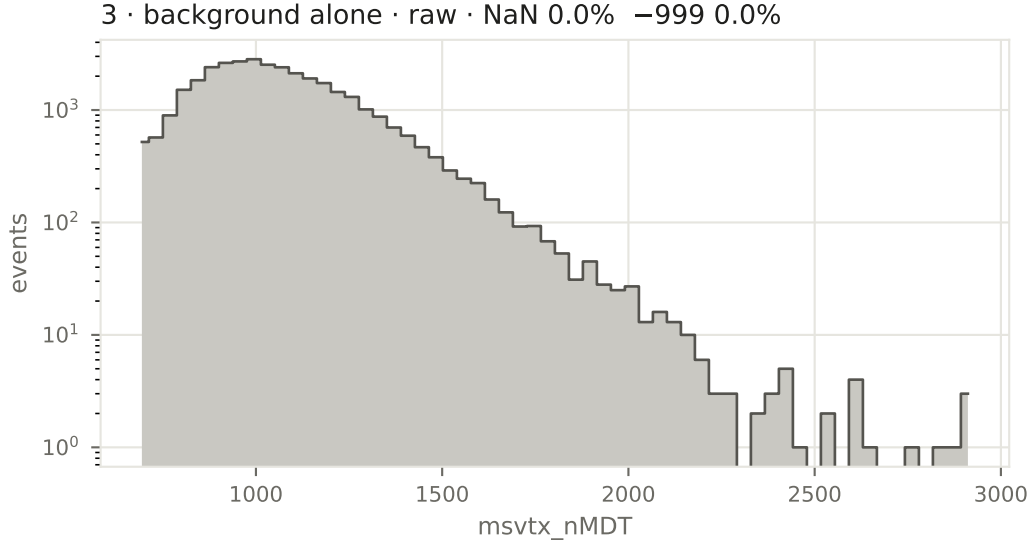
msvtx_nMDT · region: endcap · rank 11/191 by combined equal-weight D · round 1 · edge bins include under/overflow

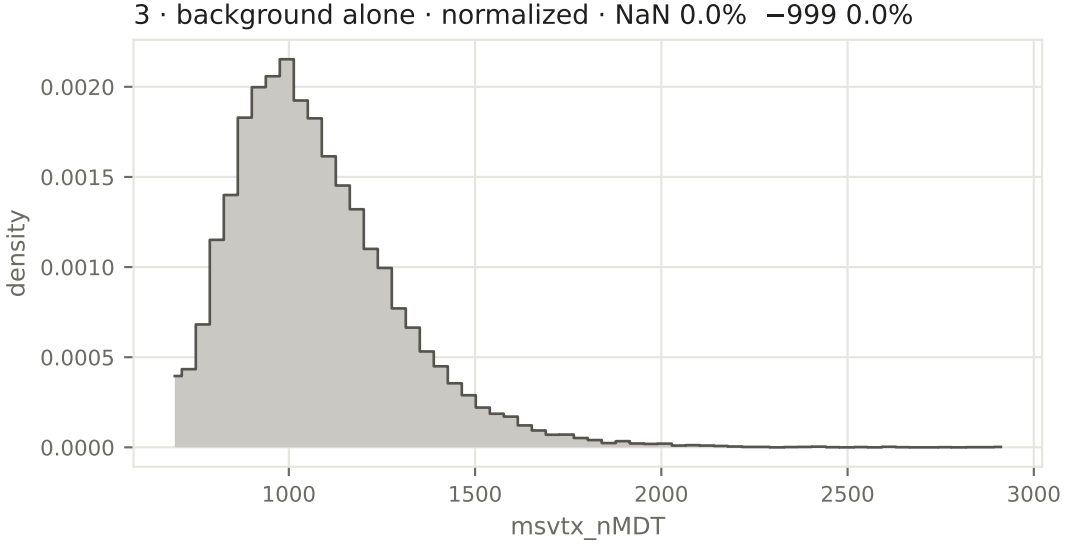




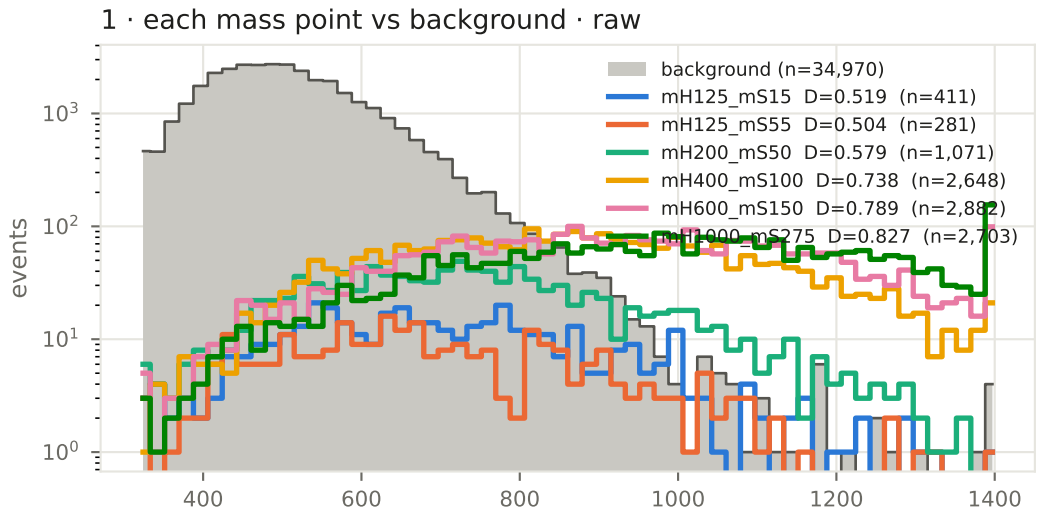


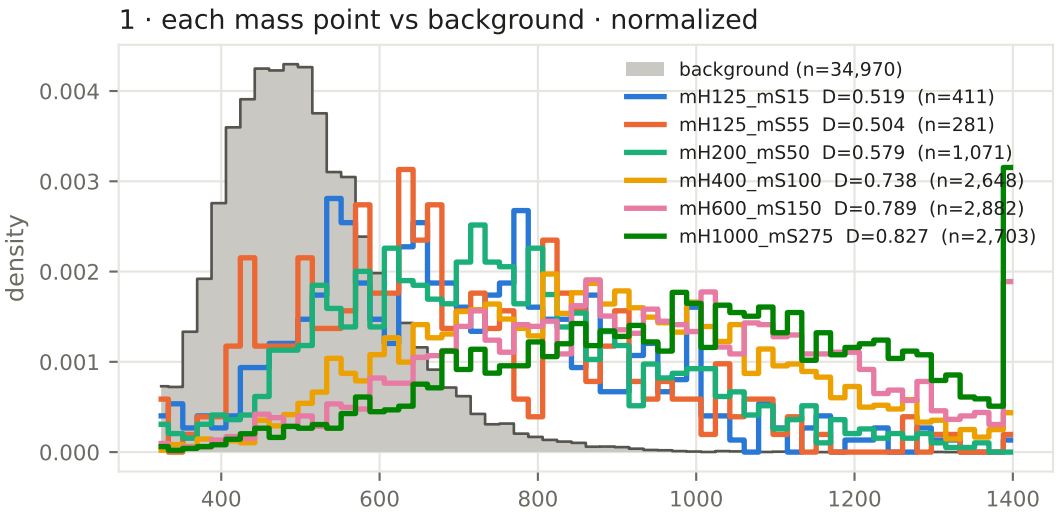


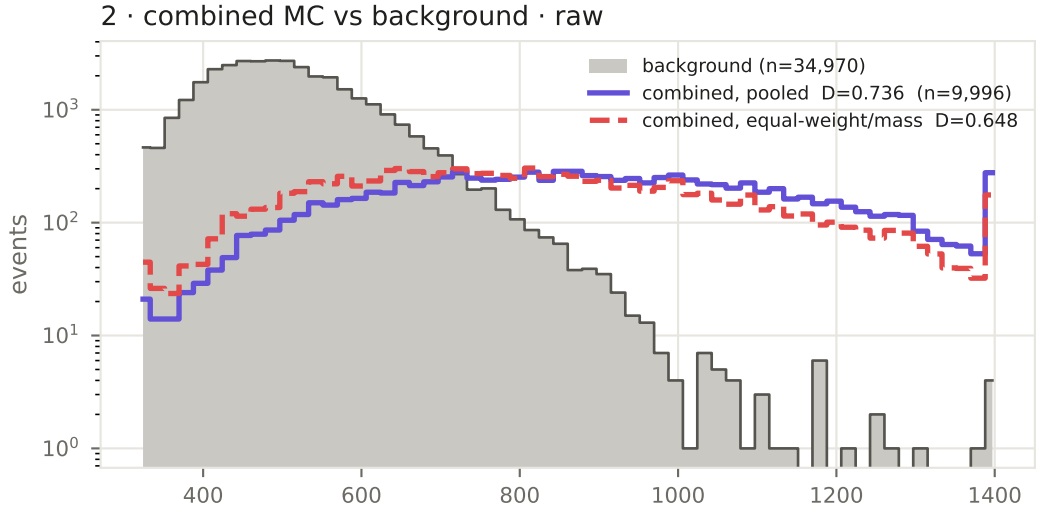


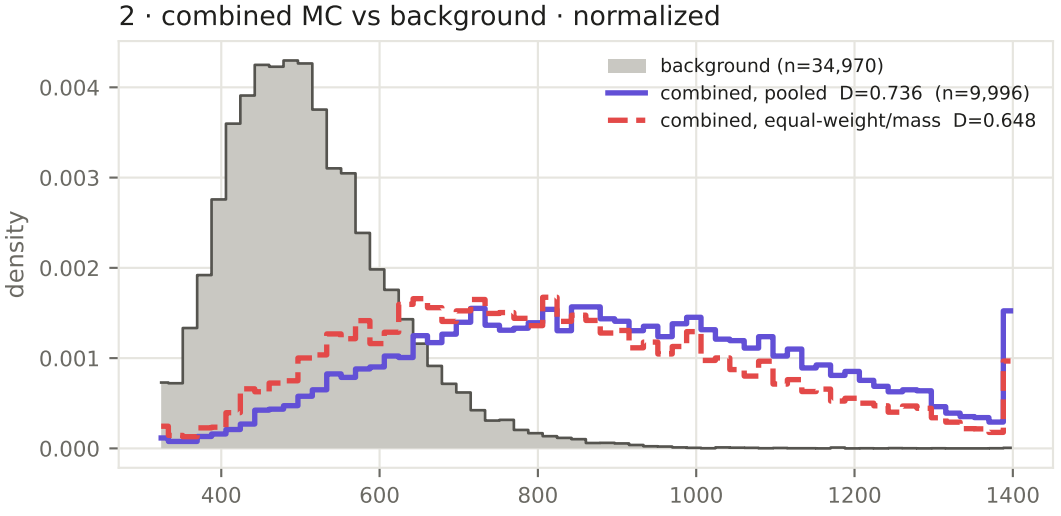


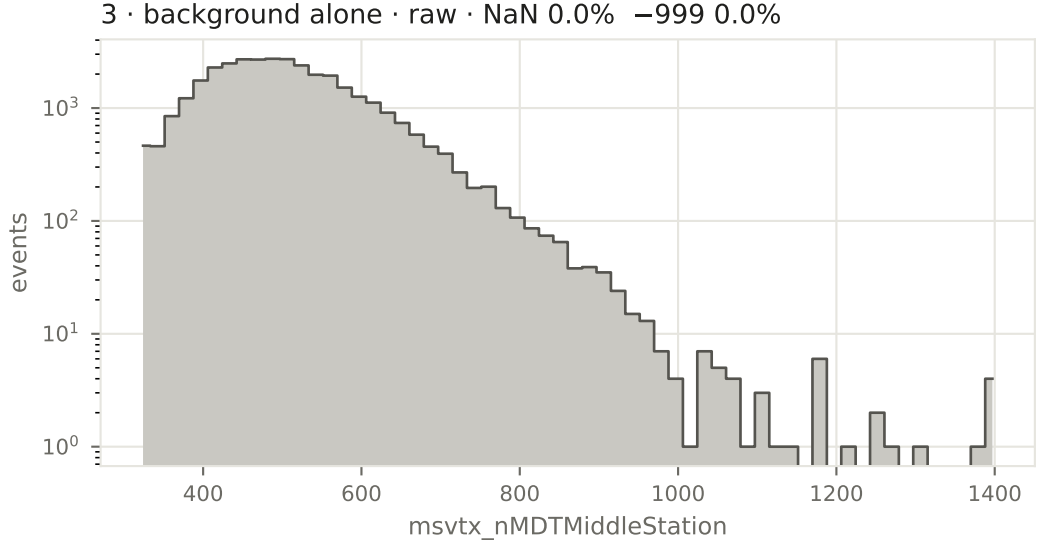
msvtx_nMDTMiddleStation · region: endcap · rank 12/191 by combined equal-weight D · round 1 · edge bins include under/overflow

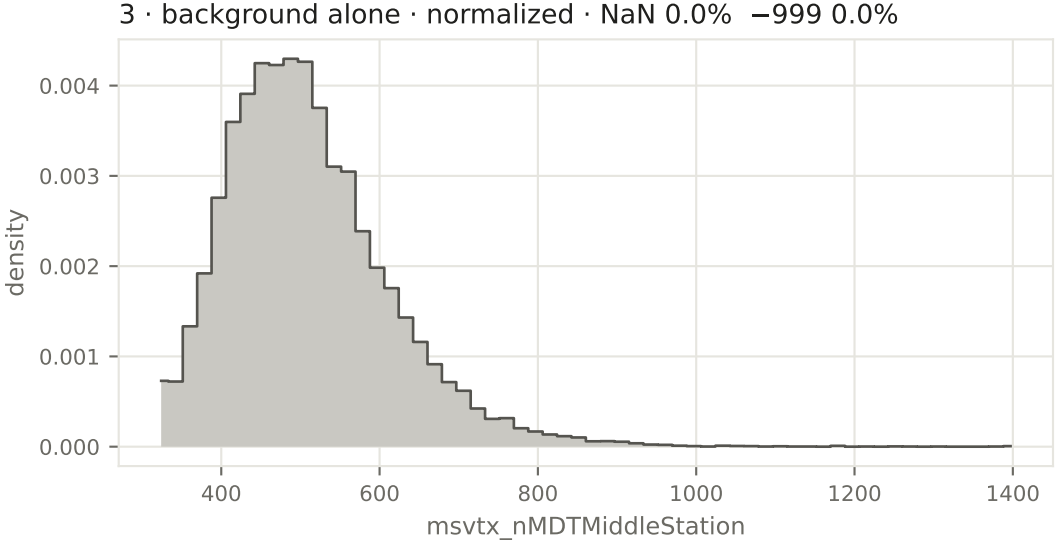




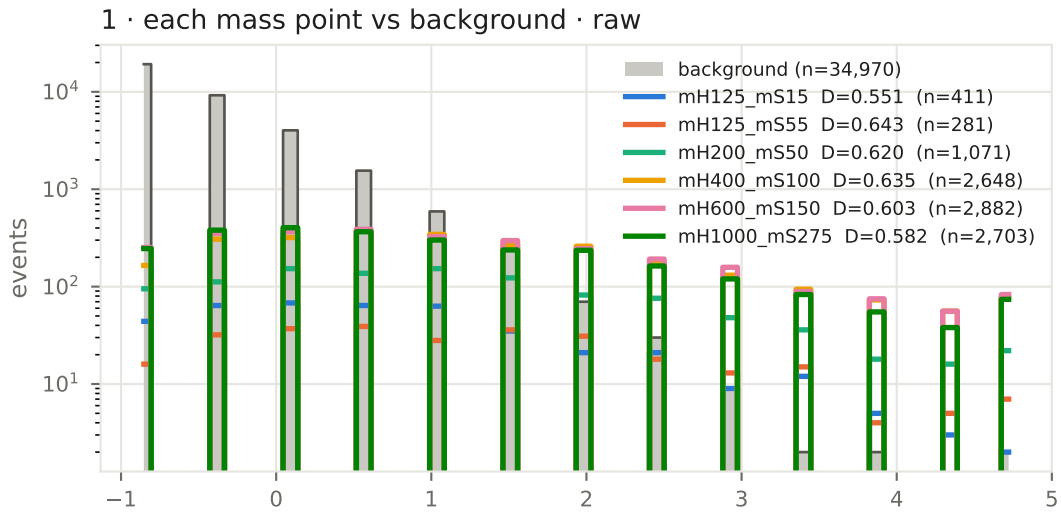


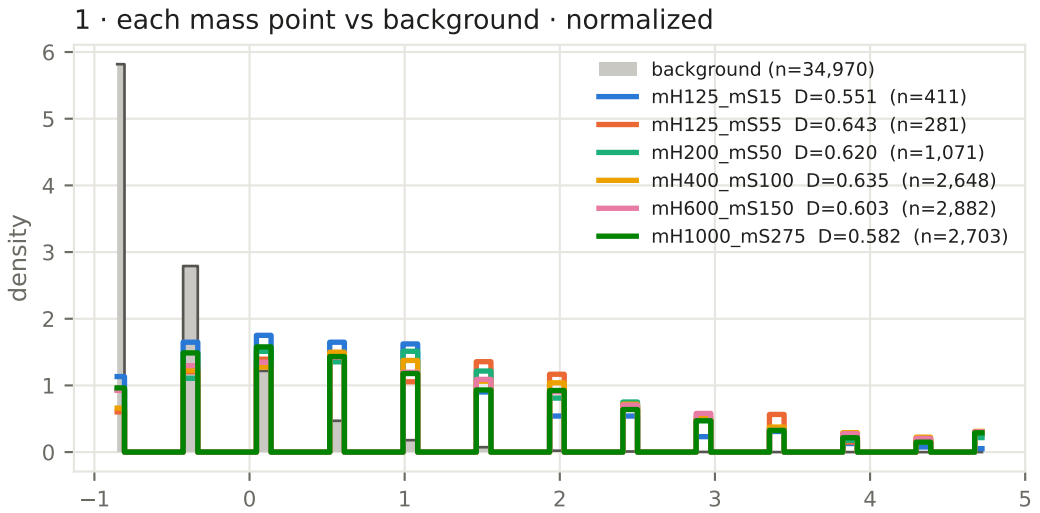


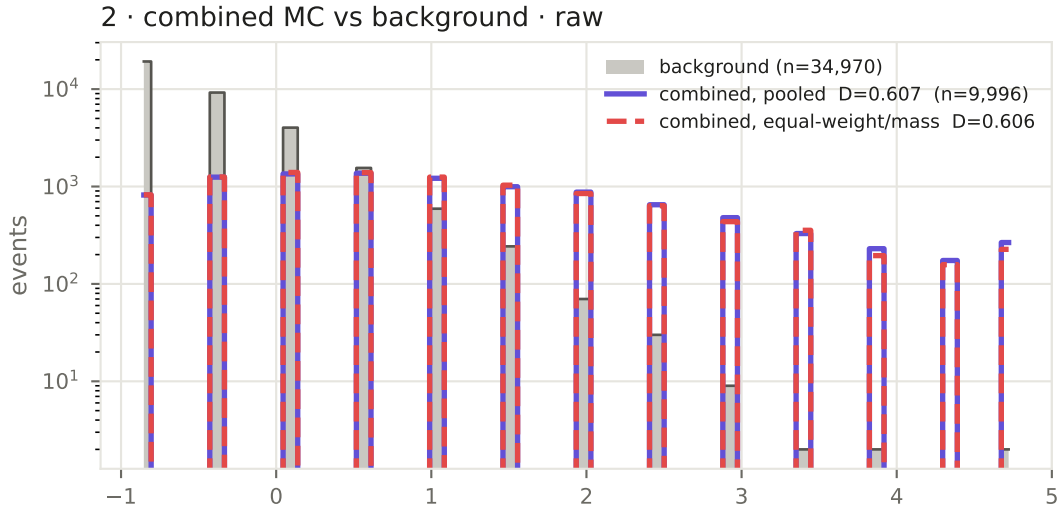


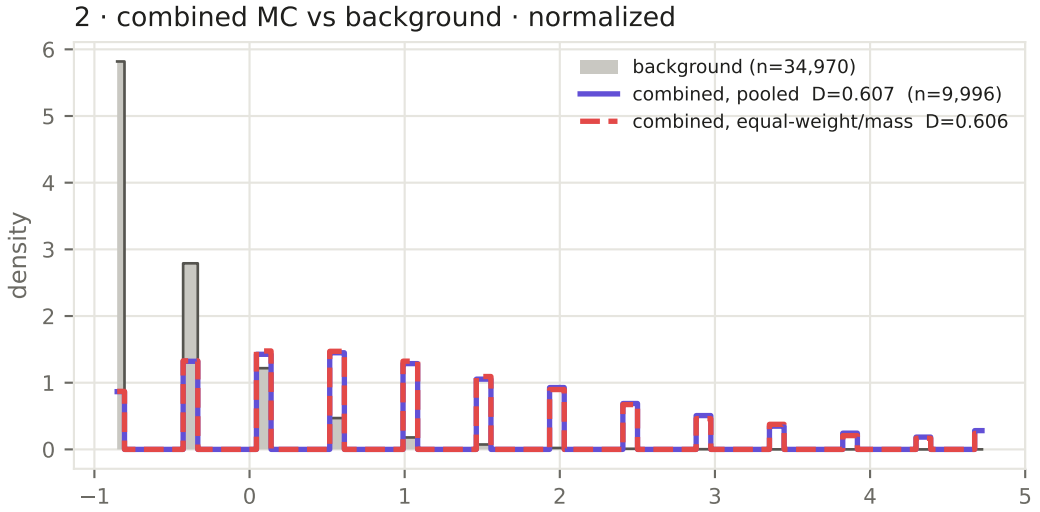


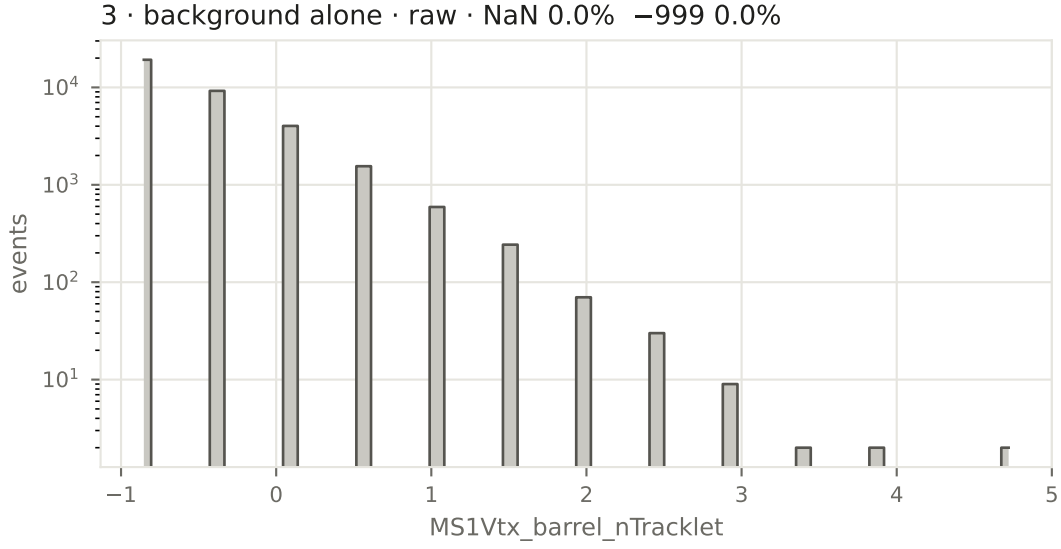
MS1Vtx_barrel_nTracklet · region: endcap · rank 13/191 by combined equal-weight D · round 1 · edge bins include under/overflow

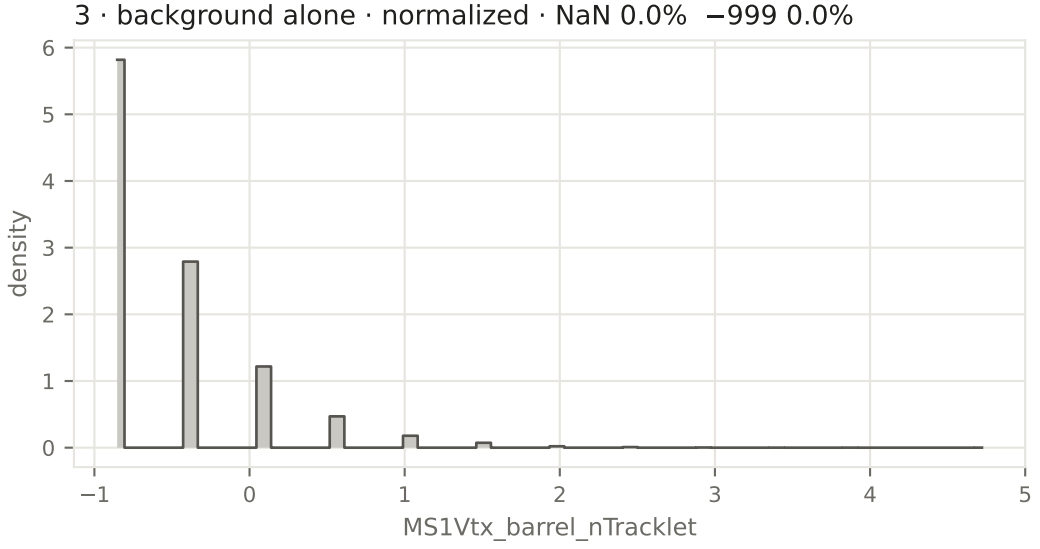




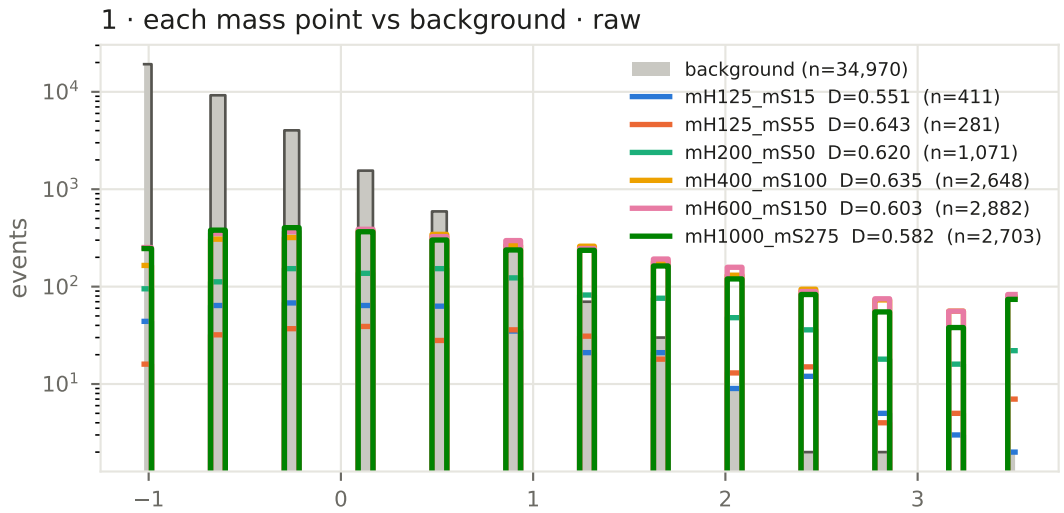


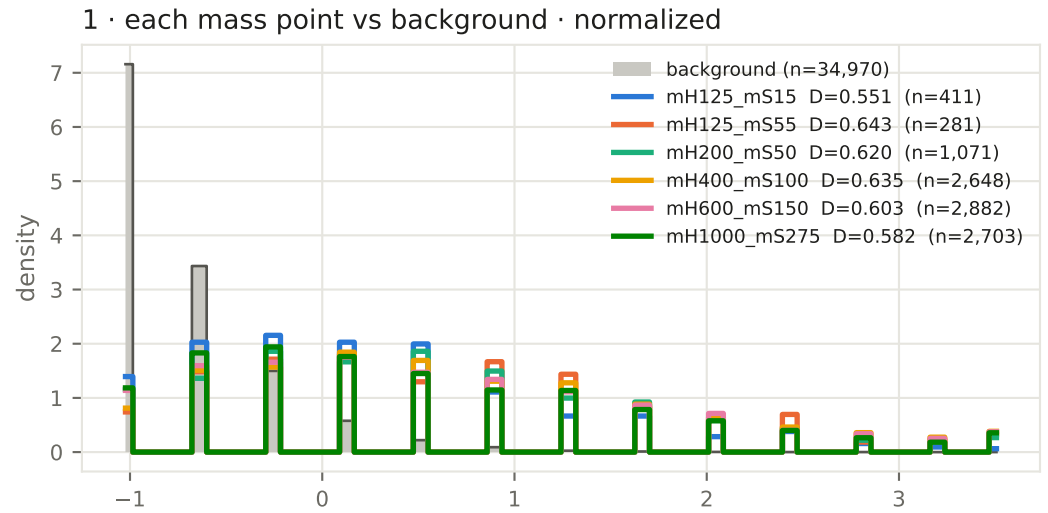


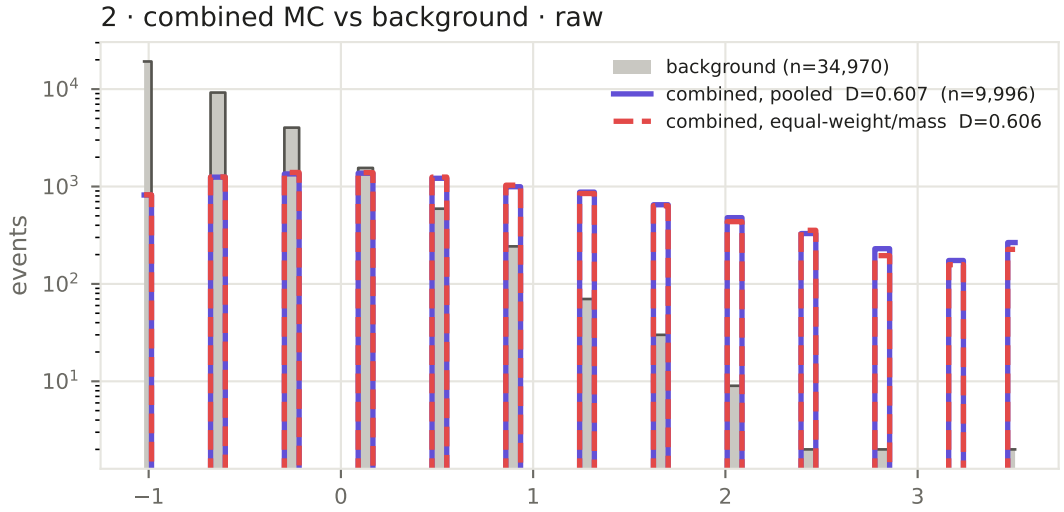


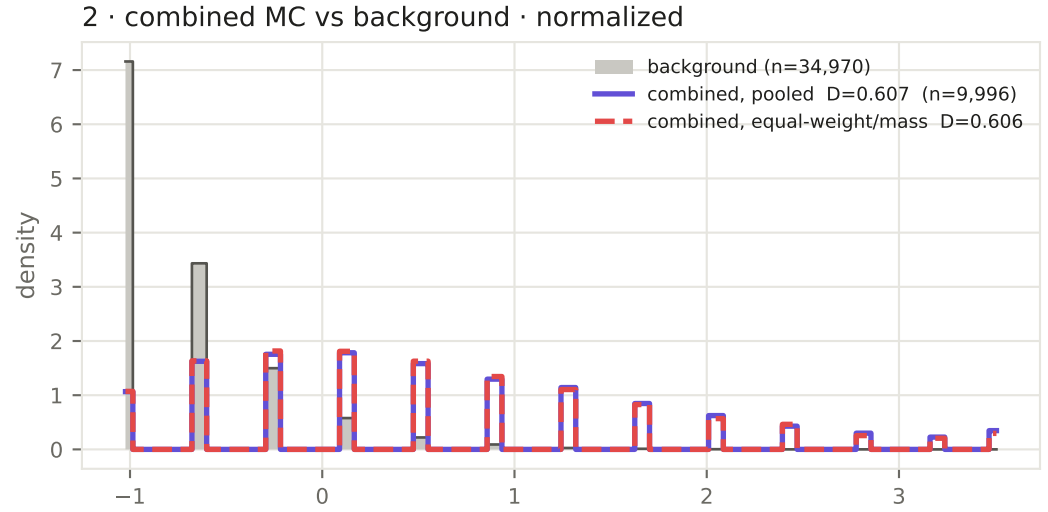


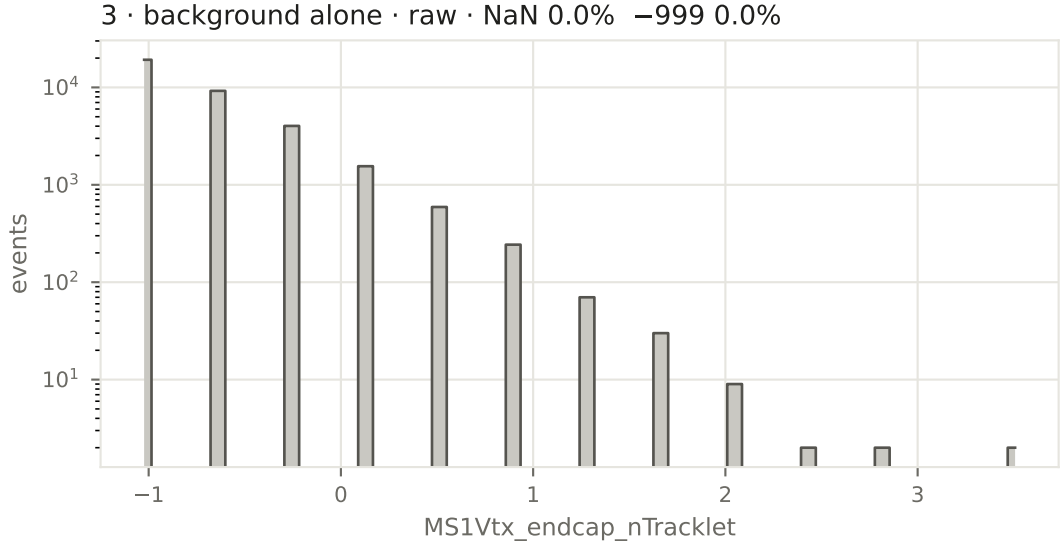
MS1Vtx_endcap_nTracklet · region: endcap · rank 14/191 by combined equal-weight D · round 1 · edge bins include under/overflow

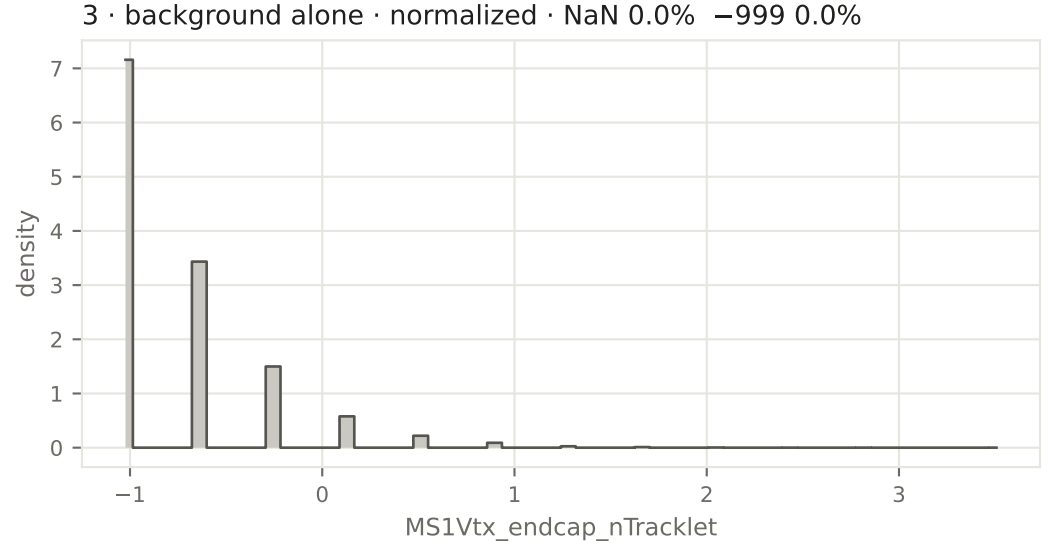




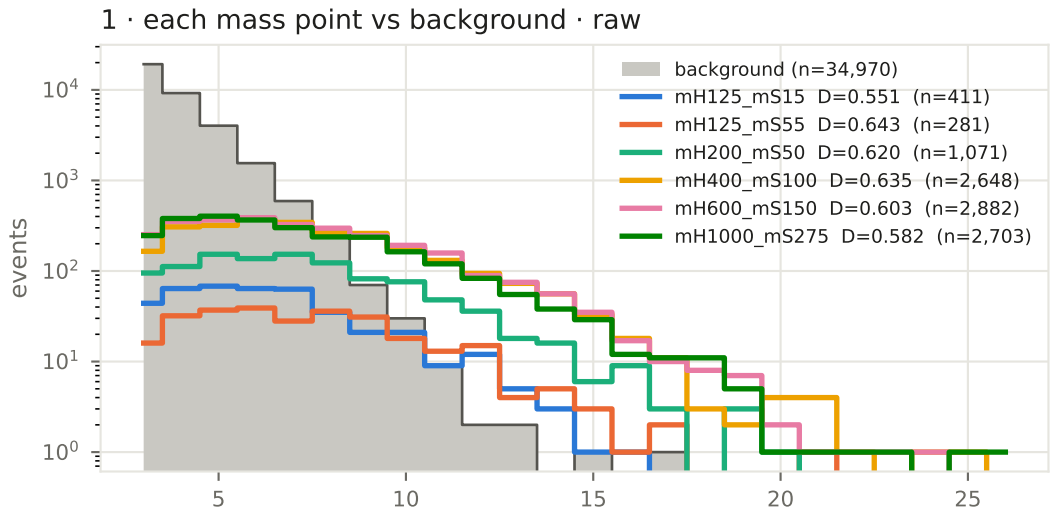


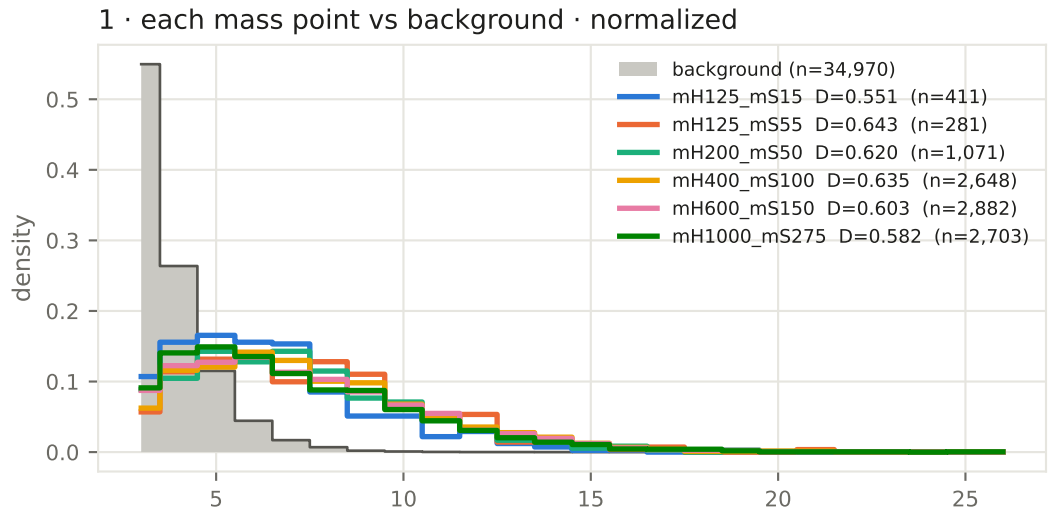


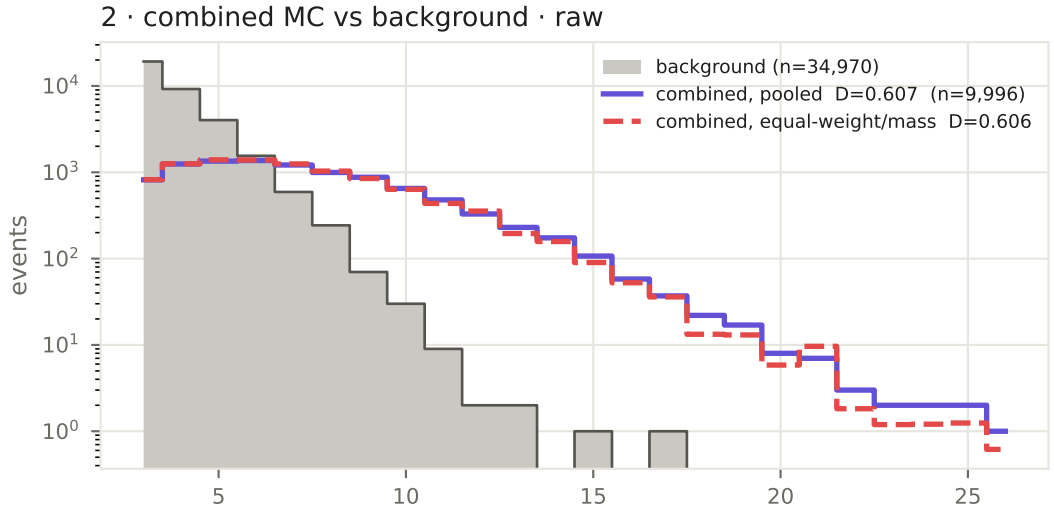


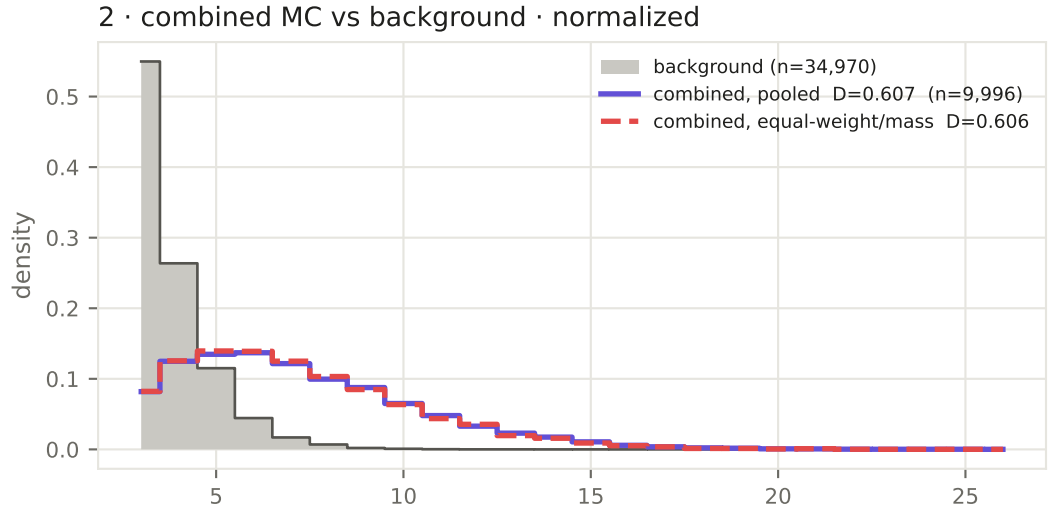


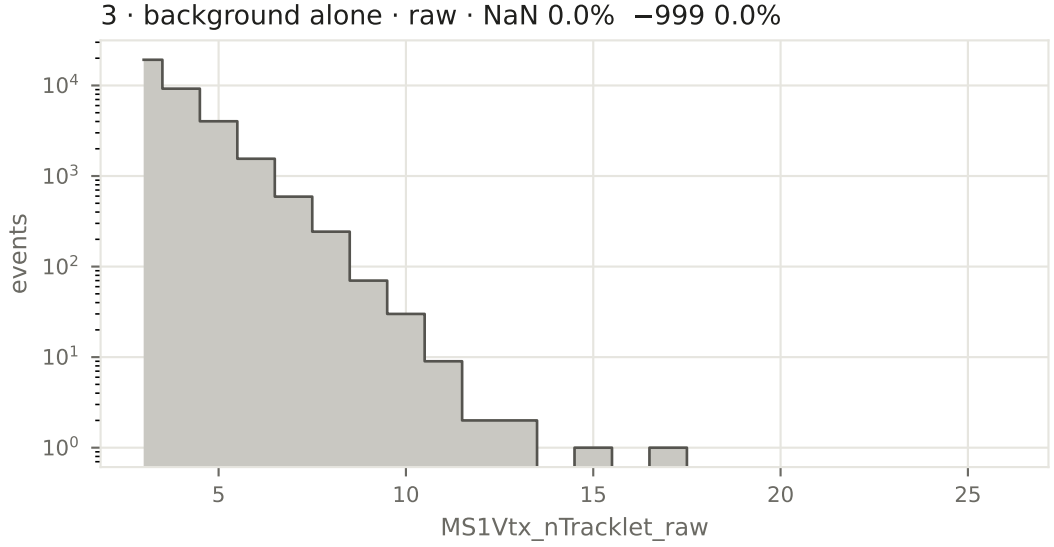
MS1Vtx_nTracklet_raw · region: endcap · rank 15/191 by combined equal-weight D · round 1 · edge bins include under/overflow

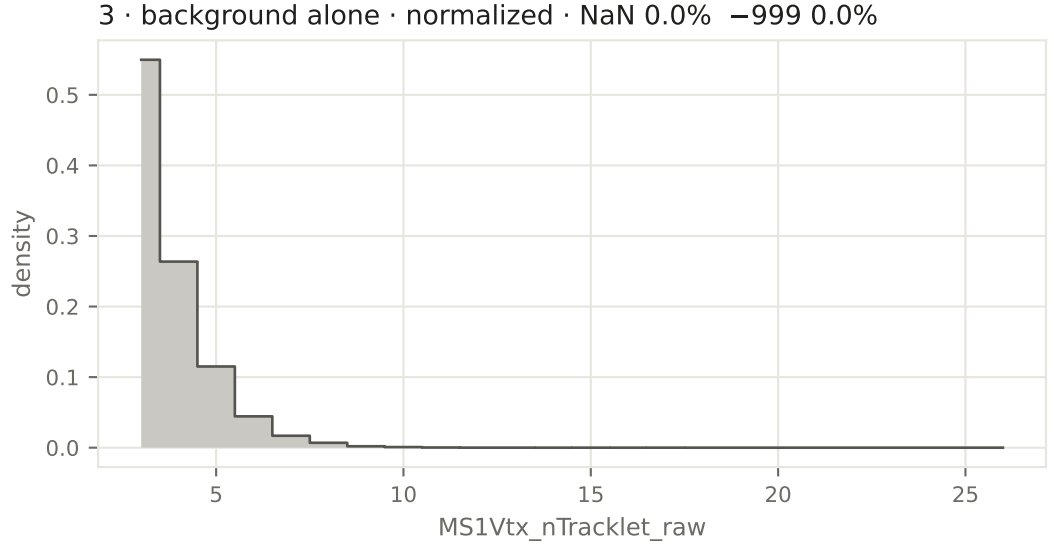


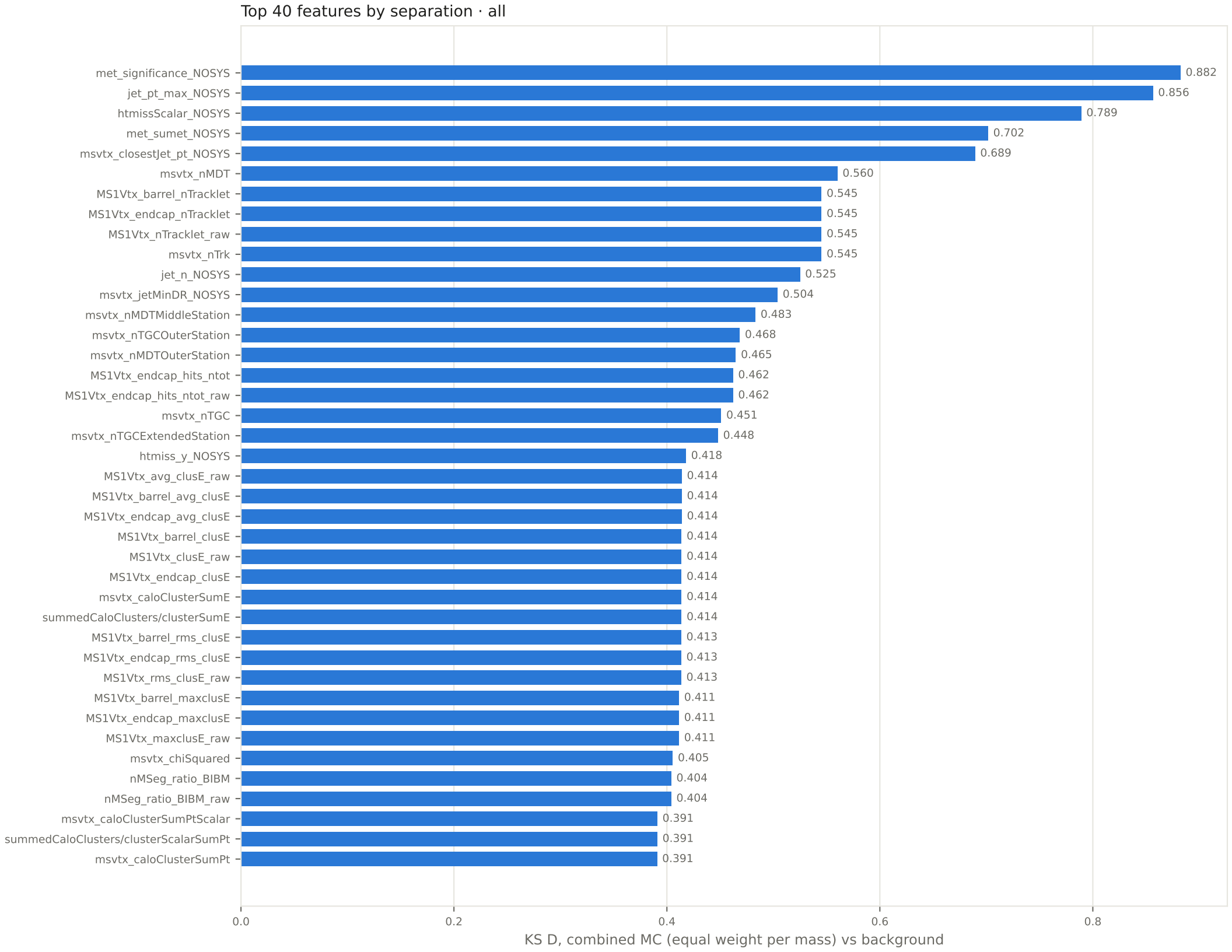






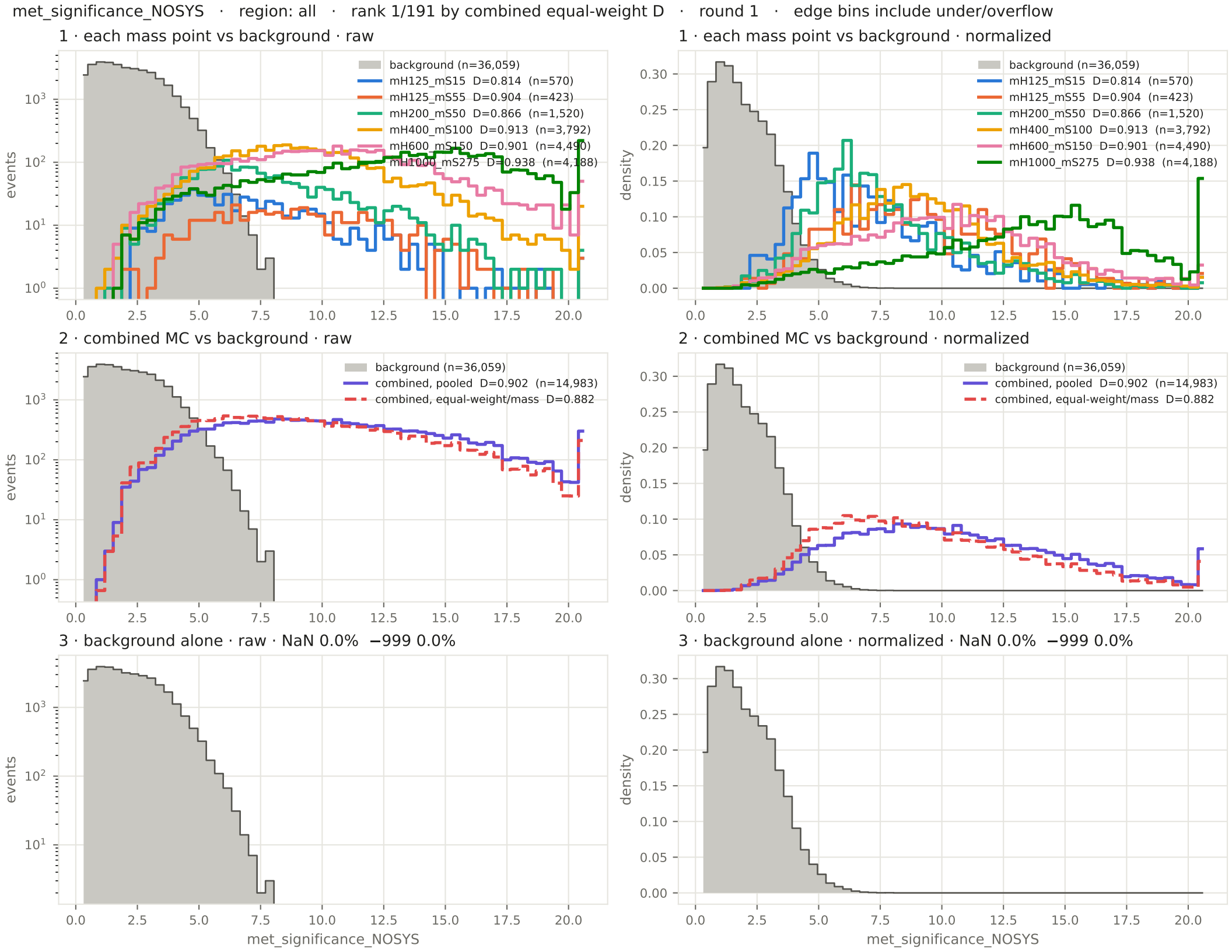




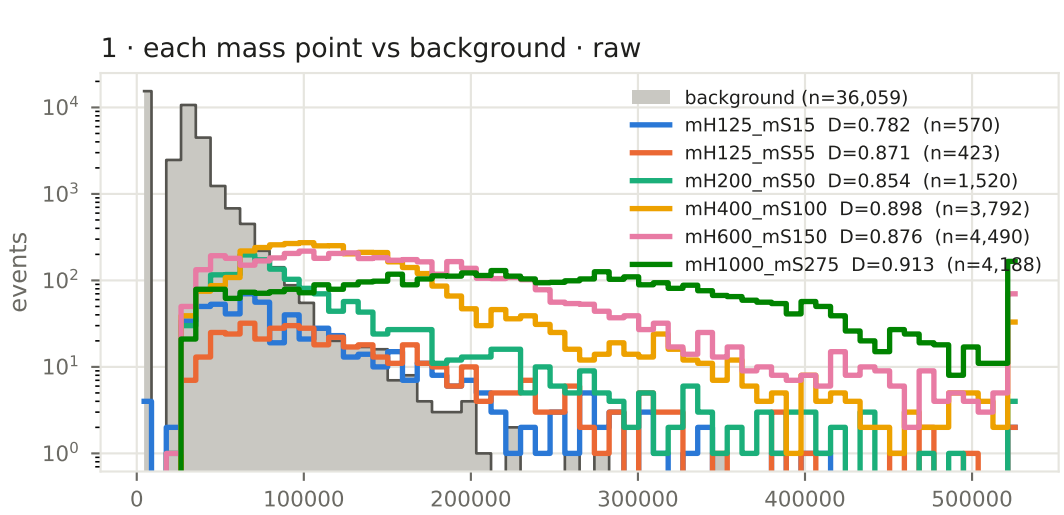

Top 40 features by separation · all
met_significance_NOSYS 0.882
jet_pt_max_NOSYS 0.856
htmissScalar_NOSYS 0.789
met_sumet_NOSYS 0.702
msvtx_closestJet_pt_NOSYS 0.689
msvtx_nMDT 0.560
MS1Vtx_barrel_nTracklet 0.545
MS1Vtx_endcap_nTracklet 0.545
MS1Vtx_nTracklet_raw 0.545
msvtx_nTrk 0.545
jet_n_NOSYS 0.525
msvtx_jetMinDR_NOSYS 0.504
msvtx_nMDTMiddleStation 0.483
msvtx_nTGCOuterStation 0.468
msvtx_nMDTOuterStation 0.465
MS1Vtx_endcap_hits_ntot 0.462
MS1Vtx_endcap_hits_ntot_raw 0.462
msvtx_nTGC 0.451
msvtx_nTGCExtendedStation 0.448
htmiss_y_NOSYS 0.418
MS1Vtx_avg_clusE_raw 0.414
MS1Vtx_barrel_avg_clusE 0.414
MS1Vtx_endcap_avg_clusE 0.414
MS1Vtx_barrel_clusE 0.414
MS1Vtx_clusE_raw 0.414
MS1Vtx_endcap_clusE 0.414
msvtx_caloClusterSumE 0.414
summedCaloClusters/clusterSumE 0.414
MS1Vtx_barrel_rms_clusE 0.413
MS1Vtx_endcap_rms_clusE 0.413
MS1Vtx_rms_clusE_raw 0.413
MS1Vtx_barrel_maxclusE 0.411
MS1Vtx_endcap_maxclusE 0.411
MS1Vtx_maxclusE_raw 0.411
msvtx_chiSquared 0.405
nMSeg_ratio_BIBM 0.404
nMSeg_ratio_BIBM_raw 0.404
msvtx_caloClusterSumPtScalar 0.391
summedCaloClusters/clusterScalarSumPt 0.391
msvtx_caloClusterSumPt 0.391
0.0
0.2
0.4
0.6
0.8
KS D, combined MC (equal weight per mass) vs background

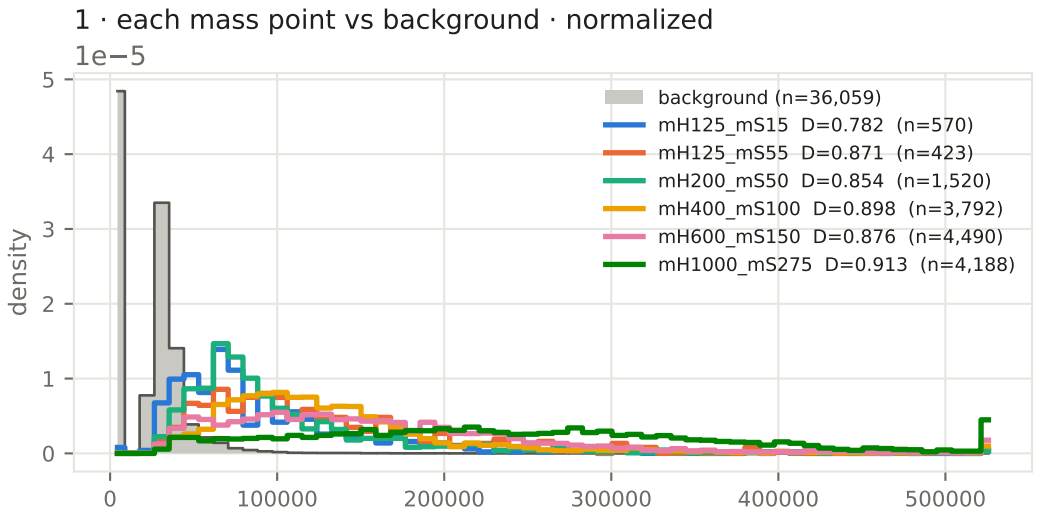
Round 1 · all · top 40 by combined equal-weight KS D

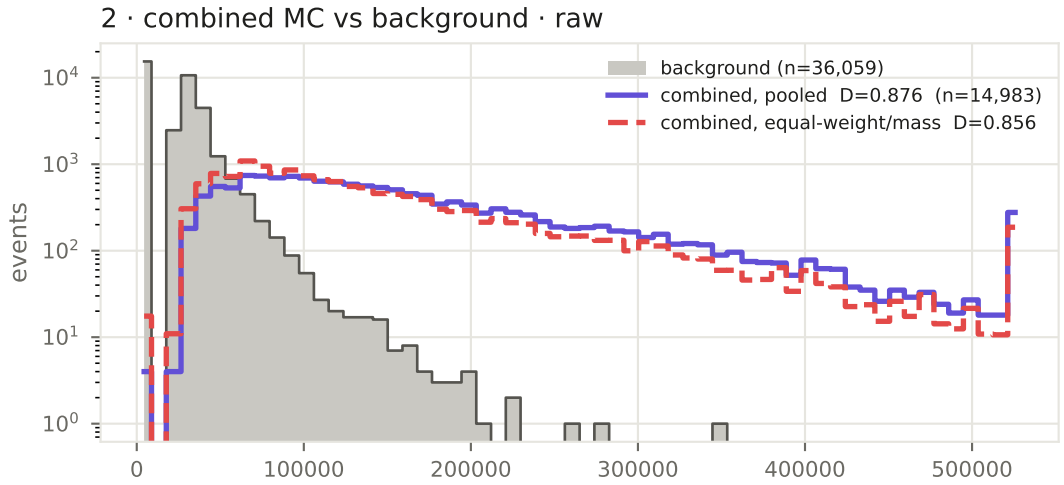
#	feature	D_equal	D_pooled	D_min	D_max	bkgNaN	bkg-999
1	met_significance_NOSYS	0.882	0.902	0.814	0.938	0.0%	0.0%
2	jet_pt_max_NOSYS	0.856	0.876	0.782	0.913	0.0%	0.0%
3	htmissScalar_NOSYS	0.789	0.792	0.789	0.868	0.0%	0.0%
4	met_sumet_NOSYS	0.702	0.753	0.563	0.856	0.0%	0.0%
5	msvtx_closestJet_pt_NOSYS	0.689	0.703	0.640	0.719	0.0%	0.0%
6	msvtx_nMDT	0.560	0.666	0.366	0.783	0.0%	0.0%
7	MS1Vtx_barrel_nTracklet	0.545	0.548	0.502	0.569	0.0%	0.0%
8	MS1Vtx_endcap_nTracklet	0.545	0.548	0.502	0.569	0.0%	0.0%
9	MS1Vtx_nTracklet_raw	0.545	0.548	0.502	0.569	0.0%	0.0%
10	msvtx_nTrk	0.545	0.548	0.502	0.569	0.0%	0.0%
11	jet_n_NOSYS	0.525	0.572	0.422	0.659	0.0%	0.0%
12	msvtx_jetMinDR_NOSYS	0.504	0.525	0.480	0.548	0.0%	0.0%
13	msvtx_nMDTMiddleStation	0.483	0.573	0.321	0.670	0.0%	0.0%
14	msvtx_nTGCOuterStation	0.468	0.505	0.371	0.532	0.0%	0.0%
15	msvtx_nMDTOuterStation	0.465	0.543	0.326	0.634	0.0%	0.0%
16	MS1Vtx_endcap_hits_ntot	0.462	0.511	0.335	0.547	0.0%	0.0%
17	MS1Vtx_endcap_hits_ntot_raw	0.462	0.511	0.335	0.547	0.0%	0.0%
18	msvtx_nTGC	0.451	0.498	0.354	0.527	0.0%	0.0%
19	msvtx_nTGCExtendedStation	0.448	0.490	0.351	0.519	0.0%	0.0%
20	htmiss_y_NOSYS	0.418	0.422	0.364	0.447	0.0%	0.0%
21	MS1Vtx_avg_clusE_raw	0.414	0.421	0.396	0.429	0.0%	0.0%
22	MS1Vtx_barrel_avg_clusE	0.414	0.421	0.396	0.429	0.0%	0.0%
23	MS1Vtx_endcap_avg_clusE	0.414	0.421	0.396	0.429	0.0%	0.0%
24	MS1Vtx_barrel_clusE	0.414	0.415	0.411	0.420	0.0%	0.0%
25	MS1Vtx_clusE_raw	0.414	0.415	0.411	0.420	0.0%	0.0%
26	MS1Vtx_endcap_clusE	0.414	0.415	0.411	0.420	0.0%	0.0%
27	msvtx_caloClusterSumE	0.414	0.415	0.411	0.420	0.0%	0.0%
28	summedCaloClusters/clusterSumE	0.414	0.415	0.411	0.420	0.0%	0.0%
29	MS1Vtx_barrel_rms_clusE	0.413	0.420	0.399	0.428	0.0%	0.0%
30	MS1Vtx_endcap_rms_clusE	0.413	0.420	0.399	0.428	0.0%	0.0%
31	MS1Vtx_rms_clusE_raw	0.413	0.420	0.399	0.428	0.0%	0.0%
32	MS1Vtx_barrel_maxclusE	0.411	0.414	0.394	0.419	0.0%	0.0%
33	MS1Vtx_endcap_maxclusE	0.411	0.414	0.394	0.419	0.0%	0.0%
34	MS1Vtx_maxclusE_raw	0.411	0.414	0.394	0.419	0.0%	0.0%
35	msvtx_chiSquared	0.405	0.490	0.494	1.000	0.0%	0.0%
36	nMSeg_ratio_BIBM	0.404	0.410	0.381	0.427	0.0%	0.0%
37	nMSeg_ratio_BIBM_raw	0.404	0.410	0.381	0.427	0.0%	0.0%
38	msvtx_caloClusterSumPtScalar	0.391	0.392	0.378	0.403	0.0%	0.0%
39	summedCaloClusters/clusterScalarSumPt	0.391	0.392	0.378	0.403	0.0%	0.0%
40	msvtx_caloClusterSumPt	0.391	0.392	0.377	0.403	0.0%	0.0%

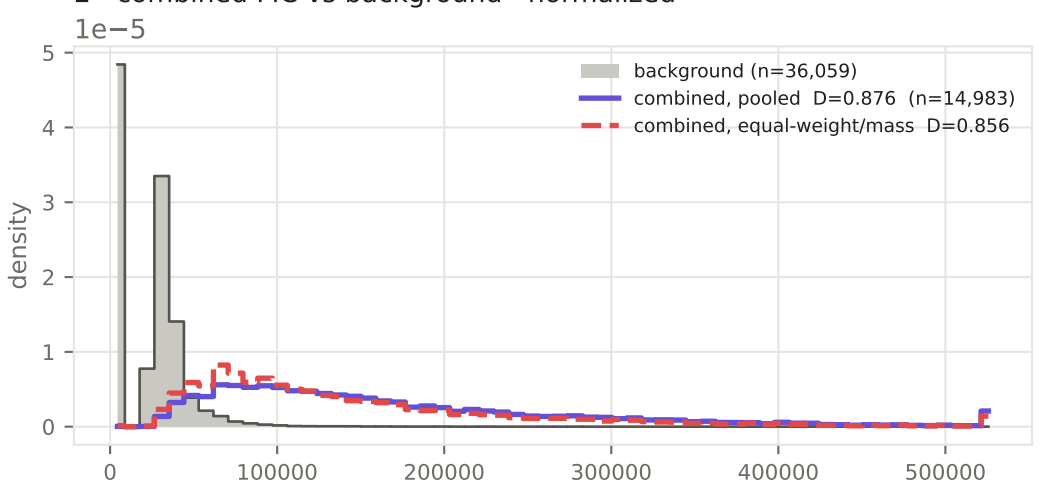

met_significance_NOSYS · region: all · rank 1/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
background (n=36,059)
mH125_mS15 D=0.814 (n=570)
mH125_mS55 D=0.904 (n=423)
mH200_mS50 D=0.866 (n=1,520)
mH400_mS100 D=0.913 (n=3,792)
mH600_mS150 D=0.901 (n=4,490)
mH1000_mS275 D=0.938 (n=4,188)
combined, pooled D=0.902 (n=14,983)
combined, equal-weight/mass D=0.882
events
density
met_significance_NOSYS

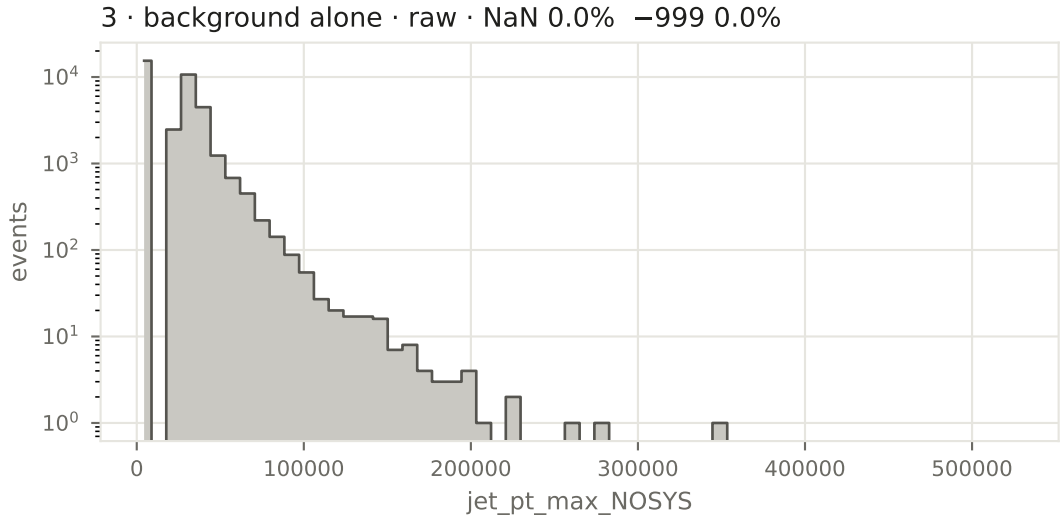
jet_pt_max_NOSYS · region: all · rank 2/191 by combined equal-weight D · round 1 · edge bins include under/overflow

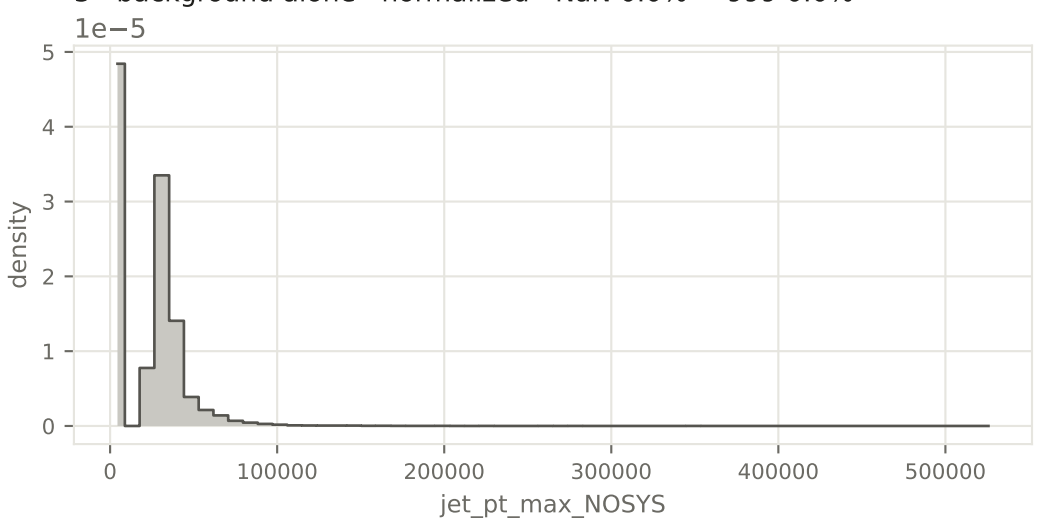






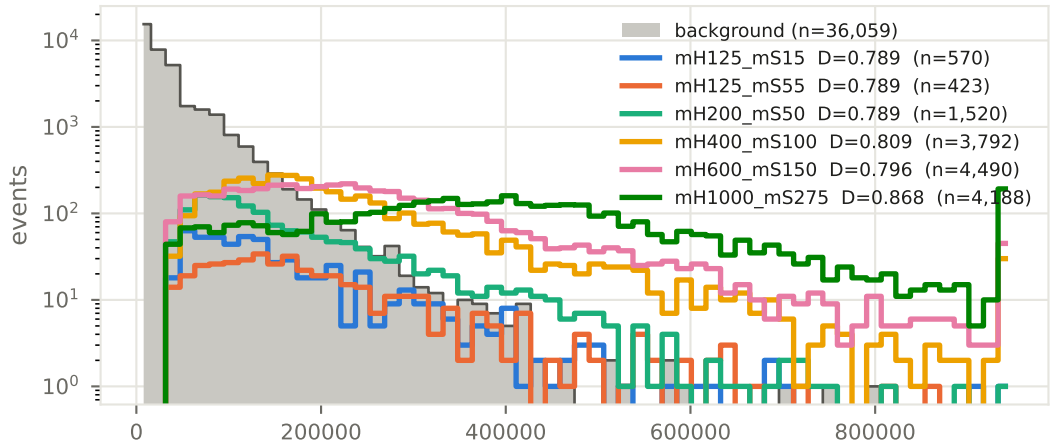




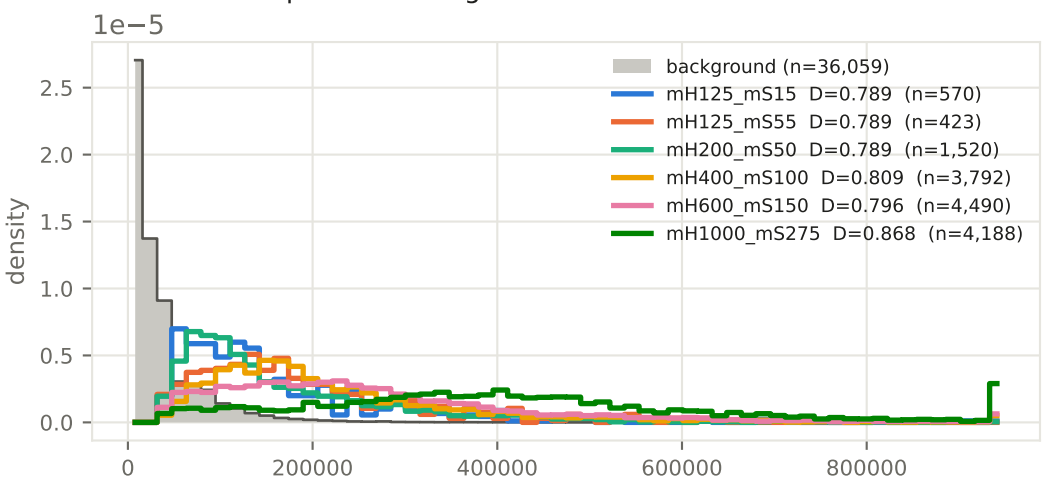


htmissScalar_NOSYS · region: all · rank 3/191 by combined equal-weight D · round 1 · edge bins include under/overflow

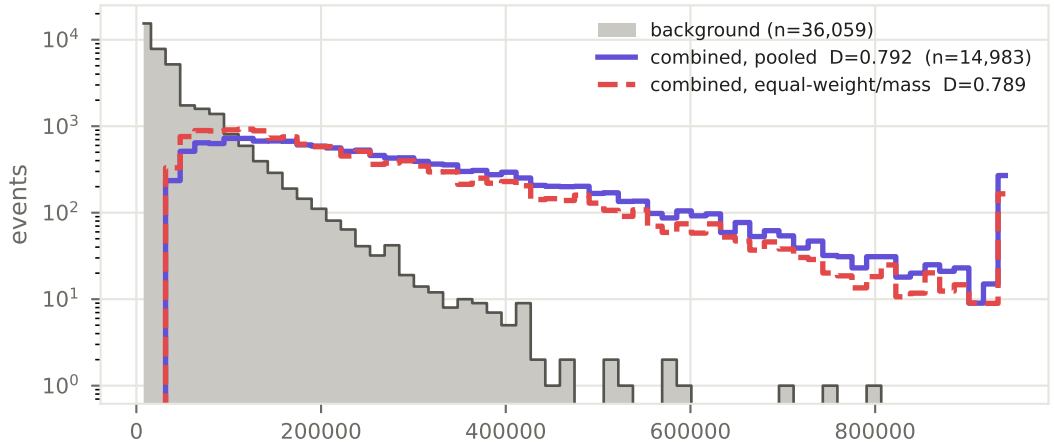
1 · each mass point vs background · raw



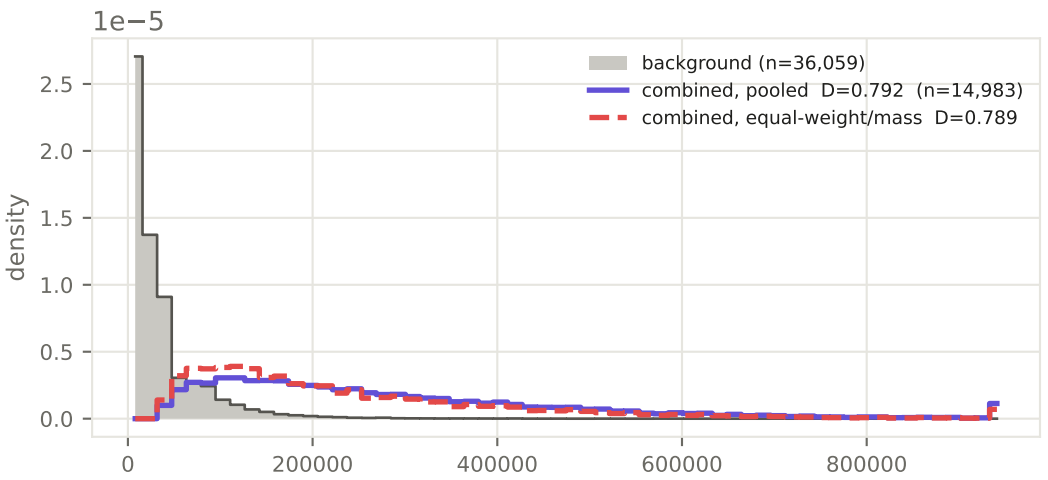
1 · each mass point vs background · normalized



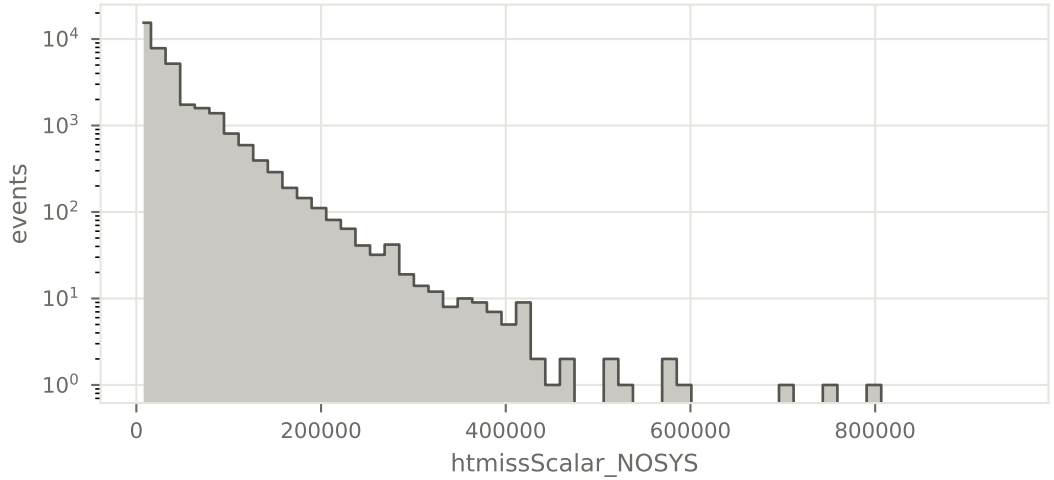
2 · combined MC vs background · raw



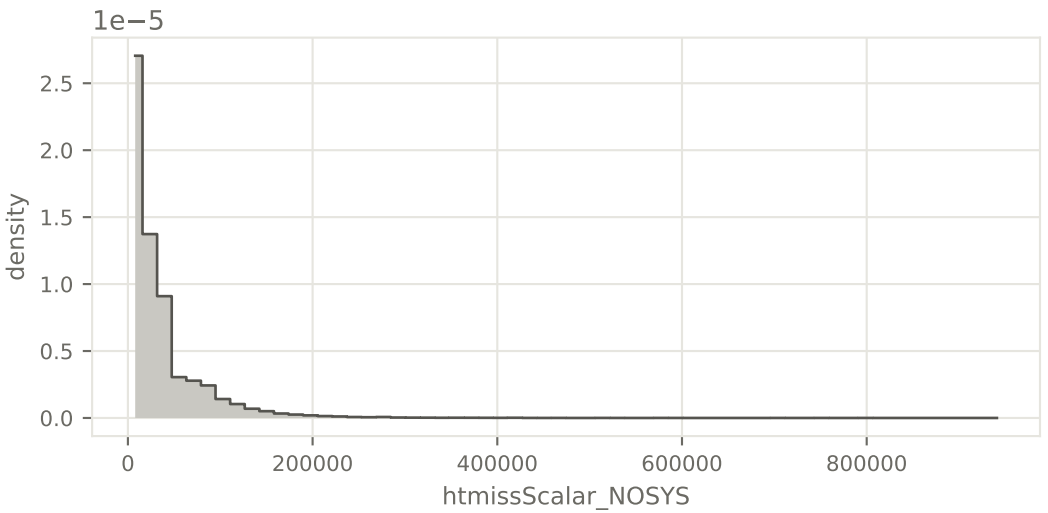
2 · combined MC vs background · normalized



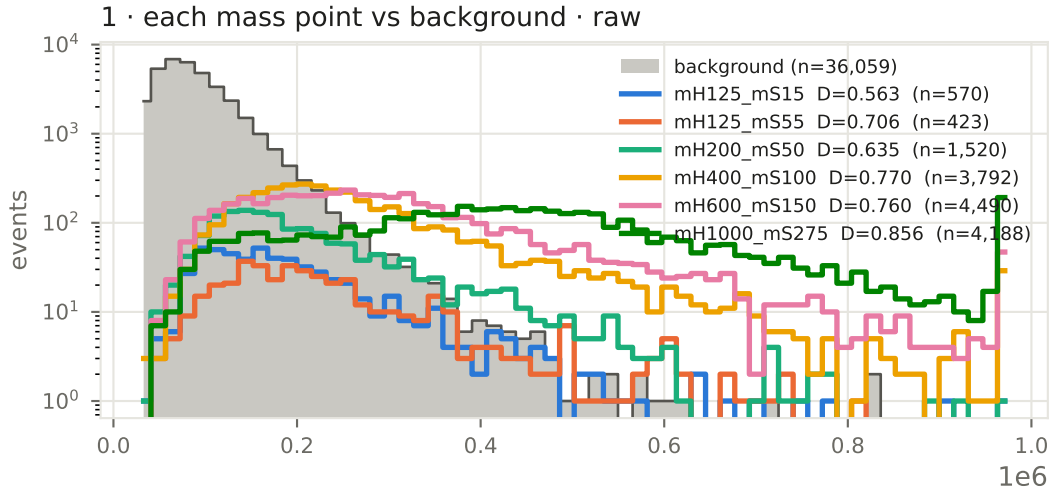
3 · background alone · raw · NaN 0.0% −999 0.0%

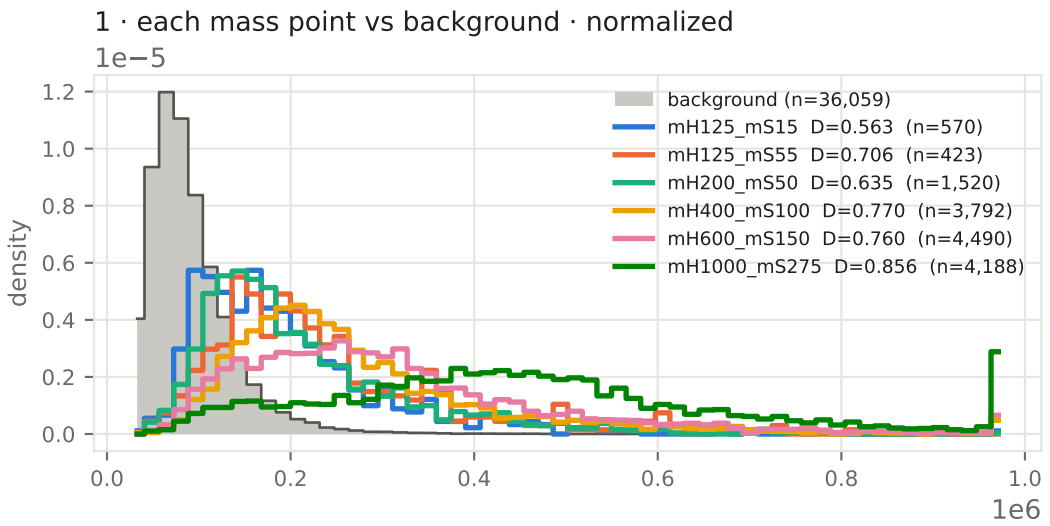


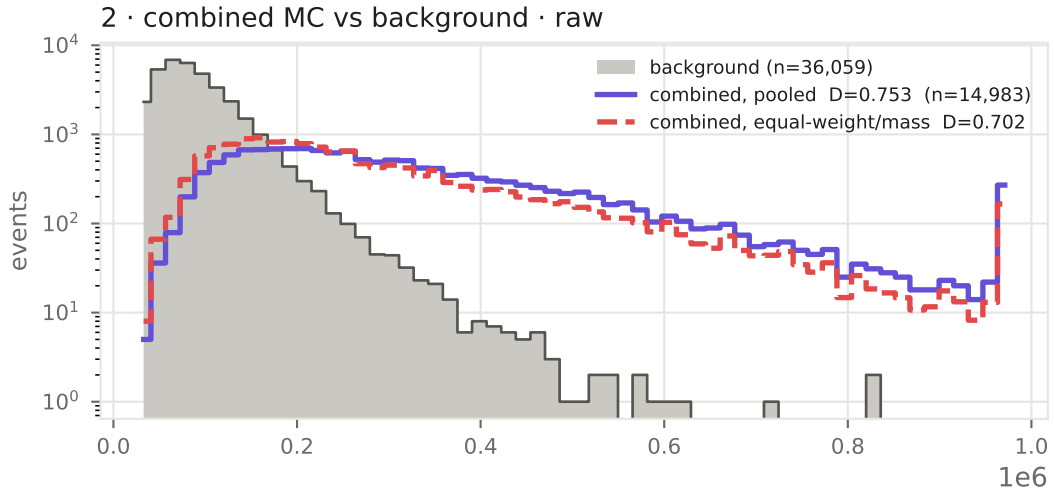
3 · background alone · normalized · NaN 0.0% −999 0.0%

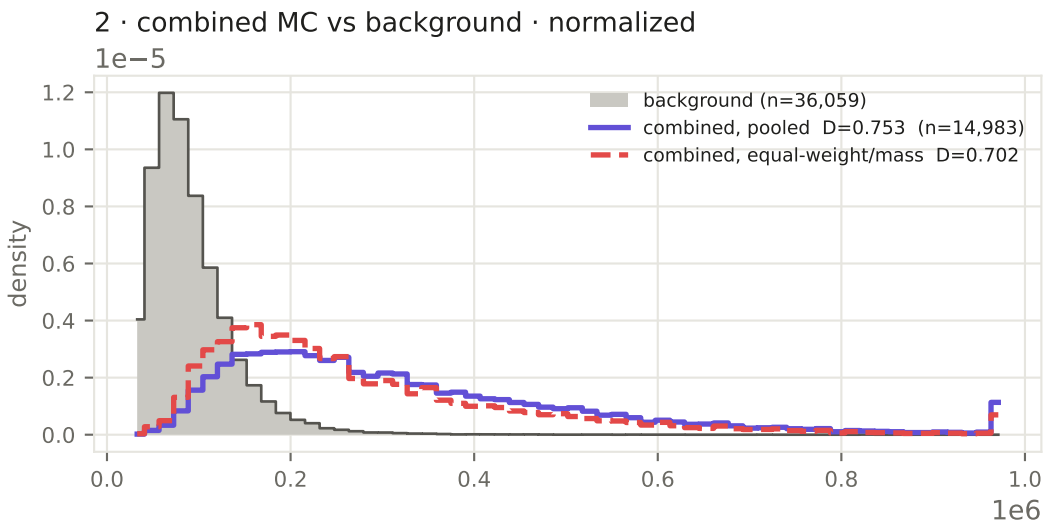


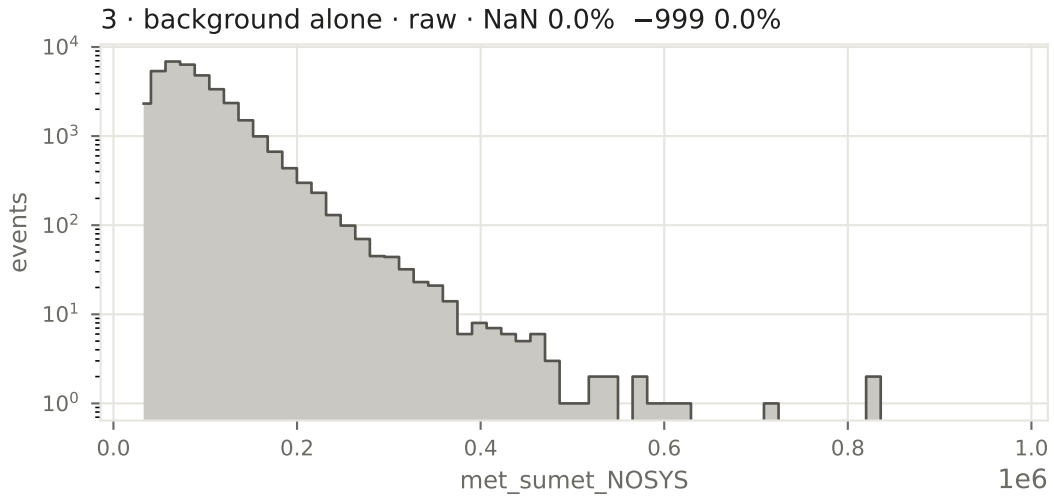
met_sumet_NOSYS · region: all · rank 4/191 by combined equal-weight D · round 1 · edge bins include under/overflow

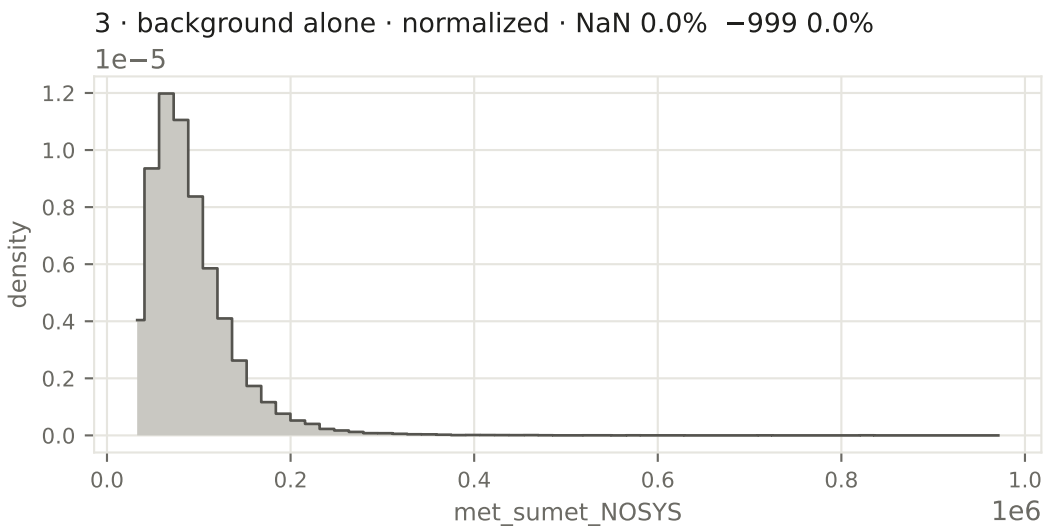




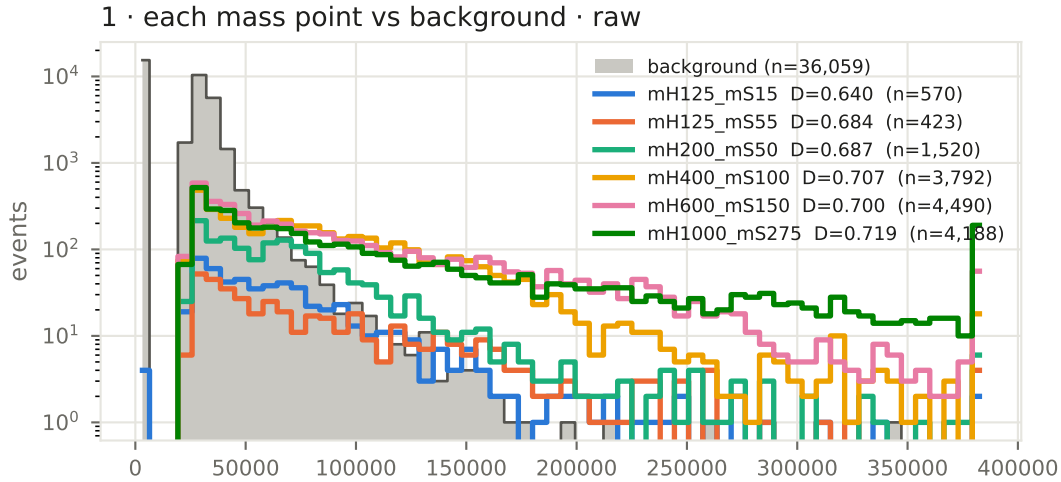


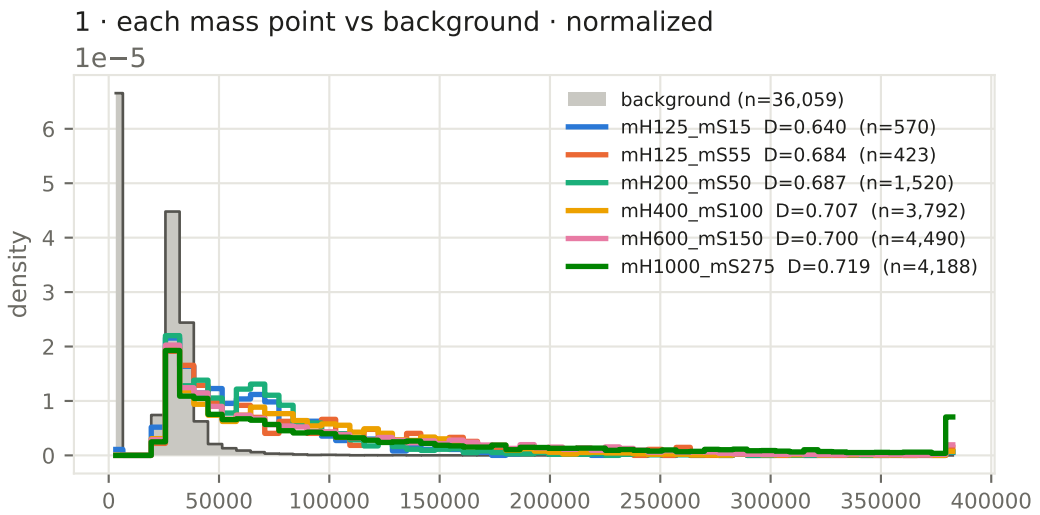


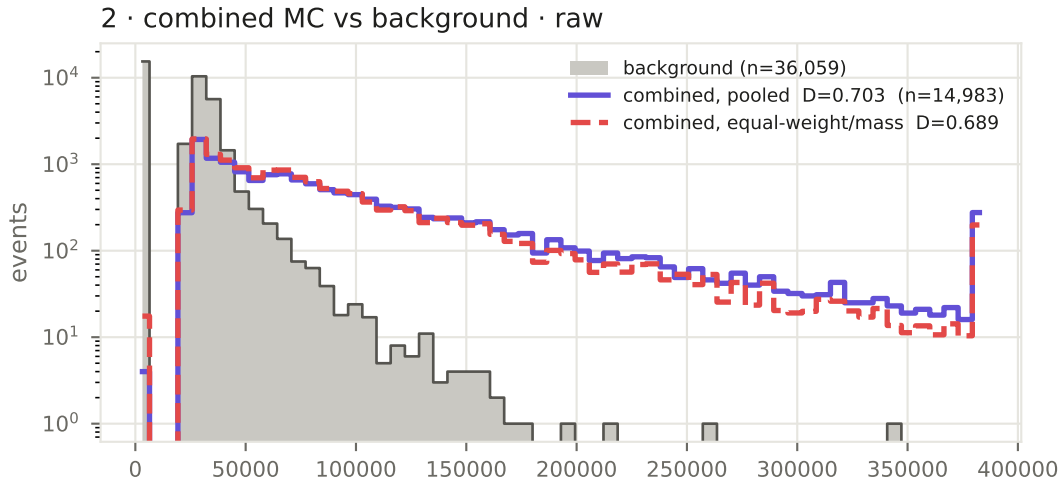


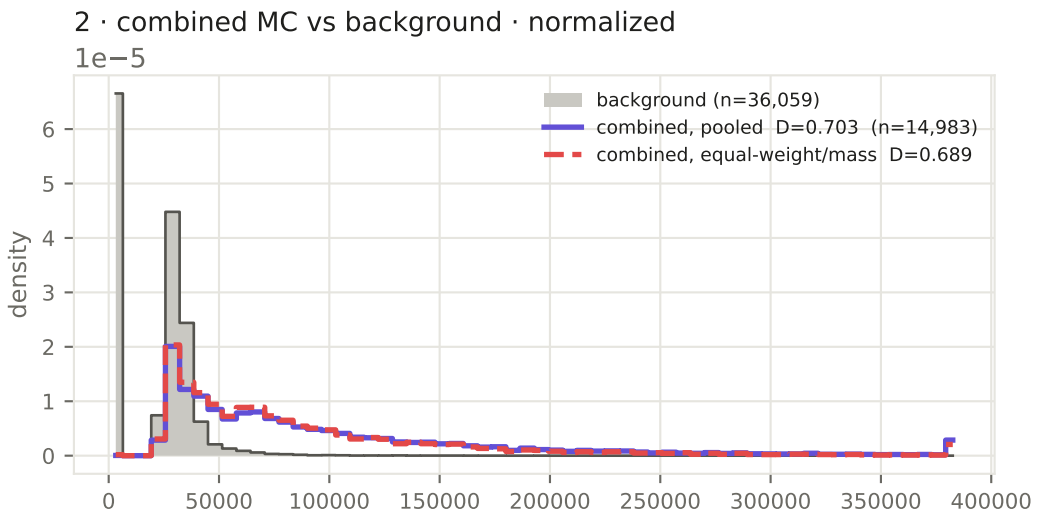


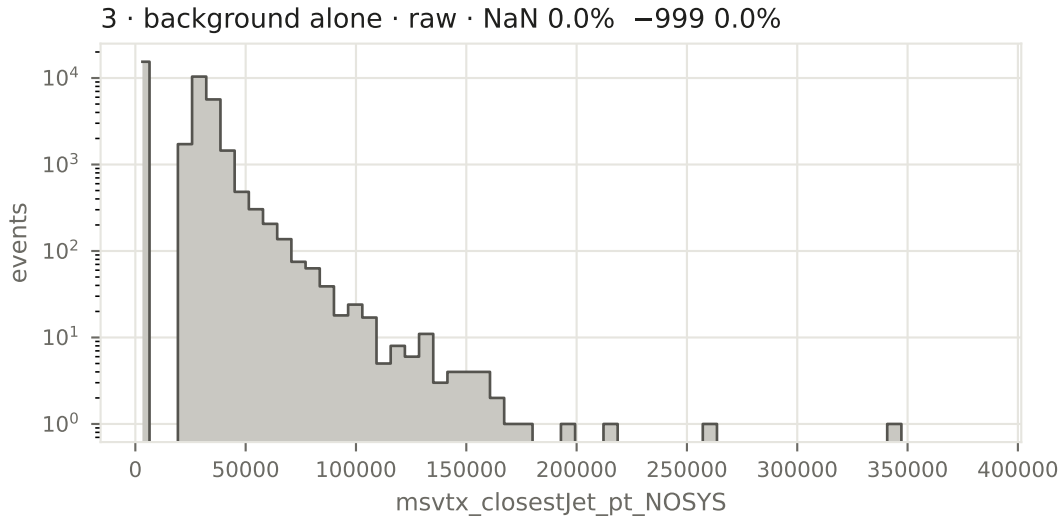
msvtx_closestJet_pt_NOSYS · region: all · rank 5/191 by combined equal-weight D · round 1 · edge bins include under/overflow

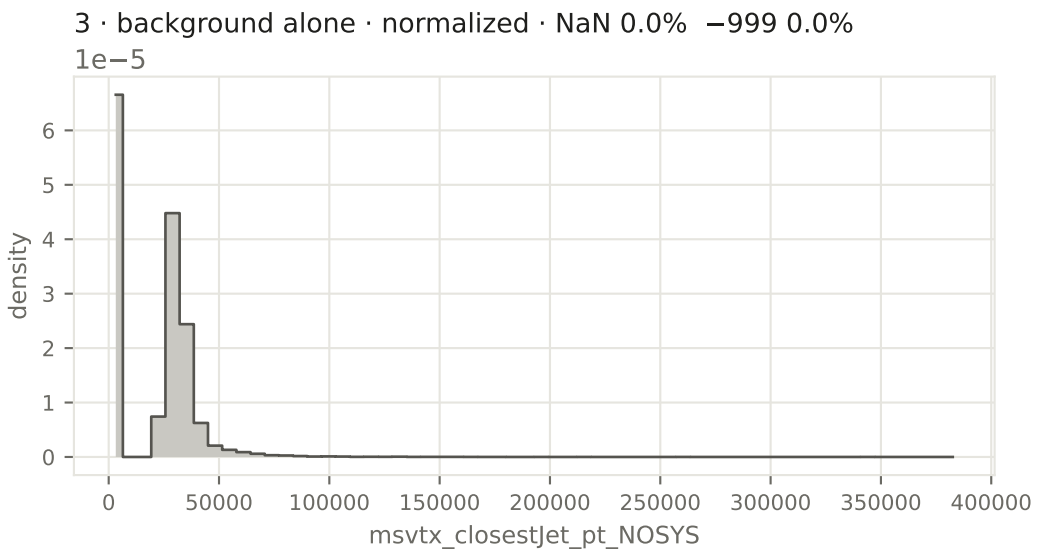






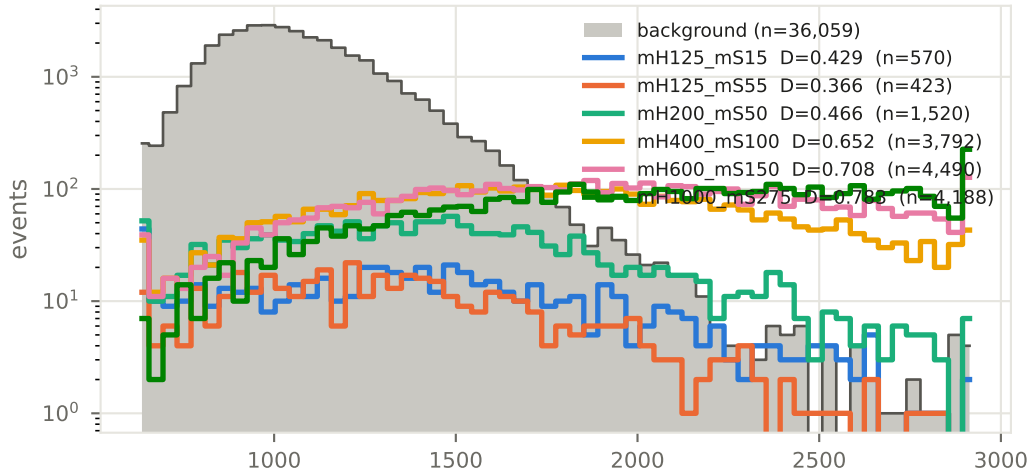




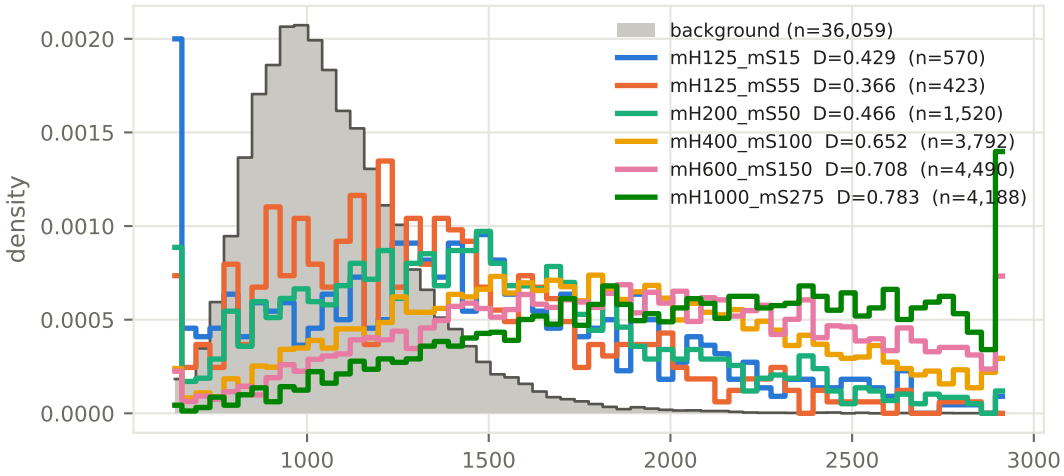


msvtx_nMDT · region: all · rank 6/191 by combined equal-weight D · round 1 · edge bins include under/overflow

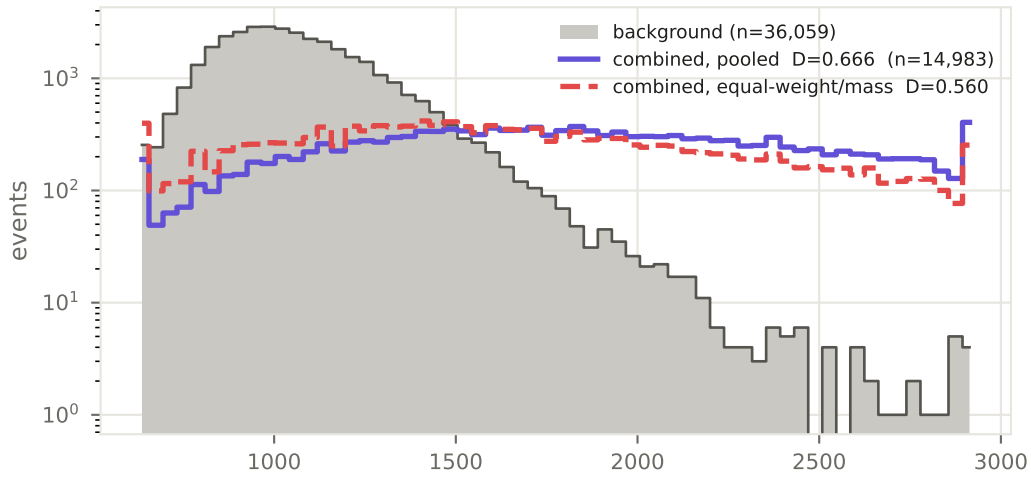
1 · each mass point vs background · raw



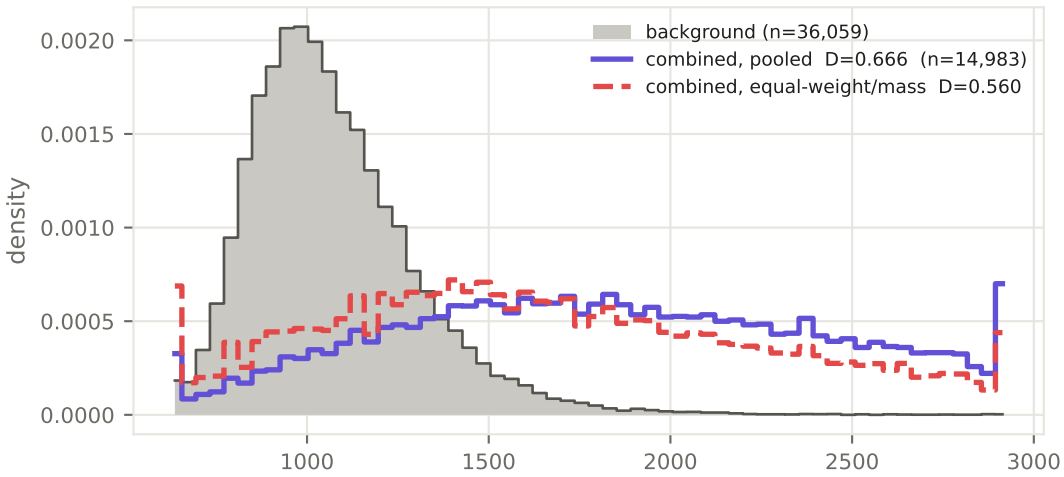
1 · each mass point vs background · normalized



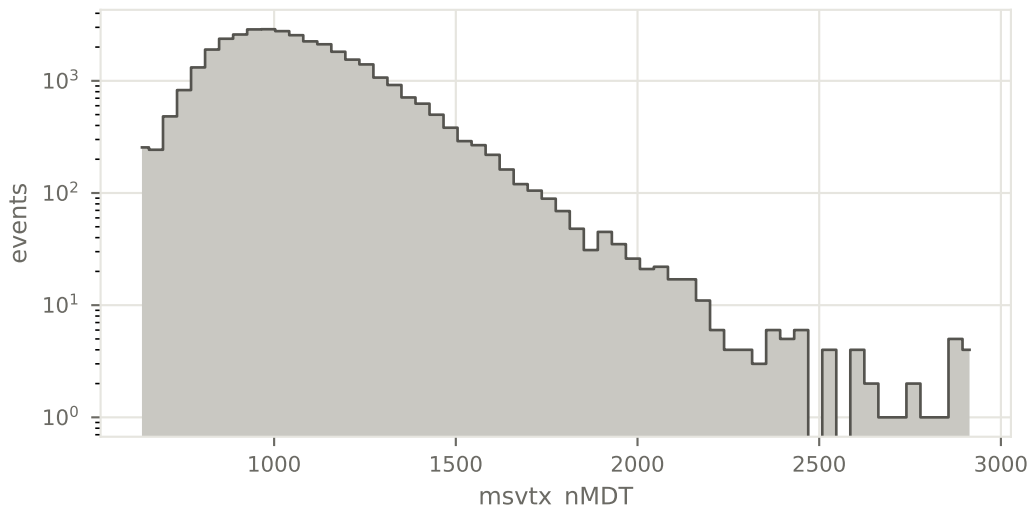
2 · combined MC vs background · raw



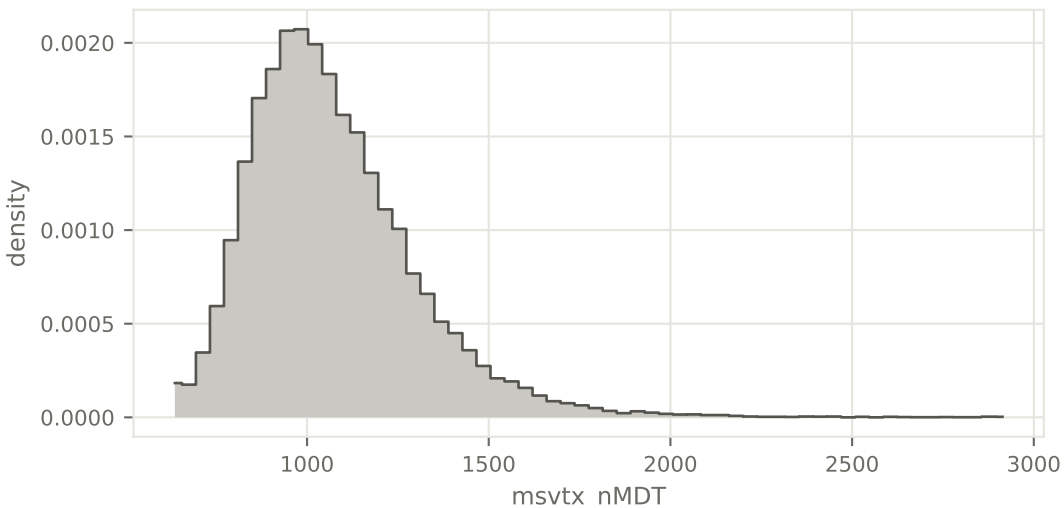
2 · combined MC vs background · normalized



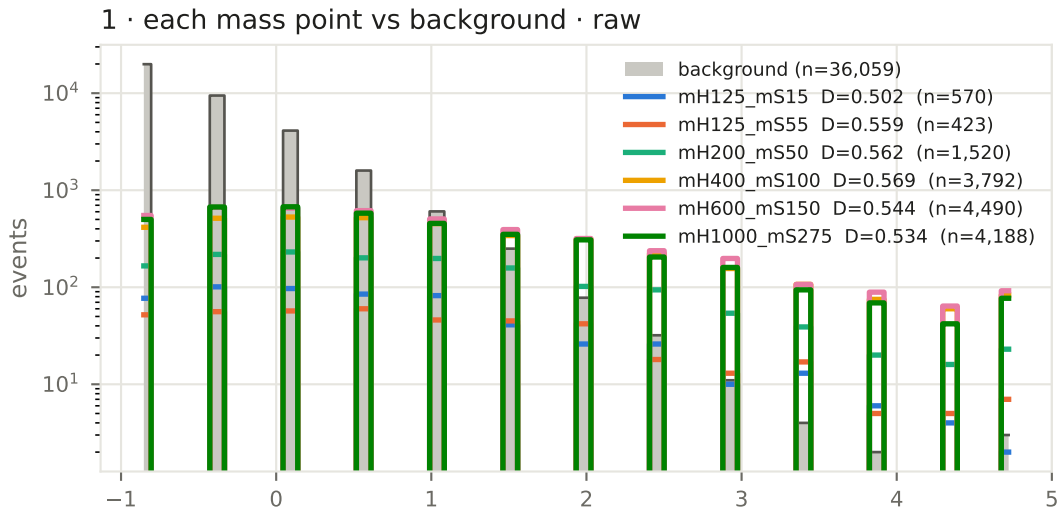
3 · background alone · raw · NaN 0.0% −999 0.0%

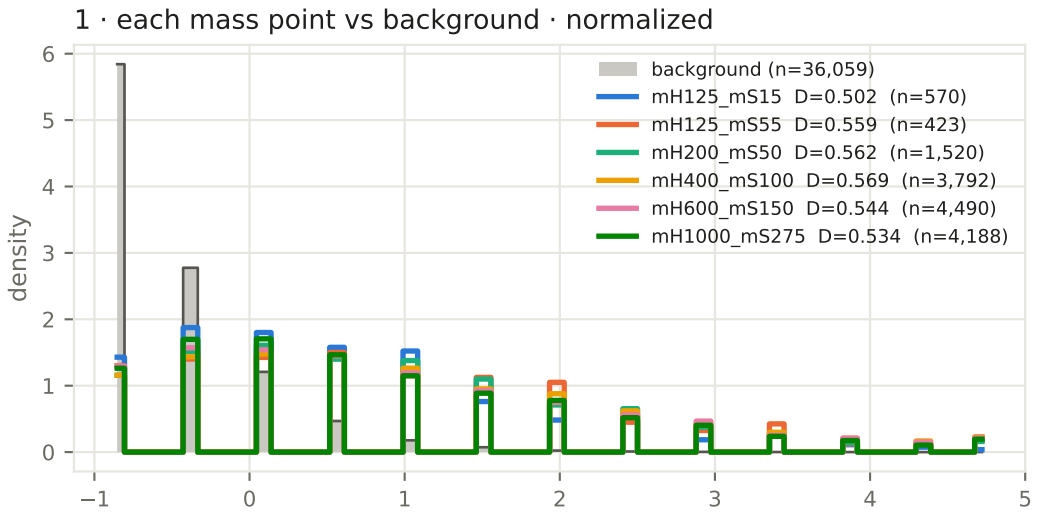


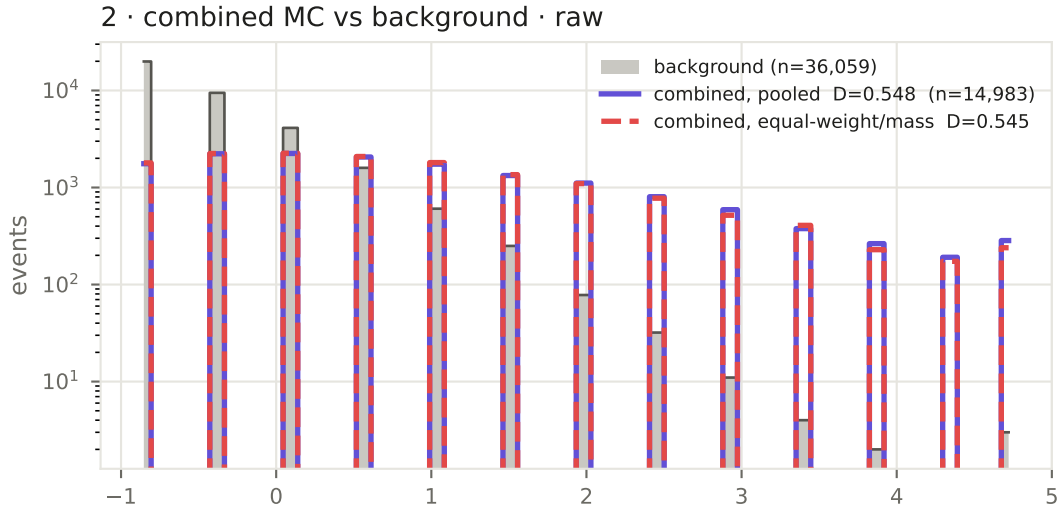
3 · background alone · normalized · NaN 0.0% −999 0.0%

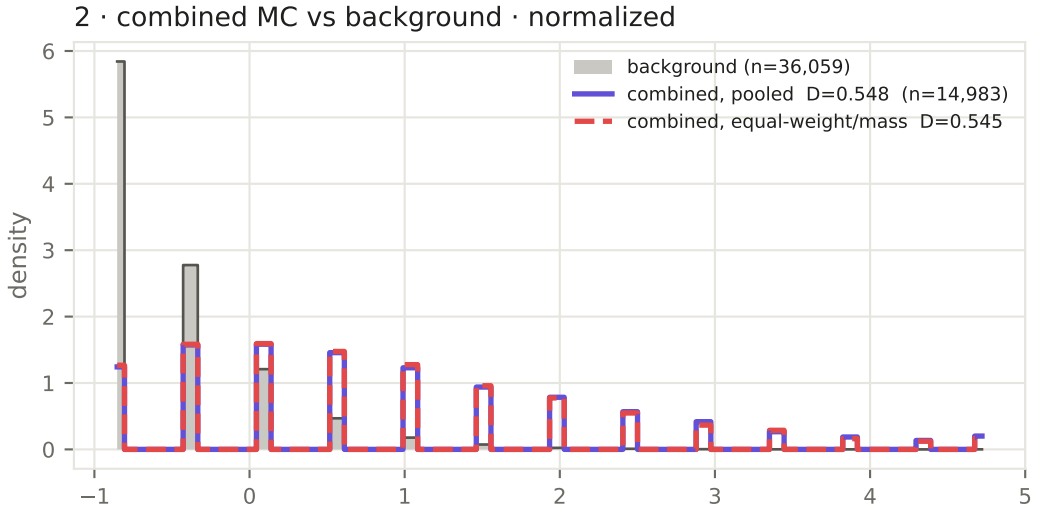


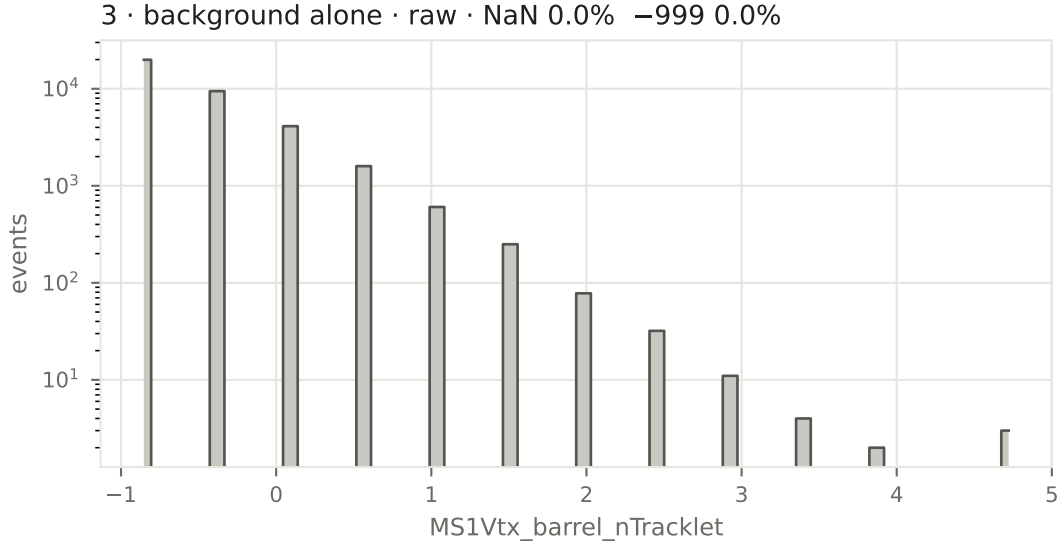
MS1Vtx_barrel_nTracklet · region: all · rank 7/191 by combined equal-weight D · round 1 · edge bins include under/overflow

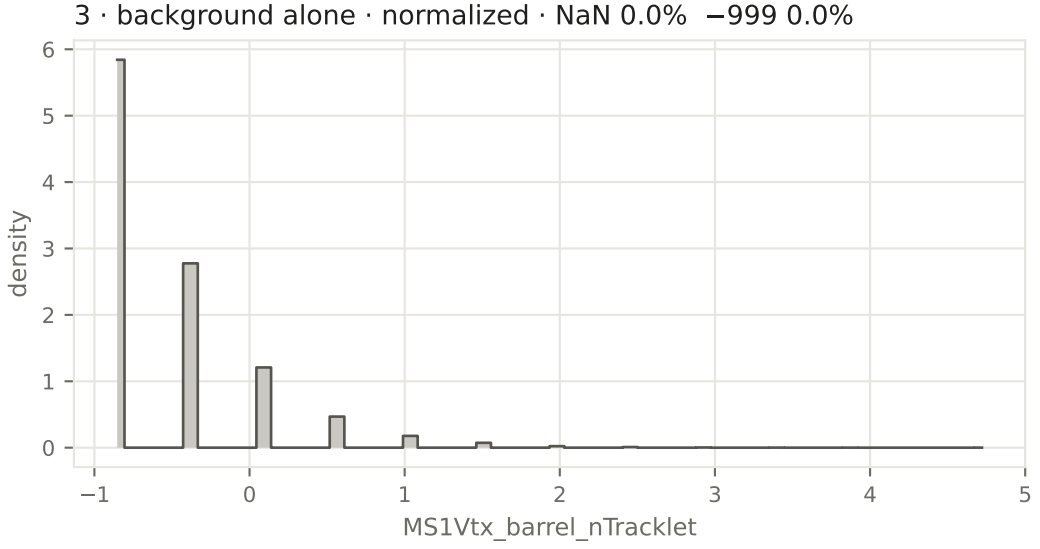






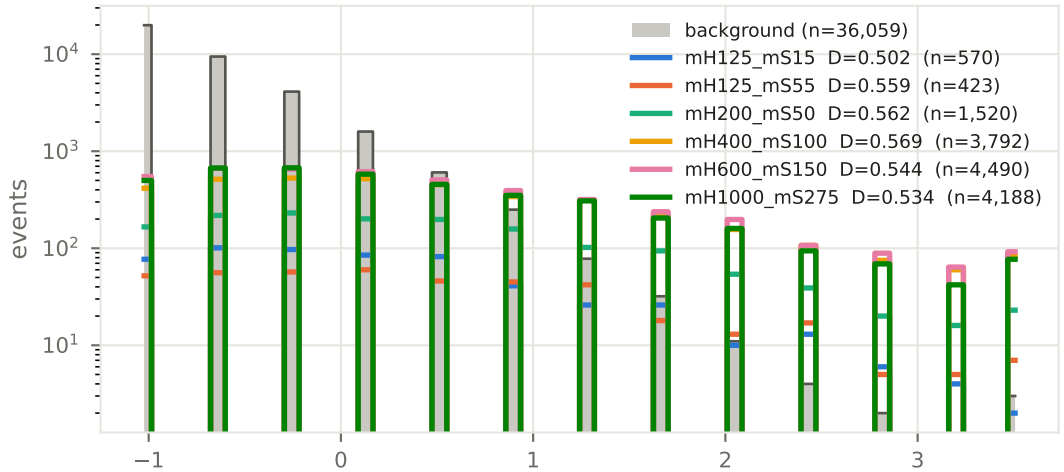




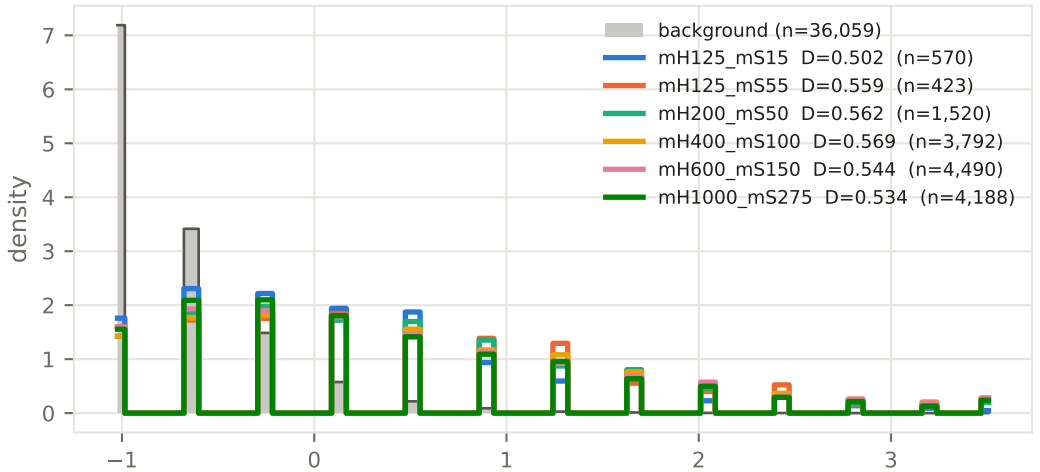


MS1Vtx_endcap_nTracklet · region: all · rank 8/191 by combined equal-weight D · round 1 · edge bins include under/overflow

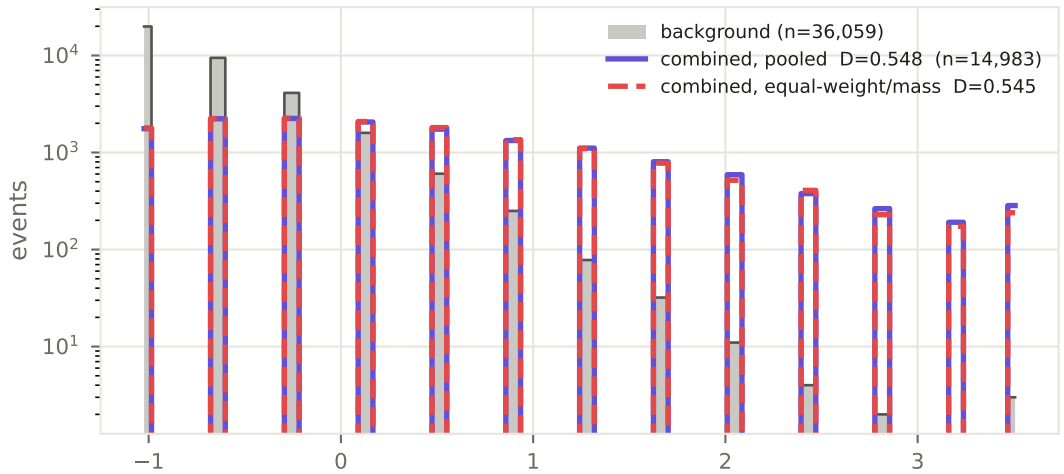
1 · each mass point vs background · raw



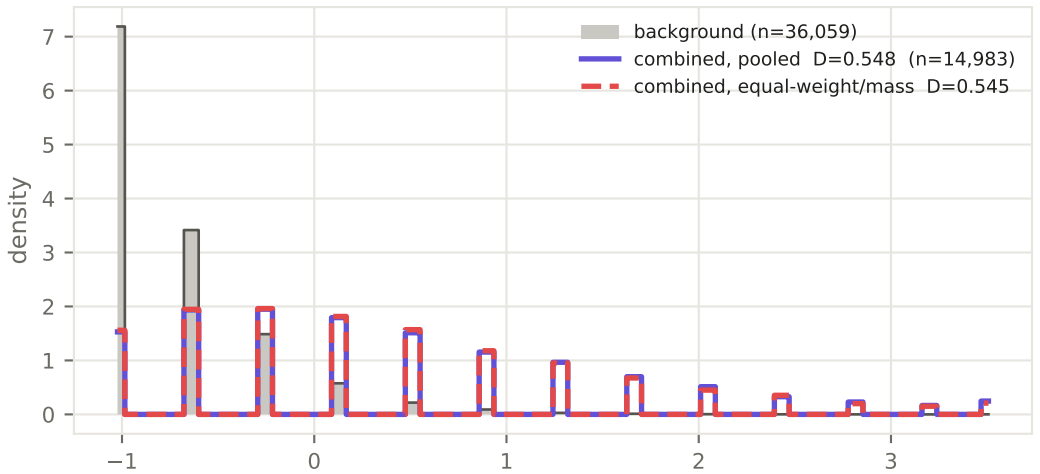
1 · each mass point vs background · normalized



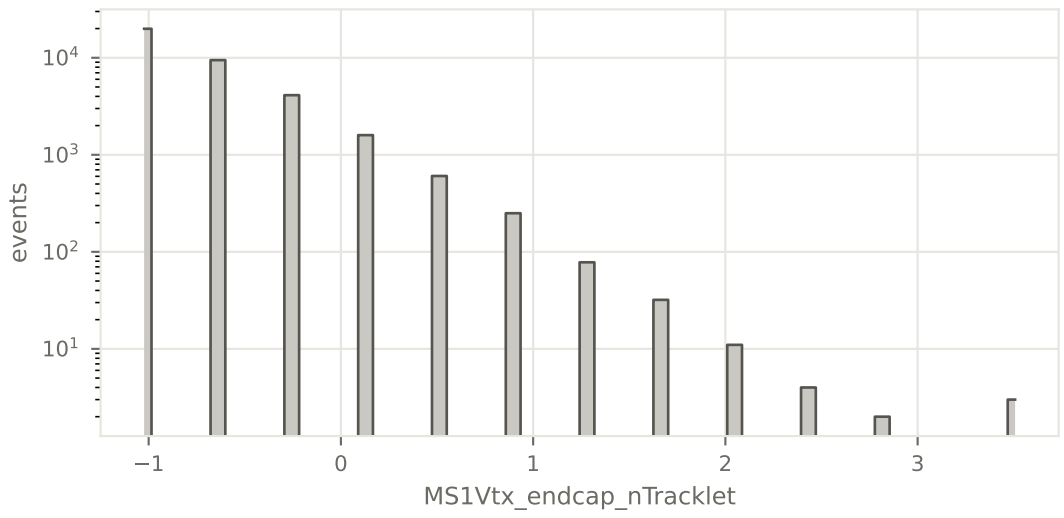
2 · combined MC vs background · raw



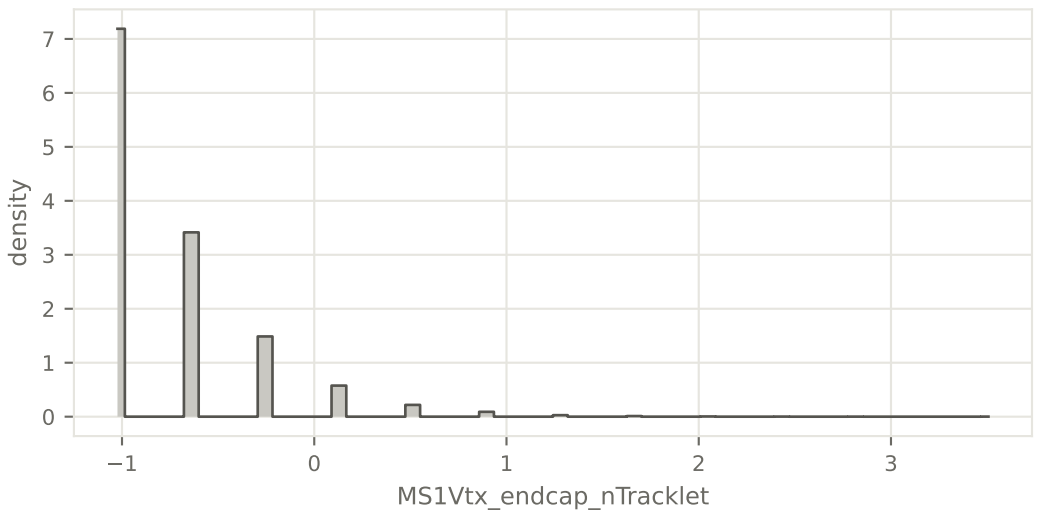
2 · combined MC vs background · normalized



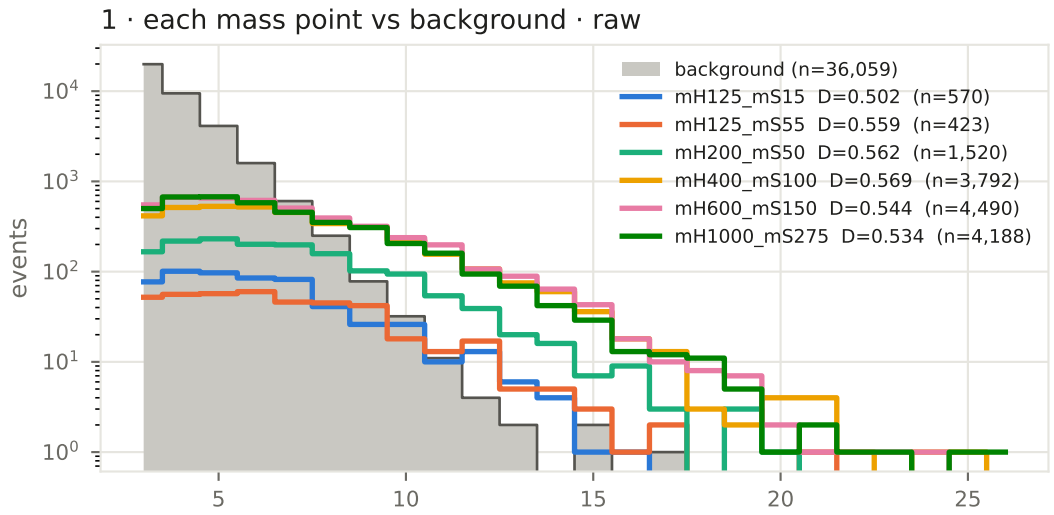
3 · background alone · raw · NaN 0.0% −999 0.0%

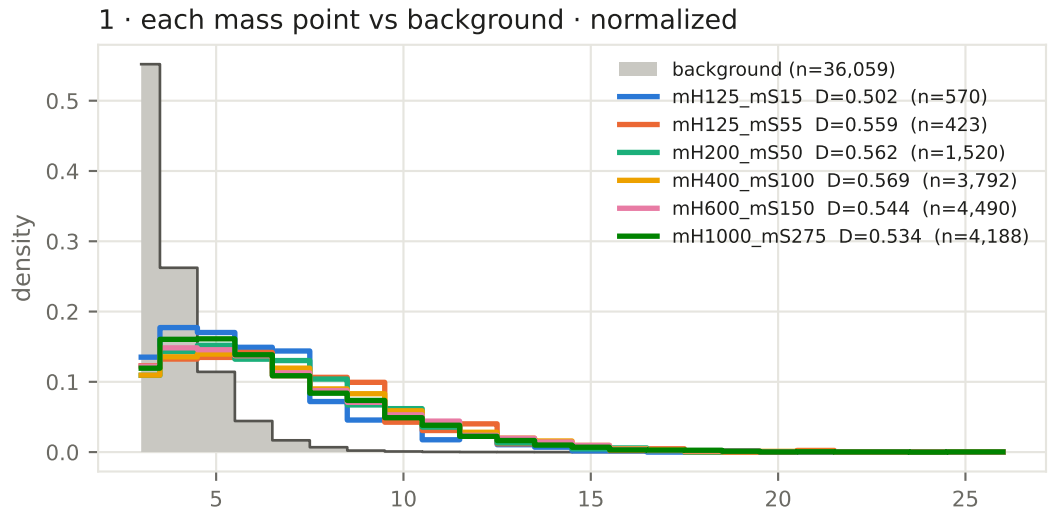


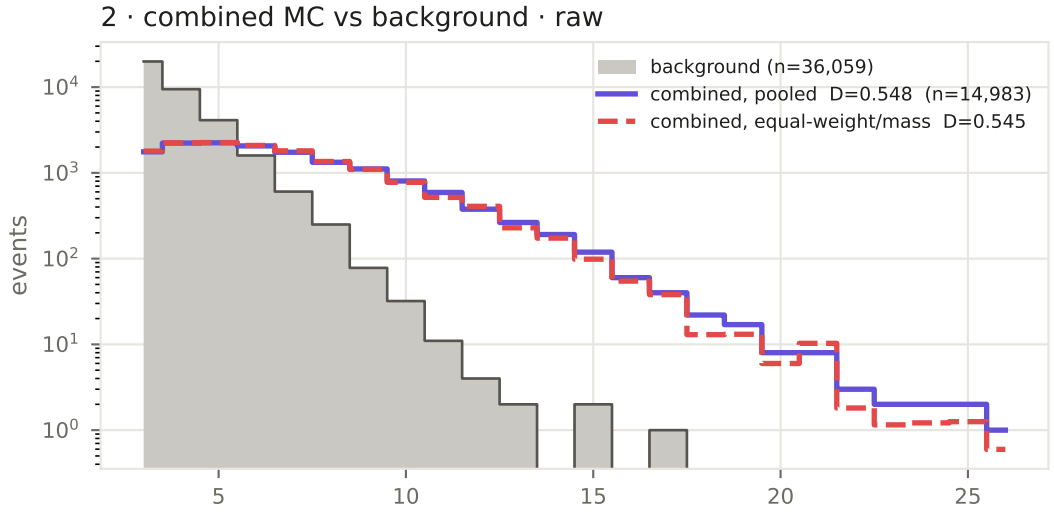
3 · background alone · normalized · NaN 0.0% −999 0.0%

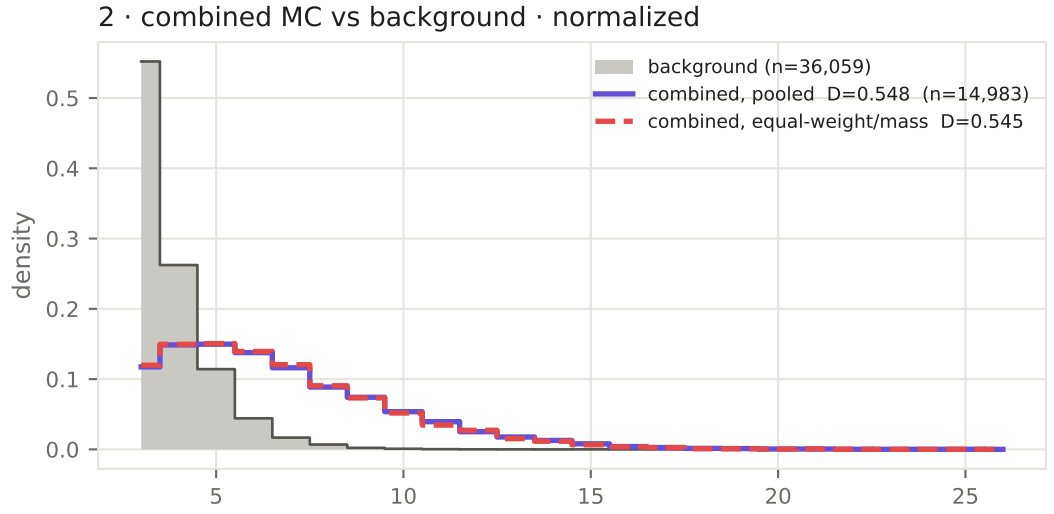


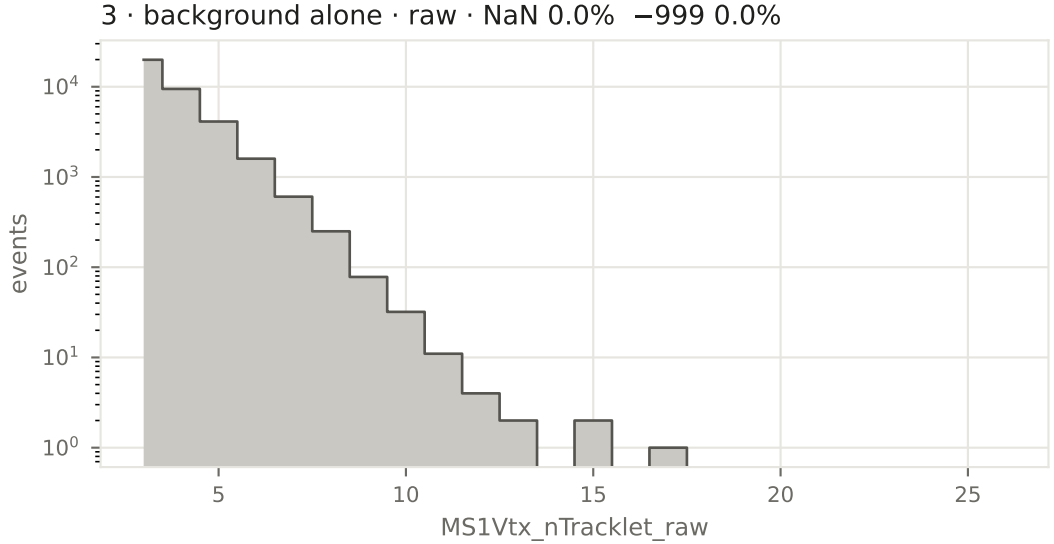
MS1Vtx_nTracklet_raw · region: all · rank 9/191 by combined equal-weight D · round 1 · edge bins include under/overflow

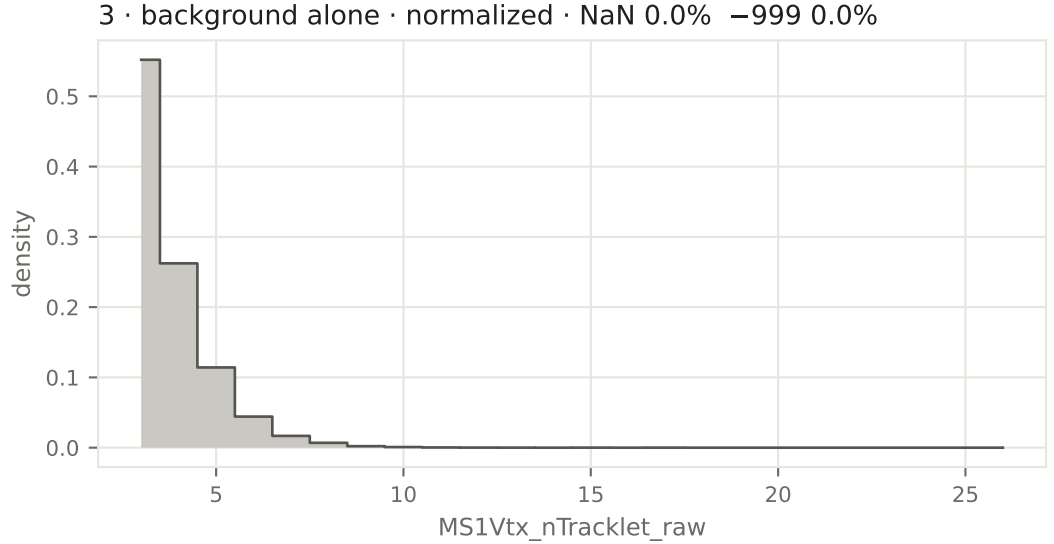


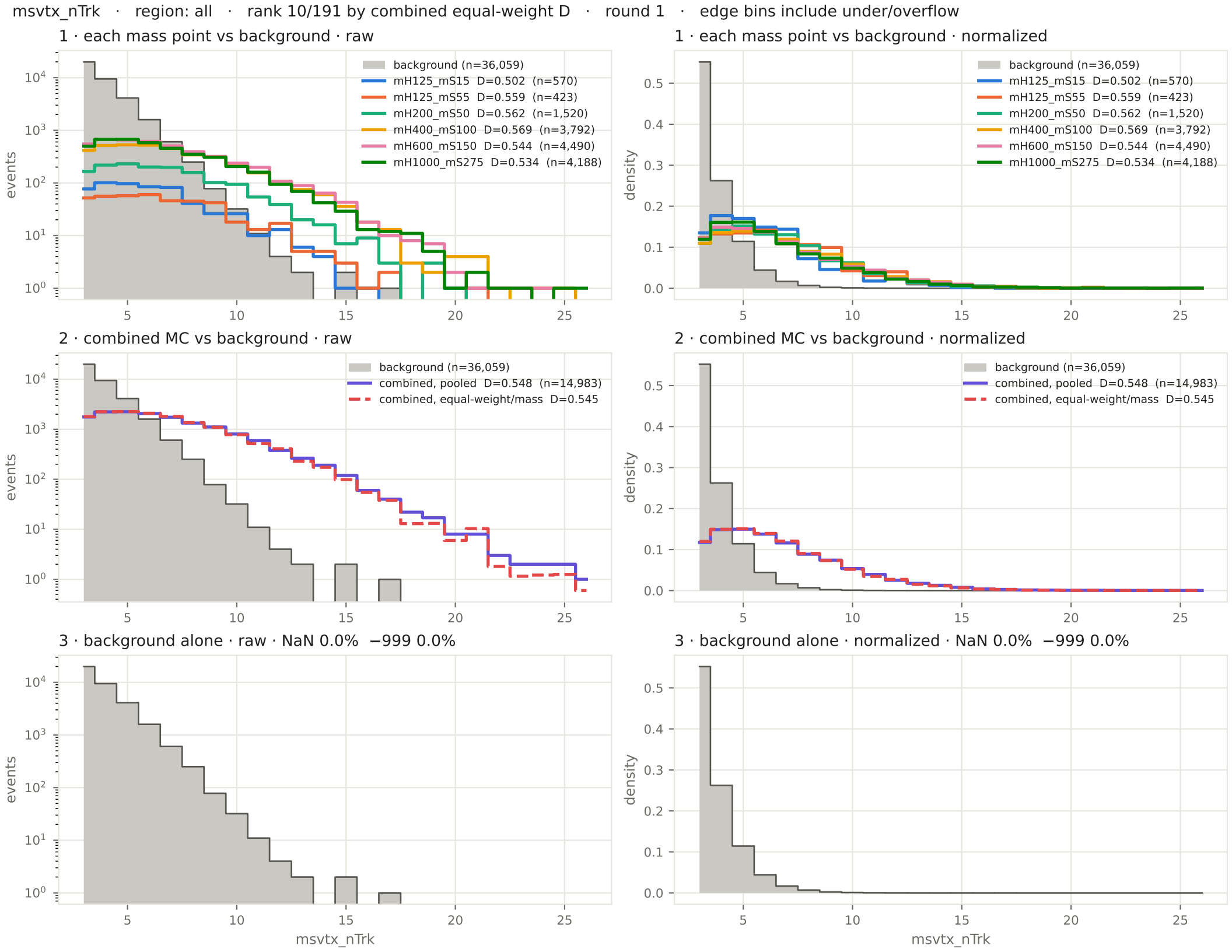


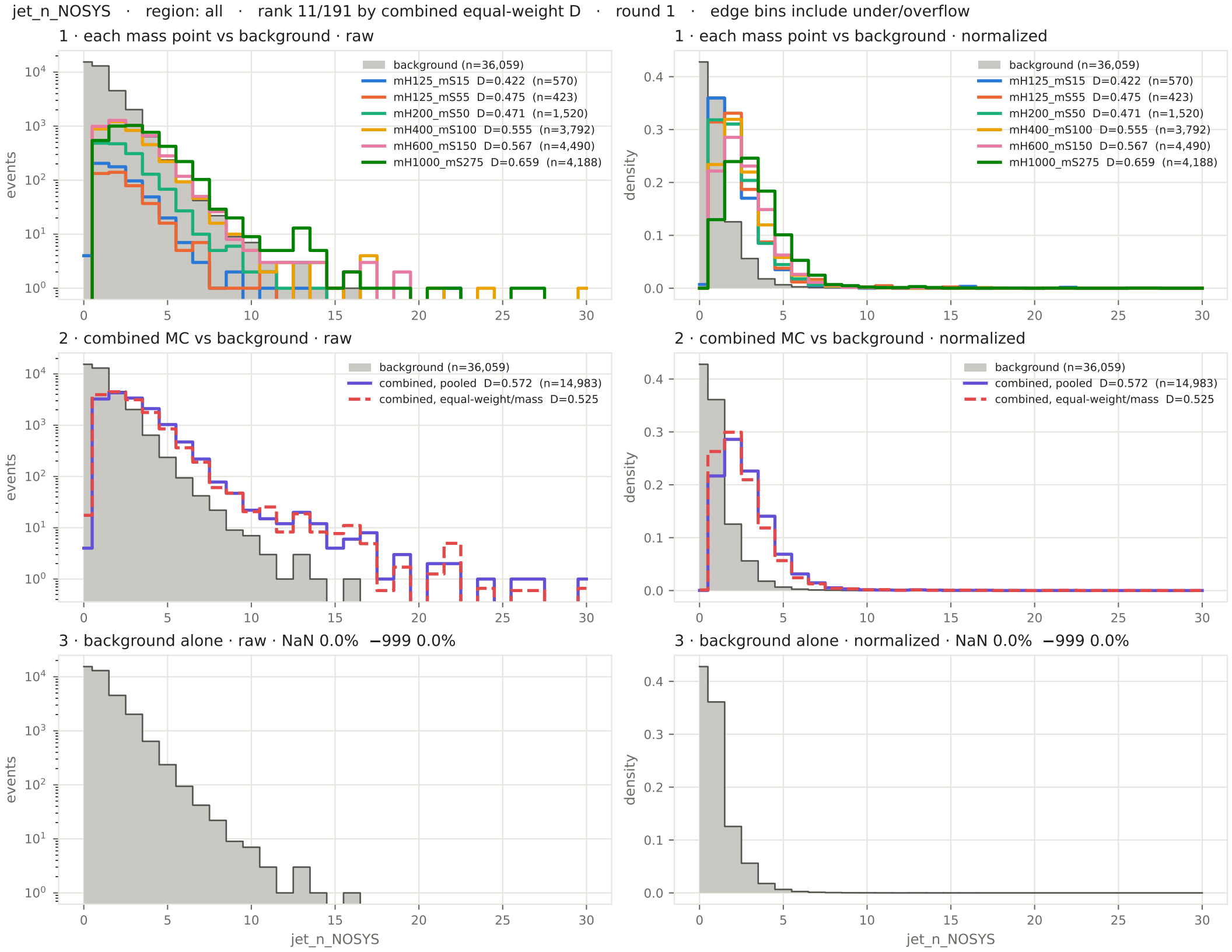




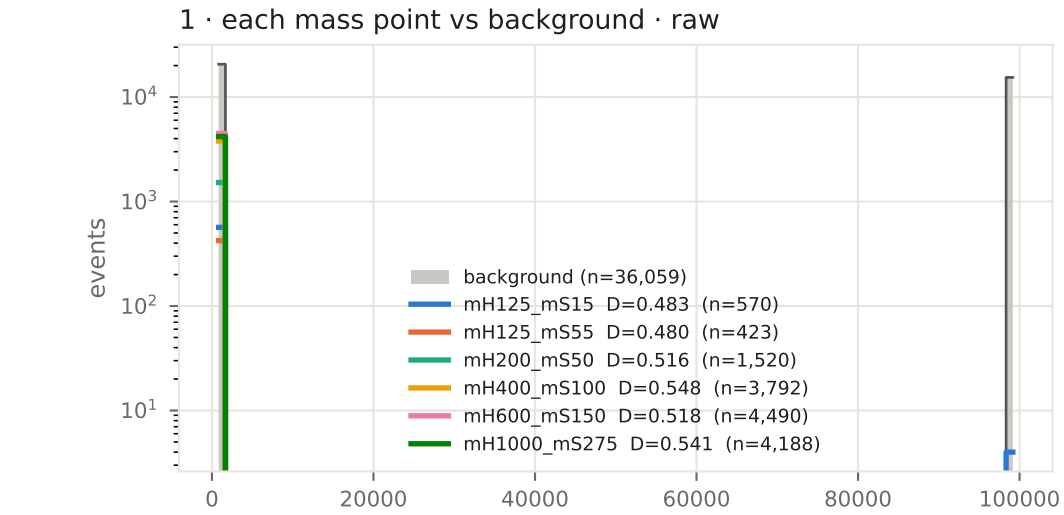


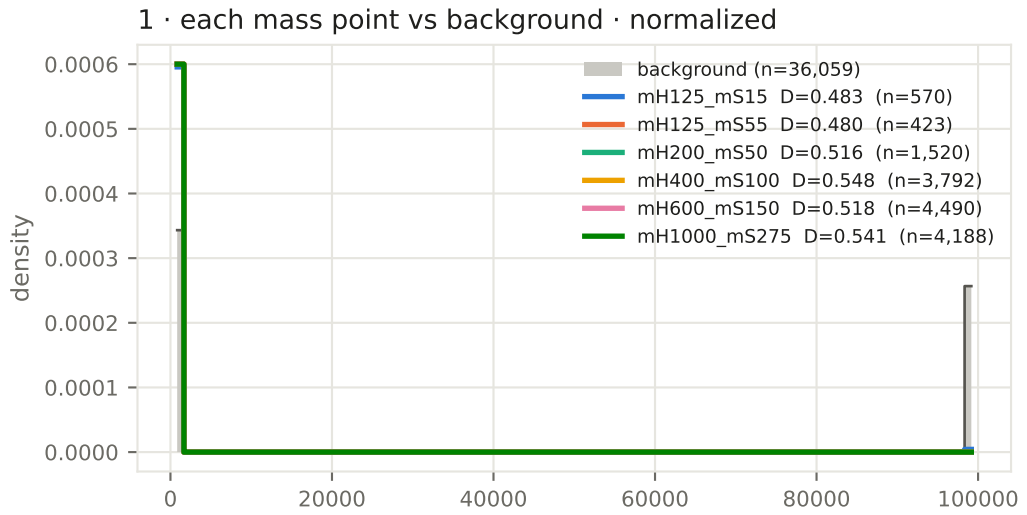


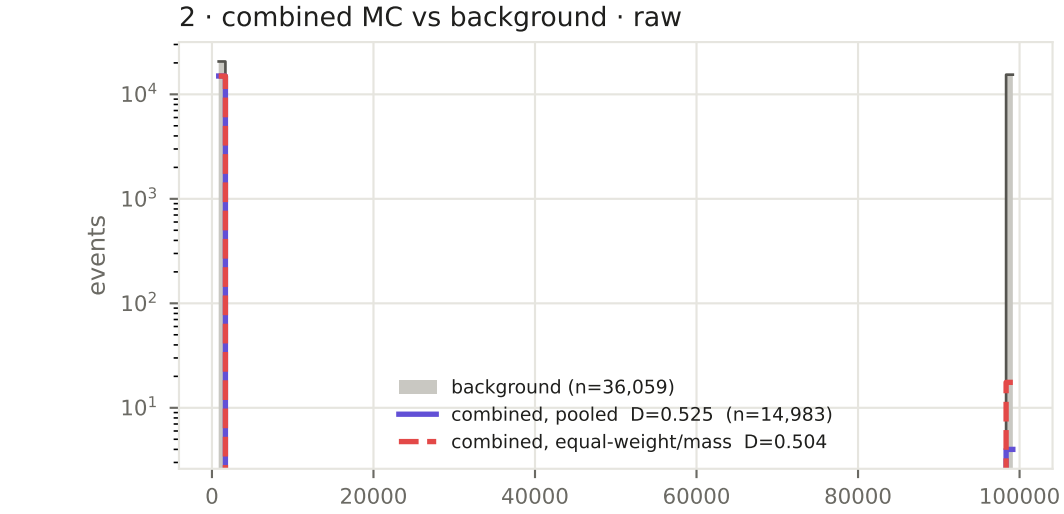

msvtx_nTrk · region: all · rank 10/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
background (n=36,059)
mH125_mS15 D=0.502 (n=570)
mH125_mS55 D=0.559 (n=423)
mH200_mS50 D=0.562 (n=1,520)
mH400_mS100 D=0.569 (n=3,792)
mH600_mS150 D=0.544 (n=4,490)
mH1000_mS275 D=0.534 (n=4,188)
events
1 · each mass point vs background · normalized
density
2 · combined MC vs background · raw
combined, pooled D=0.548 (n=14,983)
combined, equal-weight/mass D=0.545
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
msvtx_nTrk

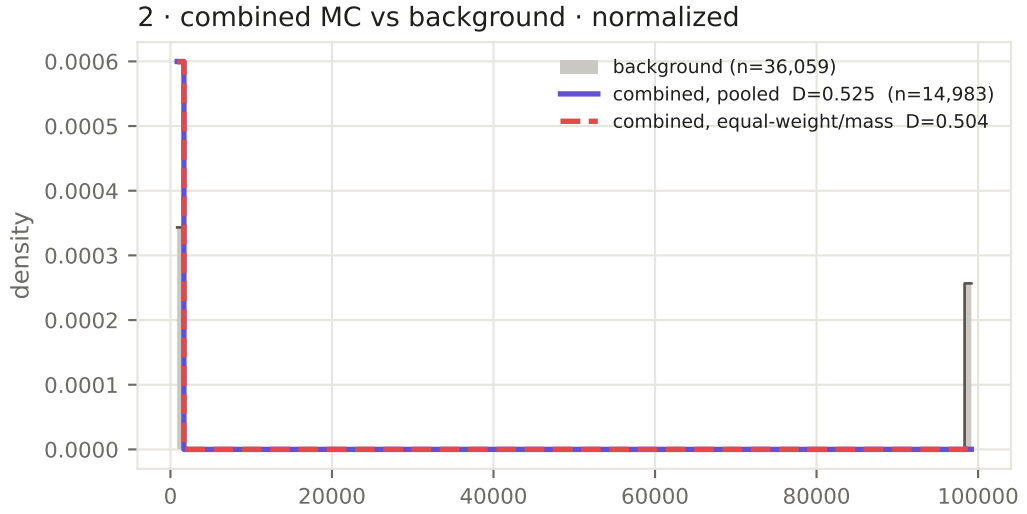

jet_n_NOSYS · region: all · rank 11/191 by combined equal-weight D · round 1 · edge bins include under/overflow
1 · each mass point vs background · raw
1 · each mass point vs background · normalized
2 · combined MC vs background · raw
2 · combined MC vs background · normalized
3 · background alone · raw · NaN 0.0% −999 0.0%
3 · background alone · normalized · NaN 0.0% −999 0.0%
background (n=36,059)
mH125_mS15 D=0.422 (n=570)
mH125_mS55 D=0.475 (n=423)
mH200_mS50 D=0.471 (n=1,520)
mH400_mS100 D=0.555 (n=3,792)
mH600_mS150 D=0.567 (n=4,490)
mH1000_mS275 D=0.659 (n=4,188)
combined, pooled D=0.572 (n=14,983)
combined, equal-weight/mass D=0.525
events
density
jet_n_NOSYS

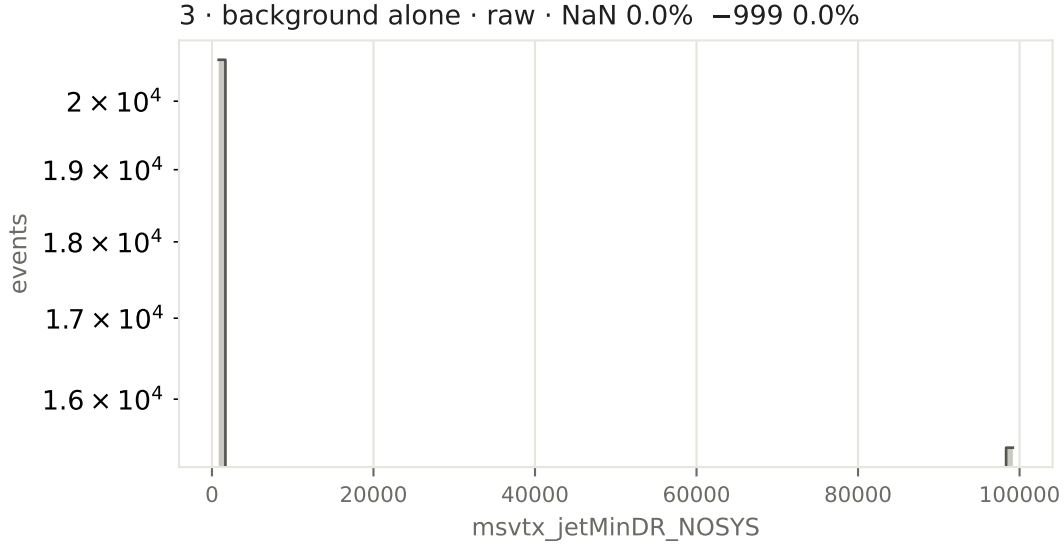
msvtx_jetMinDR_NOSYS · region: all · rank 12/191 by combined equal-weight D · round 1 · edge bins include under/overflow

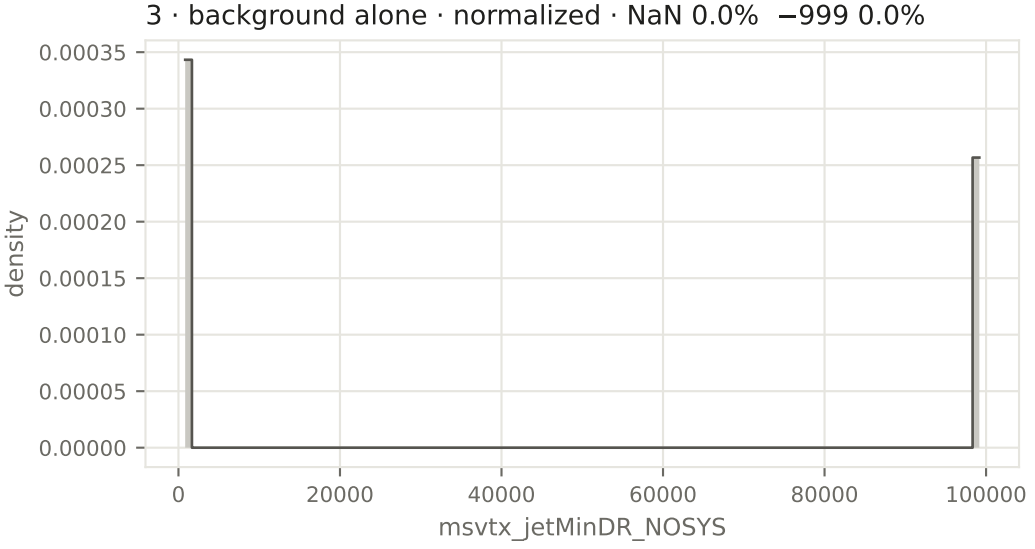




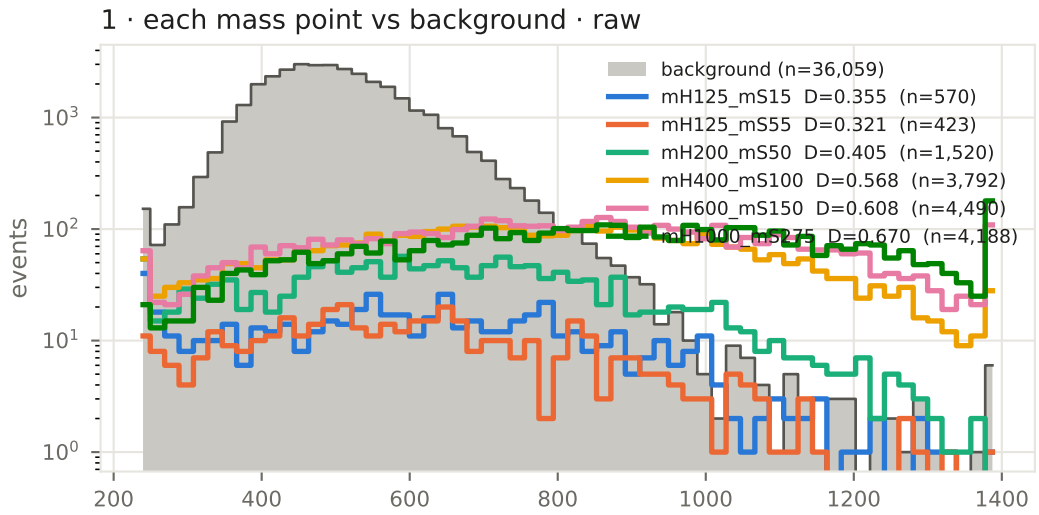


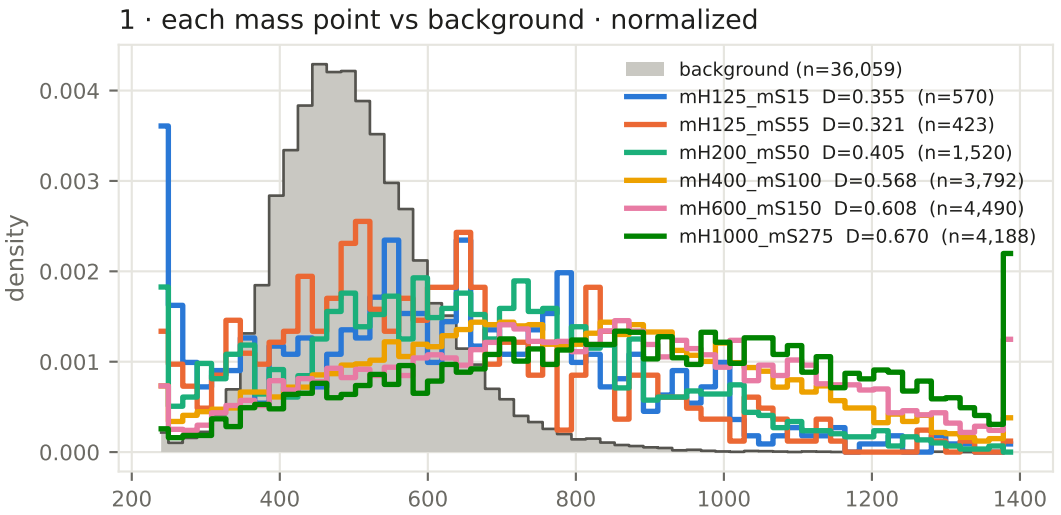


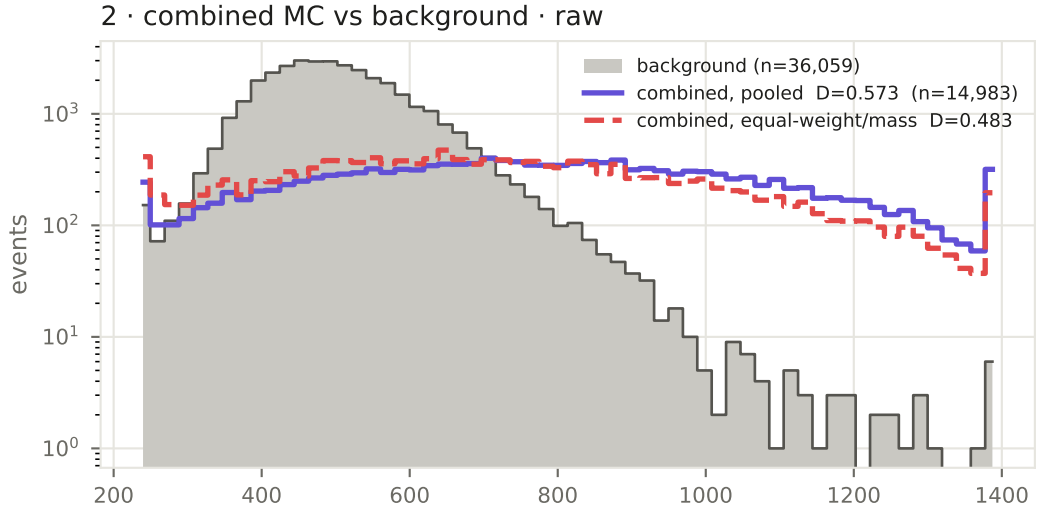


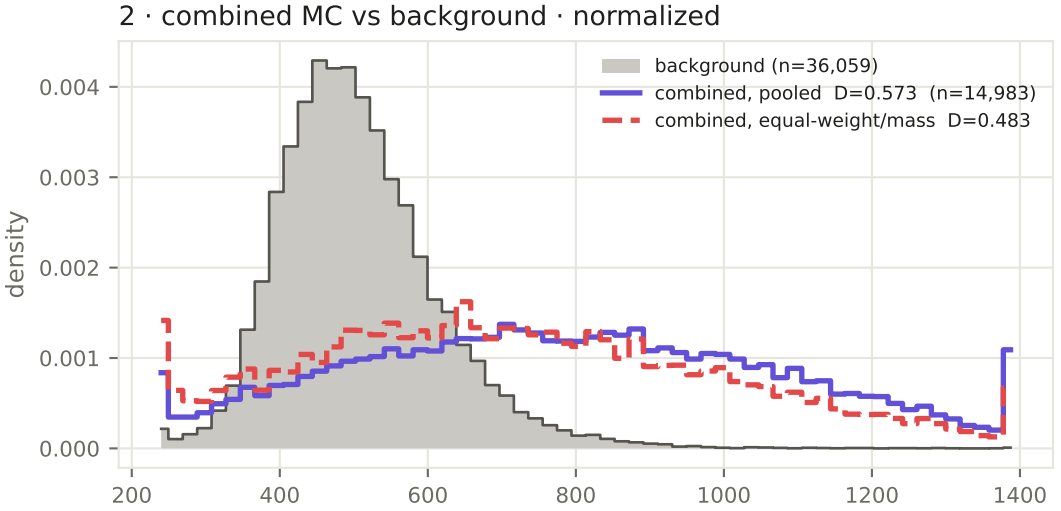


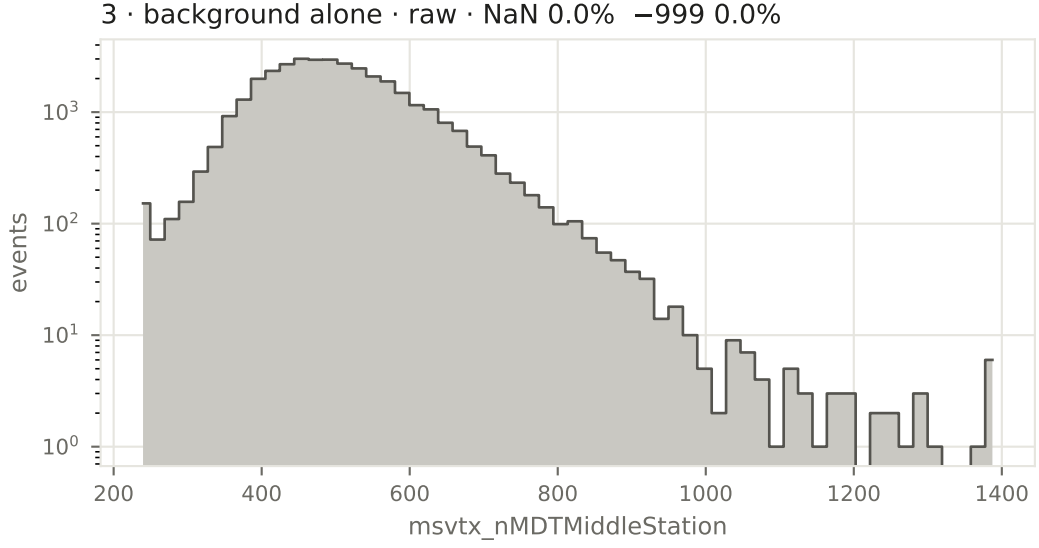
msvtx_nMDTMiddleStation · region: all · rank 13/191 by combined equal-weight D · round 1 · edge bins include under/overflow

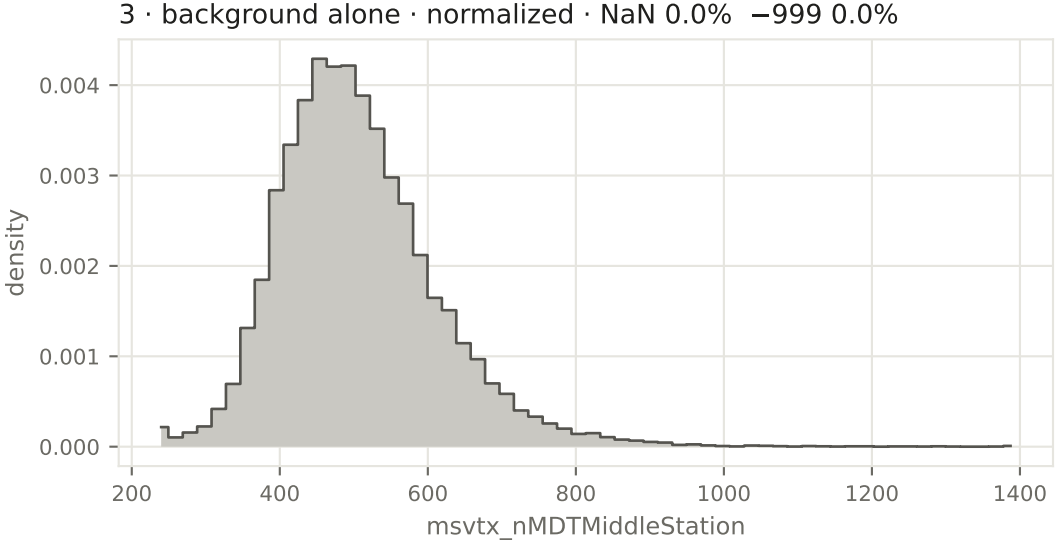




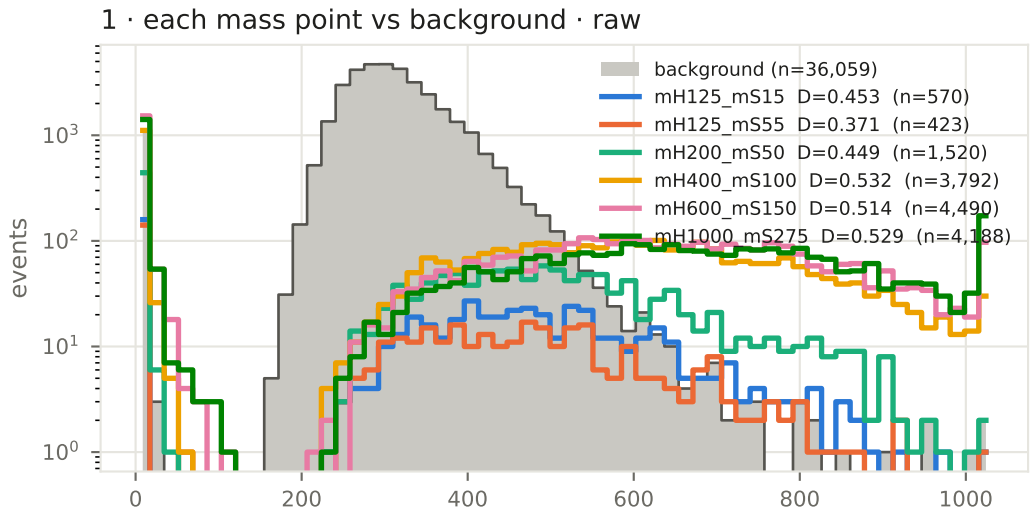


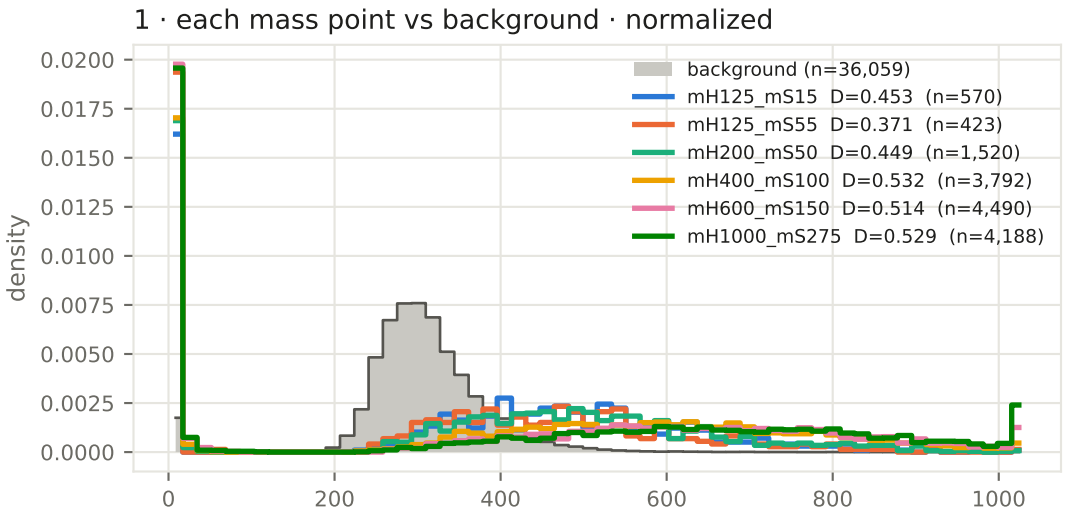


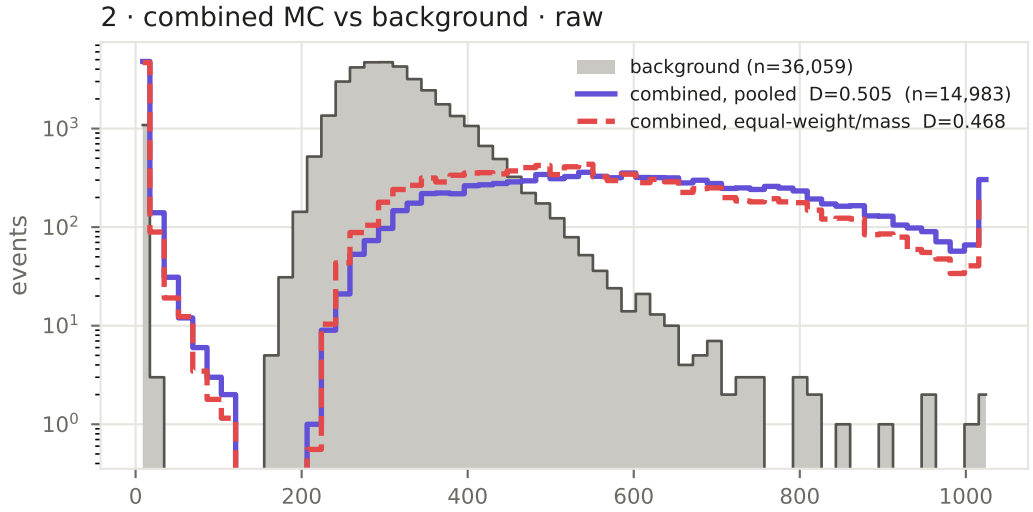


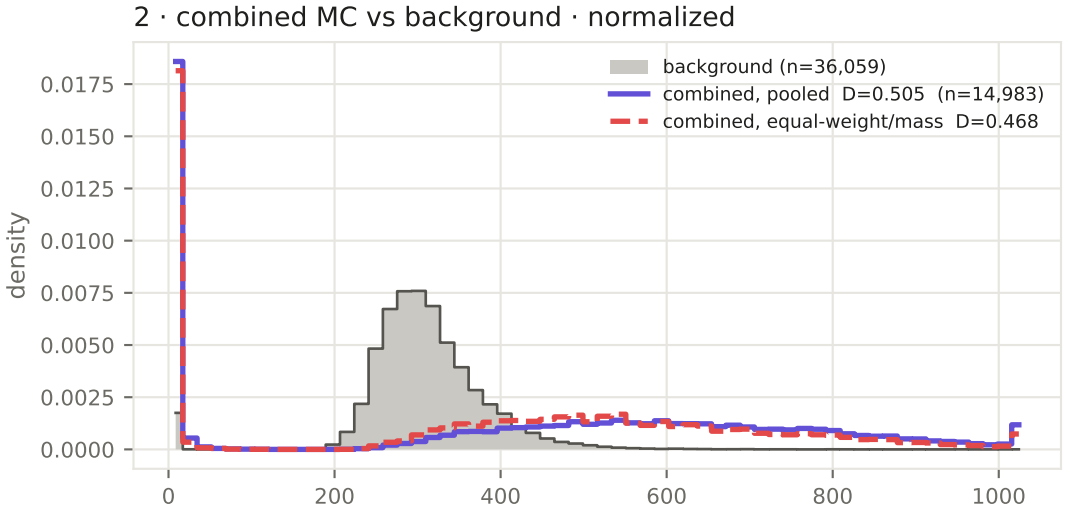


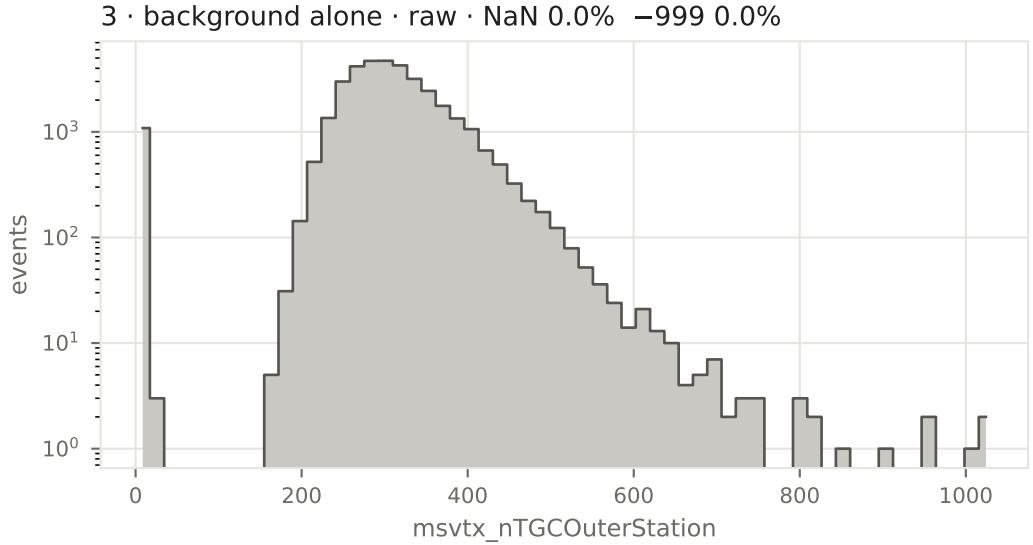
msvtx_nTGCOuterStation · region: all · rank 14/191 by combined equal-weight D · round 1 · edge bins include under/overflow

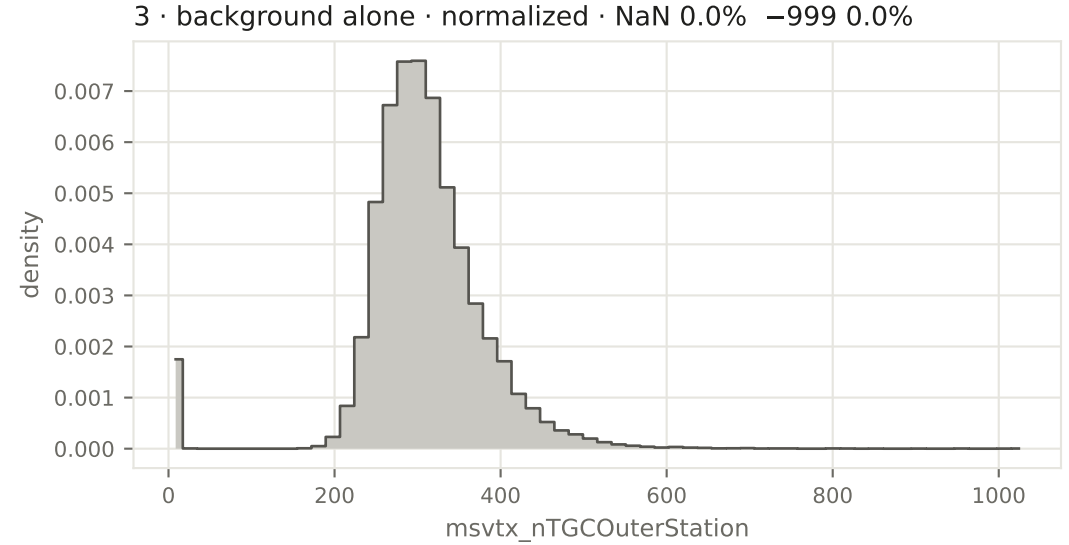






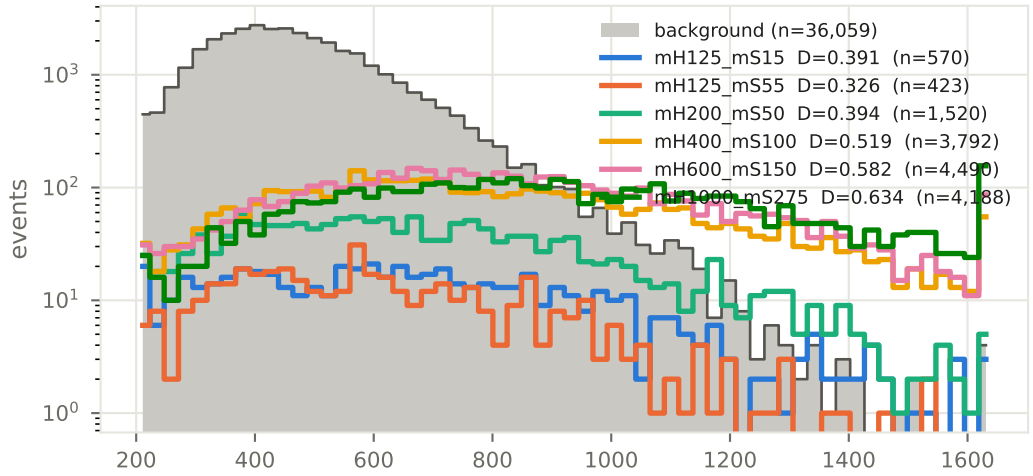




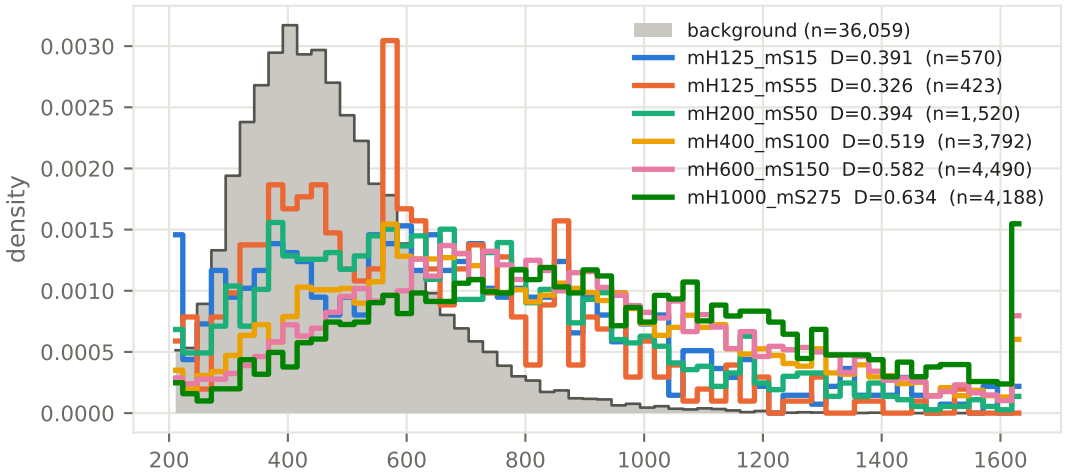


msvtx_nMDTOuterStation · region: all · rank 15/191 by combined equal-weight D · round 1 · edge bins include under/overflow

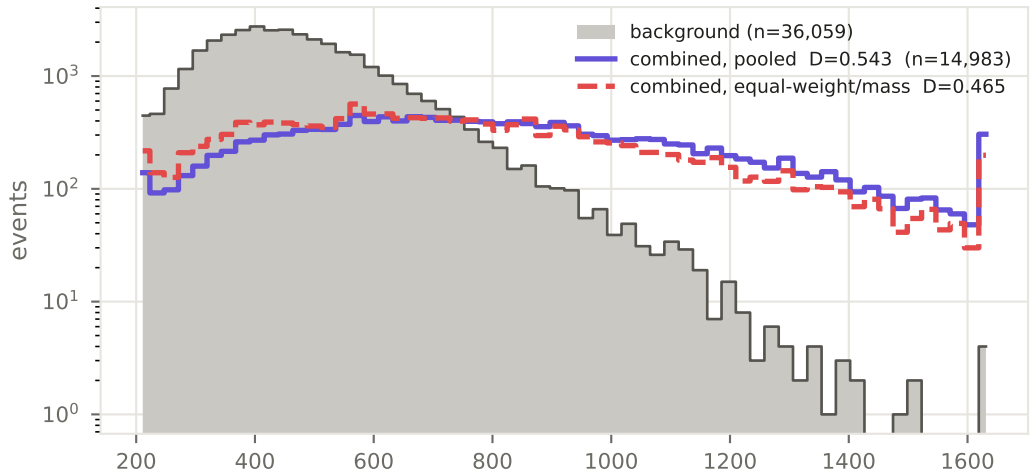
1 · each mass point vs background · raw



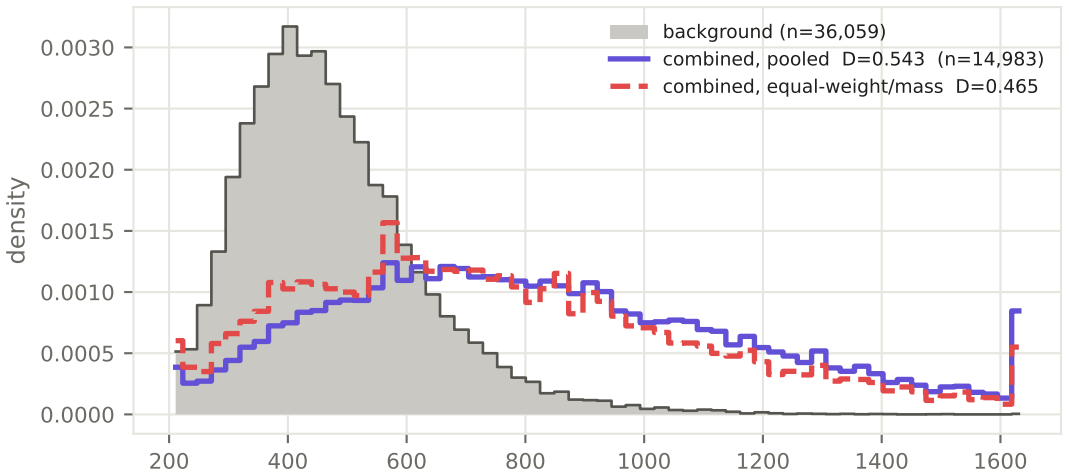
1 · each mass point vs background · normalized



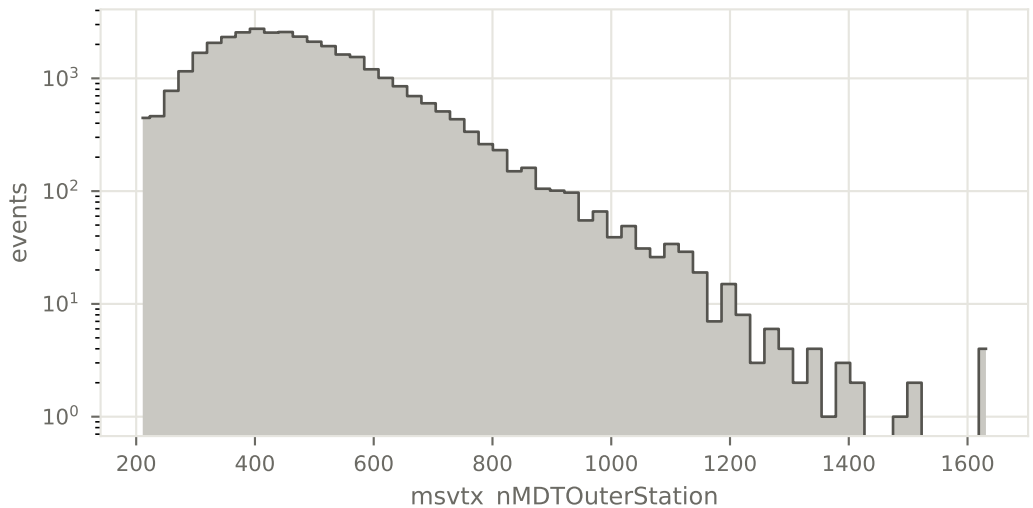
2 · combined MC vs background · raw



2 · combined MC vs background · normalized



3 · background alone · raw · NaN 0.0% −999 0.0%



3 · background alone · normalized · NaN 0.0% −999 0.0%

